

V.I. ZUBOV

METHODS OF A.M. LYAPUNOV AND THEIR APPLICATION



**METHODS OF A. M. LYAPUNOV
AND THEIR APPLICATION**

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METHODS OF A. M. LYAPUNOV AND THEIR APPLICATION

by

V. I. ZUBOV

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ПРЕДИСЛОВИЕ

Создателем теории устойчивости движения является выдающийся математик, академик А. М. Ляпунов (1857-1918 г.г.). Его труды явились отправной точкой для широкого круга исследований по проблеме устойчивости. Это в полной мере относится и к настоящей монографии, русское издание которой было приурочено к 100-летию со дня рождения А. М. Ляпунова и открывалось портретом этого замечательного ученого.

Годы, прошедшие со времени выхода в свет русского издания, ознаменовались дальнейшим развитием методов Ляпунова и расширением сферы их применения. Следует отметить плодотворное соединение идей из теории устойчивости движения и из теории автоматического управления, которое наметилось в последнее время, что привело к параллельному быстрому развитию в области теории оптимального управления и в области теории устойчивости движения. Одним из примеров такого рода может служить глубокая по своим результатам работа Кальмана.

Я надеюсь, что настоящее издание послужит дальнейшему развитию математических методов исследования нелинейных систем.

Ленинград, февраль 1964 г.

В. И. Зубов

FOREWORD*

The creator of the theory of the stability of motion was the distinguished mathematician Academician A. M. Lyapunov (1857–1918). His works were the point of departure for extensive investigations of the problem of stability. This, in full measure, pertains also to the present monograph, the Russian edition of which was timed to coincide with the hundredth anniversary of the birth of A. M. Lyapunov and opened with a portrait of this remarkable scholar.

The years which have passed since the time of the appearance of the Russian edition are marked with a further development of the methods of Lyapunov and the extension of the spheres of their application. We must note the fruitful combination of ideas from the theory of the stability of motion and from the theory of automatic regulation which has accumulated recently and which has led to a rapid development in the area of the theory of optimal regulation and in the area of the theory of stability of motion. One example of this sort is the work of Kalman which is characterized by profound results.

I hope that the present edition will serve to further develop the mathematical methods of investigating non-linear systems.

Leningrad,
February 1964

V. I. ZUBOV

* Written expressly for this English edition.

FOREWORD

This monograph by V. I. Zubov is directly allied in its contents to the celebrated work by A. M. Lyapunov, "General Problem of Stability of Motion." Let us start with a brief mention of the principal results of this work. A. M. Lyapunov reduces the problem of stability of motion to an investigation of the stability of the trivial solution $x_1 = x_2 = \dots = x_n = 0$ of the system of ordinary differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t), \quad (s = 1, \dots, n), \quad (1)$$

where the right members are power series in the variables $x_1, \dots, x_n$ without the free term, converging in a neighborhood of

$$x_1 = x_2 = \dots = x_n = 0.$$

If the right members, i.e., the coefficients of the above-mentioned series, do not contain t , then A. M. Lyapunov speaks of a steady-state motion (autonomous system). In a general investigation of the stability of the trivial solution of (1), he employed two methods developed by him: expansion of the solutions of the system (1) in terms of a special type, and the use of the functions $V(x_1, \dots, x_n, t)$ which together with their derivatives with respect to t , calculated by virtue of (1):

$$W(x_1, \dots, x_n, t) = \frac{dV}{dt} = \frac{\partial V}{\partial t} + \sum_{s=1}^n \frac{\partial V}{\partial x_s} f_s,$$

have certain properties, as functions of $(x_1, \dots, x_n, t)$. Such functions, when constructed, give sufficient conditions for stability or instability. In the construction of functions of this kind, Lyapunov used the general theorem that he proved concerning the solution

of systems of partial differential equations that do not satisfy the conditions of the well known theorem of S. V. Kovalevskaya. The functions V , about which we spoke before, are called *Lyapunov functions* and the theorem just mentioned will be called the *auxiliary theorem*.

In his works Lyapunov has established the conditions under which the linear terms of Eq. (1) alone solve the problem of stability, and has considered many important particular cases, when this problem can be solved only by an examination of higher-order terms (so-called doubtful cases). In one of the doubtful cases of an autonomous system (1), a method is given for constructing periodic solutions. Upon further investigations systems of differential equations were dispensed with, and only the points P were considered in an n -dimensional space that moved with varying t in accordance with definite laws — so-called *dynamical systems*. Such systems were considered further not in n -dimensional Euclidean space $(x_1, \dots, x_n)$ but in an arbitrary metric space.

In the first chapter of this monograph there are considered dynamical systems in metric space and the problem of stability (in different meanings) of invariant sets, i.e. sets consisting of trajectories. In the application of Lyapunov's second method, the functions V are replaced by functionals. An investigation is made of the behavior of the trajectory in a neighborhood of an invariant set in the presence of stability and under certain conditions the entire region of asymptotic stability is characterized with the aid of a corresponding functional. This chapter is purely theoretical in nature and contains no constructions of the corresponding functionals.

In the second chapter, the results of the first chapter are applied to systems of differential equations of type (1). In particular, the case is considered when the right members are homogeneous functions of $(x_1, \dots, x_n)$. Necessary and sufficient conditions for asymptotic stability are found for this case.

Next, for the case of analytic right members of Equations (1) and of autonomous systems, the doubtful case is considered, when the characteristic equation of first approximation has k zero and k pairs of identical pure imaginary roots with simple elementary divisors, and the remaining roots have a negative real part. The condition for the existence of k -holomorphic integrals is given, which leads to a sufficiency condition for the stability of the zero solution. Also

given is a sufficiency condition for asymptotic stability. In the absence of zero roots, a family of bounded solutions is constructed (analogous to the family of periodic solutions of A. M. Lyapunov, which were mentioned above). The author considers also the case when the right members of Eqs. (1) contain t .

The third chapter is connected essentially with a generalization of Lyapunov's auxiliary theorem concerning systems of partial differential equations and the application of his first method. The result mentioned leads, in a natural manner, to a generalization of the well known results of Briot and Bouquet and also those of Poincaré concerning the construction of solutions of differential equations in a neighborhood of singular points. Autonomous systems (1) are considered, the right members of which are holomorphic and contain no linear terms. Under certain additional assumptions, integral curves are constructed, which tend to the origin as $t \rightarrow +\infty$. If the origin is a point of asymptotic stability, all the integral curves are obtained in this manner. In certain cases, necessary and sufficient conditions for asymptotic stability are given.

In the fourth chapter the concept of a dynamical system in metric space is generalized to obtain a scheme that serves to investigate the stability, in various meanings, for problems connected with partial differential equations. As in the first chapter, an analogue of the Lyapunov second method is developed. Moreover, estimates are given for the distance from a moving point to the investigated invariant set. In applications to systems of ordinary equations this yields new results.

In the fifth chapter, the results of the preceding chapter are applied to partial differential equations.

In his foreword to "General Problem of Stability of Motion" Lyapunov wrote: "My only purpose in this work is to explain what I succeeded to do towards solving the problem I formulated and what may serve as a starting point for further research of similar character."

These words were justified to a great extent. Hundreds of papers directly connected with Lyapunov's research have been published. In the present monograph by V. I. Zubov these investigations receive a substantial generalization and development.

Academician V. I. SMIRNOV

FROM THE AUTHOR

The purpose of the present edition is to acquaint the reader with new results obtained in the theory of stability of motion, and also to summarize certain researches by the author in this field of mathematics. It is known that the problem of stability reduces not only to an investigation of systems of ordinary differential equations but also to an investigation of systems of partial differential equations. The theory is therefore developed in this book in such a manner as to make it applicable to the solution of stability problems in the case of systems of ordinary differential equations as well as in the case of systems of partial differential equations.

For the reader's benefit, we shall now list briefly the contents of the present monograph.

This book consists of five chapters.

In Sections 1-5 of Chapter I we give the principal information connected with the concept of metric space, and also explain the meaning of the terms which will be used below. Sections 6 and 7 are preparatory and contain examples of dynamical systems in various spaces. In Section 8 we define the concept of dynamical systems in metric space, and also give the principal theorems from the book [5] of Nemytsky and Stepanov. In Sections 9-10 we give the principal definitions, connected with the concept of stability in the sense of Lyapunov of invariant sets of a dynamical system, and also investigate the properties of certain stable invariant sets. In Section 11 we solve the problem of a qualitative construction of a neighborhood of a stable (asymptotically stable) invariant set. In particular, it is established that for stability in the sense of Lyapunov of an invariant set M of a dynamical system $f(p, t)$ it is necessary, and in the case of the presence of a sufficiently small compact

neighborhood of the set M it is also sufficient, that there exist no motions $f(p, t)$, $p \in M$, having α -limit points in M . The results obtained here are new even to the theory of ordinary differential equations. In Sections 12–13 we give criteria for stability and instability of invariant sets with the aid of certain functionals. These functionals are the analogue of the Lyapunov function and therefore the method developed here can be considered as a certain extension of Lyapunov's second method. All the results of these sections are local in character. We cite, for example, one of these. In order for an invariant set M to be uniformly asymptotically stable, it is necessary and sufficient that in a certain neighborhood $S(M, r)$ of M there exists a functional V having the following properties:

1. Given a number $c_1 > 0$, it is possible to find $c_2 > 0$ such that $V(P) > c_2$ for $\rho(p, M) > c_1$.

2. $V(p) \rightarrow 0$ as $\rho(p, M) \rightarrow 0$.

3. The function $V(f(p, t))$ does not increase for $f(p, t) \in S(M, r)$ and $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$ uniformly relative to $p \in S(M, \delta_1)$, where δ_1 is a certain number such that $0 < \delta_1 < r$. In Section 14 we describe one of the central theorems that has a non-local character. It is shown there that in order for an invariant open set A , containing a sufficiently small neighborhood of the set M , to be the region of asymptotic stability of the uniformly asymptotically stable and uniformly attracting set M , it is necessary and sufficient that there exist two functionals $V(p)$ and $\Phi(p)$ such that:

1. $V(p)$ is defined and continuous in A , $\Phi(p)$ is defined and continuous in R , $\Phi(p) = 0$, $p \in M$, and $-1 < V(p) < 0$ for $p \in A$; $\Phi(p) > 0$ for $\rho(p, M) \neq 0$.

2. For $\gamma_2 > 0$ it is possible to find γ_1 and α_1 such that $V(p) < -\gamma_1$, $\Phi(p) > \alpha_1$ for $\rho(p, M) > \gamma_2$.

3. V and $\Phi \rightarrow 0$ as $\rho(p, M) \rightarrow 0$.

4. $dV/dt = \Phi(1 + V)$.

5. $V(p) \rightarrow -1$ as $\rho(p, q) \rightarrow 0$, $p \in A$, $q \in \bar{A} \setminus A$, and $q \in M$.

Here, as above, p and q are elements of the space R , and $\rho(p, M)$ is the metric distance from the point p to the set M . Section 15

contains a method that makes it possible to estimate the distance from the motion to the investigated invariant set. The theorems obtained in this section can be considered as supplements to Sections 12–14. Sections 1–15 cover the contents of the first chapter, devoted to an investigation of invariant sets of dynamical systems.

In the second chapter we give a developed application of the ideas and methods of the first chapter to the theory of ordinary differential equations. In Section 1 of Chapter 2 we develop the theorem of Section 14 for stationary systems of differential equations, and it is shown thereby that the Lyapunov function V can be selected differentiable to the same order as the right members of the system. In the same section we give a representation of this function as a curvilinear integral and solve the problem of the analytic structure of the right members of the system, which right members have a region of asymptotic stability that is prescribed beforehand. In Section 2 of Chapter II we consider the case of holomorphic right members. The function V , the existence of which is established in Section 1 of this chapter, is represented in this case in the form of convergent series, the analytic continuation of which makes it possible to obtain the function in the entire region of asymptotic stability. The method of construction of such series can be used for an approximate solution of certain non-local problems together with the construction of bounded solutions in the form of series, that converge either for $t > 0$ or for $t \in (-\infty, +\infty)$. These series are obtained from the fact that any bounded solution is described by functions that are analytic with respect to t in a certain strip or half strip, containing the real half-axis. In Section 3 of Chapter II we develop the theory of equations with homogeneous right members. It is shown in particular that in order for the zero solution of the system to be asymptotically stable, it is necessary and sufficient that there exist two homogeneous functions: one positive definite W of order m , and one negative definite V of order $(m + 1 - \mu)$, such that $dV/dt = W$, where μ is the index of homogeneity of the right members of the system. If the right members of the system are differentiable, then these functions satisfy a system of partial differential equations, the solution of which can be found in closed form. This circumstance makes it possible to give a necessary and sufficient con-

dition for asymptotic stability in the case when the right members are forms of degree μ , directly on the coefficients of these forms. In Sections 4 and 5 of Chapter II we consider several doubtful cases: k zero roots and $2k$ pure imaginary roots. We obtain here many results on the stability, and also on the existence of integrals of the system and of the family of bounded solutions. In Section 6 of Chapter II the theory developed in Chapter I is applied to the theory of non-stationary systems of equations. In it are formulated theorems that follow from the results of Section 14, and a method is also proposed for the investigation of periodic solutions.

In Section 1 of Chapter III we solve the problem of the analytic representation of solutions of partial differential equations in the case when the conditions of the theorem of S. Kovalevskaya are not satisfied. The theorems obtained here are applied in Section 2 of Chapter III to systems of ordinary differential equations. This supplements the investigations of Briot and Bouquet, H. Poincaré, Picard, Horn, and others, and makes it possible to develop in Section 3 of Chapter III a method of constructing series, describing a family of O-curves for a system of equations, the expansions of the right members of which do not contain terms which are linear in the functions sought. The method of construction of such series has made it possible to give another approach to the solution of the problem of stability in the case of systems considered in Sections 3–5 of Chapter II and to formulate theorems of stability, based on the properties of solutions of certain systems of nonlinear algebraic equations. Thus, the third chapter represents an attempt at solving the problem of stability with the aid of Lyapunov's first method.

In Chapter IV we again consider metric spaces and families of transformations in them. In Section 1 of Chapter IV we introduce the concept of a general system in metric space.

A *general system* is a two-parameter family of operators from R into R , having properties similar to those found in solutions of the Cauchy problem and the mixed problem for partial differential equations. Thus, the general systems are an abstract model of these problems. We also develop here the concept of stability of invariant sets of general systems. In Section 2 of Chapter IV, Lyapunov's second method is extended to include the solution of

problems of stability of invariant sets of general systems. The theorems obtained here yield necessary and sufficient conditions. They are based on the method of investigating two-parameter families of operators with the aid of one-parameter families of functionals. We also propose here a general method for estimating the distance from the motion to the invariant set. In Section 3 of Chapter IV are given several applications of the developed theory to the Cauchy problem for systems of ordinary differential equations. Results are obtained here that are not found in the known literature.

The fifth chapter is devoted to certain applications of the developed theory to the investigation of the problem of stability of the zero solution of systems of partial differential equations in the case of the Cauchy problem or the mixed problem. In Section 1 of Chapter V are developed general theorems, which contain a method of solving the stability problem and which are orientative in character. In Sections 2-3 of Chapter V are given specific systems of partial differential equations, for which criteria for asymptotic stability are found. In Section 3 the investigation of the stability of a solution of the Cauchy problem for linear systems of equations is carried out with the aid of a one-parameter family of quadratic functionals, defined in $W_2^{(N)}$. Stability criteria normalized to $W_2^{(N)}$ are obtained here. However, the imbedding theorems make it possible to isolate those cases when the stability will be normalized in C . In the same section are given several examples of investigation of stability in the case of the mixed problem.

For a successful understanding of the entire material discussed here, it is necessary to have a knowledge of mathematics equivalent to the scope of three university courses. However, in some places more specialized knowledge is also necessary.

I obtained the circle of problems included in this monograph from personal sessions with N. P. Yerugin and V. V. Nemytsky, and also by participating in the work of the seminar under the leadership of N. P. Yerugin. The ideas and methods of solving these problems were taken mostly from the papers of A. M. Lyapunov, N. P. Yerugin, Ye. A. Barbashin, N. N. Krasovsky, V. V. Nemytsky, and Acad. S. L. Sobolev.

I take this opportunity to express my sincere gratitude to

Academician V. I. Smirnov and lecturer A. P. Tuzov for their interest in the present work.

In conclusion, let me note that in the preparation of the manuscript for press, invaluable help was rendered me by V. P. Petukhov, graduate student of the Leningrad State University, and by D. A. Vladimirov, instructor in the mathematics-mechanics faculty.

TABLE OF CONTENTS

	Page
<i>Foreword</i>	IV
<i>From the Author</i>	VIII
<i>Introduction</i>	1
<i>Chapter I Stability of Invariant Sets of a Dynamical System in Metric Space</i>	
1. Metric Space [2]. Basic Definitions	9
2. Operators and Functionals in R	10
3. Neighborhood of a Set	11
4. Compactness	12
5. Normed Linear Spaces	12
6. Dynamical Systems in n -Dimensional Euclidean Space E_n	14
7. Examples of Dynamical Systems in Functional Spaces	17
8. Dynamical Systems in Metric Space	20
9. Formulation of the Problem of Stability of Invariant Sets. Principal Definitions	22
10. Uniformly Asymptotically Stable and Uniformly Attracting Invariant Sets of a Dynamical System $f(p, t)$	27
11. Qualitative Description, from the Point of View of the Lyapunov Stability, of a Neighborhood of an Invariant Set	31
12. Necessary and Sufficient Conditions for Stability	41
13. Necessary and Sufficient Conditions for Instability.	48
14. Necessary and Sufficient Conditions for the Existence of Uniformly Asymptotically Stable and Uniformly Attracting Invariant Sets.	52
15. Method of Estimating the Distance from the Motion $f(p, t)$ to a Certain Invariant Set M with the Aid of Functionals	61

	Page
<i>Chapter II Investigation of the Problem of Stability of Motion for Systems of Ordinary Differential Equations</i>	67
1. Stationary Systems of Differential Equations	67
2. Case of Analytic Right Members of the System (2.1)	87
3. Systems of Differential Equations with Homogeneous Right Members	104
4. Case of k Zero Roots	130
5. Case of Several Pairs of Pure Imaginary Roots	144
6. System of Non-Stationary Differential Equations	152
 <i>Chapter III Investigation of a Neighborhood of the Trivial Solution of the Differential Equations with the Aid of Lyapunov's First Method</i>	 166
1. Auxiliary Theorems from the Theory of Partial Differential Equations	166
2. Representation of Solutions of Systems of Ordinary Differential Equations in a Neighborhood of a Singular Point	173
3. Analytic Representation of the O-Curves of a Special System of Differential Equations	179
 <i>Chapter IV Investigation of the Stability of Invariant Sets of General Systems.</i>	 196
1. General Systems. Basic Definitions.	196
2. Conditions for Stability and Instability of an Invariant Set M of a General System in Metric Space	202
3. Certain Applications to Non-Stationary Systems of Ordinary Differential Equations	224
 <i>Chapter V Solution of the Stability Problem for a System of Partial Differential Equations</i>	 235
1. Some General Propositions	235
2. Stability of Solutions of Quasi-Linear Systems of Special Type	241
3. Stability of the Trivial Solution of a System of Partial Differential Equations.	248
 <i>Bibliography</i>	 259
<i>Index</i>	262

INTRODUCTION

The theory of stability of motion as a branch of mathematical science was developed in the papers of the great Russian scientist Academician A. M. Lyapunov [1] during the end of the past century. In recent years a vigorous growth has been observed in this theory, owing to the requirements of progressing technology and particularly the requirements of automatic control. The development of the theory of stability of motion followed two paths: first, the expansion of the circle of problems, and, second, the creation of new and reinforcement of the already known methods of research. Let us give a brief description of the subject and of the principal methods of this theory.

Let

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t), \quad (s = 1, \dots, n) \quad (1)$$

be a given system of differential equations, the right members of which are defined in the region $\sum_{s=1}^n x_s^2 \leq H, t \geq 0$, and satisfy there certain conditions, which guarantee the existence of a system of continuously differentiable functions $x_s = x_s(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0)$, ($s = 1, \dots, n$), satisfying system (1) and becoming converted, when $t = t_0$, into the quantities $x_s^{(0)}$,

$$\sum_{s=1}^n x_s^{(0)^2} < H, t_0 \geq 0.$$

Let us assume that $f_s(0, \dots, 0, t) \equiv 0$ when $t \geq 0$, so that the system (1) has the trivial solution $x_1 = 0, \dots, x_n = 0$.

DEFINITION 1. The trivial solution of a system (1) is called *stable in the sense of Lyapunov* if for any $\varepsilon > 0$ and a given $t_0 \geq 0$ it is

possible to find a $\delta > 0$ such that when $\sum_{s=1}^n x_s^{(0)} < \delta^2$ we shall have $\sum_{s=1}^n x_s^2(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0) < \varepsilon$, when $t_0 \leq t$.

If, furthermore, $\sum_{s=1}^n x_s^2(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0) \rightarrow 0$ when $t \rightarrow +\infty$, then the trivial solution of system (1) is called *asymptotically stable*.

DEFINITION 2. A single-valued continuous function $V(x_1, \dots, x_n, t)$ defined in the region $\sum_{s=1}^n x_s^2 \leq H_1$, $t \geq T$ is called *positive definite* if the following conditions are satisfied:

- 1) $V(0, \dots, 0, t) \equiv 0$;
- 2) There exists a function $V_1(x_1, \dots, x_n)$ such that $V_1(0, \dots, 0) = 0$ and $V_1 > 0$ when $\sum_{s=1}^n x_s^2 \leq H_1$, with

$$V(x_1, \dots, x_n, t) \geq V_1(x_1, \dots, x_n).$$

If condition 1) is satisfied, $V(x_1, \dots, x_n, t)$ is called *negative definite* if

$$V(x_1, \dots, x_n, t) \leq -V_1(x_1, \dots, x_n).$$

DEFINITION 3. We say that the function

$$V(x_1, \dots, x_n, t)$$

admits an *infinitesimally upper limit* if $V(x_1, \dots, x_n, t) \rightarrow 0$ when $\sum x_s^2 \rightarrow 0$ uniformly for $t \geq T$.

DEFINITION 4. The *derivative* of the function $V(x_1, \dots, x_n, t)$ along the integral curve

$$(x_1(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0), \dots, x_n(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0))$$

of the system (1) is the expression

$$\frac{dV}{dt}(x_1(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0), \dots, x_n(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0), t).$$

If the function V is continuously differentiable in all its arguments, then the total derivative of the function V , by virtue of the system (1), is the expression

$$\frac{dV}{dt} = \frac{\partial V}{\partial t} + \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i = W(x_1, \dots, x_n, t).$$

As was shown by Lyapunov, the general problem of stability of motion relative to certain quantities can be reduced in a large number of cases to a solution of the problem of the stability of the

trivial solution of the system (1). To solve the problem of the stability of the trivial solution of system (1), Lyapunov proposed two methods. Let us describe the two methods briefly.

THEOREM 1. *If the system of differential equations (1) is such that it is possible to find a positive definite function V , the derivative of which, W , calculated by virtue of system (1), satisfies the inequality $W \leq 0$, then the trivial solution of the system (1) is stable.*

THEOREM 2. *The trivial solution of the system (1) will be asymptotically stable if the conditions of Theorem 1 are satisfied and furthermore the function W is negative definite, while the function F admits an infinitesimally small upper limit.*

THEOREM 3. *If the system (1) is such that it is possible to find a function V , that would have by virtue of these equations a positive definite derivative W , and furthermore would admit an infinitesimally small upper limit and would be such that for any $t > T$ (T is a sufficiently large quantity) it can be made positive by a suitable choice of the quantities x_s ($s = 1, \dots, n$) which numerically can be as small as desired, then the trivial solution is unstable.*

THEOREM 4. *If system (1) is such that one can find a bounded function V for sufficiently small $|x_s|$ and $t > T$, the derivative of which would be reduced, by virtue of the system (1), to the form*

$$\frac{dV}{dt} = \lambda V + W,$$

where λ is a positive constant and $W \geq 0$, and if in the latter case the function V obtained is such that for any $t \geq T$ it can be made a quantity of the same sign as W by a suitable choice of the quantities x_s ($s = 1, \dots, n$) which numerically are as small as convenient, then the trivial solution of the system (1) is unstable.

These four theorems comprise the principal theoretical content of the second Lyapunov method in its initial form. The function $V(x_1, \dots, x_n, t)$, which resolves the problem of the stability of the trivial solution of the system (1), is customarily called the *Lyapunov function*. In connection with these theorems, a series of questions was raised, namely:

- 1) Are the conditions formulated in these theorems necessary as well as sufficient?
- 2) Is it possible to find with the aid of the Lyapunov function

the region of variation of the initial values $x_1^{(0)}, \dots, x_n^{(0)}, t_0$, such that

$$\sum_{s=1}^n x_s^2(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0) \rightarrow 0 \text{ as } t \rightarrow +\infty?$$

3) Is it possible to indicate with the aid of the Lyapunov function various estimates of the functions $x_s(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0)$, corresponding to sufficiently small $|x_1^{(0)}|, \dots, |x_n^{(0)}|$?

These principal problems have been and are the object of research being carried out by many great mathematicians in the USSR and abroad. In the present work, the second Lyapunov method is described as applied to the study of the problem of stability of invariant sets of dynamical systems and two-parameter families of transformations in metric space.

For the sake of greater clarity in the exposition, let us note here the following: if a Lyapunov function $V(x_1, \dots, x_n, t)$ satisfies the conditions of Theorem 1 and furthermore admits an infinitesimally small upper limit, then for any $\varepsilon > 0$ it is possible to find a number $\delta > 0$, which is independent of $t_0 \geq T$, such that when $\sum_{s=1}^n x_s^{(0)^2} < \delta$ there will hold, for any $t_0 \geq T$,

$$\sum_{s=1}^n x_s^2(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0) < \varepsilon \text{ for } t \geq t_0.$$

Actually, without loss of generality, we can take $\varepsilon > 0$ and $\varepsilon < H_1$ and set $\lambda = \inf V(x_1, \dots, x_n, t)$ for $\sum x_i^2 = \varepsilon$.

In view of the presence of an infinitesimally small upper limit for the function $V(x_1, \dots, x_n, t)$, it is possible to find a $\delta > 0$ such that when $\sum_{s=1}^n x_s^2 < \delta$ we obtain $V(x_1, \dots, x_n, t) < \lambda$, $t \geq T$. By assumption, the total derivative of the function V , by virtue of system (1), is not positive, and therefore

$$V(x_1, \dots, x_n, t) \leq V(x_1^{(0)}, \dots, x_n^{(0)}, t_0) < \lambda \text{ for } t \geq t_0,$$

where

$$t_0 \geq T \text{ and } \sum_{s=1}^n x_s^{(0)^2} < \delta.$$

Here the arguments $x_1, \dots, x_n$ of the function V are functions that satisfy system (1), i.e.,

$$x_i = x_i(t, x_1^{(0)}, \dots, x_n^{(0)}, t_0).$$

We shall therefore have $\sum_{s=1}^n x_s^2 < \varepsilon$ for $t \geq t_0$. Thus, the trivial

solution of the system (1) is stable in this case, and, furthermore, for each $\varepsilon > 0$, the number $\delta > 0$ can be chosen independent of $t_0 \geq T$. Such a stability will be called *uniform* [41] *with respect to* t_0 . From now on, wherever we speak of the stability of the trivial solution of a system of non-stationary equations, we shall assume that the stability is meant to be uniform relative to t_0 , and the term "instability" will denote a fact contradicting uniform stability with respect to t_0 . It must be noted that in the case when the right members of the system do not depend explicitly on the time, the stability of the trivial solution leads to a stability which is uniform relative to $t_0 > -\infty$.

Let us now dwell on a brief description of the Lyapunov first method. Let us assume that the right members of the system (1) are expanded in powers of the quantities x_s , and that the expansions converge for sufficiently small $|x_s|$, ($s = 1, \dots, n$) and all $t \geq 0$. The first method of solving the stability problem consists of constructing a general solution in the form of series that converge when $t \geq 0$, and the form of the series is used to establish the stability or instability. Let us explain this method in somewhat greater detail.

We consider the system of equations

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si}x_i + X_s(x_1, \dots, x_n). \quad (2)$$

We assume that the functions X_s are expanded in powers of the numbers $x_1, \dots, x_n$, and the series converge for sufficiently small $|x_s|$ and do not contain terms linear in $x_1, \dots, x_n$. We denote by λ_i ($i = 1, \dots, n$) the roots of the equation $|P - \lambda E| = 0$, where $P = \|p_{ik}\|$. Let $\text{Re } \lambda_i < 0$ when $i \leq k$. Then, as was shown by Lyapunov, the system (2) has a family of solutions in the form of the series

$$x_s = \sum_{m_1 + \dots + m_k = 1}^{+\infty} K_s^{(m_1, \dots, m_k)}(t) \alpha_1^{m_1} \dots \alpha_k^{m_k} e^{t \sum m_i \lambda_i},$$

which converge when $|\alpha_i| < \alpha$ and $t > 0$, where $\alpha > 0$ is a constant. [The functions $K_s^{(m_1, \dots, m_k)}(t)$ are, generally speaking, polynomials in t .] It follows from this that when $k = n$, the trivial solution of the system (2) will be asymptotically stable regardless of the

and $\lambda_1, \dots, \lambda_k$ are the roots of the equation

$$\begin{vmatrix} q_{11} - \lambda, & q_{12}, & \dots, & q_{1k} \\ q_{21}, & q_{22} - \lambda, & \dots, & q_{2k} \\ \cdot & \cdot & \cdot & \cdot \\ q_{k1}, & q_{k2}, & \dots, & q_{kk} - \lambda \end{vmatrix} = 0,$$

and if furthermore the real parts of all the κ_s differ from zero and all have the same sign, and the quantities κ_s and λ_j are not related by any equations of the type

$$m_1\kappa_1 + m_2\kappa_2 + \dots + m_n\kappa_n = \lambda_j, \quad (j = 1, \dots, k),$$

where all the m_s ($s = 1, \dots, n$) are non-negative integers that satisfy the condition $\sum_{s=1}^n m_s > 0$, then one can always find one definite system of holomorphic functions $z_1, \dots, z_k$ in the variables $x_1, \dots, x_n$ satisfying equations (3) and vanishing when $x_1 = \dots = x_n = 0$.

In the present work we consider the problem of stability of doubtful cases also, namely the case when among the quantities λ_i there are k with zero real parts, or when among the quantities λ_i there are $2k$ pure imaginary ones and the remaining have negative real parts. This investigation is carried out with the aid of the second method and with the aid of certain auxiliary theorems, analogous to the Lyapunov auxiliary theorem. Attention is paid in this work to the construction of solutions and to their estimates.

The theorems of the second method, obtained in the investigation of invariant sets in metric spaces, are applied to the investigation of the problem of stability of ordinary differential equations and systems of partial differential equations.

STABILITY OF INVARIANT SETS OF A DYNAMICAL SYSTEM IN METRIC SPACE

1. Metric space [2]. Basic definitions

DEFINITION 1. A set R of elements p , the nature of which is immaterial, will be called a *metric space* if to any two elements $p \in R$ and $q \in R$ there corresponds a real non-negative number $\rho(p, q)$, satisfying the following conditions:

- 1) $\rho(p, q) = 0$ if and only if $p = q$;
- 2) $\rho(p, q) = \rho(q, p)$;
- 3) $\rho(p, q) \leq \rho(p, z) + \rho(z, q)$ for any $z \in R$.

The number $\rho(p, q)$ is usually called the *metric distance* or simply the *distance* from the element p to the element q , and the elements of the space R are called *points*.

DEFINITION 2. A point $p \in R$ is called a *limit* of the sequence $p_1, \dots, p_n, \dots, p_n \in R$, if $\rho(p_n, p) \rightarrow 0$ as $n \rightarrow +\infty$. If the sequence $p_n \in R$, $n = 1, 2, \dots$, has a limit, then this sequence, as can be readily verified, has the following properties:

- 1) Any infinite subsequence of the sequence $p_1, \dots, p_n, \dots$, also has a limit.
- 2) The limit is unique, i.e. if $p_n \rightarrow p$ and $p_n \rightarrow q$, then $p = q$.

DEFINITION 3. The set of all points $p \in R$, having the property $0 < \rho(p_0, p) < r$, is called the *r-neighborhood* of the point p_0 . Here, r is a positive number.

DEFINITION 4. The set $X \subset R$ is called *open* in R if it follows from $p_0 \in X$ that there exists a neighborhood of the point p_0 such that any point p from it is contained in X .

DEFINITION 5. The point p_0 is called a *limit point* of the set $X \subset R$ if there exists a sequence $p_1, \dots, p_n, \dots, p_n \in X$, such that $p_n \rightarrow p_0$ as $n \rightarrow +\infty$.

DEFINITION 6. A set $X \subset R$ which contains all its limit points is called *closed* in R . It is easy to show that the set $Y \subset R$, which is the complement of the open set $X \subset R$ relative to the entire space R is closed in R , and conversely.

DEFINITION 7. The set obtained from X by adjoining all its limit points is called the *closure* of X and is denoted by $\bar{X}$.

DEFINITION 8. The *boundary* of an open set A is the set $\bar{A} \setminus A$.

DEFINITION 9. The sequence $p_n \in R$ is called *convergent in itself* if for any $\varepsilon > 0$ one can find a natural number N_ε such that when $n > N_\varepsilon$ and $m > N_\varepsilon$ we have $\rho(p_n, p_m) < \varepsilon$.

It is clear that a sequence having a limit converges in itself.

DEFINITION 10. The space R is called *complete* if any sequence $p_n \in R$, which converges in itself, has a limit. It is easy to see that any closed set X , contained in a complete space R , is in turn a complete space (relative to the distance induced from R).

2. Operators and functionals in R

Let X and Y be two sets contained in R .

DEFINITION 11. If a rule is given that causes a certain element $q \in Y$ to correspond to each element $p \in X$, one says that an *operator* F is defined in the set X such that $q = Fp$. The set X is called the *domain of definition* or simply the *domain* of the operator F .

DEFINITION 12. The operator F is called *continuous* at the point $p_0 \in X$ if for any $\varepsilon > 0$ it is possible to find a number $\delta > 0$ such that when $\rho(p_0, p) < \delta$, $p \in X$, we have $\rho(Fp, Fp_0) < \varepsilon$. One also considers a family of operators, which depends on the parameter τ , which in turn is an element of the set Z , the nature of which is immaterial.

We shall henceforth investigate the properties of such families of operators, for which the set Z is a subset of the set of real numbers, or a subset of the set of pairs of real numbers.

DEFINITION 13. If a rule is given, by which there corresponds to each element $p \in X$ a real number $V(p)$, one says that a *functional* V is defined on the set X .

In analogy with Definition 12, we can define the concept of *continuity of a functional V at the point $p_0 \in X$* .

One also considers families of functionals, which depend on a

certain parameter $\tau \in Z$. It is clear that $V_\tau(p)$, for a fixed $p \in X$, will give a functional on the set Z . Further, we consider families of functionals, which depend on the real parameter τ , so that for fixed $p \in X$ the expression $V_\tau(p)$ defines a real function of the real argument τ . These families of functionals will be used later to study certain properties of families of operators. Such an approach to a study of properties of families of operators, defined in a metric space R , constitutes the principal content of the second Lyapunov method, taken in abstract form. Later on, however, we shall give a more accurate description of this method.

3. Neighborhood of a set

Let us consider a set $X \subset R$ and let p and q be two points such that $p \in R$ and $q \in X$. The quantity $\rho(p, q)$ is bounded from below for any $q \in X$, and therefore $\inf_{q \in X} \rho(p, q)$ exists.

DEFINITION 14. We call $\inf_{q \in X} \rho(p, q)$ the *distance* from the point p to the set X and we denote it by $\rho(p, X)$. If the set X is closed, it is easy to verify that there exists a point $\bar{q} \in X$ such that $\rho(p, X) = \rho(p, \bar{q})$.

It is clear that $\rho(p, X)$ is a functional defined for $p \in R$. Let us show that this functional is continuous at any point $p_0 \in R$. In fact, let us take any point $p \in R$ and $q \in X$; we then have

$$\rho(p, q) \leq \rho(p, p_0) + \rho(p_0, q);$$

hence

$$\inf_{q \in X} \rho(p, q) \leq \inf_{q \in X} \rho(p_0, q) + \rho(p, p_0),$$

or

$$\rho(p, X) \leq \rho(p_0, X) + \rho(p, p_0).$$

Analogously, we obtain from $\rho(p_0, q) \leq \rho(p, p_0) + \rho(p, q)$ that

$$\rho(p_0, X) \leq \rho(p, p_0) + \rho(p, X).$$

Combining these two inequalities, we obtain

$$|\rho(p, X) - \rho(p_0, X)| \leq \rho(p, p_0),$$

which leads to the foregoing statement.

DEFINITION 15. The r -neighborhood of a set $X \in R$ is the set of all points $p \in R$, which have the property $0 < \rho(p, X) < r$, where $r > 0$ is a certain number. Henceforth this neighborhood will be denoted by $S(X, r)$.

DEFINITION 16. The set $X \subset R$ is called *bounded* if $\sup_{q \in X} \rho(p, q) < +\infty$ for a certain $p \in R$.

It must be noted that from the boundedness of the set $S(X, r)$ it does not follow that the set X is bounded.

4. Compactness

DEFINITION 17. A set $M \subset R$ is called *compact* in R if every infinite subset $Y \subset M$ contains an infinite convergent sequence.

If the limit of any sequence so selected is contained in M , then the set M is called *compact in itself*. If the space R is itself compact, then it is easy to establish that it is complete. It is also possible to establish the following: a continuous functional, defined on a set that is compact in itself, is bounded and attains its least upper bound and its greatest lower bound.

DEFINITION 18. The set $Y \subset R$ is called an ε -net of the set $X \subset R$ if for any $p \in X$ it is possible to find a $q \in Y$ such that $\rho(p, q) < \varepsilon$.

THE HAUSDORFF THEOREM. *In order for the set $X \subset R$ to be compact, it is necessary, and in the case of a complete R it is also sufficient, that for any $\varepsilon > 0$ there exist a finite ε -net of the set X .*

Let us note that a compact set is bounded.

DEFINITION 19. The metric space R is called *separable* if there exists a denumerable set of elements $G \subset R$ such that no matter what $\varepsilon > 0$ and $p \in R$ are chosen, there is a $q_\varepsilon \in G$ such that $\rho(p, q_\varepsilon) < \varepsilon$. From the Hausdorff theorem and from Definition 17 it follows directly that a compact space is separable.

5. Normed linear spaces

DEFINITION 20. A set G is called an *additive group* if:

- 1) To any $x, y \in G$ there corresponds a $z \in G$, which we denote by $z = x + y$;
- 2) $x + (y + z) = (x + y) + z$;

- 3) There exists a right-hand zero element θ , such that $x + \theta = x$;
 4) There exists a right-hand inverse element $(-x)$, such that $x + (-x) = \theta$.

A group G is called *Abelian* if $x + y = y + x$ for all $x, y \in G$.

DEFINITION 21. A set L is called a *linear system* if it is an additive Abelian group and, furthermore, if multiplication of the elements of this set by complex numbers is defined, i.e., if it follows from $x \in L$ and λ , a complex number, that $\lambda x \in L$, then the foregoing multiplication is associative and distributive.

DEFINITION 22. The set E is called a *normed linear space* if E is a linear system and if to each element $x \in E$ there corresponds a non-negative number $\|x\|$ — the *norm*. The following conditions are satisfied here:

- a) $\|x\| = 0$ if and only if $x = \theta$.
- b) $\|\alpha x\| = |\alpha| \|x\|$, where α is a complex number.
- c) $\|x + y\| \leq \|x\| + \|y\|$.

If we set $\rho(x, y) = \|x - y\|$, then we define a metric in the set E . Thus, a normed linear space is a particular case of a metric space. If the normed linear space E is complete, it is called a *Banach space* or a *type B space*.

Let us give several examples of normed linear spaces and note their principal properties.

- 1) Let us consider the space of continuous functions $\varphi(x_1, \dots, x_n)$ defined in E_n such that

$$\sup_{X \in E_n} |\varphi| < +\infty.$$

The space obtained from the aggregate of these functions as a result of introducing the norm

$$\|\varphi\| = \sup_{X \in E_n} |\varphi|$$

is a linear space of type *B*. Here, $X = (x_1, \dots, x_n)$ is a point in E_n .

- 2) Let us consider the aggregate of functions $\varphi(x_1, \dots, x_n)$, defined in a finite region Ω of the space E_n , summable over Ω together with $|\varphi|^p$, $p > 1$. We set

$$\|\varphi\| = \left\{ \int_{\Omega} |\varphi|^p d\Omega \right\}^{1/p}.$$

We can show [3] that this equality defines a norm. The corresponding space is denoted by L_p .

3) The space of n -dimensional complex vectors $X = (x_1, \dots, x_n)$, in which we set

$$\|X\| = \sqrt{\sum_{i=1}^n x_i \bar{x}_i},$$

is the simplest example of a type B space.

6. Dynamical systems in n -dimensional Euclidean space E_n

Let us consider a system of n differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n), \quad (s = 1, \dots, n). \quad (1.1)$$

We assume that the functions $f_1(x_1, \dots, x_n), \dots, f_n(x_1, \dots, x_n)$ are defined for $-\infty < x_s < +\infty$ and are continuous there. If, furthermore, these functions satisfy the Lipschitz condition relative to the variables $x_1, \dots, x_n$ in any finite region g of the space E_n , i.e.

$$|f_s(\bar{x}_1, \dots, \bar{x}_n) - f_s(x_1, \dots, x_n)| < L_g \left(\sum_{i=1}^n |x_i - \bar{x}_i| \right)$$

$$(L_g = \text{constant}, s = 1, \dots, n);$$

$$(x_1, \dots, x_n) \in g, (\bar{x}_1, \dots, \bar{x}_n) \in g,$$

then the system (1.1) determines, as is known [4], a family of integral curves in E_n .

For the sake of brevity we shall denote the integral curve passing through $X^{(0)} = (x_1^{(0)}, \dots, x_n^{(0)})$ for $t = 0$ by $X = X(t, X^{(0)})$, where X is the vector $(x_1, \dots, x_n)$. Generally speaking, not every integral curve $X = X(t, X^{(0)})$ of the system (1.1) can be defined for all values of the argument t . However, there exist conditions on the functions $f_s(x_1, \dots, x_n)$, $s = 1, \dots, n$, which when satisfied make any integral curve of the system (1.1) defined for all $t \in (-\infty, +\infty)$. One such condition is the boundedness of the right members of the system (1.1). In fact, let

$$|f_s(x_1, \dots, x_n)| < m, \quad (s = 1, \dots, n) \quad (1.2)$$

for $X \in E_n$. We shall show that any integral curve of the system (1.1) is defined in this case for all $t \in (-\infty, +\infty)$. We shall show, for example, that the integral curve $X = X(t, X^{(0)})$ is defined for $t > 0$. Two cases are possible: either this integral curve is bounded for $t \geq 0$, or else it is not bounded for $t \geq 0$. In the former case successive application of the existence theorem [5] shows that the integral curve is defined for all $t \geq 0$. In the second case let us assume, to the contrary, that the integral curve is defined only for $t \leq T < +\infty$. Then, as follows from the equation

$$x_s = x_s^{(0)} + \int_0^t f_s(x_1, \dots, x_n) d\tau,$$

we obtain the estimate

$$|x_s(t, X^{(0)}) - x_s^{(0)}| < mT, \quad (s = 1, \dots, n),$$

which contradicts our assumption that the integral curve is bounded. It is easy to see that conditions (1.2) can always be realized by introducing into the system (1.1) a new independent variable, for example in accordance with the formula

$$d\tau = dt(1 + \sum_{i=1}^n f_i^2(x_1, \dots, x_n)).$$

It is clear that any integral curve $\bar{X} = \bar{X}(\tau, X^{(0)})$ of the newly-obtained system

$$\frac{dx_s}{d\tau} = \frac{f_s(x_1, \dots, x_n)}{1 + \sum_{i=1}^n f_i^2(x_1, \dots, x_n)} \quad (1.1')$$

will be defined for all values of $\tau \in (-\infty, +\infty)$. However, geometrically the integral curves of the systems (1.1) and (1.1') will coincide.

We note further that if $\bar{X}(\tau, X^{(0)})$ is an integral curve, then any function

$$\bar{X}(\tau + c, X^{(0)}),$$

where c is a real constant, also satisfies the system (1.1'). Thus, it becomes possible to define in the space E_n , with the aid of the system (1.1), a family of integral curves $\bar{X} = \bar{X}(\tau, X^{(0)})$, which have the following properties:

a) For any finite value $X^{(0)}$, the vector function $\bar{X}(\tau, X^{(0)})$ is defined for all $\tau \in (-\infty, +\infty)$, and moreover $\bar{X}(0, X^{(0)}) = X^{(0)}$;

b) The vector function $\bar{X}(\tau, X^{(0)})$ is continuous in the aggregate of its arguments, which follows from the theorem of continuity on the initial conditions [1];

c) For all values of τ and τ_1 , we have the equality

$$\bar{X}(\tau + \tau_1, X^{(0)}) = \bar{X}(\tau, \bar{X}(\tau_1, X^{(0)})). \quad (1.3)$$

Actually, on the basis of the remark made above, the left half of (1.3) is the solution of the system (1.1'). With this we have, for $\tau = 0$, the following equality

$$\bar{X}(\tau_1, X^{(0)}) = \bar{X}(0, \bar{X}(\tau_1, X^{(0)})),$$

which is valid by virtue of item "a," and then the uniqueness theorem [1] leads to the equality (1.3) for any $\tau \in (-\infty, +\infty)$ and for any $\tau_1 \in (-\infty, +\infty)$.

The vector function of two arguments $\bar{X}(\tau, X^{(0)})$ can be considered as a one-parameter family of operators, defined in E_n , or else as a one-parameter group of transformations of the space E_n into itself. Actually, the elements of the group are the transformations $\bar{X}(\tau, X^{(0)})$, which correspond to a given τ , $X^{(0)} \in E_n$. To each pair of such transformations

$$\bar{X}(\tau_1, X^{(0)}) \text{ and } \bar{X}(\tau_2, X^{(0)})$$

there corresponds their superposition, i.e. the sum

$$\bar{X}(\tau_1, \bar{X}(\tau_2, X^{(0)})) = \bar{X}(\tau_1 + \tau_2, X^{(0)}).$$

It is clear that the operation of addition of transformations is associative. Furthermore, there is a zero element (identical transformation $\bar{X}(0, X^{(0)}) = X^{(0)}$), and to each transformation $\bar{X}(\tau, X^{(0)})$ there corresponds the inverse transformation $\bar{X}(-\tau, X^{(0)})$.

A one-parameter group of transformations of the space E_n into itself, having the properties "a," "b," and "c," is called a *dynamical system* in E_n and is investigated independent of the specific type of functions $f_s(x_1, \dots, x_n)$ that give rise to the particular dynamical system.

7. Examples of dynamical systems in functional spaces

Let us consider the linear partial differential equation

$$\frac{\partial u}{\partial t} = au + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i}, \quad (1.4)$$

where $a, b_1, \dots, b_n$ are real constants. We form the space Φ of continuously differentiable functions $\varphi(x_1, \dots, x_n)$ defined for $X \in E_n$ and such that $\varphi(x_1, \dots, x_n) \rightarrow L_\varphi$ as $\|X\| \rightarrow +\infty$, where L_φ is a finite real constant. We put $\|\varphi\| = \sup_{X \in E_n} |\varphi|$.

It is clear that for any function $\varphi \in \Phi$ one can find a solution of Eq. (1.4) $u(\varphi, t)$, with the property $u(\varphi, 0) = \varphi(x_1, \dots, x_n)$. It is easy to verify that

$$u(\varphi, t) = e^{at} \varphi(x_1 + b_1 t, \dots, x_n + b_n t). \quad (1.5)$$

The aggregate of solutions (1.5) of Eq. (1.4) for $\varphi \in \Phi$ is a one-parameter family of transformations of the space Φ into itself, generated by the one-parameter family of operators $u_t(\varphi) = u(\varphi, t)$. This family of transformations has the following properties:

a) $u(\varphi, t)$ is continuous at any point (φ, t) in both its arguments φ and t , i.e., for any $\varepsilon > 0$ it is possible to find quantities $\delta_1 > 0$ and $\delta_2 > 0$ such that when

$$|t_1 - t_2| < \delta_1 \quad \text{and} \quad \|\varphi_1 - \varphi_2\| < \delta_2$$

we have

$$\|u(\varphi_1, t_1) - u(\varphi_2, t_2)\| < \varepsilon.$$

In fact,

$$\begin{aligned} u(\varphi_1, t_1) - u(\varphi_2, t_2) &= e^{at_1} \varphi_1(x_1 + b_1 t_1, \dots, x_n + b_n t_1) - \\ &- e^{at_2} \varphi_2(x_1 + b_1 t_2, \dots, x_n + b_n t_2) = e^{at_1} [\varphi_1(x_i + b_i t_1) - \\ &- \varphi_2(x_i + b_i t_1)] + (e^{at_1} - e^{at_2}) [\varphi_2(x_i + b_i t_1)] + \\ &+ e^{at_2} [\varphi_2(x_i + b_i t_1) - \varphi_2(x_i + b_i t_2)]. \end{aligned} \quad (1.6)$$

Here $\varphi(x_1 + b_1 t, \dots, x_n + b_n t) = \varphi(x_i + b_i t)$. We note that it follows from $|L_\varphi| < +\infty$ that any function $\varphi \in \Phi$ is uniformly continuous over the entire space E_n and is bounded there.

As a result of the uniform continuity of the function φ_2 , it is possible to find for any quantity $\varepsilon > 0$ a quantity $\delta_1 > 0$ such

that when $|t_1 - t_2| < \delta_1$ we have

$$e^{at_1} |\varphi_2(x_1 + t_1 b_1) - \varphi_2(x_1 + t_2 b_1)| < \varepsilon.$$

Furthermore, as a result of the boundedness of the function φ_2 , we can choose $\delta_2 > 0$ so small that when $|t_1 - t_2| < \delta_2$ we have

$$(e^{at_1} - e^{at_2}) \|\varphi_2\| < \varepsilon.$$

It is also possible to find a quantity $\gamma > 0$ such that when $\|\varphi_1 - \varphi_2\| < \gamma$ we have

$$e^{at_1} \|\varphi_1 - \varphi_2\| < \varepsilon.$$

Let us put $\delta = \min(\delta_1, \delta_2)$. Estimating the right member of Eq. (1.6), we obtain

$$\|u(\varphi_1, t_1) - u(\varphi_2, t_2)\| < 3\varepsilon \text{ for } |t_1 - t_2| < \delta \text{ and } \|\varphi_1 - \varphi_2\| < \gamma.$$

It must be noted here that the estimate obtained is not uniform with respect to t_1 and φ , i.e.

$$\delta_1 = \delta_1(t_1, \varphi), \delta_2 = \delta_2(t_1, \varphi).$$

b) By direct verification it is easy to show that

$$u(u(\varphi, t_1), t_2) = u(\varphi, t_1 + t_2).$$

This also follows from the fact that any function $u(\varphi, t + C)$, where C is a real constant, is also a solution of Eq. (1.4) and item c).

c) For any function $\varphi(x_1, \dots, x_n) \in \Phi$ there exists only one solution $u(\varphi, t)$, which is uniformly differentiable with respect to all its arguments $(x_1, \dots, x_n, t)$, of Eq. (1.4), determined by formula (1.5), and such that $u(\varphi, 0) = \varphi$.

The one-parameter family of operators $u(\varphi, t)$ with the properties "a," "b," and "c," will be called a *dynamical system in the functional space* Φ . It is clear that this family of operators for all $t \in (-\infty, +\infty)$ gives a group of transformations of the space Φ into itself.

Let us give another example of a dynamical system in functional space.

We consider the system of equations

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial t} = \frac{\partial u}{\partial x}, \quad u|_{t=0} = \varphi(x), \quad v|_{t=0} = \psi(x). \quad (1.7)$$

Let us form the space Ψ of pairs of functions $\varphi(x)$, $\psi(x)$, $x \in (-\infty, +\infty)$, each of which is continuously differentiable and has finite limits as $x \rightarrow +\infty$ and $x \rightarrow -\infty$. In the set of functions Ψ we introduce a norm, setting

$$\|(\varphi, \psi)\| = \sup |\varphi| + \sup |\psi|, \quad x \in (-\infty, +\infty).$$

It is easy to verify that the pair

$$\begin{aligned} u(\varphi, \psi, t) &= \frac{\varphi(x-t) + \varphi(x+t) + \psi(x+t) - \psi(x-t)}{2} \\ v(\varphi, \psi, t) &= \frac{\psi(x-t) + \psi(x+t) + \varphi(x+t) - \varphi(x-t)}{2} \end{aligned} \quad (1.8)$$

is a solution of the system (1.7), satisfying the Cauchy initial conditions $u(\varphi, \psi, 0) = \varphi(x)$, $v(\varphi, \psi, 0) = \psi(x)$.

It is easy to see that (1.8) yields a one-parameter family of transformations of the space Ψ into itself. It is possible to establish directly the following properties of this family of transformations:

- a) $\left. \begin{array}{l} u(\varphi, \psi, t) \\ v(\varphi, \psi, t) \end{array} \right\}$ are defined for all $t \in (-\infty, +\infty)$ for any pair $(\varphi, \psi) \in \Psi$. Here $(u(\varphi, \psi, 0), v(\varphi, \psi, 0)) = (\varphi, \psi)$.
- b) It can be verified directly that

$$\begin{aligned} u(u(\varphi, \psi, t_1), v(\varphi, \psi, t_1), t_2) &= u(\varphi, \psi, t_1 + t_2), \\ v(u(\varphi, \psi, t_1), v(\varphi, \psi, t_1), t_2) &= v(\varphi, \psi, t_1 + t_2). \end{aligned}$$

c) In view of the presence of limits as $x \rightarrow +\infty$ and $x \rightarrow -\infty$ for the functions that are elements of the space Ψ , it is possible to state that $\varphi(x)$ and $\psi(x)$ are uniformly continuous for $x \in (-\infty, +\infty)$, if $(\varphi, \psi) \in \Psi$. Using this we can establish, in analogy with the example given above, that $u(\varphi, \psi, t)$, $v(\varphi, \psi, t)$ are continuous functions of their arguments (φ, ψ) and t .

The family of transformations of the space Ψ into itself, having properties "a," "b," and "c," is called a *dynamical system in the space Ψ* .

It is possible to study various properties of dynamical systems in the space Φ or in the space Ψ regardless of the equations that give rise to these dynamical systems.

8. *Dynamical systems in metric space*

DEFINITION. A *dynamical system in a metric space* R is a one-parameter family of operators F_t , $t \in (-\infty, +\infty)$, defined in R and having the following properties:

a) On any element $p \in R$ the family of operators F_t assumes values $F_t(p) \in R$ for $t \in (-\infty, +\infty)$, so that $F_0(p) = p$.

b) The value of the operator $F_t(p)$ is a continuous function in the aggregate (t, p) , i.e., for any $\varepsilon > 0$ it is possible to find quantities $\delta_1 > 0$ and $\delta_2 > 0$ such that $\rho(F_{t_1}(p), F_{t_2}(q)) < \varepsilon$ for

$$\rho(p, q) < \delta_1 \text{ and } |t_1 - t_2| < \delta_2.$$

Here

$$\delta_1 = \delta_1(t_1, p_1), \quad \delta_2 = \delta_2(t_1, p_1).$$

c) $F_{t_1}(F_{t_2}(p)) = F_{t_1+t_2}(p)$.

In the literature it is customary to denote $F_t(p)$ by $f(p, t)$. The last property of the family of operators makes it possible to consider the dynamical system as a one-parameter group of transformations of the space R into itself.

Actually, the elements of this group are the transformations $f(p, t)$, $p \in R$. Corresponding to two such transformations $f(p, t_1)$ and $f(p, t_2)$ is their sum

$$f(f(p, t_1), t_2) = f(p, t_1 + t_2).$$

This operation of addition of transformations is associative. Moreover, there is a zero element of the transformation group $f(p, 0) = p$ (identical transformation). Furthermore, to each transformation $f(p, t)$ of the space R into itself one can find an inverse $f(p, -t)$, so that

$$f(f(p, t), -t) = f(p, 0) = p.$$

The function $f(p, t)$ for fixed p we shall call a *motion*. The set of all points $f(p, t)$, $t \in (-\infty, +\infty)$, where p is fixed, will be called the *trajectory* of the dynamical system.

If $f(p, t) = p$ when $t \in (-\infty, +\infty)$ then the point p will be called a *rest point* of the dynamical system.

If $f(p, t + \tau) = f(p, t) \neq p$, then the motion $f(p, t)$ will be called *periodic with period* τ .

THEOREM 1. [5]. (On the continuous dependence of the motion on the initial conditions.) *Consider the motion $f(p, t)$. For any $\varepsilon > 0$ (no matter how small) and for any $T > 0$ (no matter how large) we can find a quantity $\delta > 0$ so small that when $\rho(p, q) < \delta$ we have $\rho(f(p, t), f(q, t)) < \varepsilon$ for $t \in [0, T]$.*

PROOF. We assume that the theorem is not true. Then there exist two sequences $q_n \rightarrow p$, $t_n \in [0, T]$ such that for a certain $\varepsilon > 0$ and $T > 0$ we have $\rho(f(p, t_n), f(q_n, t_n)) \geq \varepsilon$. We choose from the bounded sequence t_n a converging sequence: $t_n \rightarrow t_0$ as $n \rightarrow +\infty$, $t_0 \in [0, T]$. We then have $q_n \rightarrow p$, $t_n \rightarrow t_0$, but

$$\rho(f(p, t_n), f(q_n, t_n)) \geq \varepsilon,$$

which contradicts the property of continuity of the function $f(p, t)$ at the point (p, t_0) . This proves the theorem completely.

DEFINITION. The set $M \subset R$ is called *invariant* with respect to a dynamical system $f(p, t)$ if it consists of the trajectories of this dynamical system, i.e., if from $p \in M$ there follows $f(p, t) \in M$, $t \in (-\infty, +\infty)$.

THEOREM 2. [5]. *The closure of the invariant set M is an invariant set.*

PROOF. We take a point $p \in \bar{M}$. If it turns out thereby that $p \in M$, then $f(p, t)$ is also $\in M$ for $t \in (-\infty, +\infty)$, and this means also $\in \bar{M}$. If on the other hand $p \in \bar{M} \setminus M$ then there exists a sequence $p_1, \dots, p_n, \dots \in M$ such that $p_n \rightarrow p$. Then also $f(p_n, t) \rightarrow f(p, t)$. Consequently, $f(p, t) \in \bar{M}$ for $t \in (-\infty, +\infty)$. It is clear in this case that $f(p, t) \in \bar{M} \setminus M$.

COROLLARY. *The boundary of an open invariant set consists of (complete) trajectories of the dynamical system $f(p, t)$.*

DEFINITION. A point q is called an ω -limit point of the motion $f(p, t)$ if there exists a sequence of numbers $t_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that $f(p, t_n) \rightarrow q$ as $n \rightarrow +\infty$. The point q is called an α -limit point if there exists a sequence $\bar{t}_n \rightarrow -\infty$ such that $f(p, \bar{t}_n) \rightarrow q$ as $n \rightarrow +\infty$.

THEOREM 3. [5]. *If the motion $f(p, t)$ has ω -limit (α -limit) points, then the set of all its ω -limit (α -limit) points is invariant and closed.*

PROOF. We denote the set of all ω -limit points of the motion $f(p, t)$ by Ω . Let us show that Ω is invariant. Let $q_0 \in \Omega$; then there

exists a sequence $t_n \rightarrow +\infty$ as $n \rightarrow +\infty$ such that

$$p_n = f(p, t_n) \rightarrow q_0 \text{ as } n \rightarrow +\infty.$$

It is clear that $f(p_n, t) \rightarrow f(q_0, t)$ as a result of the continuity of $f(p, t)$ in the arguments t and p . Then the point $f(q_0, t) \in \Omega$ for any $t \in (-\infty, +\infty)$, since there exists a sequence $t_n + t$ such that $f(p, t_n + t) \rightarrow f(q_0, t)$. Thus the set Ω is invariant. Let us show that it is closed. In fact, let q be a limit point of the set Ω . Then there exists a sequence $q_n \in \Omega$, $q_n \rightarrow q$ as $n \rightarrow +\infty$. Using a property of the metric distance, we find

$$\rho(f(p, t), q) \leq \rho(q, q_n) + \rho(q_n, f(p, t)).$$

For any $\varepsilon > 0$ it is possible to find a natural number n_ε such that

$$\rho(q, q_n) < \frac{\varepsilon}{2} \text{ for } n \geq n_\varepsilon.$$

Since $q_n \in \Omega$, it is possible to find a t_ε such that

$$\rho(q_n, f(p, t_\varepsilon)) < \frac{\varepsilon}{2}.$$

Choosing the sequence $\varepsilon_i \rightarrow 0$, we can construct a sequence τ_i such that $\tau_i \rightarrow +\infty$ and $f(p, \tau_i) \rightarrow q$ which means that Ω is closed in R .

REMARK. If the motion $f(p, t)$ for $t \geq 0$ remains in a certain set G which is compact in R , then this motion has ω -limit points and the set Ω of ω -limit points is contained in $\bar{G}$.

9. Formulation of the problem of stability of invariant sets

Principal definitions

DEFINITION 1. An invariant set M closed in R is called *stable in the sense of Lyapunov* if for any $\varepsilon > 0$ it is possible to find a quantity $\delta > 0$ such that when $\rho(p, M) < \delta$ we have $\rho(f(p, t), M) < \varepsilon$ for $t \geq 0$. If furthermore

$$\rho(f(p, t), M) \rightarrow 0$$

as $t \rightarrow +\infty$, then the closed invariant set M is called *asymptotically stable*. Let us clarify this definition by means of the following examples.

1. We consider the system of differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n), \quad (s = 1, \dots, n) \text{ and } f_s(0, \dots, 0) = 0. \quad (1.9)$$

Assume that the aggregate of such equations defines a dynamical system in E_n ; then the point $X = 0$ is a rest point of this dynamical system. $X = 0$ is a closed invariant set. According to Definition 1 (see Introduction) it will be stable in the sense of Lyapunov, if for any $\varepsilon > 0$ it is possible to find a $\delta > 0$ such that for $\|X^{(0)}\| < \delta$ we have $\|X(t, X^{(0)})\| < \varepsilon$ for $t \geq 0$. Thus, Definition 1 coincides in this case verbatim with the definition of the stability in the sense of Lyapunov of the trivial solution of the system (1.9). Let us assume furthermore that the system (1.9) has a periodic solution

$$X = X(t, Y_0) = X(t + \omega, Y_0); \quad 0 < \omega < +\infty.$$

This periodic solution of the system (1.9) forms a closed invariant set of the dynamical system $X(t, X^{(0)})$. This invariant set, according to Definition 1, will be stable if for any $\varepsilon > 0$ it is possible to find a quantity $\delta > 0$ such that when

$$\inf_{t \in [0, \omega]} \|X^{(0)} - X(t, Y_0)\| < \delta$$

we have

$$\inf_{\tau \in [0, \omega]} \|X(t, X^{(0)}) - X(\tau, Y_0)\| < \varepsilon \text{ for } t \geq 0.$$

2. Let us examine the system of partial differential equations

$$\frac{\partial u_s}{\partial t} = f_s\left(x_1, \dots, x_k, u_1, \dots, u_n, \frac{\partial u_i}{\partial x_j}\right) \quad (1.10)$$

$$(s = 1, \dots, n; j = 1, \dots, k; i = 1, \dots, n).$$

We assume that the right members of the system (1.10) are analytic in a certain region G of space, in which the arguments of its right members vary. We assume also that in a certain functional space E the aggregate of these equations defines a dynamical system with a rest point $U = 0$, so that $U = 0$ is a solution of the system (1.10). According to Definition 1, for the solution $U = 0$ to be stable it is necessary that for any $\varepsilon > 0$ one be able to find a

quantity $\delta > 0$ such that when $\|\varphi\| < \delta$ we have $\|U(\varphi, t)\| < \varepsilon$, $t \in (0, +\infty)$, $\varphi \in E$.

3. Let us consider the system

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n). \quad (1.11)$$

We assume that the functions $f_s(X, t)$ are defined in the region $\|X\| \leq h$, $t \geq 0$ and are continuously differentiable there. Consequently, under these assumptions a single integral curve of the system (1.11) passes through any point (X, t_0) of this region. Let $f_s(0, \dots, 0, t) \equiv 0$ for $s = 1, \dots, n$. The trivial solution of the system (1.11) is called *stable in the sense of Lyapunov* [1] *uniformly relative to $t_0 \geq 0$* if for any $\varepsilon > 0$ it is possible to find a quantity $\delta > 0$ such that when $\|X^{(0)}\| < \delta$ we have

$$\|X(t, X^{(0)}, t_0)\| < \varepsilon \text{ for } 0 \leq t - t_0 \leq t.$$

Let us show that we can construct a dynamical system in the space (X, t) , corresponding to Eqs. (1.11), such that from the stability of the trivial solution of the system (1.11) there follows the stability of the invariant set $X = 0$, and conversely. For this purpose we set

$$\begin{aligned} \bar{f}_s(x_1, \dots, x_n, t) = & \begin{cases} f_s(x_1, \dots, x_n, t) & \text{for } t \geq 1 \text{ and } |X| \leq h_1 < h, \\ 0 & \text{for } t \leq -1 \text{ and } |X| < +\infty, \\ \frac{3f_s - 2f'_s}{4}(t+1)^2 \frac{f'_s(X, 1) - f_s(X, 1)}{4}(t+1)^3 & \dots \dots \dots \text{for } t \in (-1, +1) \text{ and } |X| \leq h_1. \end{cases} \end{aligned}$$

We continue the functions $\bar{f}_s$, retaining the continuous differentiability in the region $t \geq -1$, $|X| \leq h_1$, in such a way that $\bar{f}_s \equiv 0$ when $|X| \geq h$. The system of equations

$$\frac{dx_s}{dt} = \bar{f}_s(x_1, \dots, x_n, t) \quad (1.12)$$

has a solution $X = X(t, X^{(0)}, t_0)$, which depends continuously on the initial conditions $(X^{(0)}, t_0)$ and which is defined for all real

values of $X^{(0)}, t_0$. Let us make in this system the following change of independent variable:

$$ds = dt \sqrt{1 + \sum_{s=1}^n f_s^2}.$$

Then the system

$$\frac{dx_k}{ds} = \frac{f_k(x_1, \dots, x_n, t)}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \quad (k = 1, \dots, n), \quad \frac{dt}{ds} = \frac{1}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \quad (1.13)$$

will define in the space (X, t) a dynamical system

$$X = X(s, X^{(0)}, t_0), \quad t = t(s, X^{(0)}, t_0), \quad (1.14)$$

for which the axis $X = 0$ is an invariant set, since it coincides with the motion $X = 0$ and $t = s$, $s \in (-\infty, +\infty)$. We shall show that from the stability of the trivial solution of the system (1.11), which is uniform in $t_0 \geq 0$, there follows the stability of the invariant set $X = 0$ of the dynamical system (1.14).

By virtue of the stability of the zero solution of the system (1.11), which is uniform in $t_0 \geq 0$, it is possible to find a quantity $\delta > 0$ such that

$$|X(t, X^{(0)}, t_0)| < \varepsilon \quad \text{for } 0 \leq t_0 \leq t$$

for $|X^{(0)}| < \delta$. For the obtained $\delta > 0$ we can find a quantity $\delta_1 > 0$ by virtue of the continuity of the dependence of the motion of the dynamical system (1.14) on the initial conditions so small that

$$|X(s, X^{(0)}, t_0)| < \delta \quad \text{for } |X^{(0)}| < \delta_1 \text{ and } s \in [0, S], \quad t_0 = -1.$$

Here the quantity S can be chosen so large as to make $t(S, X^{(0)}, -1) \geq 1$. Let us show that the determined quantity $\delta_1 > 0$ corresponds to the selected value of ε , in accordance with Definition 1. In fact, for $|X^{(0)}| < \delta_1$ we have, when $t_0 \leq -1$

$$|X| = |X^{(0)}| < \varepsilon \quad \text{as long as } t(s, X^{(0)}, t_0) \leq -1.$$

When $t(s, X^{(0)}, t_0) \geq -1$ the same inequality holds by virtue of the choice of the quantity δ_1 .

Thus, the quantity δ_1 is the one sought. The converse of this state-

ment is obvious. It is also easy to show that the trivial solution of the system (1.11) is asymptotically stable if and only if the invariant set $X = 0$ of the dynamical system (1.14) is asymptotically stable.

DEFINITION 2. A closed invariant set M of a dynamical system $f(p, t)$ is called *unstable in the sense of Lyapunov* if there exists a number $\varepsilon > 0$ such that for any $\delta > 0$ it is possible to find at least one point $p \in S(M, \delta)$ such that for a certain $t > 0$ we have $\rho(f(p, t), M) \geq \varepsilon$. The meaning of this definition can be readily explained with the foregoing examples.

DEFINITION 3. If the closed invariant set M of a dynamical system $f(p, t)$ is asymptotically stable, then the set A of all points $p \in R$ and $p \in M$, having the property

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

is called the *region of asymptotic stability* of M .

THEOREM 4. The region of asymptotic stability of the invariant set M of a dynamical system $f(p, t)$ is an open set, containing a sufficiently small neighborhood of M .

PROOF. Let the invariant set M be asymptotically stable. Then for any $\varepsilon > 0$, it is possible, according to Definition 1, to find a quantity $\delta > 0$ such that for $\rho(p, M) < \delta$ we have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t \geq 0$$

and

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

from which it follows that any point in the set $S(M, \delta)$ is contained in A . Thus, A contains a sufficiently small neighborhood of the set M . Let us now show that the set A is open. Let $p \in A$. We take $\varepsilon > 0$ such that $S(M, \varepsilon) \subset A$. For the chosen $\varepsilon > 0$ it is possible, according to Definition 1, to find a quantity $\delta > 0$, for which in turn it is possible to define a quantity $T > 0$ such that

$$\rho(f(p, T), M) < \frac{1}{2}\varepsilon.$$

The quantity T exists by virtue of the fact that $p \in A$. From the theorem on the continuous dependence of the quantities T and $\varepsilon_1 > 0$ on the initial conditions, it is possible to choose $\gamma > 0$ such that for $\rho(p, q) < \gamma$ we have

$$\rho(f(p, T), f(q, T)) < \varepsilon_1.$$

By virtue of the continuity of the functional $\rho(p, M)$ (see Sec. 3) the quantity ε_1 can be chosen so small that $\rho(f(q, T), M) < \delta$. Then

$$\rho(f(\bar{q}, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

where $\bar{q} = f(q, T)$. As a result of the property 3 of the dynamical system, $f(\bar{q}, t) = f(q, t + T)$, and from this we have

$$\rho(f(q, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Consequently, $q \in A$ for $\rho(p, q) < \gamma$. This completes the proof of Theorem 4.

DEFINITION 4. The *boundary* of the region of asymptotic stability is the non-empty set of all points $q \in \bar{A} \setminus A$ and $q \in M$. As follows from Theorem 2, the set $(\bar{A} \setminus A) \setminus (\bar{A} \cap M)$ is invariant. Consequently, the boundary of the region of asymptotic stability is also an invariant set. It is the task of the present chapter to perform the following: 1) find the conditions for stability of a certain invariant set M of a dynamical system $f(p, t)$; 2) provide a method for finding the boundary of the region of asymptotic stability; 3) give methods that permit an estimate of the function $\rho(f(p, t), M)$, $t \geq 0$. These problems will henceforth be considered as the principal problems.

10. Uniformly asymptotically stable and uniformly attracting invariant sets of a dynamical system $f(p, t)$

DEFINITION 1. An asymptotically stable closed invariant set M of a dynamical system $f(p, t)$ is called *uniformly asymptotically stable* if it is possible to find a quantity $\delta_1 > 0$ such that for any $H > 0$ there exists a $T(H) > 0$, having the following property:

$$\rho(f(p, t), M) < H$$

when $\rho(p, M) < \delta_1$ and for $t \geq T$. Let us analyze the meaning of this definition by using a specific example. We consider the equation $\dot{x} = g(t)x$. The function $g(t)$ is defined for $t \in (-\infty, +\infty)$ and is continuous there,

$$g(t) \leq 0, \quad \int_{t_0}^t g(t) dt \rightarrow -\infty \text{ as } t \rightarrow +\infty.$$

We introduce a new independent variable using the formula $d\tau = dt$. The system

$$\begin{aligned}\frac{dx}{d\tau} &= g(t)x, \\ \frac{dt}{d\tau} &= 1\end{aligned}$$

defines the dynamical system

$$\left. \begin{aligned}x &= x_0 e^{\int_{t_0}^{t_0+\tau} g(\theta) d\theta} \\ t &= \tau + t_0\end{aligned} \right\} \quad (1.15)$$

in the plane (x, t) .

The axis $x = 0$ is an asymptotically stable invariant set of this dynamical system.

In order for this invariant set to be uniformly asymptotically stable, it is necessary and sufficient in this case that

$$\int_{t_0}^{t_0+\tau} g(\theta) d\theta \rightarrow -\infty \text{ as } \tau \rightarrow +\infty$$

uniformly relative to $t_0 \in (-\infty, +\infty)$. If, for example,

$$g(t) = -\frac{1}{1+|t|}, \text{ then } x = \frac{x_0 t_0}{t_0 + \tau + 1} \text{ and } t = t_0 + \tau, t_0 \geq 0, \tau \geq 0.$$

It is clear that in this case $|x| \rightarrow 0$ as $\tau \rightarrow +\infty$, but not uniformly relative to t_0 .

THEOREM 5. *If a closed invariant set M of a dynamical system $f(p, t)$ is uniformly asymptotically stable, then there exists for $t \in (-\infty, +\infty)$ a continuous function $L(t)$, which decreases strictly monotonically from $+\infty$ to zero, as t increases from $-\infty$ to $+\infty$, and such that $\rho(f(p, t), M) \leq L(t)$ when $t \geq 0$ and $\rho(p, M) < \delta_1$, where δ_1 is the quantity given in Definition 1 of this section.*

PROOF. Let us set

$$\lambda(t) = \sup_{p \in S(M, \delta_1)} \rho(f(p, t), M).$$

From Definition 1 of the present section it follows that the function $\lambda(t)$ is defined for $t \geq 0$, $\lambda(t) > 0$ and $\lambda(t) \rightarrow 0$ as $t \rightarrow +\infty$. $\lambda(t) \leq \varepsilon_1$ for $t \in (0, +\infty)$, where $\varepsilon_1 < +\infty$. For the quantity $\varepsilon_1/2^k$ it is possible to find a quantity t_k such that when $t \geq t_k$ we have $\lambda(t) < \varepsilon_1/2^k$. Here we can choose the quantities t_k such that $t_k \rightarrow +\infty$. Let us set

$$l_k(t) = \frac{2\varepsilon_1 t_{k+1} - \varepsilon_1(t_k + t)}{2^k(t_{k+1} - t_k)}$$

for $t \in [t_k, t_{k+1}]$ and $k \geq 1$.

It is clear that $l_k(t) \geq \lambda(t)$ when $t \in [t_k, t_{k+1}]$. We shall consider the function $l_1(t)$ which is defined for $t \in (-\infty, t_2)$. Let us put $L(t) = l_k(t)$ for $t \in [t_k, t_{k+1}]$. Here $k > 1$ and $L(t) = l_1(t)$ for $t \in (-\infty, t_2)$. It is easy to see that the function $L(t)$ satisfies all the conditions of Theorem 5.

DEFINITION 2. An asymptotically stable closed invariant set M of a dynamical system $f(p, t)$ is called *uniformly attracting* if it is possible to find a number $\delta_2 > 0$ such that for any $0 < h < \delta_2$, there exist numbers $T > 0$ and $\alpha > 0$ such that when $h < \rho(p, M) < \delta_2$ and $t \in [0, T]$ we have $\rho(f(p, t), M) > \alpha$. Let us clarify this definition by means of the example given above. In order for the asymptotically stable closed invariant set $x = 0$ of the dynamical system (1.15) to be uniformly attracting, it is necessary and sufficient that $\int_{t_0}^{t_0+\tau} g(\theta) d\theta$ be uniformly bounded from below relative to $t_0 \in (-\infty, +\infty)$ for any finite $\tau > 0$.

Let us set $g(t) = -|t|$. Then for $t_0 \geq 0$ and $\tau \geq 0$

$$x = x_0 e^{-\tau t_0 - \frac{1}{2}\tau^2},$$

$$t = \tau + t_0.$$

It is clear that $|x| \rightarrow 0$ as $t_0 \rightarrow +\infty$, no matter what the value of $\tau > 0$. This means that the invariant set $x = 0$ is not uniformly attracting.

THEOREM 6. An asymptotically stable closed invariant set M of a dynamical system $f(p, t)$, having a sufficiently small compact neighborhood, is uniformly asymptotically stable and uniformly attracting.

PROOF. Let us show first that M is uniformly asymptotically stable. Let us assume the opposite. Then, no matter what $\delta_1 > 0$

is chosen, there exists at least one number $H > 0$ such that it is possible to find two sequences p_n and t_n such that

$$\rho(p_n, M) < \delta_1 \text{ and } \rho(f(p_n, t_n), M) \geq H, t_n \rightarrow +\infty.$$

The quantity δ_1 can be chosen to be so small that it becomes possible to select from the sequence $p_1, \dots, p_n, \dots$ a converging subsequence p_{n_k} . Let $p_{n_k} \rightarrow p_0$ as $k \rightarrow +\infty$.

It is clear that $p_0 \in A$ — the region of asymptotic stability of the set M . It is therefore possible to find a quantity $T(H)$ such that $\rho(f(p_0, T), M) < \delta(H)$, where the quantity $\delta > 0$ is chosen with respect to H in accordance with Definition 1 of Section 9. For the quantity $T(H)$ and $r_2 > 0$, by virtue of the theorem on continuity with respect to the initial conditions, we can find a quantity $r_1 > 0$ such that when $\rho(p_0, q) < r_1$ we have $\rho(f(p_0, t), f(q, t)) < r_2$ when $t \in [0, T]$. Using the continuity of the functional $\rho(p, M)$ it is possible to choose the quantity r_2 so small as to obtain

$$\rho(f(q, T), M) < \delta(H).$$

Then

$$\rho(f(q, t), M) < H \text{ for } \rho(p_0, q) < r_1 \text{ and } t \geq T.$$

Let us choose, furthermore, the index k such as to make $t_{n_k} > T$ and $\rho(p_0, p_{n_k}) < r_1$; we then have, on the one hand, $\rho(f(p_{n_k}, t_{n_k}), M) \geq H$ and, by assumption, on the other, $\rho(f(p_{n_k}, t_{n_k}), M) < H$. The contradiction thus obtained proves the theorem.

Let us now show that M is uniformly attracting. Let us assume the opposite, i.e., no matter what δ_2 is chosen, there exists at least one quantity $h > 0$ such that for any quantity $T > 0$ we obtain

$$\inf_{\substack{t \in [0, T] \\ h < \rho(p, M) < \delta_2}} \rho(f(p, t), M) = 0.$$

In this case it is possible to find sequences p_n and t_n , α_n such that $t_n \in [0, T]$, $h < \rho(p_n, M) < \delta_2$, $\alpha_n > 0$ and $\alpha_n \rightarrow 0$, $n \rightarrow +\infty$ and $\rho(f(p_n, t_n), M) \leq \alpha_n$. By assumption, the set M has a sufficiently small compact neighborhood, and therefore the quantity δ_2 can be considered so small that it is possible to choose from the sequence p_n a convergent subsequence p_{n_k} . Let us assume that $p_{n_k} \rightarrow p_0$ and $t_{n_k} \rightarrow t_0 \in [0, T]$; then

$$f(p_{n_k}, t_{n_k}) \rightarrow f(p_0, t_0).$$

As a result of this and by virtue of the continuity of the functional $\rho(p, M)$ it is possible to state that $\rho(f(p_0, t_0), M) = 0$; consequently, $f(p_0, t_0) \in M$, which is impossible, since $\rho(p_0, M) \geq h > 0$ and M is invariant, i.e. it follows from $f(p_0, t_0) \in M$ that

$$f(f(p_0, t_0), t) \in M.$$

When $t = -t_0$, we obtain $p_0 \in M$.

The contradiction obtained shows that the assumption made above is incorrect. This proves the theorem completely.

REMARK. This theorem shows that necessary and sufficient conditions that a closed invariant set M be uniformly asymptotically stable and uniformly attracting (see Definition 2 of Section 10) are at the same time the necessary and sufficient conditions for asymptotic stability of the closed invariant set M , which has a sufficiently small compact neighborhood.

11. Qualitative description, from the point of view of the Lyapunov stability, of a neighborhood of an invariant set

THEOREM 7. A necessary and sufficient condition that a closed invariant set M be stable in the sense of Lyapunov is that the quantity

$$\delta(\varepsilon) = \inf_{\substack{t \leq 0 \\ \rho(p, M) = \varepsilon}} \rho(f(p, t), M)$$

be positive for arbitrary sufficiently small $\varepsilon > 0$.

PROOF. *Necessity.* It follows from the definition of the stability in the sense of Lyapunov of the invariant set M that there exists a function $\bar{\delta}(\varepsilon)$ for which the following two conditions are satisfied:

- 1) $\bar{\delta}(\varepsilon) > 0$ for arbitrary sufficiently small $\varepsilon > 0$,
- 2) $\rho(f(p, t), M) < \varepsilon$ for $t \geq 0$ provided $\rho(p, M) < \bar{\delta}(\varepsilon)$.

We shall show that the function $\delta(\varepsilon)$ indicated in the formulation of Theorem 7 satisfies the inequality $\delta(\varepsilon) \geq \bar{\delta}(\varepsilon)$ for arbitrary sufficiently small $\varepsilon > 0$. Let us assume the contrary; then there exists at least one sufficiently small $\bar{\varepsilon} > 0$ such that $\delta(\bar{\varepsilon}) < \bar{\delta}(\bar{\varepsilon})$ holds. It is easy to show, on the basis of this inequality, that there exist a point $\bar{p}$ and a moment of time $\bar{t}$, $\rho(\bar{p}, M) = \bar{\varepsilon}$, $\bar{t} < 0$, such that the point $\bar{Q} = f(\bar{p}, \bar{t})$ satisfies the condition $\rho(\bar{Q}, M) < \delta(\bar{\varepsilon})$ and $\rho(f(\bar{Q}, -\bar{t}), M) = \bar{\varepsilon}$. The last equality contradicts the

stability property, according to which we should have

$$\rho(f(\bar{Q}, t), M) < \bar{\varepsilon} \text{ for } t \geq 0.$$

The contradiction thus obtained shows that the quantity $\delta(\varepsilon)$ is positive for arbitrary sufficiently small $\varepsilon > 0$ in the presence of stability.

Sufficiency. We shall now show that if the condition of Theorem 7 is satisfied, then the closed invariant set M will be stable in the Lyapunov sense. We take an arbitrary function $\bar{\delta}(\varepsilon)$ satisfying the following two conditions:

1) $\bar{\delta}(\varepsilon)$ is defined for arbitrary sufficiently small $\varepsilon > 0$ and is positive,

2) $\bar{\delta}(\varepsilon) < \delta(\varepsilon)$ for all those ε for which the quantity $\delta(\varepsilon)$ is defined.

We shall show that $\bar{\delta}(\varepsilon)$ meets the requirements for any prescribed ε according to the definition of the stability in the sense of Lyapunov of the set M . In other words, $\rho(f(p, t), M) < \varepsilon$ for $t \geq 0$ provided $\rho(p, M) < \bar{\delta}(\varepsilon)$. In fact, suppose there exists at least one point $\bar{p}$ and at least one moment $\bar{t} > 0$ such that $\rho(\bar{p}, M) < \bar{\delta}(\varepsilon)$, and $\rho(f(\bar{p}, t), M) = \varepsilon$ for some ε which is sufficiently small and positive; then the point $\bar{Q} = f(\bar{p}, \bar{t})$ will possess the property that $\rho(\bar{Q}, M) = \varepsilon$, $\rho(f(\bar{Q}, -\bar{t}), M) < \bar{\delta}(\varepsilon)$, but, then, for this ε we must necessarily have that $\delta(\varepsilon) < \bar{\delta}(\varepsilon)$, which contradicts the choice of the function $\bar{\delta}(\varepsilon)$. The contradiction thus obtained shows that the set M will be stable in the sense of Lyapunov when the condition of Theorem 7 is satisfied.

REMARK 1. The condition of Theorem 7 is included in the inequality $\delta(\varepsilon) > 0$ for arbitrary sufficiently small $\varepsilon > 0$. This condition is equivalent to the absence of motions $f(p, t)$, $\rho(p, M) = \varepsilon$, having α -limit points in M with one additional condition consisting in that the set M has a sufficiently small compact neighborhood which does not contain entire trajectories of the dynamical system under consideration.

In fact, suppose there exists a sufficiently small compact neighborhood of M which does not contain entire trajectories of the dynamical system $f(p, t)$. If there do not exist motions $f(p, t)$ such that $p \in M$ and $f(p, t)$ does not have α -limit points situated in M , then $\delta(\varepsilon) > 0$ for arbitrary sufficiently small $\varepsilon > 0$. We denote

by $\delta(p, \varepsilon)$ the function defined by the equality

$$\inf_{t \leq 0} \rho(f(p, t), M) = \delta(p, \varepsilon),$$

where p satisfies the equality $\rho(p, M) = \varepsilon$. Then, by virtue of the absence of α -limit points in the set M for any motion $f(p, t)$, we shall have $\delta(p, \varepsilon) > 0$ for $\varepsilon > 0$. Clearly,

$$\delta(\varepsilon) = \inf_{\substack{p \\ \rho(p, M) = \varepsilon}} \delta(p, \varepsilon).$$

If $\delta(\varepsilon) = 0$ for some $\varepsilon > 0$, then there exists a sequence of points p_N for which the following two conditions are satisfied:

- 1) $\delta(p_N, \varepsilon) \rightarrow 0, \quad N \rightarrow \infty$
- 2) $\rho(p_N, M) = \varepsilon, \quad N = 1, 2, \dots$

An arbitrary semi-trajectory $f(p_N, t), t \leq 0$, does not remain in the ε -neighborhood of the set M since, by virtue of the assumption of the compactness of this neighborhood, there must necessarily exist in it at least one entire trajectory belonging to the α -limit set of such semi-trajectory and contained entirely in the neighborhood mentioned earlier. Thus, an arbitrary motion $f(p_N, t)$ leaves the set $\rho(p, M) < \varepsilon$ after a finite time and, consequently, there exists a sequence t_N such that $t_N < 0$ and $\rho(f(p_N, t_N), M) = \delta(p_N, \varepsilon)$. Further, the sequence t_N certainly possesses the property that $t_N \rightarrow -\infty$, for, otherwise there would exist a bounded sub-

sequence of this sequence and then, by virtue of the compactness of the chosen neighborhood of the set M there would exist a motion which, starting at the point $p_0, \rho(p_0, M) = \varepsilon$, would reach the set M in a finite time, which contradicts the invariance of this set. It follows from the properties of the sequence t_N and the compactness of the chosen neighborhood that there exists a point Q_0 for which the following two conditions are satisfied:

- 1) $\rho(Q_0, M) = \varepsilon,$
- 2) $\rho(f(Q_0, t), M) \leq \varepsilon$ for $t \leq 0$.

Q_0 is one of the limit points of the sequence Q_N . This semi-trajectory cannot leave the closure of the ε -neighborhood of the set M since otherwise motions starting in a sufficiently small neighborhood of Q_0 would also leave the above-mentioned closure of the neighbor-

hood of Q_0 in finite time, which is impossible because among such motions there is a motion passing through points p_N , which negative semi-trajectories remain an arbitrarily long time in the ε -neighborhood.

Thus, the assumption that $\delta(\varepsilon) = 0$ for $\varepsilon > 0$ leads to the existence of a motion $f(Q_0, t)$ such that $\rho(f(Q_0, t), M) \leq \varepsilon$ for $t \leq 0$ and $\rho(Q_0, M) = \varepsilon$. This situation contradicts the assumption according to which entire trajectories of the dynamical system are absent in this neighborhood.

The converse assertion is obvious: if $\delta(\varepsilon) > 0$ for $\varepsilon > 0$, then $\delta(p, \varepsilon) > 0$, and, consequently, there are no motions whose limit points lie in the set M .

Taking the assertion just proved as point of departure, one can give a modified formulation of Theorem 7, namely: *A necessary and sufficient condition for a closed invariant set M , having a sufficiently small compact neighborhood not containing entire trajectories, to be stable in the sense of Lyapunov is that there do not exist motions $f(p, t)$, $p \in M$, having α -limit points situated in M .* Subsequent theorems will make use of precisely this modified formulation of Theorem 7.

REMARK 2. In the Russian edition of this book, 1957, Theorem 7 was formulated incorrectly, which fact was pointed out to the author by Solomon Lefschetz. This prompted the present correction. The author is very happy to have the opportunity here to once again express his profound gratitude to Solomon Lefschetz for his attentive and kindly regard for the author and his work.

THEOREM 8. *In order for a closed invariant set M of a dynamical system $f(p, t)$ with a sufficiently small compact neighborhood to be asymptotically stable, it is necessary and sufficient to satisfy the following conditions:*

1) *There exists no motion $f(p, t)$ which has α -limit points in M , $p \in M$;*

2) *There exists a sufficiently small neighborhood of the set M , which does not contain trajectories of the dynamical system $f(p, t)$.*

PROOF. *Necessity.* Let the closed invariant set M be asymptotically stable; then it is simply stable. Therefore Condition 1 is satisfied by virtue of Theorem 7. We shall show that Condition 2 is also satisfied. Suppose, to the contrary, no matter what neigh-

neighborhood of M is taken, that it contains at least one point $\bar{p}$ such that the motion $f(\bar{p}, t)$ remains in this neighborhood for $t \in (-\infty, +\infty)$. We take any sequence $t_n \rightarrow -\infty$. Then from the sequence $p_n = f(\bar{p}, t_n)$ it is possible to choose a convergent subsequence p_{n_k} , by virtue of the compactness of the sufficiently small neighborhood of the set M . Let $p_{n_k} \rightarrow p_0$. Condition 1 is satisfied, and therefore we have

$$\inf_{t < 0} \rho(f(\bar{p}, t), M) = \delta(\bar{p}) > 0.$$

For the quantity $\varepsilon = \delta(\bar{p})$, according to Definition 1 of Section 9, it is possible to find a certain quantity $\delta > 0$. For this δ we can find a quantity $T > 0$ such that $\rho(f(p_0, T), M) < \delta$. Next, according to the theorem on continuity of the initial conditions, one can find for T and $r_1 > 0$ a quantity $r_2 > 0$ such that when $\rho(p_0, q) < r_2$ we obtain

$$\rho(f(p_0, T), f(q, T)) < r_1.$$

As a result of the continuity of the functional $\rho(p, M)$, the quantity r_1 can be chosen so small as to make

$$\rho(f(q, T), M) < \delta \text{ for } \rho(p_0, q) < r_2.$$

For the quantities r_2 and T it is possible to find a value of n_k so large that

$$T + t_{n_k} < 0 \text{ and } \rho(p_0, p_{n_k}) < r_2.$$

We then have, on the one hand,

$$\rho(f(p_{n_k}, T), M) < \delta$$

and, on the other hand,

$$f(p_{n_k}, T) = f(\bar{p}, T + t_{n_k}).$$

Therefore

$$\rho(f(p_{n_k}, T), M) \geq \delta(\bar{p}).$$

The resultant contradiction shows that the assumption made is incorrect. Consequently, Condition 2 is satisfied. Thus, the necessity of the conditions is fully proved.

Sufficiency. Let Conditions 1 and 2 be satisfied. Let us show that M is asymptotically stable. As shown in Theorem 7, the set

M is stable when Condition 1 is satisfied. Consequently, for any sufficiently small $\varepsilon > 0$ it is possible to find a corresponding $\delta > 0$ such that when $\rho(p, M) \leq \delta$ we have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t \geq 0.$$

Let us show that in this case we also have that

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Suppose this is not so. Then there exists a point $\bar{p}$ such that

$$\rho(\bar{p}, M) \leq \delta \text{ and } \rho(f(\bar{p}, t), M) \not\rightarrow 0$$

as $t \rightarrow +\infty$. Then by virtue of the compactness of the sufficiently small ε -neighborhood of the set M it is possible to find a sequence $t_n \rightarrow +\infty$ such that

$$p_n = f(\bar{p}, t_n) \rightarrow q \text{ as } n \rightarrow +\infty, q \in M \text{ and } \rho(q, M) \leq \delta.$$

The motion $f(q, t)$ consists entirely of ω -limit points of the motion $f(\bar{p}, t)$. Consequently, the distance $\rho(f(q, t), M) \leq \varepsilon$ for $t \in (-\infty, +\infty)$. And then in the ε -neighborhood of the set M there is contained a trajectory of the dynamical system $f(p, t)$, which contradicts Condition 2. Thus, no matter what the point p may be, $\rho(p, M) < \delta$, we have

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

This completes the proof of the theorem.

LEMMA. *In order for a closed invariant set M with a sufficiently small compact neighborhood that does not contain a trajectory of the dynamical system $f(p, t)$ not to have an α -limit point for a certain motion $p \in M$, it is necessary and sufficient that there exists no motion $f(q, t)$ such that $q \in M$, and $\rho(f(q, t), M) \rightarrow 0$ as $t \rightarrow -\infty$.*

PROOF. *Necessity.* If there exists a point p , $p \in M$, such that $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow -\infty$, then there exists a sequence $t_n \rightarrow -\infty$ such that $\rho(p_n, M) \rightarrow 0$ as $n \rightarrow +\infty$, $p_n = f(p, t_n)$. From the sequence $p_1, \dots, p_n, \dots$, by virtue of the compactness of the sufficiently small neighborhood of the set M , it is possible to choose a converging subsequence $p_{n_k} \rightarrow q$. We then have

$$q \in M, \rho(q, f(p, t_{n_k})) \rightarrow 0 \text{ as } n_k \rightarrow +\infty.$$

Consequently, the point q is an α -limit point of the motion $f(p, t)$. Thus, if there exists a motion $f(p, t)$, $p \in M$, having the property $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow -\infty$, then this motion has α -limit points contained in M . The necessity of this condition can be considered proved.

Sufficiency. Let us assume that there exists no motion $f(p, t)$, $p \in M$, $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow -\infty$. We then show that there exists no motion $f(\bar{p}, t)$ which has α -limit points in M . Suppose, to the contrary, that there exists a point $\bar{p} \in M$, such that certain of the α -limit points of the motion $f(\bar{p}, t)$ are contained in M . Then for any sufficiently small $\varepsilon > 0$, the equality $\rho(f(\bar{p}, t), M) = \varepsilon$ is realized no matter how large the values of $|t|$. Let us take two sequences t_n and t'_n , $t'_n < t_n$, such that

$$t_n \rightarrow -\infty, \rho(f(\bar{p}, t_n), M) = \varepsilon, \rho(f(\bar{p}, t'_n), M) = \varepsilon,$$

and

$$\rho(f(\bar{p}, t), M) < \varepsilon \text{ for } t \in (t_n, t'_n),$$

and

$$\inf_{t \in [t_n, t'_n]} \rho(f(\bar{p}, t), M) = \beta_n \rightarrow 0 \text{ as } n \rightarrow +\infty.$$

Let us set $p_n = f(\bar{p}, t_n)$. As a result of the compactness of the sufficiently small neighborhood of the set M , the sequence t_n can be considered chosen such that $p_n \rightarrow p_0$ as $n \rightarrow +\infty$. Using Theorem 1 concerning the continuous dependence on the initial conditions, we can show that $\rho(f(p_0, t), M) \leq \varepsilon$ for $t \leq 0$. We then show that the motion $f(q, t)$, where q is an α -limit point for the motion $f(p_0, t)$, $q \in M$, has the property $\rho(f(q, t), M) \leq \varepsilon$ for $t \in (-\infty, +\infty)$, which leads us to a contradiction with the condition of the lemma, since $\varepsilon > 0$ was chosen from the very beginning as small as desired.

In carrying out the program described above, let us first show that

$$t'_n - t_n \rightarrow -\infty \text{ as } n \rightarrow +\infty.$$

Let, to the contrary, $t'_n - t_n \rightarrow -\infty$. Furthermore, let the following bounds hold: $t'_n - t_n \geq N$, $N > -\infty$. We choose the quantities τ_n as so to have

$$\rho(f(\bar{p}, t_n + \tau_n), M) = \beta_n.$$

It is clear that τ_n satisfies the inequalities $N \leq \tau_n \leq 0$. As a result

of this, we can choose from the sequence τ_n a converging subsequence. We shall assume that τ_n is from the outset such that $\tau_n \rightarrow t_0$. We then have from $p_n \rightarrow p_0$, $\tau_n \rightarrow t_0$ that

$$f(p_n, \tau_n) \rightarrow f(p_0, t_0) = q_0.$$

It follows from this that, on the one hand, $q_0 \in M$, and on the other hand $f(q_0, -t_0) = p_0$, $p_0 \notin M$, which is impossible, for the set M is invariant. Thus, the assumption $t'_n - t_n \geq N$ is incorrect. Furthermore, it is not satisfied even for any infinite subsequence $t'_{n_k} - t_{n_k}$. Consequently, $t'_n - t_n \rightarrow -\infty$. Let us show further that

$$\rho(f(p_0, t), M) \leq \varepsilon \text{ for } t \leq 0.$$

In fact, if there exists a finite quantity $T < 0$ such that

$$\rho(f(p_0, T), M) > \varepsilon,$$

then for sufficiently large values of n , by virtue of the continuity with respect to the initial conditions, we have

$$\rho(f(p_n, T), M) > \varepsilon,$$

which is impossible because

$$\rho(f(p_n, t), M) \leq \varepsilon \text{ for } t \in [0, t'_n - t_n].$$

Thus

$$\rho(f(p_0, t), M) \leq \varepsilon \text{ for } t \leq 0.$$

As a result of this and by virtue of the compactness of the sufficiently small neighborhood of the set M , there exists a point $q \in M$, which is an α -limit point for the motion $f(p_0, t)$. The motion $f(q, t)$ consists of α -limit points of the motion $f(p_0, t)$, and consequently

$$\rho(f(q, t), M) \leq \varepsilon \text{ for } t \in (-\infty, +\infty),$$

which contradicts the assumption made in the theorem concerning the absence of trajectories of the dynamical system in a sufficiently small neighborhood. This completes the proof of the lemma.

THEOREM 9. *In order for a closed invariant set M with a sufficiently small compact neighborhood to be asymptotically stable, it is necessary and sufficient that the following conditions be satisfied:*

- 1) *There exists no motion $f(p, t)$, $p \notin M$, such that*

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow -\infty;$$

2) *There exists a sufficiently small neighborhood of the set M , containing no trajectories of the dynamical system $f(p, t)$.*

PROOF. It follows from the lemma that conditions 1 and 2 of Theorem 9 are satisfied if and only if Conditions 1 and 2 of Theorem 8 are satisfied. Thus, Theorem 9 represents a different formulation of Theorem 8.

It should be noted here that Theorem 9 allows us to make specific conclusions concerning the presence of asymptotic stability of invariant sets of dynamical systems, defined by a system of differential equations.

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n) \quad (1.16)$$

We assume that the functions $f_s(x_1, \dots, x_n)$ are defined in E_n and satisfy there certain conditions, which guarantee the existence of a unique solution $X = X(t, X^{(0)})$, which is continuous in $X^{(0)}$, t and which passes through the point $X^{(0)}$ and $f_s(0, \dots, 0) = 0$ when $t = 0$.

THEOREM 10. *In order for the trivial solution of the system (1.16) to be asymptotically stable, it is necessary and sufficient to satisfy the following conditions:*

- 1) *There exists no integral curve of the system (1.16) $X = X(t, X^{(0)})$, $|X^{(0)}| \neq 0$ such that $|X| \rightarrow 0$ as $t \rightarrow -\infty$;*
- 2) *There exists a sufficiently small neighborhood of the point $X = 0$, which contains no integral curve*

$$X = X(t, X^{(0)}), t \in (-\infty, +\infty); |X^{(0)}| \neq 0.$$

PROOF. By changing the independent variable in accordance with the formula

$$ds = dt \sqrt{1 + \sum_{i=1}^n f_i^2}$$

we set up the system of equations

$$\frac{dx_k}{ds} = \frac{f_k(x_1, \dots, x_n)}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \quad (k = 1, \dots, n). \quad (1.17)$$

It is clear that the system (1.17) defines in the space E_n a dynamical system $X = X(s, X^{(0)})$, for which the point $X = 0$ is a closed invariant set. Applying Theorem 9 to this dynamical system and considering the set M to coincide with $X = 0$, we can establish without difficulty the correctness of Theorem 10.

Let us examine the system (1.16) for $n = 2$. Let us assume that the point $x_1 = x_2 = 0$ is in this case an isolated equilibrium position for the system (1.16).

THEOREM 11. *In order for the point $x_1 = x_2 = 0$, which is an isolated equilibrium position for the system (1.16) for $n = 2$, to be asymptotically stable, it is necessary and sufficient to satisfy the following conditions:*

- 1) *There exists no integral curve of the system (1.16) $X = X(t, X^{(0)})$, $|X^{(0)}| \neq 0$, such that $|X| \rightarrow 0$ as $t \rightarrow -\infty$, $n = 2$,*
- 2) *There exists at least one integral curve $X = X(t, X^{(0)})$ of the system (1.16) for $n = 2$ such that $|X| \rightarrow 0$ as $t \rightarrow +\infty$.*

PROOF. Let us consider the dynamical system defined by the system of differential equations (1.17) for $n = 2$. We shall show that when Condition 2 of Theorem 11 is satisfied, there exists a sufficiently small neighborhood of the point $x_1 = x_2 = 0$, which contains no trajectories of the dynamical system $X = X(s, X^{(0)})$, $s \in (-\infty, +\infty)$. Let us assume the opposite, that in any sufficiently small neighborhood of $x_1 = x_2 = 0$ there exist trajectories of the dynamical system. Then there exists a periodic motion, which includes this point [5]. This follows from the fact that the point $x_1 = x_2 = 0$ is an isolated position of equilibrium for the system (1.17) and from Condition 1 of Theorem 11. However, the presence of the periodic motion in as small a neighborhood of the point $x_1 = x_2 = 0$ as desired contradicts Condition 2 of Theorem 11 for then the uniqueness is violated, namely: the integral curve $X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$ would intersect all such periodic solutions, which is impossible. Thus, there exists a sufficiently small neighborhood of the point $x_1 = x_2 = 0$, containing no trajectories $X = X(s, X^{(0)})$, $s \in (-\infty, +\infty)$. It follows from this that the satisfaction of Conditions 1 and 2 of Theorem 11 entails the satisfaction of Conditions 1 and 2 of Theorem 10, from which follows the correctness of Theorem 11.

This purely qualitative analysis of the behavior of the motions

of the dynamical system in a neighborhood of the invariant set allows us, for certain systems of differential equations of type (1.16), to give specific sufficient criteria for asymptotic stability of the trivial solution, and also of its instability. The sufficient condition for the instability of the trivial solution of this system (1.16) was given for the first time in reference [6] and involves the presence of an integral curve $X = X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow -\infty$.

12. Necessary and sufficient conditions for stability

THEOREM 12. *In order for a closed invariant set M to be stable in the sense of Lyapunov, it is necessary and sufficient that in a sufficiently small neighborhood $S(M, r)$ of the set M there exist a specified functional E with the following properties:*

1) *For any sufficiently small quantity $c_1 > 0$, it is possible to find a quantity $c_2 > 0$ such that $V(p) > c_2$ for $p \in S(M, r)$ and $\rho(p, M) > c_1$.*

2) *For any quantity $\gamma_2 > 0$ as small as desired, it is possible to find a quantity $\gamma_1 > 0$ so small that when $\rho(p, M) < \gamma_1$ we have $V(p) \leq \gamma_2$.*

3) *$V(f(p, t))$ is a non-increasing function for $t \geq 0$, for $p \in S(M, r)$, for all t , as long as $f(p, t) \in S(M, r)$.*

PROOF. *Necessity.* Let the closed invariant set M be stable in the sense of Lyapunov. Let us show that the conditions of the Theorem 12 are satisfied. We choose a certain $\varepsilon > 0$. According to Definition 1 of Section 9, there corresponds to it a certain quantity $\delta > 0$ such that when $\rho(p, M) < \delta$ we have $\rho(f(p, t), M) < \varepsilon$ for $t \geq 0$. Following reference [7], we set

$$V(p) = \sup_{t \geq 0} \rho(f(p, t), M).$$

This defines the functional $V(p)$ in $S(M, \delta)$. The functional $V(p)$ satisfies Condition 1, since $V(p) \geq \rho(p, M)$, from which it follows that when $\rho(p, M) > c_1$ and $\rho(p, M) < \delta$, we obtain

$$V(p) > c_2, \quad c_1 = c_2.$$

Let us show that Condition 2 is satisfied. For the quantity $\gamma_2 > 0$, one can find, in accordance with Definition 1, a quantity

$\gamma_1 > 0$ such that for $\rho(p, M) < \gamma_1$ we obtain

$$\rho(f(p, t), M) < \gamma_2 \text{ for } t \geq 0.$$

Consequently

$$\sup_{t \geq 0} \rho(f(p, t), M) \leq \gamma_2,$$

and then $V(p) \leq \gamma_2$ for $\rho(p, M) < \gamma_1$. This shows that the functional $V(p)$ satisfies Condition 2. We show further that $V(p)$ satisfies Condition 3. Let the point $p \in S(M, \delta)$; then for all values of $t \in [0, T)$ we have $f(p, t) \in S(M, \delta)$. Consequently, the value of the functional is defined at any point $f(p, \bar{t})$, $\bar{t} \in [0, T)$. It is clear that

$$\begin{aligned} V(f(p, \bar{t})) &= \sup_{t \geq 0} \rho(f(p, \bar{t}), t), M) = \\ &= \sup_{t \geq 0} \rho(f(p, t + \bar{t}), M) = \sup_{t \geq \bar{t}} \rho(f(p, t), M). \end{aligned}$$

Thus

$$V(f(p, \bar{t})) = \sup_{t \geq \bar{t}} \rho(f(p, t), M) \leq \sup_{t \geq 0} \rho(f(p, t), M) = V(p).$$

Consequently $V(f(p, \bar{t})) \leq V(p)$ for any sufficiently small $\bar{t} > 0$. This proves completely the necessity of Conditions 1–3 of Theorem 12.

Sufficiency. Let there exist a functional V , with the properties 1, 2 and 3 in a certain neighborhood of the set $S(M, r)$. Let us show that the closed invariant set M is stable in the sense of Lyapunov. We take $\varepsilon > 0$ and $\varepsilon < r$ and set, following Lyapunov,

$$\lambda = \inf V(p) \text{ for } \rho(p, M) = \varepsilon.$$

By virtue of the Property 1, we have $\lambda > 0$. On the basis of property 2 it is possible to find for the quantity λ another quantity $\delta > 0$ such that for $\rho(p, M) < \delta$ we have $V(p) < \lambda$. Let us show that the quantity $\delta > 0$ thus found corresponds to the chosen $\varepsilon > 0$ in accordance with Definition 1 of Section 9, i.e. when $\rho(p, M) < \delta$ we obtain

$$\rho(f(p, t), M) < \varepsilon \text{ for } t \geq 0.$$

Let us assume the opposite, namely let there exist a point $p \in S(M, \delta)$ such that at a certain finite value $\bar{t} > 0$ the equality $\rho(f(p, \bar{t}), M) = \varepsilon$

holds. Then

$$V(f(p, t)) \geq \lambda.$$

But by virtue of Property 3,

$$V(f(p, t)) \leq V(p) < \lambda.$$

We thus obtain a contradiction, which shows that the assumption we made is incorrect and consequently the invariant set M is stable in the sense of Lyapunov.

THEOREM 13. *In order for a closed invariant set M of a dynamical system $f(p, t)$ to be asymptotically stable in the sense of Lyapunov, it is necessary and sufficient that there exist in a certain sufficiently small neighborhood $S(M, r)$, $r > 0$, of the set M a functional V , having all the properties 1–3, indicated in the Theorem 12, and furthermore*

$$V(f(p, t)) \rightarrow 0 \text{ as } t \rightarrow +\infty$$

for any motion

$$f(p, t) \in S(M, r) \text{ for } t \in [0, +\infty).$$

PROOF. Necessity. Let M be asymptotically stable. Then the set M is stable in the sense of Lyapunov and consequently in a certain sufficiently small neighborhood $S(M, \delta)$ it is possible to construct a functional $V(p)$ which satisfies conditions 1–3 of Theorem 12. By virtue of the asymptotic stability of M , all the motions $f(p, t)$ which begin in the δ_1 -neighborhood of the set M remain in $S(M, \delta)$ when $t \geq 0$. Let $f(p, t)$ be one of these motions. Let us show that

$$V(f(p, t)) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

For $\varepsilon' > 0$ we can find a quantity $T > 0$ such that

$$\rho(f(p, t), M) < \varepsilon' \text{ for } t \geq T.$$

The existence of the quantity T , having the necessary property, follows from the fact of the asymptotic stability.

It is clear that

$$V(f(p, t)) = \sup_{\tau \geq 0} \rho(f(p, t + \tau), M).$$

It follows from

$$\rho(f(p, T + \tau), M) < \varepsilon' \text{ for } \tau \geq 0$$

that

$$V(f(p, t)) \leq \epsilon' \text{ for } t \geq T.$$

Thus, $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$. This completes the proof of the necessity of the conditions.

Sufficiency. Let the conditions of Theorem 13 be satisfied. Let us prove that the invariant set M is asymptotically stable. From the satisfaction of the conditions of Theorem 13 it follows that in the neighborhood $S(M, r)$ there exists a functional $V(p)$, satisfying Conditions 1–3 of Theorem 12. Consequently, the set M is stable in the sense of Lyapunov, i.e. for any $\epsilon > 0$ it is possible to find a quantity $\delta > 0$ such that when $\rho(p, M) < \delta$ we have $\rho(f(p, t), M) < \epsilon$. Let us show that the quantity δ can at the same time be chosen such as to make $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow +\infty$ and for $\rho(p, M) < \delta$. In fact, for the value of $\delta > 0$ obtained, we construct by means of the process indicated in Theorem 12 (as for ϵ) a quantity δ_1 such that when $\rho(p, M) < \delta_1$ we have $\rho(f(p, t), M) < \delta$ at $t \geq 0$. It is clear that $V(f(p, t))$ is defined for $t \in [0, +\infty)$ for any $p \in S(M, \delta_1)$. Let us show that the quantity δ_1 is the one sought. In fact, assume that this is not so, i.e., there exists at least one point $p \in S(M, \delta_1)$ such that $\rho(f(p, t), M) > \gamma_1 > 0$ for $t \geq 0$. We then have $V(f(p, t)) > \gamma_2 > 0$ in accordance with property 1, which contradicts the condition $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$. This completes the proof of the theorem.

REMARK. In proving the theorem we used the following readily-established fact: if the set M is stable, the condition $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow +\infty$ is equivalent to $\rho(f(p, t), M) > c_1 > 0$ for $t \in [0, +\infty)$.

THEOREM 14. *In order for a closed invariant set M to be uniformly asymptotically stable, it is necessary and sufficient that in some neighborhood $S(M, r)$ of the set M there exist a functional V , satisfying all the conditions of Theorem 13, and, furthermore, that there exist a quantity $\delta_1 > 0$ such that $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$ uniformly relative to $p \in S(M, \delta_1)$.*

PROOF. Necessity. Let the set M be uniformly asymptotically stable. We shall show that the conditions of theorem 14 are satisfied. As follows from Definition 1 of Section 10, the invariant set M will be asymptotically stable. Therefore there exists a functional V , satisfying the conditions of Theorem 13. According to Definition 1

of Section 10, there exists a quantity $\delta_1 > 0$ such that for any quantity $H > 0$ it is possible to find a quantity $T(H) > 0$ such that

$$\rho(f(p, t), M) < H \text{ for } t \geq T(H)$$

for $p \in S(M, \delta_1)$. From the definition of the functional we have

$$V(f(p, t)) = \sup_{\tau \geq 0} \rho(f(p, t + \tau), M).$$

When $p \in S(M, \delta_1)$, we have

$$V(f(p, t)) = \sup_{\theta > t} \rho(f(p, \theta), M) < H \text{ for } t \geq T(H).$$

Thus, $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$, uniformly relative to $p \in S(M, \delta_1)$.

Sufficiency. As follows from Theorem 13, when the conditions of Theorem 14 are satisfied, the set M is asymptotically stable. Therefore it is possible to find, for $\varepsilon > 0$ and $\varepsilon < r$, a quantity $\delta > 0$ such that when $\rho(p, M) < \delta$ we have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t > 0,$$

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Let us assume that M is non-uniformly asymptotically stable. Then there exists at least one quantity $H > 0$ such that no matter what value $T > 0$ is taken, there exists a point $p \in S(M, \delta)$ such that $\rho(f(p, T), M) \geq H$. According to Condition 1 of Theorem 12, for this H it is possible to find a $c_2 > 0$ such that for $\rho(p, M) \geq H$ we have $V(p) > c_2$ so that we get $V(f(p, T)) > c_2$, which contradicts the uniform tending to zero of $V(f(p, t))$ for $p \in S(M, \delta_1)$ since the quantity $\delta > 0$ can be chosen so small as to make $\delta < \delta_1$. This completes the proof of the theorem.

THEOREM 15. *In order for a closed invariant set M of a dynamical system $f(p, t)$ to be uniformly attracting, it is necessary and sufficient that all the conditions of Theorem 13 be satisfied and that furthermore, for any quantity $h > 0$, it be possible to find two quantities $T > 0$ and $\alpha > 0$ such that*

$$\rho(p, M) \geq hV(f(p, t)) > \alpha \text{ for } t \in [0, T].$$

PROOF. *Necessity.* Let the set M be uniformly attracting. Then, according to Definition 2 of Section 10, the set M is asymptotically

stable. Consequently, there exists in a certain neighborhood $S(M, \tau)$ of the set M a functional V satisfying all the conditions of Theorem 13.

From the Definition 2 of Section 10 it follows that for any $h > 0$ it is possible to find quantities $T > 0$ and $\alpha > 0$ such that for $h \leq \rho(p, M) \leq \delta_2$, where δ_2 is the certain quantity corresponding by Definition 1 of Section 10 to a certain $\varepsilon_2 > 0$, we get

$$\rho(f(p, t), M) \geq \alpha \text{ for } t \in [0, T].$$

The functional V satisfies condition 1 of Theorem 12. Consequently, for the quantity α one can find a certain quantity $c_2 > 0$ such that $V(p) > c_2$ for $\rho(p, M) > \alpha$. Consequently, $V(f(p, t)) \geq c_2$, since

$$\rho(f(p, t), M) \geq \alpha \text{ for } t \in [0, T].$$

This proves the necessity of the condition.

Sufficiency. Let us assume that the conditions of the theorem are satisfied. We shall show that the set M is uniformly attracting. In fact, it follows from the conditions of the theorem that the set M is asymptotically stable. Therefore, for any $\varepsilon > 0$ it is possible to find a quantity $\delta > 0$ such that for $\rho(p, M) < \delta$ we have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t > 0.$$

Let us assume that the set M is not uniformly attracting. Then there exists at least one quantity $0 < h < \delta$ such that no matter what the chosen value of $T > 0$ may be, one can find a sequence p_n , $n \rightarrow +\infty$, having the property that

$$h < \rho(p_n, M) < \delta, \text{ but } \inf_{t \in [0, T]} \rho(f(p_n, t), M) = 0 \quad (n = 1, 2, \dots).$$

Consequently, for any $c_1 > 0$ it is possible to find two quantities $\tau \in [0, T]$ and $n_k > 0$ such that

$$\rho(f(p_{n_k}, \tau), M) < c_1,$$

and then, according to Condition 2 of Theorem 12, we have

$$V(f(p_{n_k}, \tau)) < c_2,$$

where the quantity c_2 is as small as desired, provided c_1 is sufficiently small. Thus, we obtained a contradiction. Consequently,

the assumption made above is incorrect. This completes the proof of the theorem.

Let us explain the connection between Theorems 12 and 13 and the sufficiency criteria for the stability and asymptotic stability of the trivial solution of the system (1.11), uniform in t_0 (see Introduction).

As shown in Section 9, by using system (1.11) we can construct in the space (X, t) a dynamical system

$$\begin{aligned} X &= X(s, X^{(0)}, t_0), \\ t &= t(s, X^{(0)}, t_0), \end{aligned}$$

which is defined by the system of equations (1.14) and (1.13), such that the problem of the stability of the trivial solution of the system (1.11) becomes equivalent to the problem of the stability of the invariant set $X = 0$.

THEOREM 16. *In order for the invariant set $X = 0$ of the dynamical system (1.14) to be stable, it is necessary and sufficient that in a certain region $0 < |X| < r$, $t \in (-\infty, +\infty)$ there exist the function $V(X, t)$ with the following properties:*

- 1) *For any sufficiently small quantity $c_1 > 0$ it is possible to find a quantity $c_2 > 0$ such that $V(X, t) > c_2$ for $|X| > c_1$;*
- 2) *For any quantity $\gamma_2 > 0$, no matter how small, it is possible to find a quantity $\gamma_1 > 0$ so small that for $|X| < \gamma_1$ we have $V(X, t) < \gamma_2$;*
- 3) *The function $V(X(s, X^{(0)}, t_0), t(s, X^{(0)}, t_0))$ is non-increasing for all $s > 0$ as long as $|X(s, X^{(0)}, t_0)| < r$.*

Condition 3 of Theorem 16 means that the function

$$V(X(s, X^{(0)}, t_0), t(s, X^{(0)}, t_0))$$

is differentiable along the integral curves of the system (1.12) for almost all s , and that furthermore the following inequality holds:

$$\frac{dV}{ds} \leq 0$$

(from the Lebesgue theorem on the differentiability of a monotonic function [34]).

The statement is true with respect to the function

$$V(X(t, X^{(0)}, t_0), t),$$

for which $dV/dt \leq 0$ almost everywhere for $t \geq t_0$ as long as $|X(t, X^{(0)}, t_0)| < r$. In the sufficiency criterion for stability of the trivial solution of the system (1.11), proposed by Lyapunov, Condition 3 of Theorem 16 was replaced by a more stringent one, namely: it is assumed that the function $V(x_1, \dots, x_n, t)$ has continuous partial derivatives with respect to all its arguments and the function W is not positive:

$$W = \frac{\partial V}{\partial t} + \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i \leq 0. \quad (1.18)$$

However, in the course of Lyapunov's proof, condition (1.18) is necessary only for the purpose of satisfying condition 3 of Theorem 16. Thus, the Lyapunov sufficient criterion, as shown in Theorem 16, is at the same time also the necessary one.

The conditions of asymptotic stability of the trivial solution of the system (1.11), given by Lyapunov, are, as shown by analogous analysis, only sufficient. However, in the proof it is merely necessary to satisfy conditions 1-3 of Theorem 16 in order that in addition

$$V(X(s, X^{(0)}, t_0), t(s, X^{(0)}, t_0)) \rightarrow 0 \text{ as } s \rightarrow +\infty$$

for all $(X^{(0)}, t_0)$ such that $|X(s, X^{(0)}, t_0)| < r$. Thus, the Lyapunov sufficient conditions are essentially also necessary.

13. Necessary and sufficient conditions for instability

Let us consider a certain functional $V(p)$, defined for $p \in M_1 \subset R$.

DEFINITION. If the quantity $V(f(p, t)) - V(p)$, is defined for all sufficiently small t and if

$$\lim_{t \rightarrow 0} \frac{V(f(p, t)) - V(p)}{t} = W(p) = \frac{dV(p)}{dt}$$

exists, it is called the *derivative of the functional V* at the point p along the motion $f(p, t)$.

THEOREM 17. *In order for a closed invariant set M of a dynamical system $f(p, t)$ to be unstable, it is necessary and sufficient that there exist at a certain neighborhood $S(M, r)$ of the set M a functional V having the following properties:*

- 1) *The functional V is bounded in $S(M, r)$.*

2) In as small a neighborhood of the set M as desired, there exists at least one point p such that $V(p) > 0$.

3) For any point $p \in S(M, r)$ the following equation holds:

$$\frac{dV}{dt} = \lambda V + W,$$

where $\lambda > 0$ and W is a non-negative functional defined in $S(M, r)$.

PROOF. Necessity. Let the set M be unstable. Then there exists at least one quantity $\varepsilon > 0$ such that in an arbitrarily small neighborhood $S(M, \delta)$ of the set M there exists a point p such that for a certain $t \geq 0$ we have

$$\rho(f(p, t), M) = \varepsilon.$$

Let us take a point $p \in S(M, \varepsilon)$. Two cases are possible:

- 1) either $\rho(f(p, t), M) < \varepsilon$ for $t \geq 0$, or
- 2) there exists a quantity $t(p)$ such that

$$\rho(f(p, t(p)), M) = \varepsilon$$

and

$$\rho(f(p, t), M) < \varepsilon$$

for $t \in [0, t(p))$. In the first case, let us set $V(p) = 0$, and in the second $V(p) = e^{-t(p)}$. Thus, the functional $V(p)$ is defined at any point of the set $S(M, \varepsilon)$ and is bounded there: $V(p) \leq 1$.

As noted above, the points p of the second type exist in any neighborhood $S(M, \delta)$ of the set M , no matter how small. Therefore the functional constructed above satisfies the Condition 2. Let us now show that the functional V also satisfies Condition 3. In fact, if the point $p \in S(M, \varepsilon)$ belongs to the former type, then $V(f(p, t)) \equiv 0$ as long as $f(p, t) \in S(M, \varepsilon)$, and therefore $dV/dt = V$ along any such motion.

If the point p is of the second type, then

$$V(f(p, t)) = e^{t-t(p)},$$

from which it follows that $dV/dt = V$. Thus, Condition 3 is satisfied along any motion $f(p, t) \in S(M, \varepsilon)$ for $\lambda = 1$ and $W = 0$.

Sufficiency. Let there exist in a certain neighborhood of a closed invariant set M a functional V , having the properties 1–3. We shall show that the set M is unstable. Let, to the contrary,

the set M be stable. Then, for any $\varepsilon > 0$ according to Definition 1 of Section 9 it is possible to find a quantity $\delta > 0$ such that

$$\rho(f(p, t), M) < \varepsilon < r \text{ for } t \geq 0$$

when $\rho(p, M) < \delta$. Then, on any motion $f(p, t)$, the function $V(f(p, t))$ is bounded for $t \geq 0$ and $p \in S(M, \delta)$ since this motion lies in the region where the functional V is defined. According to Condition 2, there exists a point p in $S(M, \delta)$ such that $V(p) > 0$. The function $V(f(p, t))$ satisfies the linear differential equation

$$\frac{d}{dt} V(f(p, t)) = \lambda V(f(p, t)) + W(f(p, t))$$

:

with initial condition

$$V(f(p, 0)) = V(p).$$

Integrating this equation and then going over to the inequality, we obtain $V(f(p, t)) \geq V(p)e^{\lambda t}$ for $t \geq 0$, which contradicts the boundedness of the function $V(f(p, t))$. The contradiction obtained shows that the set M is unstable.

Let us apply Theorem 17 to finding the instability conditions for the trivial solution of the system (1.11).

THEOREM 18. *In order for the trivial solution of the system (1.11) to be unstable in the sense of Lyapunov, it is necessary and sufficient that there exist a function $V(X, t)$, defined in the region $t \geq T$, $0 < |X| \leq h$ (the region $B(T, h)$), having the following properties:*

- 1) *The function $V(X, t)$ is bounded in the region $B(T, h)$.*
- 2) *There exist points $(X, t) \in B(T, h)$, ($|X|$ as small as desired), such that $V(X, t) > 0$.*

$$3) \quad \frac{d}{dt} V(X(t, X^{(0)}, t_0), t) = \lambda V(X(t, X^{(0)}, t_0)) + W(X(t, X^{(0)}, t_0), t),$$

$\lambda > 0$, $W \geq 0$, defined in the region $B(T, h)$.

[As stated in the Introduction, the term "instability" denotes here a property opposite to stability that is uniform with respect to $t_0 \geq T$.]

PROOF. For the system (1.11), as shown above, it is possible to

construct a dynamical system

$$X = X(s, X^{(0)}, t_0), \quad t = t(s, X^{(0)}, t_0)$$

in the space (X, t) . In order for the invariant set $X = 0$ of such a system to be unstable, it is necessary and sufficient that there exist a function $V(X, t)$, defined in the region $B(-\infty, R)$, having the following properties:

- 1) $V(X, t)$ is bounded in $B(-\infty, R)$.
- 2) There exist points (X, t) , $|X|$ as small as desired, $t \geq T$, such that $V(X, t) > 0$.
- 3) $\frac{dV}{ds} = \lambda V + W$.

Furthermore, without loss of generality, $V(X, t)$ can be considered non-negative.

Conditions 1 and 3 follow from Theorem 17, and Condition 2 can be established with the aid of Theorem 17 and the definition of instability.

Along the integral curve $X = X(t, X^{(0)}, t_0)$, which remains within the region $B(T, h)$ for certain $t \geq t_0$, we have

$$\frac{dV}{dt} = \frac{dV}{ds} \sqrt{1 + \sum_{i=1}^n f_i^2};$$

hence

$$\frac{dV}{dt} = \lambda V + W_1,$$

where W_1 is the non-negative function

$$W_1 = W \sqrt{1 + \sum_{i=1}^n f_i^2} + \lambda V \left(\sqrt{1 + \sum_{i=1}^n f_i^2} - 1 \right).$$

Thus, the function $V(X, t)$, which exists by virtue of Theorem 17 for a dynamical system, will satisfy all conditions of Theorem 18. This completes the proof of the theorem. Let us note that Lyapunov gave a sufficiency criterion for instability, which contains all the conditions of Theorem 18. However, he required the existence of partial derivatives of $V(X, t)$ with respect to the variables $x_1, \dots, x_n, t$. This requirement was necessary for him only so that the derivative of the function $V(X, t)$ exist along the integral curves

of the system (1.11), i.e.

$$\frac{dV(X(t, X^{(0)}, t_0), t)}{dt}.$$

Thus, Lyapunov's second theorem on instability contains essentially the necessary and sufficient conditions for the instability of the trivial solution of the system (1.11).

The necessity of the conditions of the second theorem on instability for stationary systems was first established by N. N. Krasovsky [8]. He also investigated the condition for the inversion of Lyapunov's first theorem on instability for the case of stationary equations.

REMARK. In proving the theorems contained in Sections 12 and 13, we used nowhere essentially the property of continuity of the dynamical system $f(p, t)$ in both arguments. The group property of the dynamical system was used only for $t \geq 0$, and therefore the results obtained here hold for more general families of transformations of R into itself than a dynamical system.

In Chapter IV we shall consider such general systems, for which the results formulated here remain in force.

14. *Necessary and sufficient conditions for the existence of uniformly asymptotically stable and uniformly attracting invariant sets*

In this section we give necessary and sufficient conditions for an open invariant set A , containing a sufficiently small neighborhood of a closed invariant set M , to be the region of asymptotic stability of a uniformly asymptotically stable and uniformly attracting invariant set M . These conditions make it possible to find in principle the boundary of the region of asymptotic stability.

THEOREM 19. [9]. *In order for an open invariant set $A \subset R$, containing a certain neighborhood of the closed invariant set $M \subset R$ to be the region of asymptotic stability of a uniformly asymptotically stable and uniformly attracting set M , it is necessary and sufficient that there exist two functionals $V(p)$ and $\Phi(p)$ having the following properties:*

1) *The functional $V(p)$ is defined and continuous in A , and the functional $\Phi(p)$ is defined and continuous in R .*

2) $-1 < V(p) < 0$ for $p \in A$; $\Phi(p) > 0$ for $p \in R$, $\rho(p, M) \neq 0$ and $\Phi(p) = 0$ for $p \in M$.

3) For any sufficiently small $\gamma_2 > 0$ it is possible to find quantities γ_1 and α_1 such that

$$V(p) < -\gamma_1 \text{ for } \rho(p, M) \geq \gamma_2,$$

and

$$\Phi(p) > \alpha_1 \text{ for } \rho(p, M) \geq \gamma_2.$$

4) The functionals V and Φ tend to 0 as $\rho(p, M) \rightarrow 0$.

5) If there exists a point $q \in M$, $q \in \bar{A} \setminus A$, then $\lim_{\rho(p, q) \rightarrow 0} V(p) = -1$.

$$6) \left(\frac{dV}{dt} \right)_{t=0} = \Phi(p)(1 + V(p)).$$

PROOF. *Necessity.* In accordance with the conditions of the theorem, the invariant set M is uniformly asymptotically stable. Therefore there exists a number $\delta_1 > 0$ and a function $L(t)$ that is continuous and strictly monotonically decreasing from $+\infty$ to zero, having the following property

$$\rho(f(p, t), M) \leq L(t) \text{ for } \rho(p, M) < \delta_1$$

(see Theorem 5). Let us examine the function

$$\Phi(x) = xe^{-L^{-1}(x)},$$

where $L^{-1}(x)$ is the inverse of $L(t)$, and $x \geq 0$. It is easy to see that the function $\Phi(x)$ is positive and continuous, and that $\Phi(x) \rightarrow 0$ as $x \rightarrow 0$. If $x > \alpha_1 \geq 0$, then $\Phi(x) > \alpha_2(\alpha_1) > 0$. Thus, the functional $\Phi(\rho(p, M)) = \Phi(p)$ will have the required properties. Let us set

$$V(p) = e^{-\int_0^{+\infty} \Phi(f(p, t)) dt} - 1, \quad (1.19)$$

where $\Phi(f(p, t)) = \Phi(\rho(f(p, t), M))$. It follows from (1.19) that

$$1 + V(f(p, t)) = (1 + V(p)) e^{\int_0^t \Phi(f(p, t')) dt'}. \quad (1.20)$$

Let us show this. From (1.19) we have

$$1 + V(p) = e^{-\int_0^t \Phi(f(p, t')) dt' - \int_t^{+\infty} \Phi(f(p, t')) dt'};$$

hence

$$(1 + V(p))e^{\int_0^t \Phi(j(p, t')) dt'} = e^{-\int_0^{+\infty} \Phi(j(p, t+\tau)) d\tau} = e^{-\int_0^{+\infty} \Phi(j(p, t), \tau) d\tau} = 1 + V(j(p, t)).$$

Here we use Property 3 of the dynamical system (see Definition 1).

Thus, the functional $V(p)$ is formally defined by means of (1.20) along the entire motion $j(p, t)$. Let us show now that the integral $\int_0^\infty \Phi(j(p, t')) dt'$ converges for $p \in A$ so that the formula (1.19) defines a certain functional for $p \in A$. If $p \in A$, it is possible to find a quantity $T(p) \geq 0$ such that

$$\rho(j(p, t), M) < \delta_1 \text{ for } t \geq T(p).$$

Let us break the integral $\int_0^{+\infty} \Phi(j(p, t')) dt'$ up into two:

$$\int_0^\infty \Phi(j(p, t')) dt' = \int_0^T \Phi(j(p, t')) dt' + \int_T^{+\infty} \Phi(j(p, t')) dt'.$$

In the integral $\int_T^{+\infty}$, we change variables of integration, $t' = \tau + T$, and obtain

$$\int_T^{+\infty} \Phi dt' = \int_0^{+\infty} \Phi(j(p, \tau + T)) d\tau = \int_0^{+\infty} \Phi(j(p, T), \tau) d\tau.$$

Since $\rho(p_1, M) < \delta_1$, where $p_1 = j(p, T)$, we have

$$L(t) > \rho(j(p_1, t), M) \text{ for } t \geq T(p) \quad (1.21)$$

(see Theorem 5). From (1.21) it follows that

$$t < L^{-1}(\rho(j(p_1, t), M)),$$

from which we obtain

$$\begin{aligned} \rho(j(p_1, t), M) e^{-t} &> \rho(j(p_1, t), M) e^{-L^{-1}(\rho(j(p_1, t), M))} = \\ &= \Phi(j(p_1, t)). \end{aligned} \quad (1.22)$$

Applying (1.22) to an estimate of the integral

$$\int_0^\infty \Phi(j(p_1, \tau)) d\tau,$$

we obtain

$$\int_0^{\infty} \Phi(f(p_1, \tau)) d\tau < \int_0^{+\infty} \rho(f(p_1, t), M) e^{-t} dt < \delta_1.$$

Thus the convergence of $\int_0^{+\infty} \Phi(f(p, t')) dt'$ is proved. It is clear that the continuity of the functional

$$\ln(1 + V(p)) = V_1(p)$$

implies the continuity of the functional $V(p)$ for $p \in A$. It follows from (1.19) that

$$V_1(p) = - \int_0^{+\infty} \Phi(f(p, t')) dt'.$$

Let us show the continuity of the functional $V_1(p)$ for $p \in A$.

$$|V_1(p) - V_1(q)| = \left| \int_0^{+\infty} (\Phi(f(p, t)) - \Phi(f(q, t))) dt \right|. \quad (1.23)$$

Since $p, q \in A$, it is possible to find a quantity $T > 0$ such that

$$\int_T^{+\infty} \Phi(f(p, t)) dt < \varepsilon_1 \quad \text{and} \quad \int_T^{+\infty} \Phi(f(q, t)) dt < \varepsilon_1,$$

where ε_1 is a certain positive quantity. For the quantities T and $\varepsilon_1 > 0$ it is possible to choose $\varepsilon_2 > 0$ (Theorem 1) such that for $\rho(p, q) < \varepsilon_2$ we have

$$\rho(f(p, t), f(q, t)) < \varepsilon_2 \quad \text{for } t \in [0, T].$$

Furthermore, ε_2 can be chosen so small that

$$|\Phi(f(p, t)) - \Phi(f(q, t))| \leq \frac{\varepsilon_1}{T} \quad \text{for } t \in [0, T],$$

and then we obtain from (1.23) that

$$|V_1(p) - V_1(q)| \leq 3\varepsilon_1 \quad \text{for } \rho(p, q) < \varepsilon_2. \quad (1.24)$$

Thus, it is possible to find, for given ε_1 , an γ_1 such that (1.24) is satisfied, which means that the functional $V(p)$ is continuous in A .

Let us show that it is possible for the quantity $\gamma_2 > 0$ to find a quantity $\gamma_1 > 0$ such that $V(p) < -\gamma_1$ for $\rho(p, M) > \gamma_2$. Since the set M is uniformly attracting, then for any quantity $c_2 > 0$ it is possible to find $T > 0$ and $\alpha > 0$ such that

$$\rho(f(p, t), M) > \alpha \text{ for } t \in [0, T] \text{ and } \rho(p, M) > c_2.$$

Then it is possible to find a quantity $\beta > 0$ such that

$$\Phi(f(p, t)) \geq \beta \text{ for } t \in [0, T].$$

It follows from (1.19) that

$$V(p) < e^{-\int_0^T \beta dt} - 1 < -c_1,$$

where $c_1 > 0$. Let us now show that $V(p) \rightarrow 0$ as $\rho(p, M) \rightarrow 0$.

For a sufficiently small $\varepsilon_1 > 0$ we find, in accordance to Definition 1 of Section 9, a $\delta(\varepsilon_1) > 0$ such that

$$\int_0^{+\infty} \Phi dt \leq \varepsilon_1 \text{ for } \rho(p, M) < \delta(\varepsilon_1),$$

by virtue of (1.22). Thus,

$$V(p) > e^{-\varepsilon_1} - 1 \text{ as } \rho(p, M) < \delta(\varepsilon_1),$$

where ε_1 is sufficiently small, and consequently

$$V(p) \rightarrow 0 \text{ as } \rho(p, M) \rightarrow 0.$$

Let us now show that

$$\lim V(p) = -1 \text{ as } \rho(p, q) \rightarrow 0,$$

where $q \in \bar{A} \setminus A$ and $q \in M$.

Let us take $\varepsilon > 0$ and define for it $\delta(\varepsilon) > 0$, in accordance with Definition 1 of Section 9. Let us further choose a sequence of points p_n , such that $p_n \in A$, $p_n \rightarrow q$ and $\rho(p_n, M) > \delta(\varepsilon)$ for all n . We denote by τ_n a quantity having the following properties:

$$\rho(f(p_n, t), M) > \frac{\delta(\varepsilon)}{2} \text{ for } t \in [0, \tau_n) \text{ and } \rho(f(p_n, \tau_n), M) = \frac{\delta}{2}.$$

Let us show that $\tau_n \rightarrow +\infty$ as $n \rightarrow +\infty$.

Let us assume that $\tau_n < \tau < +\infty$ for $n = 1, 2, \dots$; for the quantities τ and $r_2 > 0$ it is possible to find a quantity $r_1 > 0$ such that for $\rho(q, q_1) < r_1$ we have $\rho(f(q, t), f(q_1, t)) < r_2$ for $t \in [0, \tau]$. Let us note that $\rho(f(q, \tau), M) \geq \delta$; in the opposite case $q \in A$. Therefore the quantity r_2 can be chosen so small that

$$\rho(f(q_1, t), M) > \frac{\delta}{2}, \quad t \in [0, \tau].$$

We choose the number N so large that for $n > N$ we have $\rho(p_n, q) < r_1$; then

$$\rho(f(p_n, \tau_n), M) > \frac{\delta}{2} \quad \text{for } n > N,$$

which is impossible. Thus, the sequence τ_n is unbounded, and then it cannot have a bounded subsequence, which follows from the argument given above.

Thus, $\tau_n \rightarrow +\infty$ as $n \rightarrow +\infty$. From (1.20) we have

$$1 + V(f(p_n, \tau_n)) = (1 + V(p_n)) e^{\int_0^{\tau_n} \Phi(f(p_n, t')) dt'}. \quad (1.25)$$

The quantity $\varepsilon > 0$, and with it also $\delta(\varepsilon)$, can be chosen so small that

$$V(p) > -\frac{1}{2} \quad \text{for } \rho(p, M) = \frac{\delta(\varepsilon)}{2}$$

(according to Property 4 of the present theorem). And at the same time we have

$$V(p) < -c_1 < 0.$$

From (1.25) we have

$$\frac{1}{2} e^{-\int_0^{\tau_n} \Phi(f(p_n, t')) dt'} < 1 + V(p_n) < (1 - c_1) e^{-\int_0^{\tau_n} \Phi(f(p_n, t')) dt'}. \quad (1.26)$$

Under the assumptions made, the following inequality holds

$$\Phi(f(p_n, t')) > \alpha > 0 \quad \text{for } t' \in [0, \tau_n]. \quad (1.27)$$

From and from (1.26) it follows that $1 + V(p_n) \rightarrow 0$ as $n \rightarrow +\infty$. Thus the necessity is proved.

Let us note now that $V(f(p, t)) \rightarrow 0$ as $t \rightarrow +\infty$ strictly monotonically for $p \in A$.

From (1.20) we have

$$\left. \frac{dV(f(p, t))}{dt} \right|_{t=0} = \Phi(p)(1 + V(p)). \quad (1.28)$$

It follows from (1.28) and (1.19) that $V(f(p, t))$ increases strictly monotonically from -1 to 0 under strict monotonic variation of t from $-\infty$ to $+\infty$. This proves the necessity of the condition.

Sufficiency. Let us show that the set M is asymptotically stable in the sense of Lyapunov. For $\varepsilon > 0$ we find $\lambda = \sup_{\rho(p, M) = \varepsilon} V(p)$. For this λ we can find a $\delta > 0$ such that for $\rho(p, M) < \delta$ we have $V(p) > \lambda$ (Property 3). From Condition 6 of the present theorem it follows that

$$V(f(p, t)) > V(p) \text{ for } t > 0.$$

If $\rho(p, M) < \delta$, then it is necessary to have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t \geq 0$$

because $V(f(p, t)) > \lambda$ for $t > 0$.

Let us show now that $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow +\infty$ and for $\rho(p, M) < \delta$. In fact, for $\varepsilon' > 0$ and $\varepsilon' < \delta$, it is possible to find a $\delta(\varepsilon')$ in accordance with the above rule.

Let us assume that $\rho(f(p, t), M) > \delta'$ for $t \geq 0$, and then

$$\Phi(f(p, t)) > \beta(\delta'). \quad (1.29)$$

Applying (1.29) to Condition 6, we obtain

$$1 + V(f(p, t)) \geq (1 + \lambda)e^{\beta t}, \quad (1.30)$$

from which it follows that $V(f(p, t)) \rightarrow +\infty$ as $t \rightarrow +\infty$, which is impossible. [Condition (6) along the motion $f(p, t)$ becomes a linear differential equation, which defines, for any $p \in A$, the function $V(f(p, t)) = (1 + V(p)) \exp(\int_0^t \Phi(f(p, u)) du) - 1$.] Thus,

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

where $\rho(p, M) < \delta$, since there exists a $T(\varepsilon') > 0$ such that $\rho(f(p, t), M) < \varepsilon'$ for $t > T(\varepsilon')$. At the same time, $\rho(f(p, t), M)$ tends uniformly to zero with respect to p , $\rho(p, M) < \delta$, since inequality (1.30) holds for any p that has the property $\rho(p, M) < \delta$.

Let us show further that the set M is uniformly attracting. For the quantity $L > 0$ it is possible to find an $\varepsilon > 0$ such that

$$\Phi(p) < L \text{ for } \rho(p, M) < \varepsilon$$

by virtue of the fact that

$$\Phi(p) \rightarrow 0 \text{ as } \rho(p, M) \rightarrow 0.$$

Let us take $0 < h < \delta(\varepsilon)$. It is possible to find for given h a $c_1(h)$ such that when $\rho(p, M) > h$ we have

$$V(p) < -c_1(h), \quad c_1(h) > 0.$$

We choose $\tau > 0$ such that $(1 - c_1(h))e^{L\tau} = 1 - \gamma$, where $\gamma > 0$. Let us consider the set of points M such that

$$h < \rho(p, M) < \delta. \quad (1.31)$$

From Condition 6, we have for the points of the set (1.31) that

$$1 + V(f(p, t)) \leq (1 + V(p))e^{L\tau} \text{ for } t \in [0, \tau];$$

hence

$$V(f(p, t)) < -\gamma \text{ for } t \in [0, \tau].$$

This means that there exists an $\alpha > 0$ such that

$$\rho(f(p, t), M) \geq \alpha > 0 \text{ for } t \in [0, \tau],$$

from which it follows that the set M is uniformly attracting. Let us show further that the open set A is a region of asymptotic stability. Let $p \in A$; then from condition 6 we obtain the relation (1.20).

Let us show that

$$\rho(f(p, t), M) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

In fact, suppose this is not so; then

$$\rho(f(p, t), M) > \delta(\varepsilon) \text{ for } t > 0,$$

and then

$$\Phi(f(p, t)) > \beta(\delta) \text{ for } t > 0. \quad (1.32)$$

Applying (1.32) to (1.20), we obtain $1 + V(p) = 0$, which is impossible. Thus, for $p \in A$, $\rho(f(p, t), M) \rightarrow 0$ as $t \rightarrow +\infty$. Let $p_1 \in R$ and $p_1 \notin M$. Let us show that, for $\rho(f(p_1, t), M) \rightarrow 0$ as

$t \rightarrow +\infty$, we have $p_1 \in A$. In fact, since

$$\rho(f(p_1, t), M) \rightarrow 0,$$

then on the trajectory $f(p_1, t)$ there is a point p_2 such that $\rho(p_2, M) < \delta(\epsilon)$, and then the entire trajectory $f(p_1, t) \in A$, which follows from the invariance of A . This completes the proof of the theorem.

REMARK 1. If it is possible to make in a dynamical system $f(p, t)$ the permissible transformation

$$s = \int_0^t \psi(f(p, t')) dt',$$

where $\psi(p)$ is a certain continuous positive functional, defined in R , such that the closed invariant set M of the newly obtained dynamical system $F(p, s)$ is uniformly asymptotically stable and uniformly attracting, then it is possible to give for the dynamical system $f(p, t)$ the necessary and sufficient condition under which the invariant set A , containing a certain neighborhood of the set M , is the region of asymptotic stability of the uniformly asymptotically stable and uniformly attracting set M of the dynamical system $F(p, s)$.

REMARK 2. For any $\lambda \in (0, 1)$ the equation $1 + V(p) = \lambda$ determines a set of points which is a section of the open invariant set A , i.e. any trajectory that lies in A , intersects only once the set defined by the equation $1 + V = \lambda$ for a fixed λ , as follows from (1.20).

REMARK 3. The aggregate S of the points q (if they exist) having the property $V(p) \rightarrow -1$ as $p \rightarrow q$ makes up the boundary of the region of asymptotic stability. Thus, with the aid of the functional $V(p)$ it is always possible to solve in principle the problem of finding the boundary of the region of asymptotic stability.

If the set of boundary points of the region of asymptotic stability is empty, then the set M is called *asymptotically stable in the large*.

As follows from the proof of necessity, in this case we have

$$V(p_n) \rightarrow -1,$$

where p_n is a certain sequence, having the following properties:

- 1) $p_n \in A$, $\rho(p_n, M) > \delta(\epsilon)$;

2) $\tau_n \rightarrow +\infty$ as $n \rightarrow +\infty$ (see proof of necessity of Theorem 19).

If the sequence p_n does not contain a subsequence that converges to a point lying on the boundary of A , then it does not contain any convergent subsequence what soever. In the case of the presence of compactness of a sufficiently small neighborhood of the set M , and in the presence of stability in the large, $\tau_n \rightarrow +\infty$ if and only if the sequence p_n does not contain any converging subsequence.

15. *Method of estimating the distance from the motion $f(p, t)$ to a certain invariant set M with the aid of functionals*

The theorems formulated in Sections 12–14 make up the basic theoretical content of Lyapunov's second method as applied to the investigation of the stability of invariant sets of dynamical systems, defined in a metric space. However, Lyapunov's second method makes it also possible to give a method, developed in the present section, of estimating the distance from the motion $f(p, t)$ to the invariant set M .

Let us consider the closed invariant set M . We assume that outside of this set there is defined a functional $V(p)$ having the following property. There exist two functions $m_1(x)$ and $m_2(x)$, defined for $x \geq 0$, continuous, strictly monotonically increasing from zero to $+\infty$ for $x \in [0, +\infty)$, such that

$$m_1(\rho(p, M)) \leq V(p) \leq m_2(\rho(p, M)). \quad (1.33)$$

If there is known a function $V(f(p, t))$ for a certain p , then using inequalities (1.33) it is always possible to estimate the function $\rho(f(p, t), M)$. In fact, from inequality (1.33) we have

$$m_1(\rho(f(p, t), M)) \leq V(f(p, t)) \leq m_2(\rho(p, t), M)).$$

From this we obtain

$$\rho(f(p, t), M) \leq m_1^{-1}(V(f(p, t)))$$

and

$$m_2^{-1}(V(f(p, t))) \leq \rho(p, t, M).$$

Consequently, we have

$$m_2^{-1}(V) \leq \rho(j, M) \leq m_1^{-1}(V). \quad (1.34)$$

Thus, if the functional $V(p)$ satisfies the Condition (1.33), then it is possible to obtain the estimate (1.34) with the aid of this functional. Henceforth, we shall denote everywhere by $V(p)$ a functional satisfying inequalities (1.33). Let us assume that in a certain neighborhood $S(M, r)$ of the set M there are defined two functionals V_1 and W , having the following properties:

1) The functional V_1 satisfies the inequalities

$$a_1 V^{l_1} \leq V_1 \leq a_2 V^{l_2} \quad \text{for } p \in S(M, r), \quad (1.35)$$

where a_i, l_i ($i = 1, 2$) are positive constants.

2) The functional W satisfies the inequalities

$$-b_1 V^{k_1} \leq W \leq -b_2 V^{k_2} \quad \text{for } p \in S(M, r), \quad (1.36)$$

where b_i, k_i ($i = 1, 2$) are certain positive constants.

3) The function $V_1(f(p, t))$ is continuously differentiable with respect to t for $f(p, t) \in S(M, r)$ and, furthermore, the following equality holds

$$\frac{dV_1}{dt} = W. \quad (1.37)$$

4) The qualities k_i and l_i are related by

$$\lambda_i = \frac{k_i}{l_i} \geq 1 \quad (i = 1, 2).$$

Let us take $\lambda > 1$. Multiplying both members of (1.37) by the function $V_1(f(p, t))^{-\lambda}$ and then integrating them from 0 to t , where the value of t is chosen such that $f(p, t) \in S(M, r)$, we obtain

$$V_1(f(p, t)) = \frac{V_1(p)}{\sqrt[\lambda-1]{1 + (1 - \lambda)V_1^{\lambda-1}(p) \int_0^t W(p, \tau) V_1^{-\lambda} d\tau}}. \quad (1.38)$$

We first choose $\lambda_1 > 1$ and $\lambda_2 > 1$. In this case, applying the

inequalities (1.35) and (1.36) to (1.38), we obtain

$$\frac{V_1(p)}{\sqrt[\lambda_1-1]{1 + (\lambda_1 - 1)V_1^{\lambda_1-1}(p)b_1a_1^{-\lambda_1 t}}} \leq V(f(p, t)) \leq \frac{V_1(p)}{\sqrt[\lambda_2-1]{1 + (\lambda_2 - 1)V_1^{\lambda_2-1}(p)b_2a_2^{-\lambda_2 t}}}. \quad (1.39)$$

Applying inequality (1.35) to inequality (1.39) we obtain

$$\frac{[a_1a_2^{-1}V^{\lambda_1}]^{1/\lambda_1}}{\sqrt[\lambda_1-1]{1 + (\lambda_1 - 1)(a_2V^{\lambda_1})^{\lambda_1-1}b_1a_1^{-\lambda_1 t}}} \leq V(f(p, t)) \leq \frac{[a_2a_1^{-1}V^{\lambda_2}]^{1/\lambda_2}}{\sqrt[\lambda_2-1]{1 + (\lambda_2 - 1)(a_1V^{\lambda_2})^{\lambda_2-1}b_2a_2^{-\lambda_2 t}}}. \quad (1.40)$$

Let us further set $\lambda_1 = 1$ and $\lambda_2 > 1$.

In this case, multiplying (1.37) by the function $V(f(p, t))^{-1}$ and integrating, we obtain

$$V_1(f(p, t)) = V_1(p)e^{\int_0^t W V^{-1} dr}. \quad (1.41)$$

Furthermore, let us apply inequalities (1.35) and (1.36) to inequality (1.41); we then obtain

$$\sqrt[\lambda_2]{a_1a_2^{-1}V^{\lambda_2}(p)e^{-b_1a_1^{-\lambda_2}}} \leq V(f(p, t)) \leq g(p, t), \quad (1.42)$$

where $g(p, t)$ denotes the right member of inequality (1.40). Now let $\lambda_1 = \lambda_2 = 1$; then, applying inequalities (1.35) and (1.36) to (1.41), we obtain

$$g_1(p, t) \leq V(f(p, t)) \leq \sqrt[\lambda_2]{a_2a_1^{-1}V^{\lambda_2}(p)e^{-b_2a_2^{-\lambda_2}}}. \quad (1.43)$$

We denoted the left member of the inequality (1.42) by $g_1(p, t)$.

THEOREM 20. *If in the neighborhood $S(M, r)$ of a closed invariant set M of a dynamical system $f(p, t)$ there exist two functionals V_1 and W , having the properties 1-4 indicated above, then the invariant set M is asymptotically stable. With this, for any motion $f(p, t) \in S(M, r)$ the following inequalities hold:*

- 1) (1.40) for $\lambda_1 > 1$ and $\lambda_2 > 1$;
- 2) (1.42) for $\lambda_1 = 1$, $\lambda_2 > 1$;
- 3) (1.43) for $\lambda_1 = \lambda_2 = 1$.

PROOF. The functional V_1 satisfies all the conditions of Theorem 12. Actually, the first and second conditions of Theorem 12 follow from (1.35) and (1.33). Condition 3 of Theorem 12 is satisfied by virtue of (1.37). Thus, for $\varepsilon < r$ it is possible to find a $\delta > 0$ such that for $\rho(p, M) < \delta$ we have

$$\rho(f(p, t), M) < \varepsilon \text{ for } t \geq 0.$$

Thus, for $p \in S(M, \delta)$ we have

$$f(p, t) \in S(M, r).$$

Therefore, as shown above, we obtain for all $t \geq 0$ along such motions the estimates (1.39) and (1.40) for $\lambda_i > 1$, the estimate (1.42) for $\lambda_1 = 1$ and $\lambda_2 > 1$, and estimate (1.43) for $\lambda_1 = \lambda_2 = 1$. Thus, Theorem 20 is fully proved.

Let us give furthermore the necessary and sufficient conditions which must be satisfied to make a closed invariant set M asymptotically stable and to cause furthermore $\rho(f(p, t), M)$ to satisfy estimates of a definite type.

THEOREM 21. *In order for the closed invariant set M to be asymptotically stable and at the same time in order for the following inequalities to hold:*

a) for $l > 1$,

$$\frac{c_1 V(p)}{\sqrt[l-1]{1 + c_2 t V(p)^{l-1}}} \leq V(f(p, t)) \leq \frac{d_1 V(p)}{\sqrt[l-1]{1 + d_2 t V(p)^{l-1}}}; \quad (1.44)$$

b) for $l = 1$,

$$\begin{aligned} p_1 V(p) e^{-\alpha t} &\leq V(f(p, t)) \leq q_1 V(p) e^{-\alpha t} \\ t > 0, \rho(f(p, t), M) &< r, \end{aligned} \quad (1.45)$$

where c_i, p_i, d_i and q_i are positive constants ($i = 1, 2$), it is necessary and sufficient that there exist in a certain neighborhood $S(M, r)$ of the set M two functionals V_1 and W , satisfying the following conditions:

$$1) \quad a_1 V(p) \leq V_1(p) \leq a_2 V(p); \quad (1.46)$$

$$\begin{aligned} 2) \quad -b_1 V(p)^l &\leq W(p) \leq -b_2 V^l(p) \\ l &\geq 1; a_i, b_i > 0; \end{aligned} \quad (1.47)$$

3) The function $V_1(f(p, t))$ is continuously differentiable for $f(p, t) \in S(M, r)$ and, furthermore, $dV_1/dt = W$.

PROOF. Necessity. Let the closed invariant set M be asymptotically stable, and moreover let estimates "a" or "b" be satisfied. Let us show that there exist two functionals V_1 and W , defined in $S(M, r)$ and satisfying the conditions 1-3. In fact, let us set $W = -V^l$. Let us take $\varepsilon > 0$, $\varepsilon < r$. For this ε we can find, according to Definition 1 of Section 9, a $\delta > 0$ such that for $\rho(p, M) < \delta$ we have $f(p, t) \in S(M, r)$ for $t \geq 0$. Consequently, for such motions either inequality "a" or "b" is satisfied. Let us set

$$V_1(p) = \int_0^{+\infty} V^l(f(p, t)) dt \quad \text{for } p \in S(M, \delta). \quad (1.48)$$

Let $l > 1$. Using inequalities (1.44) in (1.48), we obtain

$$\begin{aligned} c_1^l V^l \int_0^{\infty} (1 + c_2 t V^{l-1})^{-[l/(l-1)]} dt &\leq V_1(p) \leq \\ &\leq d_1^l V^l(p) \int_0^{\infty} (1 + d_2 t V(p)^{l-1})^{-[l/(l-1)]} dt. \end{aligned} \quad (1.49)$$

Setting

$$1 + c_2 t V^{l-1} = \theta_1, \quad 1 + d_2 t V(p)^{l-1} = \theta_2$$

in (1.49), we obtain

$$a_1^l V(p) \int_1^{\infty} \theta_1^{-[l/(l-1)]} d\theta_1 \leq V_1(p) \leq a_2^l V(p) \int_1^{\infty} \theta_2^{-[l/(l-1)]} d\theta_2. \quad (1.50)$$

From (1.50) it follows in a self-evident manner that the constructed functional $V_1(p)$ satisfies Condition 1 of Theorem 21. Let us show that this functional has also the Property 3:

$$\begin{aligned} V(f(p, t)) &= - \int_0^{+\infty} W(f(f(p, t), \tau)) d\tau = \\ &= - \int_0^{+\infty} W(f(p, t + \tau)) d\tau. \end{aligned} \quad (1.51)$$

From (1.51) we have

$$V(f(p, t)) = - \int_t^{+\infty} W(f(p, \theta)) d\theta,$$

from which it is obvious that

$$\frac{dV}{dt} = W.$$

This proves the necessity of the conditions for the case $l > 1$. The case $l = 1$ is simpler and can be obtained by repeating the above argument verbatim.

Sufficiency. In a certain neighborhood $S(M, r)$ of the set M let there exist two functionals V_1 and W satisfying the conditions 1–3 of the present theorem. These functionals are particular cases of those functionals referred to in Theorem 20, namely, if we put $l_1 = l_2 = 1$ and $k_1 = k_2 = l$ in Theorem 20, we obtain the case considered in Theorem 21. Thus, the sufficiency of conditions 1–3 follows from Theorem 20 and does not require any special proof.

In the next chapter we give various applications of the theory developed to the study of the problem of stability of motion in the case of a system of ordinary differential equations. In the last chapter we point out many applications of this theory to the investigation of the problem of stability of the trivial solution of a system of partial differential equations. We employ everywhere the method developed in Section 15 of estimating the distance from the motion to the invariant set.

CHAPTER II

INVESTIGATION OF THE PROBLEM OF STABILITY OF MOTION FOR SYSTEMS OF ORDINARY DIFFERENTIAL EQUATIONS

1. Stationary systems of differential equations

In the present section we give various instances of the application of Theorem 19 to systems of differential equations, the right members of which do not depend explicitly on time.

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n) \quad (s = 1, 2, \dots, n), \quad (2.1)$$

the right members of which are defined in E_n and are continuously differentiable there with respect to all their arguments up to order ν inclusive, $\nu \geq 0$. [The right members of the systems considered in this chapter are considered to be real functions unless stated otherwise.]

We shall assume that at $\nu = 0$ the right members of the system (2.1) are continuous and that they furthermore satisfy certain conditions, which guarantee the uniqueness of the solution $X = X(t, X^{(0)})$, which passes for $t = 0$ through the point $X^{(0)}$; thus, through each point $X^{(0)} \in E_n$ there passes only one integral curve of the system (2.1). It is known that the function $X = X(t, X^{(0)})$ will be continuously differentiable in the variables $x_1^{(0)}, \dots, x_n^{(0)}$, up to order ν inclusive. We assume further that the system (2.1) has the trivial solution, i.e. $f_s(0, \dots, 0) = 0$.

THEOREM 22. *In order for the region $A \subset E_n$, which contains a sufficiently small neighborhood of the point $X = 0$, to be a region of asymptotic stability of the trivial solution of the system (2.1), it is necessary and sufficient that there exist functions $V(X)$ and $\varphi(X)$, which have the following properties:*

1) The function $V(X)$ is defined and continuous in A , and the function $\varphi(X)$ is defined and continuous in E_n ;

2) $-1 < V(X) < 0$ for $X \in A$, $\varphi(X) > 0$ for $X \in E_n$, $|X| \neq 0$;

3) For any quantity $\gamma_2 > 0$ we can find quantities γ_1 and α_1 such that

$$V(X) < -\gamma_1 \text{ for } |X| \geq \gamma_2, \varphi(X) > \alpha_1 \text{ for } |X| \geq \gamma_2;$$

4) the functions $V(X)$ and $\varphi(X)$ tend to zero as $|X| \rightarrow 0$;

5) if $Y \in \bar{A} \setminus A$ and $|Y| \neq 0$, then $\lim_{X \rightarrow Y} V(X) = -1$, and if $|X| \rightarrow +\infty$, $X \in A$, then $V(X) \rightarrow -1$;

$$6) \left[\frac{dV(X(t, X^{(0)}))}{dt} \right]_{t=0} = \varphi(X)(1 + V(X)) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

In this connection, the function $\varphi(X)$ can always be selected such that the function $V(X)$ is continuously differentiable in all its arguments in the region A up to the order ν inclusive.

Before proceeding with the proof, we note that with the aid of the already known substitution

$$ds = dt \sqrt{1 + \sum_{i=1}^n f_i^2}$$

the system (2.1) can be transformed into

$$\frac{dx_k}{ds} = \frac{f_k}{\sqrt{1 + \sum_{i=1}^n f_i^2}} = g_k(x_1, \dots, x_n). \quad (2.2)$$

The system of Eqs. (2.2) defines a dynamical system in the space E_n . [We assume here that when $\nu = 0$ the solutions of the system (2.1) depend continuously on the initial conditions.] If the invariant set $X = 0$ of this dynamical system is asymptotically stable, then the set of all the points $X^{(0)}$, which have the property $|X(s, X^{(0)})| \rightarrow 0$ as $s \rightarrow +\infty$, $|X^{(0)}| \neq 0$ was previously called the region of asymptotic stability of the invariant set $X = 0$ and denoted by A .

It was shown that the set A is open. In this case it is also connected and is therefore a region. We shall henceforth call this region A the *region of asymptotic stability* of the trivial solution of the system (2.1).

PROOF. The point $X = 0$ has a compact neighborhood. Consequently, if the zero solution of the system (2.1) is asymptotically stable, then the invariant set $X = 0$ of the dynamical system (2.2) is a uniformly asymptotically stable and uniformly attracting invariant set. Therefore one can apply Theorem 19 in this case. Condition 6, according to Theorem 19, has the form

$$\frac{dV}{ds} = \varphi(1 + V).$$

Going over again to the independent variable t , we obtain

$$\frac{dV}{dt} = \varphi(1 + V) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

As a consequence of this, Theorem 22 will be completely proved if we establish the differential properties of the function V . On the basis of Theorem 5, there exists a function $L(s)$ which is continuous and positive for $s \in (-\infty, +\infty)$, varying strictly monotonically from $+\infty$ to 0 as $s \rightarrow +\infty$, such that $L(s) > X^2(s, X^{(0)})$ for $s \geq 0$ and $|X^{(0)}| < \delta_1$, where the quantity δ_1 corresponds by Definition 1 of Section 9 of Chapter I to a certain $\varepsilon > 0$. With this, $X = X(s, X^{(0)})$ is a dynamical system, determined by the system of Eqs. (2.2). We set

$$\varphi(x) = \int_0^x \dots \nu \dots \int_0^x \xi e^{-BL^{-1}(\xi)} (d\xi)^\nu,$$

where B is a certain positive quantity. The function $\varphi(x)$ is continuously differentiable up to order ν inclusive, and any of its derivatives $d^k \varphi / dx^k$, $k \leq \nu$, is a strictly monotonically increasing function from 0 to $+\infty$ as $x \rightarrow +\infty$, $\varphi(x)$ is defined for $x \geq 0$. In addition,

$$\frac{d^k \varphi}{dx^k} \leq x^{\nu+1-k} e^{-BL^{-1}(x)},$$

which can be readily established with the aid of the theorem of the mean. We set

$$V_1(X^{(0)}) = \int_0^{+\infty} \varphi(X^2(s, X^{(0)})) ds. \quad (2.3)$$

It is clear that the function $V(X^{(0)}) = e^{-V_1(X^{(0)})} - 1$, where the function V is the function referred to in Theorem 22. From the differentiability of the function V_1 follows the differentiability of the function V , and conversely. Let us show that the function $V_1(X^{(0)})$ is continuously differentiable in the variables $X_1^{(0)}, \dots, X_n^{(0)}$.

In fact,

$$\frac{\partial V_1}{\partial x_j^{(0)}} = 2 \int_0^{+\infty} \frac{d\varphi(X^2(s, X^{(0)}))}{dx} \times \sum_{k=1}^n x_k(s, X^{(0)}) \frac{\partial x_k(s, X^{(0)})}{\partial x_j^{(0)}} ds. \quad (2.4)$$

In order to show the convergence of the integral in formula (2.4), it is necessary to estimate the functions $\partial x_k / \partial x_j^{(0)}$. These functions satisfy a system of linear differential equations with variable coefficients, obtained by differentiating the system (2.2) with respect to the variable $x_j^{(0)}$. In fact, let us substitute the solution $X = X(s, X^{(0)})$ in the system (2.2). After differentiating the obtained identity, we have

$$\frac{d}{ds} \frac{\partial x_k}{\partial x_j^{(0)}} = \sum_{s=1}^n \frac{\partial g_k}{\partial x_s} \frac{\partial x_s}{\partial x_j^{(0)}}, \quad k = 1, 2, \dots, n. \quad (2.5)$$

For the sufficiently small $|X^{(0)}|$, the functions $\partial g_k / \partial x_s$ in the system (2.5) are bounded for all $s \geq 0$. We therefore have the estimate

$$\sqrt{\sum_{s=1}^n \left[\frac{\partial x_s}{\partial x_j^{(0)}} \right]^2} \leq c e^{ps},$$

where c and p are certain positive constants.

Let us now estimate the integral in formula (2.4):

$$\begin{aligned} 2 \left| \int_0^{+\infty} \frac{d\varphi(X^2)}{dX^2} \sum_{k=1}^n x_k(s, X^{(0)}) \frac{\partial x_k}{\partial x_j^{(0)}} ds \right| &\leq \\ &\leq \int_0^{+\infty} \left| \frac{d\varphi(X^2)}{dX^2} \right| \cdot 2 \sqrt{\sum_{k=1}^n x_k^2(s, X^{(0)})} \sqrt{\sum_{k=1}^n \left(\frac{\partial x_k}{\partial x_j^{(0)}} \right)^2} ds \leq \\ &\leq \int_0^{+\infty} (X^2)^p e^{-BL^{-1}(X^2)} \cdot 2 \sqrt{\sum_{k=1}^n x_k^2(s, X^{(0)})} \cdot c \cdot e^{ps} ds. \end{aligned}$$

Furthermore, using the properties of the function L and choosing the number B so large as to make $B > p$, we obtain

$$2 \left| \int_0^{+\infty} \frac{d\varphi(X^2)}{dX^2} \sum_{k=1}^n x_k(s, X^{(0)}) \frac{\partial x_k}{\partial x_j^{(0)}} ds \right| \leq \int_0^{+\infty} 2c(X^2)^{r+\frac{1}{2}} e^{(p-B)s} ds,$$

from which it follows that formula (2.4) defines the partial derivative of the function V with respect to the variable $x_j^{(0)}$ for sufficiently small $|X^{(0)}|$. It can be seen that V has a derivative with respect to $x_j^{(0)}$ over the entire region A , since the integral by which the derivative is defined can always be broken up into two, one of which is taken over a finite interval and certainly exists, and the other of which is taken over an infinite interval and can be estimated in an analogous manner.

Employing approximately the same argument and the same method of estimates, we can show that for sufficiently large $B > 0$ the function $V_1(X^{(0)})$, and consequently the function $V(X^{(0)})$, is continuously differentiable in all its variables up to order r inclusive in the entire region A . This therefore proves Theorem 22.

COROLLARY 1. *Let us consider the set of all points X such that $1 + V(X) = \lambda$, $X \in A$, $\lambda \in (0, 1)$. This set represents a closed surface S_λ , which bounds the region G_λ , containing the point $X = 0$. With this, the surface S_λ , for any $\lambda \in (0, 1)$, forms a section of the region A , i.e. any integral curve of the system (2.1) $X = X(t, X^{(0)})$ intersects S_λ for $X^{(0)} \in A$ only once, from the outside of the region G_λ to the inside. The surface S_0 is the boundary of the region A , and S_1 coincides with the point $X = 0$. The surface consists of complete integral curves of the system (2.1) [10]. Thus, the equation $V = -1$ gives the entire surface S_0 . This makes it possible to state that the equation of the boundary of the region of asymptotic stability of the trivial solution (2.1) has been obtained in principle. [The set of points X such that $1 + V(X) > \lambda$ is open, connected and bounded. Consequently, $1 + V(X) = \lambda$ for $\lambda \in (0, 1)$ is the equation of a closed surface. The function $V(X)$ can be supplementarily defined by continuity in $\bar{A}$.]*

COROLLARY 2. *The trivial solution of the system (2.1) is called asymptotically stable in the large, if the region $\bar{A} = E_n$, i.e.*

$X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$ for any $X^{(0)}$, and the trivial solution of the system (2.1) is asymptotically stable.

Theorem 22 gives a necessary and sufficient condition for the asymptotic stability in the large. Let us formulate them separately. In order for the trivial solution of the system (2.1) to be asymptotically stable in the large, it is necessary and sufficient that there exist two functions $V_1(X)$ and $\varphi(X)$, which have the following properties:

- 1) The functions $V_1(X)$ and $\varphi(X)$ are defined in E_n and are continuous there, $V_1(0) = \varphi(0) = 0$;
- 2) $V_1 < 0$ for $X \neq 0$ and $V_1 \rightarrow -\infty$ as $|X|^2 \rightarrow +\infty$, and $\varphi(X) > 0$ when $X \neq 0$;
- 3) For any quantity $\gamma_2 > 0$ one can find a quantity γ_1 such that $\varphi(X) \geq \gamma_1$ when $|X| > \gamma_2$;

$$4) \frac{dV_1}{dt} = \varphi \sqrt{1 + \sum_{i=1}^n f_i^2} = \varphi_1(X).$$

It must be noted that necessary and sufficient conditions for asymptotic stability in the large were first given by Barbashin and Krasovsky [11]. As applied to the system (2.1), these conditions were formulated by them only for $\nu \geq 1$.

The necessary analysis of the function, resolving the problem of the asymptotic stability in the large, was first carried out also in the work by N. P. Yerugin [12].

COROLLARY 3. *If $\nu \geq 1$ and if the trivial solution of the system (2.1) is asymptotically stable, then the equation*

$$\begin{aligned} \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(x_1, \dots, x_n) = \\ = \varphi(x_1, \dots, x_n) \sqrt{1 + \sum_{i=1}^n f_i^2(x_1, \dots, x_n)} (1 + V) \end{aligned} \quad (2.6)$$

has a unique continuously differentiable solution, determined by the condition $V(0) = 0$, defined when $X \in A$ and satisfying the conditions of Theorem (22) for no function $\varphi(x_1, \dots, x_n)$. In order for Eq. (2.6) to have such a solution, it is sufficient that the function $\varphi(X)$ have the property

$$\int_0^{+\infty} \varphi(X(t, X^{(0)})) dt < +\infty$$

for sufficiently small $|X^{(0)}|$. Thus, the function φ , at which the Eq. (2.6) can be solved under the condition $V(0) = 0$, depends on the character of the decrease of the function $X(t, X^{(0)})$ as $t \rightarrow +\infty$. Therefore, if for any reason whatever one knows the approximate order of decrease of the function $X(t, X^{(0)})$, then the function $\varphi(X)$ can always be constructed. For example, when $|X(t, X^{(0)})| \leq c \cdot e^{-pt}$, $c > 0$, $p > 0$, for sufficiently small $|X^{(0)}|$ the function φ can be chosen as a form in the numbers $x_1, \dots, x_n$ or in general chosen so as to have $\varphi(X) \leq |X|^m$, $m > 0$. If, for example,

$$|X(t, X^{(0)})| \leq t^{-\alpha}, \quad \alpha > 0 \quad \text{for } t > T,$$

then m can be chosen so as to make $m\alpha > 1$, and then Eq. (2.6) will have a solution for any $\varphi \leq |X|^m$.

Let us give several examples which illustrate the foregoing results:

$$1. \quad \begin{aligned} \frac{dx}{dt} &= -x + 2x^2y; \\ \frac{dy}{dt} &= -y. \end{aligned}$$

The corresponding partial differential equation will have the form

$$\frac{\partial V}{\partial x}(2x^2y - x) + \frac{\partial V}{\partial y}(-y) = (x^2 + y^2)(1 + V),$$

which can be obtained from (2.6) by setting

$$\sqrt{1 + \sum_{i=1}^n f_i^2} \varphi(x, y) = x^2 + y^2.$$

It is easy to verify that

$$V(x, y) = e^{-(x^2/2) - (x^2/[2(1-xy)])} - 1$$

is a solution of this equation. From the form of this function it follows that the curve $xy = 1$ forms the boundary of the stability region.

2.

$$\frac{dx}{dt} = f_1(x, y) = -3x + 3a \left[\frac{x}{a} \right]^{\frac{1}{3}} \left[\frac{y}{b} \right]^{\frac{1}{3}} + 3a \left[\frac{x}{a} \right]^{\frac{1}{3}} + 3x \left[\frac{y}{b} \right]^{\frac{1}{3}},$$

$$\frac{dy}{dt} = f_2(x, y) = -3y - 3b \left[\frac{y}{b} \right]^{\frac{1}{3}} \left[\frac{x}{a} \right]^{\frac{1}{3}} + 3b \left[\frac{y}{b} \right]^{\frac{1}{3}} + 3y \left[\frac{x}{a} \right]^{\frac{1}{3}}.$$

The corresponding partial differential equation is taken to be in the form

$$\frac{\partial V}{\partial x} \cdot f_1(x, y) + \frac{\partial V}{\partial y} \cdot f_2(x, y) = 2\mu \left\{ \left[\frac{x}{a} \right]^{\frac{1}{3}} + \left[\frac{y}{b} \right]^{\frac{1}{3}} \right\} (1 + V), \quad \mu > 0.$$

Its continuous solution $V(x, y)$, determined by the condition $V(0, 0) = 0$, will have the form

$$V(x, y) = \left[1 - \left(\frac{x}{a} \right)^{\frac{1}{3}} - \left(\frac{y}{b} \right)^{\frac{1}{3}} \right]^{\mu} - 1,$$

from which it follows that the integral curve, bounding the region of stability, will be given by the equation

$$\left(\frac{x}{a} \right)^{\frac{1}{3}} + \left(\frac{y}{b} \right)^{\frac{1}{3}} = 1.$$

[The function V is not everywhere differentiable, but dV/dt is continuous in A , which agrees with Theorem 22.]

The family of sections of the region of asymptotic stability will have the form

$$\left[1 - \left(\frac{x}{a} \right)^{\frac{1}{3}} - \left(\frac{y}{b} \right)^{\frac{1}{3}} \right]^{\mu} = \lambda, \quad \lambda \in (0, 1).$$

3.

$$\left. \begin{aligned} \frac{dx}{dt} &= \frac{(1 - x^2 + y^2)2x}{(x+1)^2 + y^2} + xy \\ \frac{dy}{dt} &= \frac{1 - x^2 + y^2}{2} - \frac{4x^2y}{(x+1)^2 + y^2} \end{aligned} \right\}.$$

The corresponding partial differential equation will have the

form

$$\begin{aligned} \frac{\partial V}{\partial x} \cdot \left(\frac{(1-x^2+y^2)2x}{(1+x)^2+y^2} + xy \right) + \frac{\partial V}{\partial y} \left(\frac{1-x^2+y^2}{2} - \frac{4x^2y}{(1+x)^2+y^2} \right) = \\ = \frac{\partial V}{\partial x} \cdot f_1(x, y) + \frac{\partial V}{\partial y} \cdot f_2(x, y) = 2 \frac{(x-1)^2+y^2}{(x+1)^2+y^2} (1+V). \end{aligned}$$

In this system the rest point has the coordinates $(1, 0)$. It is easy to verify that

$$V(x, y) = - \frac{(x-1)^2+y^2}{(x+1)^2+y^2}$$

is a continuous solution, determined by the condition $V(1, 0) = 0$. Setting $V(x, y) + 1 = 0$, we find that the boundary of the region of asymptotic stability of the solution $x = 1, y = 0$ of the system is the straight line $x = 0$, and the region of the stability coincides with the right half plane. Let us find the equation for the family of cross sections, which fill the region of asymptotic stability. For this purpose we set $V(x, y) = \lambda - 1$, where $\lambda \in (0, 1)$. We then have

$$\left(x + 1 - \frac{2}{\lambda} \right)^2 + y^2 = \left(1 - \frac{2}{\lambda} \right)^2 - 1.$$

Consequently, the closed curves which fill the region of asymptotic stability are circles.

4.

$$\left. \begin{aligned} \frac{dx}{dt} &= s(x) + \sigma(y) \\ \frac{dy}{dt} &= \delta(x) \end{aligned} \right\}. \quad (2.7)$$

The right members of (2.7) are defined for any x and y and satisfy the conditions that guarantee the existence of the unique solution

$$x(t, x^0, y^0), \quad y(t, x^0, y^0)$$

for all finite values x^0 and y^0 , and has a unique rest point $x = y = 0$. Let us assume that $\delta(x)$ and $s(x)$ have signs opposite to the sign of x , and $\sigma(y)$ has the same sign as y .

Let us consider the partial differential equation corresponding to the system (2.7):

$$\frac{\partial V}{\partial x}(s + \sigma) + \frac{\partial V}{\partial y}\delta = \delta(x)s(x)(1 + V). \quad (2.8)$$

The function

$$V(x, y) = e^{\int_0^x \delta(x) dx - \int_0^y \sigma(y) dy} - 1$$

is a solution of the Eq. (2.8).

If the integrals $\int_0^x \delta(x) dx$ and $\int_0^y \sigma(y) dy$ tend to ∞ as $|x|$ and $|y| \rightarrow \infty$, then the trivial solution of the system (2.7) is asymptotically stable in the large.

In this example we used as the function $\varphi(x, y)$ the expression $s(x)\delta(x)$, from which it follows that for $x = 0$ we have $\varphi(0, y) = s(0)\delta(0) = 0$. However, this circumstance does not prevent us from applying to Eq. (2.7) the theory developed above, since it is easy to show that the Theorem 22 remains in force for Eq. (2.8), if in this case the right members of the system (2.7) are such that the solution $x(t, x^0, y^0)$, $y(t, x^0, y^0)$ is defined for $t \in (-\infty, +\infty)$. Theorem 22 makes it possible to solve the problem of the invariance of the boundary S_0 of the region A under small changes in the functions $f_s(x_1, \dots, x_n)$.

It is known that everytime when the system (2.1) is obtained in solving a practical problem, the functions $f_s(x_1, \dots, x_n)$ are known only approximately. Let us ascertain to what accuracy these facts coincide with the actual state of the matter, i.e. whether the obtained qualitative estimate is stable with respect to small changes of the functions $f_s(x_1, \dots, x_n)$. Let us consider the system of n equations

$$\frac{dx_i}{dt} = f_i(x_1, \dots, x_n) + R_i(x_1, \dots, x_n, t) \quad (i = 1, \dots, n). \quad (2.9)$$

We assume that the functions $f_i(x_1, \dots, x_n)$ are continuously differentiable and that the $R_i(x_1, \dots, x_n, t)$ are certain continuous functions such that the system (2.9) satisfies the conditions of existence and uniqueness.

THEOREM 23. *If the system (2.1) has an asymptotically stable trivial solution, having a certain region of asymptotic stability A , it is possible to indicate for the functions $R_t(x_1, \dots, x_n, t)$ an upper limit $R(X)$ such that for $|R_t(x_1, \dots, x_n, t)| < R(x_1, \dots, x_n)$ the system (2.9) will have an asymptotically stable trivial solution with the same region of asymptotic stability.*

PROOF. By virtue of the foregoing assumptions with respect to the system (2.1), there exists a Lyapunov function $V(x_1, \dots, x_n)$ having the properties

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(x_1, \dots, x_n) = W(x_1, \dots, x_n),$$

where $W(x_1, \dots, x_n)$ is a negative definite, continuously differentiable function.

$$W(x_1, \dots, x_n) < -\alpha(\beta) < 0 \text{ for } \sum_{i=1}^n x_i^2 > \beta > 0,$$

with $V(x_1, \dots, x_n) \rightarrow \infty$ as $(x_1, \dots, x_n) \rightarrow M^* \in \bar{A} \setminus A$, $M^* \neq 0$. Let us form the derivative of the function V by virtue of the system (2.9):

$$\begin{aligned} \frac{dV}{dt} &= \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(x_1, \dots, x_n) + \sum_{i=1}^n \frac{\partial V}{\partial x_i} R_t(x_1, \dots, x_n, t) = \\ &= W + \sum_{i=1}^n \frac{\partial V}{\partial x_i} R_t(X, t). \end{aligned}$$

From the Bunyakovsky (Schwarz) inequality we obtain

$$\begin{aligned} \sum_{i=1}^n \frac{\partial V}{\partial x_i} R_t(x_1, \dots, x_n, t) &\leq \sqrt{\sum_{i=1}^n \left(\frac{\partial V}{\partial x_i} \right)^2} \times \\ &\times \sqrt{\sum_{i=1}^n R_i^2} < \sqrt{n} R(X) \sqrt{\sum_{i=1}^n \left(\frac{\partial V}{\partial x_i} \right)^2}. \end{aligned}$$

We chose the function $R(x_1, \dots, x_n)$ so that

$$\begin{aligned} \sqrt{n} R(X) \sqrt{\sum_{i=1}^n \left(\frac{\partial V}{\partial x_i} \right)^2} &< \kappa(x_1, \dots, x_n) |W(x_1, \dots, x_n)|, \\ 0 &\leq \kappa \leq \frac{1}{2}. \end{aligned}$$

Let us set, for example,

$$R(x_1, \dots, x_n) = \frac{\kappa(x_1, \dots, x_n) |W(x_1, \dots, x_n)|}{\sqrt{1 + \sum_{i=1}^n \left(\frac{\partial V}{\partial x_i}\right)^2}} \sqrt{n}.$$

For such a function $R(x_1, \dots, x_n)$ the function dV/dt , calculated by virtue of the system (2.9), will be negative definite for all $x_1, \dots, x_n$:

$$\frac{dV}{dt} < -\alpha_1 < 0 \text{ for } \sum_{i=1}^n x_i^2 > \beta > 0. \quad (2.10)$$

We have thus constructed a function $V(x_1, \dots, x_n)$ for the system (2.9) having the following properties:

$$0 \leq V(x_1, \dots, x_n) < +\infty, (x_1, \dots, x_n) \in A;$$

$$V(x_1, \dots, x_n) \rightarrow +\infty \text{ as } (x_1, \dots, x_n) \rightarrow M^* \in \bar{A} \setminus A,$$

and the derivative of the function $V(x_1, \dots, x_n)$, calculated by virtue of the system (2.9), has the property (2.10).

Then, according to Theorem 22, the region A is a region of asymptotic stability of the trivial solution $x_1 = \dots = x_n = 0$ of the system (2.9).

This completes the proof of the theorem.

REMARK 1. If the functions $R_i(x_1, \dots, x_n, t)$ are such that $|R_i| < R(x_1, \dots, x_n)$ only in the region A , then the statement of the theorem remains in force.

REMARK 2. If the region A coincides with the entire space, then the trivial solution of the system (2.9) will also be asymptotically stable in the large.

The theorem proved above yields a certain method of investigating systems of differential equations. Let us clarify this with examples.

5. We consider a system of linear equations with constant coefficients:

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si} x_i \quad (s = 1, \dots, n), \text{ where } p_{si} = \text{const.} \quad (2.11)$$

Let all the roots of the equation

$$|P - \lambda E| = 0, \quad p_{ik} = \{P_{ik}\} \quad (2.12)$$

have negative real parts.

We consider further the system

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si} x_i + X_s(x_1, \dots, x_n).$$

Let us find out for which functions

$$X_s(x_1, \dots, x_n) \quad (2.13)$$

the trivial solution $x_1 = \dots = x_n = 0$ of the system (2.13) is asymptotically stable in the large. We use Theorem 23.

Let us construct for the system (2.11) the function $V(x_1, \dots, x_n)$, in the form of a positive definite quadratic form:

$$V = \sum_{i,k=1}^n a_{ik} x_i x_k, \quad a_{ik} = \text{const.},$$

$$\frac{dV}{dt} = W(x_1, \dots, x_n),$$

where the function W is a negative definite quadratic form:

$$W = \sum_{i,k=1}^n b_{ik} x_i x_k, \quad b_{ik} = \text{const.}$$

We construct the function

$$R_l(x_1, \dots, x_n) = -\alpha W(x_1, \dots, x_n) \frac{\partial V}{\partial x_l} \left[1 + \left(\frac{\partial V}{\partial x_l} \right)^2 \right]^{-1},$$

$$R_l(x_1, \dots, x_n) = -\alpha W \sum_{k=1}^n a_{lk} x_k \left[1 + \left(\sum_{k=1}^n a_{lk} x_k \right)^2 \right]^{-1},$$

where α is a certain function of $x_1, \dots, x_n$, $-l < \alpha < l$, $l < 1/2n$. If

$$|X_l(x_1, \dots, x_n)| < |R_l(X)|,$$

then the trivial solution of the system (2.13) is asymptotically stable in the large. If the functions $X_i(x_1, \dots, x_n)$ are expanded in series, then the latter inequality gives an estimate of the coeffi-

cients of these series in terms of the arbitrary quantities b_{ik} and in terms of the coefficients p_{st} of the system (2.11), where the constants b_{ik} are related by the Sylvester inequalities:

$$b_{11} < 0, \quad \begin{vmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{vmatrix} > 0, \dots, (-1)^n \begin{vmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ \cdot & \cdot & \cdot & \cdot \\ b_{n1} & b_{n2} & \dots & b_{nn} \end{vmatrix} > 0.$$

6.
$$\begin{aligned} \frac{dx}{dt} &= -x + y + x(x^2 + y^2) + \varphi(x, y), \\ \frac{dy}{dt} &= -x - y + y(x^2 + y^2) + \psi(x, y). \end{aligned} \quad (2.14)$$

The "first-approximation" system obtained by discarding the functions $\varphi(x, y)$ and $\psi(x, y)$ has a limit cycle $x^2 + y^2 = 1$. Let us construct the Lyapunov function V for the first-approximation system

$$V = -\ln(1 - x^2 - y^2),$$

$$\frac{dV}{dt} = -2(x^2 + y^2).$$

We take

$$R(x, y) = (x^2 + y^2) \left[1 + \left(\frac{\partial V}{\partial x} \right)^2 + \left(\frac{\partial V}{\partial y} \right)^2 \right]^{-1}$$

or

$$R(x, y) = (x^2 + y^2)(1 - x^2 - y^2)(1 + x^2 + y^2)^{-1}.$$

Then: if

$$|\varphi(x, y)| < |R(x, y)|, \quad |\psi(x, y)| < |R(x, y)|,$$

then the system (2.14) also has a limit cycle. We note that instead of $R(x, y)$ we can take any other function, satisfying the conditions of Theorem 23.

If the trivial solution of the system (2.1) is asymptotically stable, then any solution $X(t, X^{(0)})$ for $X^{(0)} \in A$ is defined when $t \in [0, +\infty)$. Using Theorem 22, we can ascertain under what conditions these solutions will be defined also for $t \in (-\infty, 0]$.

THEOREM 24. *In order for any integral curve $X = X(t, X^{(0)})$, $X^{(0)} \in A$ of the system (2.1) to be defined for $t \in (-\infty, +\infty)$, it is necessary and sufficient that all the conditions of the Theorem 22*

be satisfied, and, furthermore, that

$$\varphi(x_1, \dots, x_n) \sqrt{1 + \sum_{i=1}^n f_i^2} < K < +\infty.$$

PROOF. Necessity. Let all the integral curves be continuable to $t \in (-\infty, +\infty)$. Let us show that the conditions of the theorem are satisfied. In fact, the trivial solution of the system (2.1) is asymptotically stable. Therefore there exists a region A of asymptotic stability. The dynamical system $X = X(t, X^{(0)})$ is defined in this region. Therefore, in proving Theorem 22 there is no need for introducing the parameter s , since all the arguments used in its proof can be realized in terms of the independent variable t . In addition, the function φ , referred to in Theorem 22, can be chosen such that

$$\varphi \sqrt{1 + \sum_{i=1}^n f_i^2} < K < +\infty,$$

for the nature of the proof does not require that the function

$$\varphi \sqrt{1 + \sum_{i=1}^n f_i^2}$$

be unbounded.

Sufficiency. Let the conditions of Theorem 24 be satisfied. We shall show that any integral curve $X = X(t, X^{(0)})$ is defined for all $t \in (-\infty, +\infty)$. In fact, suppose this is not so, i.e. there exists at least one point $X^{(0)} \neq 0$ and $X^{(0)} \in A$ such that the integral curve $X(t, X^{(0)})$ is defined only for $t \geq T > -\infty$. Then along this curve we have

$$1 + V(X(t, X^{(0)})) = [1 + V(X^{(0)})] e^{\int_0^t \varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2} d\tau},$$

from which it follows that

$$V(X(t, X^{(0)})) > [1 + V(X^{(0)})] e^{Kt} - 1 > -1 + \varepsilon, \quad \varepsilon > 0.$$

On the other hand, the interval curve considered is unbounded for $t \leq 0$ by virtue of the non-continuability. Therefore there exists a sequence of quantities $t_k \leq 0$ such that $|X(t_k, X^{(0)})| \rightarrow +\infty$. Then $V(X(t_k, X^{(0)})) \rightarrow -1$, which contradicts the inequality ob-

tained above. Thus, any integral curve $X(t, X^{(0)})$, $X^{(0)} \in A$ is defined for all $t \in (-\infty, +\infty)$.

The next question is how the region of asymptotic instability of the trivial solution of the system (2.1) is determined by means of its right members and, conversely, how to determine from the right members of the system (2.1) the region A . Theorem 22 makes it possible to establish the analytic form of the right members of the system (2.1), having a previously specified region A .

Let there be specified any region A , containing a sufficiently small neighborhood of the point $X = 0$. The boundary of this region will be denoted by S . We assume that the boundary S is such that it is possible to construct a continuously differentiable function $V(X)$, defined in A and having the following properties:

- 1) $V(0) = 0$, $-1 < V(X) < 0$ for $X \in A$;
- 2) The equation $1 + V = \lambda$, $\lambda \in (0, 1)$, defines a closed surface S_λ , which bounds the region G_λ , defined by the condition $V > \lambda - 1$; [We assume that the S_λ do not intersect; if $x_n \in A$, $|x_n| \rightarrow +\infty$ as $n \rightarrow +\infty$, then $v(x_n) \rightarrow -1$.]
- 3) $V \rightarrow -1$ as $|X - \bar{X}| \rightarrow 0$, where $\bar{X} \in \bar{A} \setminus A$, $\bar{X} \neq 0$.

We determine the right members of the system (2.1) from the linear equations

$$\sum_{i=1}^n p_{st}(X) f_i(X) = q_s(X) \quad (s = 1, \dots, n), \quad (2.15)$$

where $p_{1t} = \partial V / \partial x_t$ and $q_1 = \varphi_1(1 + V)$. The function φ_1 is defined in E_n , $\varphi_1(0) = 0$, $\varphi_1 > \alpha > 0$ for $|X| > \beta > 0$. We assume that the functions p_{st}, q_{st} , $s \geq 2$, $i = 1, \dots, n$, are so chosen that the system (2.15) has a solution $f_s = \bar{f}_s$ such that the system of equations

$$\frac{dx_s}{dt} = \bar{f}_s(x_1, \dots, x_n) \quad (2.16)$$

has the trivial solution and satisfies the conditions for existence and uniqueness in A or in E_n if V is defined in E_n .

THEOREM 25. *The trivial solution of the system (2.16) has a region of asymptotic stability A , and, conversely, if the system of differential equations (2.1) has an asymptotically stable trivial solution with a region of asymptotic stability A , then its right members can be found from a linear system of the type (2.15).*

PROOF. The system (2.16) has the trivial solution. Moreover, there exist functions V and φ , having the same properties as in Theorem 22, since

$$\frac{dV}{dt} = \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i = \varphi(1 + V).$$

Consequently, by Theorem 22, the region A is the region of asymptotic stability of the trivial solution of the system (2.1).

Let us show that the converse also holds. Let A be the region of asymptotic stability of the trivial solution of the system (2.1), $\nu \geq 1$. Then there exist functions V and φ having the properties indicated in Theorem 22. Let us set

$$p_{sj} = 0, \quad s \neq j \text{ and } s > 1, \quad p_{ss} = 1, \quad s > 1,$$

$$q_s = f_s(x_1, \dots, x_n), \quad s > 1, \quad p_{1s} = \frac{\partial V}{\partial x_s}.$$

$$q_1 = \varphi \sqrt{1 + \sum_{i=1}^n f_i^2 (1 + V)}$$

in the system (2.15).

Here the functions V and φ are those dealt with in Theorem 22. The system (2.15) thus obtained defines for us the right members of the system (2.1).

COROLLARY. *If the function φ is bounded, then all the solutions of the system (2.16) for $X^{(0)} \in A$ will be defined for $t \in (-\infty, +\infty)$.*

Let us introduce an example for Theorem 25.

7. We consider the curve $S(x, y) = 0$, which bounds the region A : $(0, 0) \in \bar{A}$ and $S(x, y) < 0$ for $(x, y) \in A$. If $S(0, 0) \neq 0$, it can be assumed that $S(0, 0) = -1$. The general form of the systems, for which the curve $S(x, y) = 0$ is an integral curve, was derived by N. P. Yerugin [14], namely:

$$\begin{aligned} \dot{x} &= f_1(x, y, S) - \frac{\partial S}{\partial y} M(x, y), \\ \dot{y} &= \frac{\partial S}{\partial x} M(x, y) + f_2(x, y, S), \end{aligned} \tag{2.17}$$

where M is any continuous function, and $f_1 = f_2 = 0$ when $S = 0$.

We separate from this set of systems the class for which the point $x = 0, y = 0$ is asymptotically stable and the curve $S(x, y) = 0$ serves as the boundary of the region of asymptotic stability.

For this purpose we put

$$M(x, y) = -\gamma(x, y)w(x, y),$$

$$f_1(x, y, S) = S(x, y) \left[-\gamma(x, y) \frac{\partial w}{\partial y} + S(x, y) \varphi(x, y) d_1(x, y) \right],$$

$$f_2(x, y, S) = S(x, y) \left[\gamma(x, y) \frac{\partial w}{\partial x} - S(x, y) \varphi(x, y) d_2(x, y) \right],$$

where

$$\varphi(0, 0) = w(0, 0) = \frac{\partial w}{\partial x} \Big|_{x=y=0} = \frac{\partial w}{\partial y} \Big|_{x=y=0} = 0.$$

$\varphi(x, y)$ and $w(x, y)$ are positive definite functions on the entire plane, the function $\gamma(x, y)$ is an arbitrary continuously differentiable function, and the functions $d_1(x, y)$ and $d_2(x, y)$ satisfy the relation

$$d_1(x, y) \left[S(x, y) \frac{\partial w}{\partial x} - \frac{\partial S}{\partial x} w \right] - d_2(x, y) \left[S(x, y) \frac{\partial w}{\partial y} - \frac{\partial S}{\partial y} w \right] = 1,$$

and remain in all other respects arbitrary continuously differentiable functions. It is also assumed that the functions $S(x, y)$ and $w(x, y)$ are continuously differentiable for all values of x and y .

Let us consider the equation

$$\begin{aligned} & \frac{\partial V}{\partial x} \left[S(x, y) \left(-\gamma(x, y) \frac{\partial w}{\partial y} + S(x, y) \varphi(x, y) d_1(x, y) \right) + \right. \\ & \quad \left. + \frac{\partial S}{\partial y} \gamma(x, y) w \right] + \frac{\partial V}{\partial y} \left[-\frac{\partial S}{\partial x} \gamma(x, y) w(x, y) + \right. \\ & \quad \left. + S(x, y) \left(\gamma(x, y) \frac{\partial w}{\partial x} - S(x, y) \varphi(x, y) d_2(x, y) \right) \right] = \\ & = \varphi(x, y) [1 + V]. \end{aligned}$$

It has a unique continuous solution, defined by the condition

$$V(0, 0) = 0; \quad 1 + V(x, y) = e^{w(x, y)/S(x, y)}.$$

On the basis of the preceding theorems, it can be stated that the trivial solution of this system (2.17) will be asymptotically stable, and the curve $S(x, y) = 0$ will serve as the boundary of the region A . Any solution of the system (2.17), beginning in the region of asymptotic stability, will be continuable to the half-axis $t \in (-\infty, 0]$, if the function $\varphi(x, y)$ is considered to be bounded. This class of equations can be broadened considerably if we replace everywhere in the preceding formulas $S(x, y)$ by $F(x, y, S)$, where the function $F(x, y, S)$, is continuously differentiable,

$$F(0, 0, S(0, 0)) < 0 \text{ and } F(x, y, S(x, y)) = 0 \text{ for } S(x, y) = 0.$$

In the region bounded by the curve $S(x, y) = 0$, containing the origin $x = y = 0$, we have

$$F(x, y, S(x, y)) \neq 0.$$

As follows from the foregoing, it is very important to know the methods of finding the function $V(X)$, which establishes the fact of asymptotic stability of the trivial solution of the system (2.1) and which permits determining the entire region A . In Theorem 22 (see Theorem 19) the function $V(X)$ was constructed in the form

$$1 + V(X^{(0)}) = e^{-\int_0^{+\infty} \varphi(X) ds}.$$

$\int_0^{+\infty} \varphi(X) ds$ is a curvilinear integral of the first kind, taken over the positive semi-trajectory, which starts from the point $X^{(0)}$. Let us transform this into a curvilinear integral of the second kind. For this purpose we take the continuously differentiable function $H(x_1, \dots, x_n)$ and form its total derivative

$$\frac{dH}{ds} = \sum_{i=1}^n \frac{f_i}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \frac{\partial H}{\partial x_i}$$

by virtue of the system (2.2), or

$$ds = dH \left(\sum_{i=1}^n \frac{f_i}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \frac{\partial H}{\partial x_i} \right)^{-1}.$$

Replacing ds by this expression, we obtain

$$\int_0^{+\infty} \varphi(X) ds = \int_L \varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2 \left(\sum_{i=1}^n \frac{\partial H}{\partial x_i} f_i \right)^{-1}} dH. \quad (2.18)$$

The curvilinear integral is taken here along the path which is the integral curve of the system (2.1), passing through the point $X^{(0)}$ when $t \geq 0$ and defined for $t \geq 0$. An interesting case is when this integral is independent of the path of integration, since then it is easy to determine the function $V(X)$ along any ray. It is clear that this integral can always be made such by setting $H = H(V)$.

Let us assume that $H(X)$ is an arbitrary given function. Then, in order for the integral to be independent of the path of integration, it is sufficient that the function

$$\varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2 \left(\sum_{i=1}^n \frac{\partial H}{\partial x_i} f_i \right)^{-1}} = \theta(H) \quad (2.19)$$

be an arbitrary continuous function of H . This segregates a class of systems of differential equations, having the following property: for a specified function $H(X)$ and for any positive definite function $\varphi(X)$, the integral

$$\int_L \varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2} dH \left(\sum_{i=1}^n \frac{\partial H}{\partial x_i} f_i \right)^{-1}$$

is independent of the path of integration. Let the system of differential equations

$$\frac{dx_i}{dt} = f_i(x_1, \dots, x_n) \quad (i = 1, \dots, n)$$

belong to the foregoing class. Then the corresponding Lyapunov function $V(X)$, which is the solution of the Eq. (2.3), is the function $H(X)$.

From (2.19) we have

$$\theta(H) \sum_{i=1}^n \frac{\partial H}{\partial x_i} f_i = \varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

We multiply both members of this equation by the function $1 + V(X)$. Then

$$[1 + V(X)]\theta(H) \frac{dH}{dV} \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i = [1 + V(X)]\varphi(X) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

It follows from this equation that

$$[1 + V(X)]\theta(H) \frac{dH}{dV} = 1.$$

We find that the Lyapunov function $V(X)$ is the function H . If we now choose the function H to be arbitrary together with the function $\theta(H)$, then for any system (2.1) they can always be selected such that the integral (2.18) will be independent of the path of integration.

These arguments may prove to be useful in the case when we deal with any specific system of differential equations of the form (2.1).

2. Case of analytic right members of the system (2.1)

Let us first turn to a study of linear systems of differential equations with constant coefficients:

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si} x_i \quad (s = 1, \dots, n). \quad (2.20)$$

The coefficients p_{si} of the right members of the system (2.20) will be considered to be arbitrary complex numbers.

We denote by $\lambda_1, \dots, \lambda_n$ the roots of the characteristic equation $|P - \lambda E| = 0$, where $P = \{p_{ik}\}$. We consider all possible complex solutions of the system (2.20):

$$X(t, X^{(0)}) = e^{Pt} X^{(0)}. \quad (2.21)$$

It is clear that Eq. (2.21) defines a dynamical system in the complex n -dimensional space.

THEOREM 26 [16]. *If $\operatorname{Re} \lambda_i < 0$, then there exist two Hermitian forms V and W , having the following properties:*

- 1) *the forms V and $-W$ are positive definite;*
- 2) *the total derivative of the form V , calculated by virtue of Eq. (2.21), satisfies the equality $dV/dt = W$.*

PROOF. We take any negative definite Hermitian form W , $W(X) = X^* C \bar{X}$, where C is a matrix having the property that $\bar{C}^* = C$. We set

$$V(X^{(0)}) = \int_0^{+\infty} -W(X(t, X^{(0)})) dt. \quad (2.22)$$

[The bar over the symbol of the matrix denotes the complex conjugate, the asterisk indicates the transpose.]

Inserting into (2.22) the form W as calculated from the solution (2.21), we have

$$V(X^{(0)}) = - (X^{(0)})^* \int_0^{+\infty} e^{P^* t} C e^{F t} dt \bar{X}^{(0)}. \quad (2.23)$$

Let us first show that the integral in formula (2.23) converges. Let S be the matrix that transforms P into the canonical form, i.e., $SPS^{-1} = A$. We denote by

$$\lambda = \sup_{i=1, \dots, n} \operatorname{Re} \lambda_i;$$

then we obtain from formula (2.23)

$$V(X^{(0)}) = (X^{(0)})^* \int_0^{+\infty} e^{A t} B(t) dt \bar{X}^{(0)}.$$

It is easy to see that the elements of the matrix

$$B(t) = S^* S^{-1} e^{(A^* - (\lambda/2)E)t} S^* C \bar{S} e^{[\bar{A} - (\lambda/2)E]t} \bar{S}^{-1}$$

are bounded in absolute value for all $t \geq 0$. Consequently, the integral in formula (2.23) converges and therefore it defines a quadratic form $V(X^{(0)})$. Let us show now that this form is Hermitian. We set

$$D = \int_0^{+\infty} e^{P^* t} C e^{F t} dt.$$

Let us show that $\bar{D}^* = D$. In fact,

$$\bar{D}^* = \int_0^{+\infty} \overline{(e^{F t})^* C^* (e^{P^* t})^*} dt = D$$

because $\bar{C}^* = C$. From formula (2.22) it follows that the Hermitian form $V(X)$ is positive definite. Let us show now that Condition 2 is satisfied. In fact,

$$\begin{aligned} V(X(t, X^{(0)})) &= \int_0^{+\infty} -W(X(\tau, X(t, X^{(0)}))) d\tau = \\ &= \int_0^{+\infty} -W(X(\tau + t, X^{(0)})) d\tau. \end{aligned}$$

Setting $\tau + t = \theta$ in the last integral, we obtain

$$V(X(t, X^{(0)})) = \int_t^{+\infty} -W(X(\theta, X^{(0)})) d\theta.$$

Differentiating this equality, we obtain

$$\frac{dV}{dt} = W.$$

Thus, the existence of the Hermitian forms V and W , satisfying the Conditions 1 and 2, is proved. As follows from Theorem 19 (which incidentally was clear also earlier), the trivial solution of the system (2.20) is asymptotically stable in this case, and the region of asymptotic stability A coincides with the entire complex n -dimensional space.

THEOREM 27. *If among the quantities $\lambda_1, \dots, \lambda_n$ there are $k > 0$ with positive real parts, and $n - k$ with negative real parts, then there exist two Hermitian forms V and W , having the following properties:*

- 1) *W is negative definite, and the form V is alternating;*
- 2) *the total derivative of the form V calculated by virtue of the system (2.20), satisfies the equation $dV/dt = W$.*

PROOF. Let us make a linear transformation in the system (2.20) with the non-singular matrix S , $X = SY$, so as to have

$$\begin{aligned} \frac{dy_s}{dt} &= \varepsilon_{s-1}y_{s-1} + \lambda_s y_s \quad (s = 1, \dots, n), \\ \varepsilon_0 &= 0. \end{aligned} \tag{2.24}$$

The quantities ε_{s-1} are either ones or zeros. Let us take the first k equations. Using these, we construct the Hermitian form

$$V_1(y_1^{(0)}, \dots, y_k^{(0)}) = \int_0^{-\infty} -W_1(y_1(t, y_1^{(0)}, \dots, y_k^{(0)}), \dots, y_k(t, y_1^{(0)}, \dots, y_k^{(0)})) dt,$$

where W_1 is a negative definite Hermitian form in the quantities $y_1, \dots, y_k$. We shall assume that $\operatorname{Re} \lambda_j < 0$ for $j = 0, 1, \dots, k$. The form V_1 will be Hermitian negative as a result of the Theorem 26, and by virtue of the first k equations of the system (2.24) we have

$$\frac{dV_1}{dt} = W_1.$$

Let us take further the negative definite form W_2 in the quantities $y_{k+1}, \dots, y_n$ and, using the last $n - k$ equations of the system (2.24), let us construct, following Theorem 26, the Hermitian form

$$V_2 = \int_0^{+\infty} -W_2 dt.$$

It is clear that by virtue of the last $n - k$ equations of the system (2.24) we have

$$\frac{dV_2}{dt} = W_2.$$

We set $V = V_1 + V_2$ and $W = W_1 + W_2$. Then, reverting to the old variables, we obtain two Hermitian forms, satisfying all the conditions of the theorem.

REMARK. If the coefficients in the system (2.20) are real, then the forms V and W in Theorems 26 and 27 can be selected as simply quadratic with real symmetric matrices.

Let us consider now the system of differential equations

$$\begin{aligned} \frac{dx_s}{dt} &= \sum_{i=1}^n p_{si} x_i + \sum_{\substack{\sum_{k=1}^n m_k > 1 \\ k=1}} P_s^{(m_1, \dots, m_n)} x_1^{m_1} \dots x_n^{m_n} \equiv \\ &\equiv F_s(x_1, \dots, x_n) \quad (s = 1, \dots, n), \end{aligned} \quad (2.25)$$

the right members of which are holomorphic functions in $x_1, \dots, x_n$. Let us assume further that the coefficients p_{st} and $P_s^{(m_1, \dots, m_n)}$ are real and the real parts of the roots of the characteristic equation $|P - \lambda E| = 0$ are negative. Let us consider the partial differential equation

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} F_i = \varphi(X) [1 + \sum_{i=1}^n F_i^2] (1 + V), \quad (2.26)$$

corresponding to the system (2.25). Here the function $\varphi(x_1, \dots, x_n)$ is a positive definite quadratic form in $x_1, \dots, x_n$. By way of the function $\varphi(X)$ one can also take any holomorphic positive function, the lowest terms in the expansion of which in integral positive powers of the quantities $x_1, \dots, x_n$ form a positive definite form of degree $2m$, $m \geq 1$. It is easy to show (see Theorem 22) that for a function $\varphi(X)$ chosen in this manner the Equation (2.26) has a solution that satisfies all the conditions of Theorem 22.

As follows from the theorem of A. M. Lyapunov [1], with this choice of the function $\varphi(X)$, the function $V(X)$ can be obtained uniquely in the form of a power series, which converges for sufficiently small $|x_s|$, and which vanishes for $X = 0$. We shall seek a solution $V(X)$ in the form of a series

$$V(X) = V_2(X) + V_3(X) + \dots + V_m(X) + \dots, \quad (2.27)$$

where $V_m(X)$ is a homogeneous form of m -th degree in the quantities $x_1, \dots, x_n$. Inserting (2.27) into (2.26), we obtain for the definition of the form $V_m(X)$ the following system of equations

$$\left. \begin{aligned} \sum_{i=1}^n \frac{\partial V_2}{\partial x_i} \left(\sum_{k=1}^n p_{ik} x_k \right) &= \varphi(x_1, \dots, x_n), \\ \sum_{i=1}^n \frac{\partial V_m}{\partial x_i} \left(\sum_{k=1}^n p_{ik} x_k \right) &= R_m(x_1, \dots, x_n) \\ &\quad (m = 3, 4, \dots), \end{aligned} \right\} \quad (2.28)$$

(where R_m is a known form of m -th degree), provided $V_2, \dots, V_{m-1}$ have already been obtained.

From the system (2.28) we obtain successively the forms $V_2, V_3, \dots$; as follows from Theorem 26, the form $V_2(X)$ will be negative definite.

It is known [1] that the series (2.27) converges in a certain sufficiently small neighborhood of the origin. Let us make a few remarks with respect to its properties.

THEOREM 28. *If the right members of the system (2.25) are series that converge for all finite values of the quantities $x_1, \dots, x_n$, then in a sufficiently small neighborhood of the point $X^{(0)} \in A$ the function $V(X)$ can be represented in the form of a series*

$$V(X) = -1 + \sum_{\substack{\sum_{i=1}^n m_i > 0}} c_{m_1, \dots, m_n} (x_1 - x_1^{(0)})^{m_1} \dots (x_n - x_n^{(0)})^{m_n},$$

which converges when $|X - X^{(0)}|^2 \leq r^2(X^{(0)})$.

PROOF. Let us construct the solution $X(t, X^{(0)})$ of the system (2.25), which passes at the instant $t=0$ through the point $X^{(0)} \in A$. Let us make the following change in the system (2.25):

$$X = X(t, X^{(0)}) + \xi, \text{ i.e., } x_i = x_i(t, X^{(0)}) + \xi_i \quad (i = 1, \dots, n). \quad (2.29)$$

Then the system (2.25) assumes the form

$$\frac{d\xi_i}{dt} = \sum_{k=1}^n u_{ik}(t) \xi_k + \sum_{\substack{\sum_{i=1}^n m_i > 1}} U_i^{(m_1, \dots, m_n)} \xi_1^{m_1} \dots \xi_n^{m_n}.$$

The solution of this system can be represented in the form of a series [1]:

$$\xi_i = \sum_{m_1 + \dots + m_n \geq 1} \varphi_i^{(m_1, \dots, m_n)} (\xi_1^{(0)})^{m_1} \dots (\xi_n^{(0)})^{m_n},$$

where $\xi^{(0)}$ is the point through which the solution $\xi(t, \xi^{(0)})$ passes at the instant $t=0$.

Thus, $\xi^{(0)} = X - X^{(0)}$. These series converge for sufficiently small values of $\xi^{(0)}$.

Inserting (2.29) into the right member of the equation

$$1 + V(X) = e^{\int_{-\infty}^t \varphi(X(t, X)) \left(1 + \sum_{i=1}^n \int dt\right)},$$

we obtain what is required.

COROLLARY. *The function $V(x_1, \dots, x_n) = V(X)$, which is defined as a solution of (2.26) in the form of a converging series in a sufficiently small neighborhood of the origin, can be analytically continued along*

any ray, starting from the origin up to the boundary of the stability region.

We note that the choice of the function $\varphi(X)$ in Eq. (2.26) influences the region of convergence of the series (2.27), for example: the equation

$$-\sum_{i=1}^n x_i \frac{\partial V}{\partial x_i} = \varphi(X)(1 + V) \quad \text{for } \varphi = 2 \sum_{i=1}^n x_i^2$$

has a solution

$$V(X) = e^{-\sum_{i=1}^n x_i^2} - 1,$$

and when $\varphi(X)$ is equal to $\frac{\sum_{i=1}^n x_i^2}{\sqrt{(1 + \sum_{i=1}^n x_i^2)^3}}$,

$$V(X) = e^{\frac{1}{\sqrt{1 + \sum_{i=1}^n x_i^2}} - 1} - 1.$$

If we seek in this example $V(X)$ in the form of a series, then the region of its convergence in the former case will be all of space, whereas in the latter case only its bounded part.

Let us consider the first-order equation

$$\frac{dy}{dt} = f(y),$$

where $f(y)$ is an entire function, which has no real roots different from $x = 0$. The equation for the function V will have the form

$$f(y) \frac{dV}{dy} = \varphi(y)(1 + f^2)(1 + V).$$

It is easy to show that when $\varphi = f^2$, V is also obtained in the form of an entire function, and furthermore if $f(y)$ does not have a linear term in its expansion in integer powers of y , but is possible by choosing the function φ to see to it that in this case V is obtained in the form of a series.

These problems must be resolved for the case of a system of equations. The function V constructed here makes it possible to find a certain region, which is completely imbedded in A .

Let us now give a method of finding a region completely imbedded in the region A . We consider the family of surfaces $V_2(X) = -\mu$ [see (2.27)], where $\mu \in (0, +\infty)$, $\mu = \text{const}$. If the surface S is the boundary of the region A , then there is a value $\mu = \bar{\mu}$ such that the surface $V_2(X) = -\bar{\mu}$ will be tangent at a certain point to the surface S . In fact, since the family of surfaces $V_2 = -\mu$ fills the entire space, one can find a value μ , for which $V_2 = -\mu$ will intersect the surface S . We denote by $-\bar{\mu}$ the maximum value of $V_2(X)$ on a piece of the surface S , included within the surface $V_2 = -\mu$. Then $V_2(X) = -\bar{\mu}$ will indeed be tangent to S .

As shown earlier, the surface S has the equation $1 + V(X) = 0$. Then at the point of tangency $\bar{X}$ we shall have

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} F_i = 0$$

and

$$\frac{\frac{\partial V}{\partial x_1}}{\frac{\partial V_2}{\partial x_1}} = \frac{\frac{\partial V}{\partial x_2}}{\frac{\partial V_2}{\partial x_2}} = \dots = \frac{\frac{\partial V}{\partial x_n}}{\frac{\partial V_2}{\partial x_n}},$$

from which we have

$$\sum_{i=1}^n F_i \frac{\partial V_2}{\partial x_i} = 0.$$

[It is clear that the case when S is located in its entirety outside A is excluded.]

We note that $V_2(X)$ is the Lyapunov function for the system (2.25). Let us find dV_2/dt by virtue of the system (2.25).

$$\begin{aligned} W = \frac{dV_2}{dt} &= \sum_{i=1}^n \frac{\partial V_2}{\partial x_i} F_i = \varphi(X) + \\ &+ \sum_{i=1}^n \frac{\partial V_2}{\partial x_i} \left(\sum_{m_1 + \dots + m_n = 1} P_i^{(m_1, \dots, m_n)} x_1^{m_1} \dots x_n^{m_n} \right). \end{aligned}$$

The function W is positive definite. Let us find the set of points X , satisfying the equation

$$W(X) = 0,$$

and denoted by W_0 , considering that the point $X = 0$ does not belong to the set W_0 . Let us find the largest value of the function $V_2(X)$ on the set W_0 and let us denote it by $-\mu_0$. $\mu_0 > 0$, since $W(X)$ is a positive definite function.

THEOREM 29. *The surface $V_2(X) = -\mu_0$ is contained in its entirety in the region A .*

PROOF. Let us assume the contrary, that the surface $V_2(X) = -\mu_0$ is partially contained in the region A . [This follows from the property of $\bar{\mu}$.] Then there exists on a piece of the surface S a point, in which the function $V_2(X)$ attains the maximum value $-\mu_0$ and, as was shown, $W = 0$ at this point, which is impossible, since $-\mu_0$ is the largest value of $V_2(X)$ on the set W_0 . Consequently, the value $-\mu_0$ cannot be the maximum for the function V_2 on the set W_0 , which was to be proved.

We note that in our arguments the function φ is an arbitrary positive definite quadratic form. Thus

$$V_2(X) = -\mu_0 \quad (2.30)$$

is a family of curves, that depend on the parameters $a_{ik} = \varphi_{x_i x_k}$. Using this, we can formulate many criteria by which to establish the character of the stability region.

THEOREM 30. *In order for the region of stability to be bounded, it is necessary that the family of surfaces (2.30) be bounded, and in order for the stability region to be unbounded it is sufficient that this family be unbounded.*

The proof is obvious. For practical application of the foregoing method it is possible to make successful use of the family of surfaces

$$S_n = -\mu, \quad (2.31)$$

where

$$S_n = v_2(X) + \dots + v_n(X),$$

wherein it is assumed that $S_n = -\mu$ bounds the region $S_n > -\mu$ which contains the origin $X = 0$. Let us define for this family, in an analogous manner, the function $W_n(X)$, $W_n(X) = dS_n/dt$,

and the set of its zeros W_{0n} , where $X = 0$ does not belong to W_{0n} . For the family (2.31), it is possible to define the quantities $-\mu_{0n}$, where $-\mu_{0n}$ represents the maximum value of the function on the set W_{0n} . As before, it can be shown that the surface $S_n = -\mu_{0n}$ is contained in its entirety in the stability region, and the deductions formulated above can be carried over to the family (2.31). This also suggests a method of constructing a family of regions, which belong in their entirety to the region A .

The idea of constructing some region that belongs in its entirety to A , based on the use of the Lyapunov function, was employed by Lyapunov himself, and later, in various forms, also by other authors. In the present section this idea is used in a somewhat different form.

Generally speaking, as n increases the family $S_n = -\mu_{0n}$ represents more and more accurately the boundary of the region A . In this work a method is proposed for constructing a small solution in the region A and one uses in conjunction with the foregoing method an analytical procedure, which makes it possible to construct all solutions in the region A , for the investigation of certain non-local problems.

Let us proceed now to the construction [39, 40] of any solution in the region of asymptotic stability. Let us assume that a system of differential equations

$$\frac{dx_i}{dt} = f_i(x_1, \dots, x_n) \quad (2.32)$$

is given, where the functions f_i are holomorphic in a certain region G , i.e. if the point $(X_1^{(0)}, \dots, x_n^{(0)}) \in G$, then

$$\begin{aligned} f_i(x_1, \dots, x_n) = \\ = \sum_{m_1 + \dots + m_n = 0}^{+\infty} p_i^{(m_1, \dots, m_n)} (x_1 - x_1^{(0)})^{m_1} \dots (x_n - x_n^{(0)})^{m_n} \end{aligned}$$

are series that converge in the region $|x - x_i^{(0)}| < \rho$ ($i = 1, \dots, n$).

Let a certain integral curve of the system (2.32), $X(t, X^{(0)})$, remain in the bounded region G_1 , which is completely imbedded in G . Then it can be continued to all values $t \in (-\infty, +\infty)$.

No matter what the point $(x_1^{(1)}, \dots, x_n^{(1)}) \in G_1$ may be, there

are numbers M and ρ such that the function

$$F(X) = \frac{M}{1 - \frac{\sum_{i=1}^n (x_i - x_i^{(1)})}{\rho}}$$

is a majorant of the series

$$f_i(X) = \sum_{m_1 + \dots + m_n = 0}^{+\infty} P_i^{(m_1, \dots, m_n)} (x_1 - x_1^{(1)})^{m_1} \dots (x_n - x_n^{(1)})^{m_n}. \quad (2.33)$$

The system of differential equations

$$\frac{dx_i}{dt} = F(X)$$

is a majorant for (2.32).

Let the integral curve $X(t, X^{(0)})$ reach the points X_1 at the instant $t = t_1$. Then by the Cauchy theorem the solution $X(t, X^{(0)})$ can be represented in the form of series

$$x_i = x_i^{(1)} + \sum_{k=1}^{\infty} a_{ki}(X_1)(t - t_1)^k \quad (i = 1, \dots, n), \quad (2.34)$$

which converge for $|t - t_1| < h$.

Let us construct for these series the corresponding majorant function. For this purpose we find the solution of the Equation (2.33) under the assumption that at $t = t_1$ we have $x_i = x_i^{(1)}$; then $x_i - x_i^{(1)} = z$ will satisfy the equation

$$\frac{dz}{dt} = \frac{M}{\left(1 - \frac{nz}{\rho}\right)}.$$

Integrating this under the assumption that $z = 0$ for $t = t_1$, we obtain

$$z = \frac{\rho}{n} \left[1 - \sqrt{1 - \frac{2Mn}{\rho}(t - t_1)} \right].$$

From the form of this function it follows that the series (2.34)

converge when

$$|t - t_1| < \frac{\rho}{2Mn} = h$$

independently of the point $X^{(1)}$, lying on the integral curve $X(t_1, X^{(0)})$. This proves the theorem.

THEOREM 31. *If an integral curve of the system (2.32) remains in a bounded region, imbedded together with its boundary in the region of analyticity of the right members of the system (2.32), then the functions that describe this integral curve are regular in a strip of width $2h$, which contains the real axis t .*

COROLLARY. *If the integral curve $X(t, X^{(0)})$ remains in the region G_1 for $t \in [0, +\infty)$, then the functions $x_i(t, x_1^{(0)}, \dots, x_n^{(0)})$ will be regular in a half-strip of width $2h$, containing the real positive half-axis $t \in [0, +\infty)$.*

Let us assume that $X(t, X^{(0)})$ is the solution of the system (2.32) and remains in the bounded region G_1 , completely imbedded in G . Then, as was shown, the functions $x_i(t, x_1^{(0)}, \dots, x_n^{(0)})$ are regular in a strip of width $2h$. Let us map this strip conformally on the circle $|y| < 1$, putting $t = 0$, $y = 0$ into correspondence. We then obtain

$$y = \frac{e^{\pi t/h} - 1}{e^{\pi t/h} + 1}. \quad (2.35)$$

We now make the change of independent variable (2.35) in the system (2.32)

$$\frac{dx_i}{dy} \frac{dy}{dt} = f_i(x_1, \dots, x_n). \quad (2.36)$$

It is easy to see that

$$\frac{dy}{dt} = \frac{\pi}{2h} (1 - y^2) = \kappa(1 - y^2), \quad \text{where } \kappa = \frac{\pi}{2h}.$$

The solution of the system (2.36), which passes at the instant $t = 0$ through the point $X^{(0)}$, will be regular in the circle $|y| < 1$, and can therefore be represented in the form of a converging series

$$x_i = x_i^{(0)} + \sum_{k=1}^{\infty} A_{ki}(X^{(0)}) y^k \quad (i = 1, \dots, n), \quad (2.37)$$

the region of convergence of which is $|y| < 1$ or $-\infty < t < +\infty$.

Let us give the actual determination of the coefficients of these series.

Inserting these into the system (2.36) and comparing terms with equal powers of y , we obtain the following equations:

$$\kappa(k+1)A_{k+1,i} - \kappa(k-1)A_{k-1,i} = \psi_{ki} \quad \left(\begin{matrix} k = 2, 3, \dots \\ i = 1, \dots, n \end{matrix} \right),$$

where ψ_{ki} are polynomials in $A_{\mu i}$ ($\mu < k$) with coefficients which are combinations of the coefficients of the expansions of the functions $f_i(X)$ in powers of $(x_i - x_i^{(0)})$ ($i = 1, \dots, n$) and positive integers, where it is easy to see that $\psi_{1i} = f_i(x_1^{(0)}, \dots, x_n^{(0)})$. Let there exist in the system (2.36) an asymptotically stable trivial solution. We assume that the region of its asymptotic stability A is bounded and is entirely imbedded in the region G together with its boundary.

Each solution that lies in the region A can be represented by series of the type (2.37), which converge for $t \in (-\infty, +\infty)$.

THEOREM 32. *The general solution in the bounded region of asymptotic stability A of the system (2.32), which is completely imbedded together with its boundary in the region G , can be represented in the form of series (2.37), which converge for $-\infty < t < +\infty$.*

The proof is obvious. Now let the region A have an arbitrary character. Then any solution $X(t, X^{(0)})$, beginning in it, remains for $t \in [0, +\infty)$ in the bounded region G_1 and therefore the functions $X(t, X^{(0)}, Y^{(0)})$ and $Y(t, X^{(0)}, Y^{(0)})$ are regular in the half-strip of width $2h$, containing the half-axis $t \in [0, +\infty)$. We map this half-strip on the circle $|\zeta| < 1$ and make a change of the independent variable in the system (2.32). We then obtain the system

$$\frac{dx_i}{d\zeta} \frac{d\zeta}{dt} = f_i(x_1, \dots, x_n). \quad (2.38)$$

The solution of this system, passing at the instant $\zeta = 0$ through the point $X^{(0)} \in A$, can be represented in the form of the series

$$x_i = x_i^{(0)} + \sum_{k=1}^{\infty} b_{ik}(X) \zeta^k \quad (i = 1, \dots, n), \quad (2.39)$$

which converges for $|\zeta| < 1$.

A conformal mapping of the half-strip on the circle can be realized by the function

$$\zeta = \frac{\operatorname{sh} \frac{\pi t}{2h} - 1}{\operatorname{sh} \frac{\pi t}{2h} + 1};$$

hence

$$\operatorname{sh} \frac{\pi t}{2h} = \frac{1 + \zeta}{1 - \zeta}.$$

Differentiating this equation with respect to t , substituting

$$\operatorname{ch} \frac{\pi t}{2h} = \sqrt{1 + \operatorname{sh}^2 \frac{\pi t}{2h}}$$

and setting $\frac{\pi}{2h\sqrt{2}} = \omega$, we obtain

$$\frac{d\zeta}{dt} = \omega(1 - \zeta)\sqrt{1 + \zeta^2}.$$

Let us expand the right member of this equality in powers of ζ :

$$\frac{d\zeta}{dt} = \omega + \sum_{i=1}^{\infty} C_i \zeta^i, \quad (2.40)$$

where the C_i are known constants.

Substituting (2.40) into (2.38), we obtain the system

$$\frac{dx_i}{d\zeta} \left(\omega + \sum_{i=1}^{\infty} C_i \zeta^i \right) = f_i(x_1, \dots, x_n).$$

Substituting the series (2.39) in the last system, we obtain the possibility of defining successively the coefficients of these series. Let us change the independent variable in the system (2.32):

$$t = \tau + \frac{h}{\pi} \ln(1 + \sqrt{2}).$$

Then $\tau = 0$ corresponds to $\zeta = 0$. Consequently, the solution

which passes through the point $X^{(0)}$ at the instant $\tau = 0$ can be represented in the form of the series (2.39), which converges for all $t \in [0, +\infty)$.

THEOREM 33. *If the system of Eqs. (2.32) has an asymptotically stable trivial solution and its right members are holomorphic at any point $X^{(0)} \in A$, then any solution $X(t, X^{(0)})$, $X^{(0)} \in A$, can be represented by the series (2.39), which converge for $|\zeta| < 1$ or $t \in [0, +\infty)$.*

REMARK 1. The series (2.39) and (2.37) can be used not only in the region of asymptotic stability. For example, the series (2.39) can be employed for an approximate representation of the limit cycle outside the region A .

REMARK 2. The series (2.39) and (2.37) have that advantage over the series that represent the general solution in a neighborhood of the origin, constructed by A. M. Lyapunov, first that they do not require negativeness or positiveness of the real parts of all the roots of the equation

$$|P - \lambda E| = 0,$$

and, second, they converge for all finite values of $X^{(0)} \in A$, whereas the Lyapunov series converge in a sufficiently small neighborhood of the origin. Third, in the case of a bounded region A , the series converge for all $t \in (-\infty, +\infty)$.

REMARK 3. If the solution remains bounded for $t \in (-\infty, 0]$, then by using similar arguments as in the case when the solution remains bounded for $t \in [0, +\infty)$, it is possible to arrive at the same series (2.39), but which now converges only for $t \in [0, +\infty)$; consequently if the solution is bounded for all $t \in (-\infty, +\infty)$, it is possible to represent by means of these series both the positive and the negative half-trajectories.

Generally speaking, one can point out many methods of constructing a general solution in the region of asymptotic stability, and therefore the device employed here cannot be considered unique. Different forms of the general solution can be obtained by mapping strips with curved boundaries on a circle.

If, in mapping a strip of width $2h$ on the circle $|y| < 1$, we represent the point $y = 0$ by the point $t = +\infty$, we obtain a general solution in the form of some other series, which can be used in the solution of some problems with great success.

Let us now employ the theory we developed to solve the problem of finding the limit cycle.

We consider a system of two equations

$$\frac{dx_i}{dt} = f_i(x_1, x_2) \quad (i = 1, 2), \quad (2.41)$$

the right members of which are expanded in series:

$$f_i = \sum_{m_1, m_2=1}^{+\infty} P_i^{(m_1, m_2)} x_1^{m_1} x_2^{m_2}, \quad (2.42)$$

which converge for all finite values of x_1, x_2 , and all the roots of the equation $|P - \lambda E| = 0$ have negative real parts.

Let us find the function

$$S_n(x_1, x_2) = V_2(x_1, x_2) + \dots + V_n(x_1, x_2)$$

and assume that the series

$$V = V_2 + V_3 + \dots + V_n + \dots \quad (2.43)$$

converges in a certain region, containing the boundary of the region A . Then the family $S_n(x_1, x_2) = -1$, which bounds the region $S_n > -1$ that contains the point $(0, 0)$, will represent the boundary of the region A as accurately as desired.

If the limit cycle exists, then the curve $S_n(x_1, x_2) = -1$ gives its representation as accurately as desired.

If, conversely, it is found that $S_n = -1$ is bounded as $n \rightarrow +\infty$, and if in a sufficiently small neighborhood of this curve there are no rest points of the system (2.41), then the limit cycle exists and can be represented approximately by the equation $S_n = -1$.

Let us assume that the region of convergence of the series (2.43) is unknown; then, to find the limit cycle and its approximate representation, we can recommend the following method.

We plot the curve $S_n(x_1, x_2) = -\mu_{0n}$, as shown above. The region $S_n > -\mu_{0n}$, which contains the point $(0, 0)$, is completely imbedded in the region A . Let us take any point on the curve $S_n = -\mu_{0n}$. We draw from this point a negative half-trajectory, representing it in the form of the series (2.39).

If the limit cycle exists, then the trajectory is bounded and makes an infinite number of revolutions, approaching the limit cycle, and therefore the series (2.39) can represent, as accurately as desired, the limit cycle for sufficiently large values of t . Conversely, if it is found that a solution represented in the form of the series (2.39) is bounded, and the trajectory makes an infinite number of revolutions about the origin, then the limit cycle exists and these series give as accurate a representation as desired for this cycle for sufficiently large values of t . However, it is possible in this case that the boundary of the region A is a closed curve, on which the rest points of the system (2.41) lie.

We can formulate another principle, which requires a smaller number of calculations. Let us consider the curve $S_n = -1$, which bounds the region $S_n > -1$, which contains the point $(0, 0)$, and let us pick a certain point on this curve. We draw from this point a trajectory, making use of the series (2.37). If this point belongs to the bounded stability region, then the solution (2.37) tends to 0 as $t \rightarrow +\infty$ and as $t \rightarrow -\infty$ we can determine from the behavior of the series (2.37) whether the boundary is a limit cycle or not. Specifically, if the series describes a trajectory which makes an infinite number of revolutions about the origin, then the limit cycle exists. But if this point does not belong to the stability region, then, by using the series (2.39), we can find the limit cycle.

However, the second principle just described cannot always give an effective result, for it is based on the assumption that $S_n = -1$ is close to the boundary of the stability region, which can be guaranteed only in the case when the region of asymptotic stability is imbedded in the region of convergence of the series (2.43).

It has been shown that $V(x_1, x_2)$ can be expanded, in a neighborhood of each point $(x_1^{(0)}, x_2^{(0)}) \in A$, into a convergent series:

$$V = \sum_{k, l=0}^{\infty} c_{kl}(x_1 - x_1^{(0)})^k (x_2 - x_2^{(0)})^l. \quad (2.44)$$

Using this property of $V(x_1, x_2)$, we can approach the boundary of A as close as desired each time, by using analytic continuation. This analytic continuation can be performed along any ray kx_1 , starting with $(0, 0)$. In the case when a limit cycle exists, it is

possible to perform successfully the analytic continuation along the coordinate axis x_1 or x_2 .

When approaching the boundary, as was shown, the function V tends to -1 . Taking this fact into account, we can detect by means of analytic continuation the points located near the boundary of the region A .

3. Systems of differential equations with homogeneous right members

DEFINITION. A function $f(x_1, \dots, x_n)$, defined in E_n , is called *homogeneous of order* $\mu = p/q$, where p and q are natural numbers and q is odd, provided the following equality holds:

$$f(cx_1, \dots, cx_n) = c^\mu f(x_1, \dots, x_n).$$

[$c^{p/q}$ is positive for p even, and $\text{sign } c^{p/q} = \text{sign } c$ for p odd.]

Here we shall call the function $f(x_1, \dots, x_n)$ *positive-homogeneous* if $c > 0$ and μ is arbitrary.

If a homogeneous or positive-homogeneous function $f(x_1, \dots, x_n)$ is continuously differentiable over all its arguments up to order ν inclusive, $\nu \geq 1$, then it satisfies the linear partial differential equation

$$\sum_{i=1}^n \frac{\partial f}{\partial x_i} x_i = \mu f \quad [15],$$

where μ is the index of homogeneity of the function f . If $\nu \geq 2$, then the function $\partial f / \partial x_j$ is also homogeneous of order $\mu - 1$, which can be readily established by differentiating the relation $f(cx_1, \dots, cx_n) = c^\mu f(x_1, \dots, x_n)$ with respect to the variable x_j .

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = X_s^{(\mu)}(x_1, \dots, x_n). \quad (2.45)$$

[The right members will be assumed to satisfy certain uniqueness conditions at $\nu = 0$.]

We assume that the functions $X_s^{(\mu)}$ are defined in E_n and are continuously differentiable there in all their arguments to order $\nu \geq 0$ inclusive, and are homogeneous of degree μ , $\mu > 0$. We shall assume that the case $\nu = 0$ corresponds to continuous functions

$X_s^{(\mu)}$. The system (2.45) has the trivial solution $X = 0$. Let us proceed to investigate it from the point of view of its stability in the sense of Lyapunov.

THEOREM 34. *If the system of differential equations (2.45) has an integral curve $X = X(t, X^{(0)})$, then any curve of the family*

$$Y = cX(c^{\mu-1}t, X^{(0)}) \quad (2.46)$$

is also an integral curve of the system (2.45); here

$$cX(c^{\mu-1}t, X^{(0)}) = X(t, X^{(0)}). \quad (2.47)$$

PROOF. Let the function $X(t, X^{(0)})$ be a solution of the system (2.45). We then have the identity

$$\frac{dx_s(t, X^{(0)})}{dt} = X_s^{(\mu)}(x_1(t, X^{(0)}), \dots, x_s(t, X^{(0)}), \dots, x_n(t, X^{(0)})).$$

Let us calculate the derivative of the family (2.46) with respect to t . We then obtain

$$\begin{aligned} \frac{dy_s}{dt} &= c^{\mu} \frac{d}{dt} x_s(tc^{\mu-1}, X^{(0)}) = \\ &= c^{\mu} X_s^{(\mu)}(x_1(tc^{\mu-1}, X^{(0)}), \dots, x_n(tc^{\mu-1}, X^{(0)})) = X_s^{(\mu)}(cx_1, \dots, cx_n); \end{aligned}$$

hence

$$\frac{dy_s}{dt} = X_s^{(\mu)}(y_1, \dots, y_n).$$

Thus, the family of functions (2.46) forms a family of integral curves of the system (2.45). Let us show now that (2.47) actually is true. In fact, when $t = 0$ the equation (2.47) holds, and when $t \neq 0$ it will hold by virtue of the uniqueness of the solutions of (2.45).

COROLLARY. *If $X = X(t, X^{(0)})$ is an integral curve of the system (2.45) then the family of integral curves (2.46) fills a conical surface, the vertex of which lies at the point $X = 0$, and the set of directrices is the integral curve $X = X(t, X^{(0)})$.*

THEOREM 35. 1. *The trivial solution of the system (2.45) cannot be asymptotically stable at $p = 2k$, where k is a natural number.*

2. *When $\mu > 0$, $p \neq 2k$ and $\mu \neq 1$ the trivial solution of the system (2.45) can be asymptotically stable only under real perturbations.*

3. When $\mu = 1$ the trivial solution of the system (2.45) can be asymptotically stable for all complex perturbations.

[See Theorems 26, 27 and 35.]

PROOF. Consider the first case: let $p = 2k$ and let the trivial solution of (2.45) be asymptotically stable. Then there exists an integral curve of this system $X = X(t, X^{(0)})$, having the property $X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$. Let us consider the integral curve $X(t, -X^{(0)})$. By virtue of relation (2.47) we have

$$X(t, -X^{(0)}) = (-1)X((-1)^{(2k-q)/2}t, X^{(0)}) = -X(-t, X^{(0)}),$$

from which it follows that the integral curve $X(t, -X^{(0)})$ has the property $X(t, -X^{(0)}) \rightarrow 0$ as $t \rightarrow -\infty$, which contradicts the presence of asymptotic stability of the trivial solution of the system (2.45) [6]. Thus, when $p = 2k$ the trivial solution of the system (2.45) cannot be asymptotically stable.

Now consider the second case: let $p \neq 2k$ and $\mu \neq 1$. Suppose, to the contrary, that there exists a system of the type (2.45) for which the trivial solution is asymptotically stable for all complex perturbations. Let $X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$. We put $\varepsilon = \mu^{-1}\sqrt{-1}$. We consider the integral curve $X = X(t, \varepsilon X^{(0)})$. By virtue of relations (2.47) we obtain

$$\varepsilon X(-t, X^{(0)}) = X(t, \varepsilon X^{(0)}),$$

from which it follows that the integral curve considered has the property $X(t, \varepsilon X^{(0)}) \rightarrow 0$ as $t \rightarrow -\infty$. Consequently, when $p \neq 2k$ and $\mu \neq 1$ the trivial solution of any system of the form (2.45) can be asymptotically stable only for real perturbations.

Let us consider now the third case. In this case asymptotic stability is possible for all complex perturbations, as follows from relation (2.47).

COROLLARY. *If the trivial solution of the system (2.45) is stable (asymptotically stable) for sufficiently small perturbations, then all the solutions of the system (2.45) will be bounded at $t \in [0, +\infty)$, i.e., $|X(t, X^{(0)})| < K(X^{(0)}) < +\infty$ when $t \geq 0$, (asymptotic stability in the large holds).*

This conclusion was already obtained by V. S. Novoselov. If the trivial solution of the system (2.45) is asymptotically stable, then for a sufficiently broad class of systems one can show that the

following inequality holds:

$$|X(t, X^{(0)})| \leq A_1 t^{-\alpha} \text{ for } t \geq T \text{ and } |X^{(0)}| = 1, \quad (2.48)$$

where A_1 is a sufficiently large positive constant and α a sufficiently small positive constant. It is of interest to show that if the trivial solution of the system (2.45) is asymptotically stable, inequality (2.48) always holds. If this statement is considered proved, the condition (2.48) can be replaced everywhere in the following theorems by the requirement of asymptotic stability of the trivial solution of the system (2.45).

THEOREM 36. *If the system (2.45) is such that its solutions satisfy the inequality (2.48), then there exist two functions V and W , defined in E_n , and having the following properties:*

- 1) *the functions V and $-W$ are positive definite;*
- 2) *the function W is positive-homogeneous of order m , the function V is positive-homogeneous of order $m + 1 - \mu$, where m is a sufficiently large positive number;*
- 3) *the function V is continuously differentiable along the integral curves of the system (2.45), i.e., $V(X(t, X^{(0)}))$ has a continuous derivative and*

$$\frac{dV}{dt} = W.$$

PROOF. Let us set $W = -|X|^m$, where $m > 1/\alpha$ and

$$V(X^{(0)}) = \int_0^{+\infty} -W(X(t, X^{(0)})) dt. \quad (2.49)$$

Let us show that the formula (2.49) defines a certain function, defined in E_n . In fact, let $|X^{(0)}| = 1$; then for the integral curves of the system (2.45) we have the inequality (2.48). Let us estimate the integral occurring in the right member of formula (2.49), by means of (2.48). We obtain

$$V(X^{(0)}) \leq \int_0^T |X|^m dt + \int_T^{+\infty} A_1^m t^{-m\alpha} dt,$$

from which it follows that the formula (2.49) defines the function $V(X^{(0)})$ when $|X^{(0)}| = 1$.

Now let $X^{(0)}$ be any point in E_n , $X^{(0)} \neq 0$, $X^{(0)} = |X^{(0)}| Y^{(0)}$, where $Y^{(0)}$ is a unit vector, $|Y^{(0)}| = 1$. From relation (2.47) we have

$$X(t, X^{(0)}) = |X^{(0)}| X(|X^{(0)}|^{\mu-1}t, Y^{(0)}). \quad (2.50)$$

Using Eq. (2.50) and the inequality (2.48), we obtain

$$|X(t, X^{(0)})| \leq |X^{(0)}|^{1-\alpha(\mu-1)} A_1 t^{-\alpha} \text{ for } t \geq T. \quad (2.51)$$

This inequality makes it possible, as in the case when $|X^{(0)}| = 1$, to show that the integral in the right member of (2.49) converges. Thus, Eq. (2.49) determines a function defined in E_n . In this connection, $V(0) = 0$. Let us now make a substitution under the integral sign, using (2.50). We then obtain

$$\begin{aligned} V(X^{(0)}) &= \int_0^{+\infty} |X^{(0)}|^m X(|X^{(0)}|^{\mu-1}t, Y^{(0)})^m dt = \\ &= \int_0^{+\infty} |X^{(0)}|^{m+1-\mu} |X(\tau, Y^{(0)})|^m d\tau, \end{aligned}$$

from which it follows that

$$V(X^{(0)}) = |X^{(0)}|^{m+1-\mu} V(Y^{(0)}). \quad (2.52)$$

The equality (2.52) shows that the function constructed is positive-homogeneous of order $m+1-\mu$. Let us now show that the function V satisfies Condition 3. In fact, from (2.49) we have

$$V(X(t, X^{(0)})) = \int_0^{+\infty} |X(\tau, X(t, X^{(0)}))|^m d\tau = \int_0^{+\infty} |X(t + \tau, X^{(0)})|^m d\tau.$$

Setting $\theta = t + \tau$, we obtain

$$V(X(t, X^{(0)})) = \int_t^{+\infty} |X(\theta, X^{(0)})|^m d\theta.$$

Differentiating both members of the resultant equality, we obtain

$$\frac{dV}{dt} = -|X^{(0)}|^m = W.$$

Thus, the function V satisfies Condition 3.

THEOREM 37. *If the system (2.45) is such that its solutions satisfy the inequality (2.48), then the following inequalities hold:*

1) *when $\mu \in (0, 1)$, any solution $X = X(t, X^{(0)}) \neq 0$ for*

$$t \in [0, T_1], \quad T_1 = \frac{a_1}{a_2} \cdot |X^{(0)}|^{1-\mu}$$

satisfies the inequality

$${}^{1-\mu}\sqrt{a_1|X^{(0)}|^{1-\mu} - a_2 t} \leq |X(t, X^{(0)})| \leq {}^{1-\mu}\sqrt{b_1|X^{(0)}|^{1-\mu} - b_2 t}, \quad t > 0;$$

2) *when $\mu = 1$, any solution of the system (2.45) satisfies the inequalities*

$$p_1 |X^{(0)}| e^{-p_1 t} \leq |X(t, X^{(0)})| \leq q_1 |X^{(0)}| e^{-q_1 t}, \quad t > 0;$$

3) *when $\mu > 1$ and $p \neq 2k$, any solution of the system (2.45) satisfies the inequalities*

$$\frac{c_1 |X^{(0)}|}{\sqrt[\mu-1]{1 + c_2 t |X^{(0)}|^{\mu-1}}} \leq |X(t, X^{(0)})| \leq \frac{d_1 |X^{(0)}|}{\sqrt[\mu-1]{1 + d_2 t |X^{(0)}|^{\mu-1}}}, \quad t > 0,$$

where a_i, b_i, p_i, q_i, c_i and d_i ($i = 1, 2$) are certain positive constants.

PROOF. Let us show that the estimates 2) and 3) are satisfied. If the Condition (2.48) is satisfied, then, as shown in Theorem 36, there exist two functions $V(X)$ and $W(X)$, satisfying the three conditions formulated in Theorem 36. In this connection, $W(X) = -|X|^m$, and the function $V(X)$, by virtue of equality (2.52), satisfies the inequalities

$$a'_1 |X|^{m+1-\mu} \leq V(X) \leq a'_2 |X|^{m+1-\mu}, \quad (2.53)$$

where

$$a'_1 = \inf_{|X|=1} V(X), \quad a'_2 = \sup_{|X|=1} V(X).$$

Let us now turn to Theorem 20. Let us put $V(p) = |X|$, where $V(p)$ is the functional discussed in that theorem. Setting $l_1 = l_2 = m + 1 - \mu$, and $k_1 = k_2 = m$ in Theorem 20, we obtain assertion 3 from the estimate (1.40) and assertion 2 of our theorem from the estimate (1.43). Let us show now that assertion 1 is correct.

Let $\mu \in (0, 1)$. Along the integral curve $X = X(t, X^{(0)})$ of the system (2.45) we have $dV/dt = W$, where V and W are the functions

constructed in Theorem 36. Let us assume that

$$X(t, X^{(0)}) \neq 0;$$

then this defines the function

$$V^{-\lambda}(X(t, X^{(0)})), \quad \lambda > 0.$$

Let us multiply the equation $dV/dt = W$ by this function and let us integrate it from 0 to t_0 , obtaining

$$V(X(t, X^{(0)})) = \sqrt[1-\lambda]{V(X^{(0)})^{1-\lambda} + (1-\lambda) \int_0^t W(X(\tau, X^{(0)})) V^{-\lambda} d\tau}.$$

Using the inequalities for the function $V(X)$ and the form of the function $W(X)$ and putting $\lambda = m/(m+1-\mu)$, we obtain

$$\begin{aligned} \sqrt[1-\lambda]{(a_1')^{1-\lambda} |X^{(0)}|^{1-\mu} - (1-\lambda)(a_1')^{-\lambda} t} &\leq V(X(t, X^{(0)})) \leq \\ &\leq \sqrt[1-\lambda]{(a_2')^{1-\lambda} |X^{(0)}|^{1-\mu} - (1-\lambda)(a_2')^{-\lambda} t}. \end{aligned}$$

Using again the inequality which is satisfied by the function V , we find that assertion 1 of our theorem holds.

COROLLARY 1. *Theorem 37 shows that if the trivial solution of the system (2.45) is asymptotically stable and if estimate (2.48) holds for no matter how small a value of $\alpha > 0$, then actually any integral curve of the system (2.45) satisfies the inequalities indicated in items 1-3.*

COROLLARY 2. *The estimates contained in items 1-3 of Theorem 37 are exact, i.e., the method of calculation of the constants is such that in specific examples these inequalities turn into equalities, as can be shown, for example, by integrating the equation $\dot{x} = -x^\mu$, $\mu = 2k+1$, namely, the estimate obtained in accordance with items 2 and 3 is exact, i.e. the inequalities become equalities. Naturally the statement made here is not based only on this obvious example, but is based on the very nature of the method.*

COROLLARY 3. *As demonstrated by the conclusion, the estimate in the case $\mu \in (0, 1)$ holds only as long as $|X(t, X^{(0)})| \neq 0$. By virtue of the asymptotic stability of the trivial solution of the system (2.45), the integral curve, after arriving at the point $X = 0$ for $t = T$, will*

remain there for $t \geq T$, i.e. $X(t, X^{(0)}) \equiv 0$ for $t \geq T$. As can be shown by an estimate in the case $\mu \in (0, 1)$, any integral curve of the system (2.45) enters in the point $X = 0$ in a finite time $T(X^{(0)})$, so that when $t \in [0, T_1]$ the estimate given in item 1 of Theorem 37 is satisfied and when $t \geq T(X^{(0)})$ we have $X(t, X^{(0)}) \equiv 0$. This possibility was already noted in references [17, 18].

THEOREM 38. *If the system (2.45) is such that its solutions satisfy the inequality (2.48), then there exist two functions $V(X)$ and $W(X)$, satisfying all the conditions of Theorem 36, and, in addition, the function $W(X)$ can be chosen such that the function $V(X)$ will be continuously differentiable in all its arguments up to the same order ν inclusive as the right members of the Eq. (2.45). With this, the functions V and W will satisfy, when $\nu \geq 1$, the system of partial differential equations*

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} X_i^{(\mu)} = W, \quad \sum_{i=1}^n \frac{\partial V}{\partial x_i} x_i = (m - \mu + 1)V, \quad (2.54)$$

where m , the degree of homogeneity of the function W , is a sufficiently large positive number.

PROOF. Let the condition of the theorem be satisfied and let $\nu \geq 1$. Then one must have $\mu \geq 1$. If these conditions are satisfied, the assertions of Theorem 37 are correct. Therefore any integral curve of the system (2.45) satisfies either the inequality of item 2 of Theorem 37, or the inequality of item 3 for $\mu > 1$ and $p \neq 2k$. Let us put $W(X) = -|X|^m$; then, as shown in Theorem 36, there exists a function $V(X)$ defined by the formula (2.49):

$$V(X^{(0)}) = \int_0^{+\infty} |X(t, X^{(0)})|^m dt, \quad X \in E_n.$$

Let us show that the quantity m can always be chosen so large that the function V is continuously differentiable over all its arguments up to order ν inclusive. Let us differentiate formally this equation with respect to the variable $x_j^{(0)}$, obtaining

$$\frac{\partial V(X^{(0)})}{\partial x_j^{(0)}} = \int_0^{+\infty} |X(t, X^{(0)})|^{m-2} m \sum_{i=1}^n x_i(t, x_1^{(0)}, \dots, x_n^{(0)}) \frac{\partial x_i(t, X^{(0)})}{\partial x_j^{(0)}} dt. \quad (2.55)$$

We shall show that the integral in the right member of formula (2.55) converges uniformly, and this makes the formula define the partial derivative of the function V with respect to the argument x_j . For this purpose we first find those estimates, which are satisfied by the functions $\partial x_i / \partial x_j^{(0)}$. These functions satisfy a system of linear differential equations with variable coefficients of the form

$$\frac{dy_s}{dt} = \sum_{i=1}^n p_{si}(t, x_1^{(0)}, \dots, x_n^{(0)}) y_i, \quad (2.56)$$

where

$$p_{si}(t, x_1^{(0)}, \dots, x_n^{(0)}) = \frac{\partial X_s^{(\mu)}(X(t, X^{(0)}))}{\partial x_i}.$$

In fact, we shall show that this is true. Substituting the solution $X = X(t, X^{(0)})$ in the system (2.45), we obtain an identity. Differentiating this identity with respect to the variable $x_j^{(0)}$, we obtain

$$\begin{aligned} \frac{d}{dt} \frac{\partial x_s(t, x_1^{(0)}, \dots, x_n^{(0)})}{\partial x_j^{(0)}} &= \\ &= \sum_{i=1}^n p_{si}(t, x_1^{(0)}, \dots, x_n^{(0)}) \frac{\partial x_i(t, x_1^{(0)}, \dots, x_n^{(0)})}{\partial x_j^{(0)}}, \end{aligned}$$

and this means that the functions

$$\frac{\partial x_i(t, X^{(0)})}{\partial x_j^{(0)}}$$

satisfy the linear system (2.56). The coefficients p_{si} of the system (2.56) are homogeneous functions in the quantities $x_s(t, X^{(0)})$ of order $\mu - 1$, and therefore for $\mu = 1$ we have

$$|p_{si}| \leq K < +\infty \text{ for } t \geq 0.$$

When $\mu > 1$ and $p \neq 2k$ we obtain

$$|p_{si}(t, x_1^{(0)}, \dots, x_n^{(0)})| \leq |X(t, X^{(0)})|^{\mu-1} K \leq \frac{K |X^{(0)}|^{\mu-1}}{1 + d_2 t |X^{(0)}|^{\mu-1}},$$

where d_2 is the positive constant referred to in Theorem 37.

Multiplying the s -th equation of the system (2.56) by y_s and

summing over s from 1 to n , we obtain

$$|Y| \frac{d}{dt} |Y| = \sum_{i,j=1}^n \frac{p_{ij} + p_{ji}}{2} y_i y_j,$$

where

$$|Y| = \sqrt{\sum_{i=1}^n y_i^2}.$$

Using the estimates which are satisfied by the functions $p_{st}(t, X^{(0)})$, we obtain for $\mu = 1$ that

$$|Y| \frac{d}{dt} |Y| \leq K n^2 |Y|^2,$$

and for $\mu > 1$ and $p \neq 2k$ we obtain

$$|Y| \frac{d}{dt} |Y| \leq \frac{K n^2 |X^{(0)}|^{\mu-1} |Y|^2}{1 + d_2 t |X^{(0)}|^{\mu-1}},$$

from which we have in the former case that

$$|Y| \leq |Y^{(0)}| e^{K n^2 t},$$

and in the latter case that

$$|Y| \leq |Y^{(0)}| (1 + d_2 t |X^{(0)}|^{\mu-1})^{K n^2 / d_2}.$$

We now estimate the integral in the right member of (2.55) for $\mu = 1$. We obtain

$$\begin{aligned} \left| \int_0^{+\infty} |X(t, X^{(0)})|^{m-2} m \sum_{i=1}^n x_i(t, X^{(0)}) \frac{\partial x_i(t, X^{(0)})}{\partial x_j^{(0)}} dt \right| &\leq \\ &\leq \int_0^{+\infty} m |X(t, X^{(0)})|^{m-1} \sqrt{\sum_{i=1}^n \left(\frac{\partial x_i}{\partial x_j^{(0)}} \right)^2} dt \leq \\ &\leq \int_0^{+\infty} m q_1^{m-1} |X^{(0)}|^{m-1} e^{-a_1(m-1)t} e^{K n^2 t} dt. \end{aligned}$$

Hence, choosing the value of m sufficiently large, we obtain that the integral in (2.55) converges uniformly and Eq. (2.45) defines

the partial derivative of the function V with respect to the variable $x_j^{(0)}$. We shall now show that this integral is also convergent when $\mu > 1$ and $p \neq 2k$, provided m is chosen sufficiently large. In fact,

$$\begin{aligned} & \left| \int_0^{+\infty} m |X(t, X^{(0)})|^{m-2} \sum_{i=1}^n x_i(t, X^{(0)}) \frac{\partial x_i(t, X^{(0)})}{\partial x_j^{(0)}} dt \right| \leq \\ & \leq \int_0^{+\infty} m |X(t, X^{(0)})|^{m-1} \sqrt{\sum_{i=1}^n \left(\frac{\partial x_i}{\partial x_j^{(0)}} \right)^2} dt \leq \\ & \leq \int_0^{+\infty} m d_1^{m-1} |X^{(0)}|^{m-1} (1 + d_2 t |X^{(0)}|^{\mu-1})^{-[(m-1)/(\mu-1)]} \times \\ & \quad \times (1 + d_2 t |X^{(0)}|^{\mu-1})^{Kn^2/d_2} dt. \end{aligned}$$

It follows from this estimate that the value of m can be chosen so large that the integral (2.55) becomes convergent. Thus, for $\nu \geq 1$ the function V has first order partial derivatives. Let $\nu \geq 2$; in this case by a slight modification of the foregoing arguments we can show that the second derivatives of the function V exist. Here it is easy to estimate the functions $\partial^2 x_i / \partial x_j^{(0)} \partial x_k^{(0)}$ readily from the linear inhomogeneous equations which they satisfy. These equations can be obtained by repeated differentiation with respect to the variables $x_j^{(0)}$, $x_k^{(0)}$ of the system of equations (2.45). Taking this into account, we can assume that the quantity m can be chosen so large that the function V will be continuously differentiable in all its arguments up to order ν inclusive.

From Theorem 36 it follows that $dV/dt = W$, from which we have

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} X_i^{(\mu)} = W.$$

In addition, the function V is homogeneous of order $m - \mu + 1$, and therefore the relation $\sum_{i=1}^n (\partial V / \partial x_i) x_i = (m + 1 - \mu)V$ holds. Thus, the function V satisfies the system (2.54).

COROLLARY. *When $n = 2$, the function $V(x_1, x_2)$ for $\nu \geq 1$ can always be found in closed form. Actually, let us determine from the*

system of the two equations

$$\begin{aligned}\frac{\partial V}{\partial x_1} X_1^{(\mu)} + \frac{\partial V}{\partial x_2} X_2^{(\mu)} &= W, \\ \frac{\partial V}{\partial x_1} x_1 + \frac{\partial V}{\partial x_2} x_2 &= (m - \mu + 1)V\end{aligned}$$

the partial derivatives $\partial V/\partial x_1$ and $\partial V/\partial x_2$. We then obtain

$$\begin{aligned}V_{x_1} &= A(x_1, x_2)V + B(x_1, x_2); \\ V_{x_2} &= C(x_1, x_2)V + D(x_1, x_2),\end{aligned}\tag{2.54'}$$

where

$$\begin{aligned}A(x_1, x_2) &= \frac{-(m+1-\mu)X_2^{(\mu)}}{x_2X_1^{(\mu)} - x_1X_2^{(\mu)}}; \quad B(x_1, x_2) = \frac{Wx_2}{x_2X_1^{(\mu)} - x_1X_2^{(\mu)}}; \\ C(x_1, x_2) &= \frac{-(m+1-\mu)X_1^{(\mu)}}{x_1X_2^{(\mu)} - x_2X_1^{(\mu)}}; \quad D(x_1, x_2) = \frac{Wx_1}{x_1X_2^{(\mu)} - x_2X_1^{(\mu)}}.\end{aligned}$$

The first of the equations in (2.54') is linear in V , and the variable x_2 is contained in it as a parameter. Therefore its general solution will have the form

$$V = e^{\int_{x_1^{(0)}}^{x_1} A(\xi, x_2) d\xi} \left[\varphi(x_2) + \int_{x_1^{(0)}}^{x_1} B(\xi, x_2) e^{-\int_{x_1^{(0)}}^{\xi} A(\eta, x_2) d\eta} d\xi \right].$$

Let us choose the function $\varphi(x_2)$ such that the function constructed by us satisfies the second of the equations (2.54') and, in addition, so that

$$V(x_1, x_2) = 0 \quad \text{for } x_1 = x_2 = 0.$$

Let us set

$$e^{\int_{x_1^{(0)}}^{x_1} A(\xi, x_2) d\xi} = M_1 \quad \text{and} \quad M_1 \int_{x_1^{(0)}}^{x_1} B(\xi, x_2) e^{-\int_{x_1^{(0)}}^{\xi} A d\eta} d\xi = M_2.$$

Substituting the function

$$V = M_1 \varphi(x_2) + M_2$$

into the second equation of the system (2.54'), we obtain

$$\frac{d\varphi}{dx_2} = N_1 \varphi + N_2,$$

where

$$N_1 = \left[C(x_1, x_2)M_1 - \frac{\partial M_1}{\partial x_2} \right] M_1^{-1},$$

$$N_2 = \left[C(x_1, x_2)M_2 + D(x_1, x_2) - \frac{\partial M_2}{\partial x_2} \right] M_2^{-1}.$$

It is easy to establish that

$$\frac{\partial N_1}{\partial x_1} = \frac{\partial N_2}{\partial x_1} \equiv 0.$$

The general solution of the equation for the function φ will have the form

$$\varphi(x_2) = \gamma P_1(x_2) + P_2(x_2),$$

where γ is an arbitrary constant. Let us set

$$\gamma = -[P_2 + M_2 M_1^{-1}] P_1^{-1} \text{ for } x_1 = x_2 = 0.$$

Then the function

$$V(x_1, x_2) = \gamma M_1 P_1 + M_1 P_2 + M_2$$

will satisfy the system (2.54') and the condition that $V = 0$ when $x_1 = x_2 = 0$, for $n = 2$.

The function constructed here gives the necessary and sufficient condition for asymptotic stability of the trivial solution of the system (2.45), and it is asymptotically stable if and only if the function $V(x_1, x_2)$ is negative definite. It is understood here that the function $W(x_1, x_2)$ is positive definite and homogeneous of order $m > \mu - 1$.

Let us indicate now a method for constructing the function V in closed form in the case $n = 3$.

When $n = 3$, we determine the functions $\partial V / \partial x_1$ and $\partial V / \partial x_3$ from the system of partial differential equations (2.54).

We obtain

$$\frac{\partial V}{\partial x_1} = \frac{x_3(W - X_2^{(\mu)}(\partial V / \partial x_2)) - X_3^{(\mu)}(m - \mu + 1)V}{X_1^{(\mu)}x_3 - X_3^{(\mu)}x_1},$$

$$\frac{\partial V}{\partial x_3} = \frac{X_1^{(\mu)}(m - \mu + 1)V - x_1(W - X_2^{(\mu)}(\partial V / \partial x_2))}{X_1^{(\mu)}x_3 - X_3^{(\mu)}x_1}.$$

Equating the mixed derivatives, $\partial^2 V / \partial x_1 \partial x_3 = \partial^2 V / \partial x_3 \partial x_1$, and making use of the fact that the function $\partial V / \partial x_2$ is homogeneous of

degree $m - \mu$, we find an equation interrelating the derivatives of the function V with respect to x_2 for determining the function V . The variables x_1 and x_3 will occur in this equation as parameters.

In fact, differentiating the last equation with respect to x_3 and x_1 , we find:

$$\frac{\partial}{\partial x_3} \frac{\Delta_1}{\Delta} = \frac{\partial}{\partial x_1} \frac{\Delta_3}{\Delta},$$

where

$$\Delta_1 = x_3 \left(W - X_2^{(\mu)} \frac{\partial V}{\partial x_2} \right) - X_3^{(\mu)} (m - \mu + 1) V,$$

$$\Delta_3 = X_1^{(\mu)} (m - \mu + 1) V - x_1 \left(W - X_2^{(\mu)} \frac{\partial V}{\partial x_2} \right),$$

$$\Delta = X_1^{(\mu)} x_3 - X_3^{(\mu)} x_1.$$

The equation

$$\frac{\partial}{\partial x_1} \Delta_3 - \frac{\partial}{\partial x_3} \Delta_1 + \frac{\Delta_1}{\Delta} \frac{\partial}{\partial x_3} \Delta - \frac{\Delta_3}{\Delta} \frac{\partial}{\partial x_1} \Delta = 0$$

can be written in the form

$$b_0 \frac{\partial^2 V}{\partial x_2^2} + b_1 \frac{\partial V}{\partial x_2} + b_2 V + b_3 = 0.$$

Analogously, we have

$$a_0 \frac{\partial^2 V}{\partial x_1^2} + a_1 \frac{\partial V}{\partial x_1} + a_2 V + a_3 = 0$$

and

$$c_0 \frac{\partial^2 V}{\partial x_3^2} + c_1 \frac{\partial V}{\partial x_3} + c_2 V + c_3 = 0,$$

where the coefficients of these equations are determined by the following formulas:

$$b_0 = -x_2 X_2^{(\mu)};$$

$$b_1 = \frac{X_2^{(\mu)} x_2 (\partial/\partial x_2) \Delta}{\Delta} - x_2 \frac{\partial}{\partial x_2} X_2^{(\mu)};$$

$$b_2 = (m - \mu + 1) \Delta \left(\frac{\partial}{\partial x_1} \frac{X_1^{(\mu)}}{\Delta} + \frac{\partial}{\partial x_3} \frac{X_3^{(\mu)}}{\Delta} \right);$$

$$b_3 = -x_2 W \frac{(\partial/\partial x_2) \Delta}{\Delta} + x_2 \frac{\partial}{\partial x_2} W.$$

The coefficients of the remaining equations are obtained, obviously, by a permutation of the subscripts.

Let us assume that two independent solutions φ_1 , φ_2 and some particular solution φ_3 of the first of these equations have been found. Then its general solution can be written in the form

$$V = c_1\varphi_1 + c_2\varphi_2 + \varphi_3,$$

where

$$c_1 = c_1(x_1, x_3)$$

and

$$c_2 = c_2(x_1, x_3)$$

are arbitrary functions in the variables x_1 and x_3 which have yet to be determined. The determination of them can be reduced to the integration of a system of ordinary differential equation.

To this end, we substitute the expressions found in the equation, which is at our disposal, for the function considered as a function of the variable x_1 . We then obtain a system of equations of the form:

$$\begin{aligned} a_{11} \frac{\partial c_1}{\partial x_1} + a_{12} \frac{\partial c_2}{\partial x_1} + a_{13}c_1 + a_{14}c_2 + a_{15} &= 0, \\ b_{11} \frac{\partial^2 c_1}{\partial x_1^2} + b_{12} \frac{\partial^2 c_2}{\partial x_1^2} + b_{13} \frac{\partial c_1}{\partial x_1} + b_{14} \frac{\partial c_2}{\partial x_1} + \\ &+ b_{15}c_1 + b_{16}c_2 + b_{17} = 0. \end{aligned}$$

If ψ_1 , ψ_2 , ψ_3 is a set of independent solutions of this system and ψ_4 is any particular solution, then the vector (c_1, c_2) , which yields the general solution of this system, can be written as the combination

$$\gamma_1\psi_1 + \gamma_2\psi_2 + \gamma_3\psi_3 + \psi_4.$$

The quantities γ_1 , γ_2 , γ_3 are arbitrary functions of the variable x_3 , which is subject to determination from the remaining equations and the condition that

$$V = 0 \quad \text{for} \quad x_1 = x_2 = x_3 = 0.$$

Thus, for $n = 3$, the construction of the function V reduces to finding the solution of a system of ordinary differential equations. And this holds true also for any other n ; however, with increasing n , the difficulties encountered along the way in such an approach increase very rapidly.

The results obtained above make it possible to make headway in the solution of certain local and non-local problems for systems of equations more general in form than (2.45).

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = X_s^{(\mu)}(x_1, \dots, x_n) + F_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n). \quad (2.57)$$

The functions $F_s(x_1, \dots, x_n, t)$ will be assumed such that the solution $X = X(t, X^{(0)}, t_0)$ exists for all $X^{(0)}$ and $t_0 \geq 0$. The solution of the system (2.57) is called *bounded* for $t \geq t_0$ if the inequality $|X(t, X^{(0)}, t_0)| \leq K(t_0, X^{(0)}) < +\infty$ is satisfied for $t \geq t_0$.

THEOREM 39. *In order for any solution of the system (2.57) to be bounded at $t \geq t_0$ for any choice of the functions $F_s(x_1, \dots, x_n, t)$, such that*

$$|F_s(X, t)| \leq C|X|^\lambda \quad (2.58)$$

for

$$|X| \geq R, \lambda \leq \mu, \mu \geq 1,$$

where R is a sufficiently large positive quantity and C is a positive quantity which is sufficiently small when $\lambda = \mu$ in the case of $\nu \geq 1$, it is necessary and sufficient that the solution of the system (2.45) satisfy the condition (2.48).

PROOF. Necessity. Let any solution of the system (2.57) be bounded at $t \geq t_0$ for any choice of the functions $F_s(x_1, \dots, x_n, t)$ which satisfy inequalities (2.58). Let us show that the solutions of the system (2.45) satisfy the inequality (2.48).

Let us examine the system

$$\frac{dx_s}{d\tau} = x_s + X_s^{(\mu)}$$

for $\mu > 1$. The functions $F_s = x_s$ satisfy the condition (2.58). Therefore any solution of this system is bounded for $\tau \geq 0$. In addition, this system has the trivial solution, which is asymptotically stable as $\tau \rightarrow -\infty$. Let us take a sufficiently small $X^{(0)}$. Then the solution of this system $\bar{X} = \bar{X}(\tau, X^{(0)})$ is bounded for $\tau \in (-\infty, +\infty)$. Let us examine the function

$$X = z(t)\bar{X}(-\ln z, X^{(0)}), \quad (2.59)$$

where $z(t)$ is a positive function of the argument t , satisfying the equation

$$\frac{dz}{dt} = -z^\mu.$$

It is clear that

$$z = \frac{z_0}{\sqrt[\mu-1]{1 + (\mu-1)t z_0^{\mu-1}}}.$$

Differentiating both members of (2.59) with respect to the variable t , we obtain

$$\begin{aligned} \frac{dX}{dt} &= \frac{dz}{dt} \bar{X}(-\ln z, X^{(0)}) - \frac{d}{d\tau} \bar{X}(-\ln z, X^{(0)}) \frac{dz}{dt} = \\ &= -z^\mu \left(\bar{X}(-\ln z, X^{(0)}) - \frac{d}{d\tau} \bar{X}(-\ln z, X^{(0)}) \right), \end{aligned}$$

from which it follows that

$$\frac{dx_s}{dt} = z^\mu (X_s^{(\mu)}(\bar{x}_1, \dots, \bar{x}_n)) = X_s^{(\mu)}(x_1, \dots, x_n).$$

Thus, the functions (2.59) satisfy the system (2.45). In view of the boundedness of $\bar{X}(-\ln z(t), X^{(0)})$ for $t \in [0, +\infty)$, we find that the solution of the system (2.45) will satisfy the condition (2.48).

Thus, the necessity of the condition of the theorem is proved completely for $\mu > 1$.

Let us now set $\mu = 1$. Let us consider the system

$$\frac{dx_s}{dt} = cx_s + X_s^{(\mu)},$$

where c is a sufficiently small positive quantity. The function $X = e^{-ct} \bar{X}(t, X^{(0)})$, where $\bar{X}(t, X^{(0)})$ is the solution of the foregoing system, is bounded at $t \geq 0$, and will satisfy the system (2.45); consequently, the following estimate holds:

$$|X| \leq A_2 e^{-ct} \leq A_1 t^{-\alpha} \quad \text{for } |X^{(0)}| = 1 \text{ and } t \geq T.$$

Thus, in this case too the solutions of the system (2.45) satisfy the condition (2.48).

Let us note that when proving the necessity of the conditions of the theorem, we made no use anywhere of the differentiability properties of the functions $X_s^{(\mu)}$.

Sufficiency. The right members of the system (2.45) are differentiable, $\nu \geq 1$, and the solutions of this system satisfy the inequality (2.48). Consequently, according to Theorem 38 there exist two functions V and W , which have all the properties indicated in Theorem 38, and which satisfy the system of Equations (2.54). Let us calculate the total derivative of the function V by virtue of the system (2.57). We then obtain

$$\frac{dV}{dt} = \sum_{i=1}^n \frac{\partial V}{\partial x_i} (X_i^{(\mu)} + F_i(x_1, \dots, x_n, t)) = W + W_1,$$

where

$$W_1 = \sum_{i=1}^n \frac{\partial V}{\partial x_i} F_i(X, t).$$

Let us estimate the function W_1 :

$$|W_1| \leq |X|^{m-\mu+\lambda} CB \quad \text{for } |X| \geq R.$$

It is clear that there exists a quantity R_1 such that

$$W + CB |X|^{m+\lambda-\mu} < 0 \quad \text{for } |X| \geq R_1$$

because $W(x_1, \dots, x_n)$ is a homogeneous function of order $m \geq M + \lambda - \mu$, and when $\lambda = \mu$ the value of C is sufficiently small. Let us set

$$V_1 = \max_{|X|=R_1} V(X).$$

If the point $X^{(0)}$ is such that $V(X^{(0)}) > V_1$, then

$$V(X(t, X^{(0)}, t_0)) < V(X^{(0)}) \quad \text{for } t \geq t_0 \quad \text{because } \frac{dV}{dt} < 0.$$

Consequently

$$A'_1 |X(t, X^{(0)}, t_0)|^{m-\mu+1} \leq A'_2 |X^{(0)}|^{m-\mu+1},$$

where

$$A'_1 = \inf_{|X^{(0)}|=1} V(X^{(0)}), \quad A'_2 = \sup_{|X^{(0)}|=1} V(X^{(0)}).$$

Thus, if the solution of the system (2.57) begins at $V(X^{(0)}) > V_1$, then it is bounded; if it begins at a value of $X^{(0)}$ such that $V(X^{(0)}) < V_1$, then it remains in the region $V(X) \leq V_1$, because all the integral curves intersect the surface $V = V_1$ from the outside in. This completes the proof of the theorem.

COROLLARY 1. *If it is assumed that the system of differential equations (2.45) possesses the property (2.48) and its right members are continuously differentiable, it is possible to find a region G of the space E_n , in which any integral curve of the system (2.57) enters at $t \rightarrow +\infty$ and remains there.*

In fact, let us take the quantity R_1 , indicated in the proof of Theorem 39, and let us calculate

$$V_1 = \sup_{|X|=R_1} V(X).$$

The region $V(X) \leq V_1$ has the indicated properties. In fact, it was shown in the theorem that all the integral curves, beginning in it, remain there when $t \geq t_0$. Let us consider the point $X^{(0)}$ outside of this region, and let us assume that the integral curve $X = X(t, X^{(0)}, t_0)$ of the system (2.57) does not enter into the region G : $V(X) \leq V_1$. Then, along this integral curve, we have

$$V(X(t, X^{(0)}, t_0)) > V_1.$$

Let us set

$$W_2 = \sup_{V(X) \geq V_1} (W + W_1).$$

It is clear that $W_2 < 0$. Then along the indicated integral curve we have $dV/dt < W_2$; hence

$$V(X(t, X^{(0)}, t_0)) < V(X^{(0)}) + W_2(t - t_0).$$

The latter inequality shows that there exists a positive quantity T such that

$$V(X(t, X^{(0)}, t_0)) < 0 \text{ for } t - t_0 > T,$$

which is impossible, for it has been assumed that

$$V(X(t, X^{(0)}, t_0)) > V_1 \text{ for } t > t_0.$$

Thus, there exists a T_1 such that

$$V(X(t, X^{(0)}, t_0)) < V_1 \text{ for } t - t_0 \geq T_1,$$

i.e., the integral curve enters the region G and stays there.

COROLLARY 2. *If the right members of the system (2.45) are differentiable and inequality (2.48) holds for the solution of the system (2.45), then any integral curve $X(t, X^{(0)})$, which begins outside the region G , remains there for a certain time $t - t_0 \leq T(X^{(0)})$. In this time interval it will satisfy the following inequalities.*

For $\mu = 1$, $p_1 |X^{(0)}| e^{-p_1'(t-t_0)} \leq |X(t, X^{(0)}, t_0)| \leq q_1' |X^{(0)}| \cdot e^{-q_1'(t-t_0)}$.
 For $\mu > 1$ and $p \neq 2k$, any solution of the system (2.45) satisfies the inequalities

$$\frac{c_1' |X^{(0)}|}{\sqrt[\mu-1]{1 + c_{2\tau} |X^{(0)}|^{\mu-1}}} \leq |X(t, X^{(0)}, t_0)| \leq \frac{d_1' |X^{(0)}|}{\sqrt[\mu-1]{1 + d_{2\tau} |X^{(0)}|^{\mu-1}}},$$

$$V(X^{(0)}) > V_1, \quad \tau = t - t_0 \leq T_1(X^{(0)}),$$

where $T_1(X^{(0)})$ is such that $V(X(T_1(X^{(0)}), X^{(0)}, t_0)) = V_1$.

Let us prove this. When $|X| \geq R_1$, the function $W + W_1$ satisfies the inequalities

$$-b_1 |X|^m \leq W + W_1 \leq -b_2 |X|^m. \quad (2.60)$$

The function V satisfies the inequalities

$$A_1' |X|^{m+1-\mu} \leq V \leq A_2' |X|^{m+1-\mu}. \quad (2.61)$$

Along the integral curve $X = X(t, X^{(0)}, t_0)$, we have

$$V(X(t, X^{(0)}, t_0)) = \frac{V(X^{(0)})}{\sqrt[\lambda-1]{1 + (1-\lambda)V(X^{(0)})^{\lambda-1} \int_{t_0}^t (W + W_1)V^{-\lambda} d\tau}}. \quad (2.62)$$

This equality holds for all $t \geq t_0$. We shall consider this only for $t - t_0 \leq T_1(X^{(0)})$. Then, along the integral curve, the inequalities (2.60) and (2.61) will be satisfied. Applying them to (2.62) and setting

$$\lambda = \frac{m}{m+1-\mu},$$

we obtain, as in Theorem 20, the required inequality, which will hold only for $t_0 \leq t \leq t_0 + T_1(X^{(0)})$.

THEOREM 40. *If the right members of the system (2.45) are differentiable and the inequality (2.48) holds for the solutions of this system, then for all functions F_θ such that*

$$|F_\theta| \leq c |X^{(0)}|^\lambda \quad \text{for } |X| \leq h, \quad \lambda \geq \mu, \quad (2.63)$$

where h is a sufficiently small positive number (for $\lambda = \mu$, c is sufficiently small), the trivial solution of the system (2.57) will be

asymptotically stable. In this connection, any integral curve

$$X = X(t, X^{(0)}, t_0)$$

of the system (2.57), for sufficiently small $|X^{(0)}|$, will satisfy the following inequalities: for $\mu = 1$,

$$p_1' |X^{(0)}| e^{-p_1'(t-t_0)} \leq |X(t, X^{(0)}, t_0)| \leq q_1' |X^{(0)}| e^{-q_1'(t-t_0)};$$

for $\mu > 1$ and $\phi \neq 2k$,

$$\frac{c_1' |X^{(0)}|}{\sqrt[\mu-1]{1 + c_2 \tau |X^{(0)}|^{\mu-1}}} \leq |X(t, X^{(0)}, t_0)| \leq \frac{d_1' |X^{(0)}|}{\sqrt[\mu-1]{1 + d_2 \tau |X^{(0)}|^{\mu-1}}},$$

$$\tau = t - t_0 \geq 0.$$

PROOF. Let us assume that the system (2.45) has differentiable right members and that its solution satisfies the condition (2.48). Consequently, by Theorem 38, there exist two functions V and W satisfying all the conditions of Theorem 36 and the system (2.54). Let us calculate the total derivative of the function V by virtue of the system (2.57). We then obtain

$$\frac{dV}{dt} = \sum_{i=1}^n \frac{\partial V}{\partial x_i} (X_i^{(\mu)} + F_i) = W + W_1,$$

where

$$W_1 = \sum_{i=1}^n \frac{\partial V}{\partial x_i} F_i(x_1, \dots, x_n, t).$$

Let us estimate the function W_1 : $|W_1| \leq c_1 B_2 |X|^{m-\mu+\lambda}$, $m - \mu + \lambda \geq m$. When $m = \lambda$, c is sufficiently small and, therefore, in a sufficiently small neighborhood of the point $X = 0$, one can find positive numbers b_1 and b_2 such that

$$-b_1 |X|^m \leq W + W_1 \leq -b_2 |X|^m \text{ for } |X| \leq h_1 < h. \quad (2.64)$$

The function V is homogeneous and therefore it satisfies the inequalities

$$A_1' |X|^{m-\mu+1} \leq V(X) \leq A_2' |X|^{m+1-\mu};$$

therefore we can use the method used to prove Theorem 20, in which we must set $k_1 = k_2 = m$ and $l_1 = l_2 = m - \mu + 1$, and replace the functional $V(p)$, referred to in Theorem 20, by the

function $|X|$. Consequently, the trivial solution of the system (2.57) is asymptotically stable and one can obtain an estimate of the integral curve by following Theorem 20; thus, multiplying both members of the equation $dV/dt = W + W_1$ by $V^{-\lambda}$ and integrating from t_0 to t , we obtain (2.62). Applying to it the inequality (2.64) and the inequality which is satisfied by the function V , we obtain for $\lambda = m/(m + 1 - \mu)$ the estimate given in the theorem, corresponding to the case $\mu > 1$ and $p \neq 2k$. If we now multiply the equality $dV/dt = W + W_1$ by V^{-1} and integrate from t_0 to t , we obtain

$$V(X(t, X^{(0)}, t_0)) = V(X^{(0)}) e^{\int_{t_0}^t W V^{-1} d\tau} \quad (2.65)$$

Applying to (2.65) the inequality that is satisfied by the functions V and W , we obtain the required estimate for the quantity $|X(t, X^{(0)}, t_0)|$. This completes the proof of Theorem 40.

COROLLARY. *Let*

$$V_2 = \inf_{|X|=h_1} V(X);$$

the region $V \leq V_2$ lies in its entirety in the region of asymptotic stability of the trivial solution of the system (2.47).

In fact, if $V(X^{(0)}) < V_2$, we obtain for $t \geq t_0$ that

$$\frac{dV(X(t, X^{(0)}, t_0))}{dt} < 0;$$

this means that

$$V(X(t, X^{(0)}, t_0)) < V(X^{(0)}) < V_2 \text{ for } t > t_0,$$

from which it follows that the integral curve of the system (2.57), which begins in the region $V(X) < V_2$, remains there when $t \geq t_0$, and moreover

$$|X(t, X^{(0)}, t_0)| \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Thus, Theorem 40 makes it possible to find a certain region, which is contained in its entirety in the region of asymptotic stability of the trivial solution of the system (2.57). Let us now apply Theorem 22 for an investigation of the system (2.45). We put $\mu > 1$. All the solutions of the system (2.57) for any choice of the functions

$$F_s(x_1, \dots, x_n, t),$$

satisfying condition (2.58), will, in accordance with Theorem 39, be bounded for $t \geq t_0$ if and only if any solution

$$X = X(t, X^{(0)}) \text{ for } |X^{(0)}| = 1$$

of the system (2.45) satisfies the inequality (2.48). Let us put $F_s = x_s$ in the system (2.57) and thus obtain the system

$$\frac{dx_s}{dt} = x_s + X_s^{(\mu)} \quad (s = 1, \dots, n). \quad (2.66)$$

It was shown in Theorem 39 that from the boundedness of the solutions of this system it follows that the condition (2.48) is satisfied by the solutions of the system (2.45), and, conversely, from the satisfaction of the condition (2.48) by the solutions of the system (2.45), all the solutions of the system (2.66) will be bounded for $t > 0$. The trivial solution of the system (2.66) is asymptotically stable as $t \rightarrow -\infty$, and therefore if all its solutions are bounded, then the region of asymptotic stability A for the system

$$\frac{dx_s}{dt} = -x_s - X_s^{(\mu)} \quad (2.67)$$

will be bounded. If the region of asymptotic stability A of the trivial solution of the system (2.67) is bounded, then any solution of the system (2.66), which begins at a sufficiently small $|X^{(0)}|$, will be bounded for $t \in (-\infty, +\infty)$, and then, as shown in Theorem 39, any solution of the system (2.45), $X = X(t, X^{(0)})$, satisfies the condition (2.48) at $|X^{(0)}| = 1$. We thus obtain the following theorem.

THEOREM 41. *In order for any solution $X = X(t, X^{(0)})$, $|X^{(0)}| = 1$, of the system (2.45) to satisfy the inequality (2.48), it is necessary and sufficient that the region of asymptotic stability A of the trivial solution of the system (2.67) be bounded.*

COROLLARY. *Let us examine the partial differential equation*

$$\begin{aligned} \sum_{i=1}^n -\frac{\partial V}{\partial x_i} (x_i + X_i^{(\mu)}) &= \\ &= \varphi(X)(1 + V) \sqrt{1 + \sum_{i=1}^n (x_i + X_i^{(\mu)})^2}. \end{aligned} \quad (2.68)$$

It is easy to verify that any solution of the system (2.67) satisfies inequality

$$|X(t, X^{(0)})| \leq |X^{(0)}| e^{-\alpha t}, \quad \alpha \in (0, 1), \quad t \geq 0$$

(for sufficiently small $|X^{(0)}|$). For any positive definite function $\varphi(X)$ such that $\varphi(X) \leq \varphi_1(|X|)$, where $\varphi_1(|X|)$ tends to 0 strictly monotonically as $|X| \rightarrow 0$, the integral

$$\int_0^{+\infty} \varphi_1(|X^{(0)}| e^{-\alpha t}) dt$$

converges. As follows from Theorem 22, the Equation (2.68) has a solution $V(X)$, which satisfies all the conditions of the foregoing theorem. It is known (see Theorems 19 and 22) that the region A coincides with the region in which the inequalities $-1 < V(X) < 0$ are satisfied, and therefore the following theorem holds.

THEOREM 42. *In order for the solution of the system (2.45) to satisfy the conditions (2.48), it is necessary and sufficient that Eq. (2.68) have a solution $V(X)$, having the following properties: the region where the inequalities $-1 < V(X) < 0$ are satisfied, which contains the sufficiently small neighborhood of the point $X = 0$, is bounded.*

COROLLARY. *If the functions $X_g^{(u)}$ are forms of odd power, and the function $\varphi(X)$ in Eq. (2.68) is analytic, then, as was shown in Section 2 of Chapter II, the function V can be obtained in the form of series which converge for sufficiently small $|x_s|$, which can be analytically continued to the entire region A . Here in order to clarify the boundedness of the region A it is useful to employ the method proposed in Section 2 of Chapter II for constructing regions which are completely embedded in A .*

The theory developed in this section makes it possible to solve, as already noted, many local and non-local problems. In the following section we shall treat the application of this theory to the solution of the problem of stability in the small of the trivial solution of the system of differential equations in certain doubtful cases. Let us give now an example that illustrates the developed theory. Let us consider a system of two

equations:

$$\begin{aligned}\frac{dx}{dt} &= \sum_{k+l=0}^{\mu} a_{kl} x^k y^l, \\ \frac{dy}{dt} &= \sum_{k+l=0}^{\mu} b_{kl} x^k y^l, \quad \mu = 2n + 1.\end{aligned}$$

Consider the system of equations

$$\begin{aligned}\frac{\partial V}{\partial x} \sum_{k+l=\mu} a_{kl} x^k y^l + \frac{\partial V}{\partial y} \sum_{k+l=\mu} b_{kl} x^k y^l &= W^{(m)}, \\ x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} &= (m + 1 - \mu) V,\end{aligned}$$

where $W^{(m)}$ is a negative definite homogeneous function of order m . As indicated above, it is possible to find a function V satisfying this system in closed form. If the function $V(x, y)$ thus obtained is positive definite, then all the solutions of the system of the two equations will be bounded for $t \in [0, +\infty)$ (Theorem 39), since the function V will be homogeneous of order $m + 1 - \mu$, and then the conditions (2.48) are satisfied for the solutions of the system

$$\begin{aligned}\frac{dx}{dt} &= \sum_{k+l=\mu} a_{kl} x^k y^l, \\ \frac{dy}{dt} &= \sum_{k+l=\mu} b_{kl} x^k y^l.\end{aligned}$$

Consequently, the sufficient conditions of Theorem 39 are satisfied. If the system under consideration has a trivial solution ($a_{00} = b_{00} = 0$) and no other rest point exists, then the system will have a limit cycle for $a_{10} + b_{01} > 0$.

If the resultant function $V(x, y)$ is alternating and bounded for $|x| + |y| < +\infty$, then all the solutions of the system of two equations cannot be bounded. In fact,

$$\frac{dV}{dt} = \frac{\partial V}{\partial x} \sum_{k+l=0}^{\mu} a_{kl} x^k y^l + \frac{\partial V}{\partial y} \sum_{k+l=0}^{\mu} b_{kl} x^k y^l = W + W_1.$$

It is clear that for sufficiently large R we have

$$|W_1| \leq c r^{m-1} \quad \text{for } r \geq R, \quad r = \sqrt{x^2 + y^2}.$$

Let us choose R_1 so large as to obtain $W + W_1 < 0$ for $r \geq R_1$. We set

$$\inf (W + W_1) = \alpha \text{ for } r \geq R_1$$

and

$$\beta = \inf_{r=R_1} V(x, y).$$

We take a point x_0, y_0 such that $r_0 > R_1$ and $V(x_0, y_0) < 0$. We then have from $dV/dt = W + W_1$ that

$$V(t, x_0, y_0) \leq V(x_0, y_0) + \alpha t.$$

If $V(x_0, y_0) < \beta$, then

$$V(t, x_0, y_0) < \beta + \alpha t \text{ for } t \geq 0.$$

Consequently, the integral curve $x(t, x_0, y_0), y(t, x_0, y_0)$ lies outside a circle of radius R_1 . If it is assumed that it is bounded, then we obtain as a corollary that it is defined for all $t \geq 0$, and then, on the one hand, the function $V(x, y)$ should be bounded on it, and, on the other hand, it is found that $V(t, x_0, y_0) \rightarrow -\infty$ as $t \rightarrow +\infty$. Thus, the integral curve chosen is not bounded.

We gave above a condition for the existence of a limit cycle. If we use a specific form of the function $V(x, y)$ and assume that the point $x = y = 0$ is asymptotically stable as $t \rightarrow -\infty$, then, by using the method for constructing regions which belong in their entirety to A , we can indicate an annulus containing the limit cycle. N. N. Krasovsky has considered a system of type (2.45) for the case when the functions X_s are homogeneous forms in the quantities $x_1, \dots, x_n$ of order μ [19.] In that paper he solved the problem of the existence of functions which resolve the question of stability and which satisfy estimates of a given type. It is also shown in that paper that the condition for asymptotic stability of the trivial solution is equivalent to satisfaction of inequality (2.48) for $\alpha = 1/(\mu - 1)$. It is easy to establish that this result remains in force also for the system (2.45) in the case when the functions X_s are continuously differentiable. Therefore condition (2.48) is equivalent to asymptotic stability.

4. Case of k zero roots

Let us consider the system

$$\left. \begin{aligned} \frac{dx_s}{dt} &= X_s(x_1, \dots, x_k, y_1, \dots, y_n) \quad (s = 1, \dots, k) \\ \frac{dy_j}{dt} &= \sum_{i=1}^n p_{ji} y_i + Y_j(x_1, \dots, x_k, y_1, \dots, y_n) \quad (j = 1, \dots, n) \end{aligned} \right\}. \quad (2.69)$$

The functions X_s and Y_j are expanded in a series in positive integral powers of the quantities $x_1, \dots, x_k, y_1, \dots, y_n$, which converges for sufficiently small $|x_s|$, ($s = 1, \dots, k$) and $|y_j|$, ($j = 1, \dots, n$), and which does not contain terms linear in $x_1, \dots, x_k, y_1, \dots, y_n$. Let us assume further that the eigenvalues $\lambda_1, \dots, \lambda_n$ of the matrix $P = \|p_{ji}\|$ have negative real parts. We note that by a non-singular linear transformation performed on the functions sought, any system of $n + k$ differential equations with holomorphic right members can be reduced to the form (2.69) if the characteristic equation has k zero roots, to which correspond simple elementary factors.

Let us consider the system of equations:

$$\sum_{i=1}^n p_{ji} y_i + Y_j(x_1, \dots, x_k, y_1, \dots, y_n) = 0. \quad (2.70)$$

The system (2.70) determines the set of implicit functions

$$y_j = u_j(x_1, \dots, x_k) \quad (j = 1, \dots, n).$$

The functions $u_j(x_1, \dots, x_k)$ are holomorphic for sufficiently small $|x_s|$ ($s = 1, \dots, k$).

THEOREM 43. *If $X_s(x_1, \dots, x_k, u_1, \dots, u_n) \equiv 0$, then the system (2.69) has k holomorphic integrals which are independent of t .*

PROOF. Let us make, in the system (2.69), the substitution

$$y_j = u_j + \eta_j \quad (j = 1, \dots, n),$$

where the η_j are new unknown functions. Then the system (2.69) assumes the form

$$\left. \begin{aligned} \frac{dx_s}{dt} &= X_s(x_1, \dots, x_k, u_1 + \eta_1, \dots, u_n + \eta_n) \\ \frac{d\eta_j}{dt} &= \sum_{i=1}^n p_{ij}\eta_i + V_j(x_1, \dots, x_k, \eta_1, \dots, \eta_n) \end{aligned} \right\}, \quad (2.71)$$

where

$$\begin{aligned} V_j &= \sum_{i=1}^n p_{ij}u_i + Y_j(x_1, \dots, x_k, u_1 + \eta_1, \dots, u_n + \eta_n) - \\ &- \sum_{i=1}^k \frac{du_j}{dx_i} X_i(x_1, \dots, x_k, u_1 + \eta_1, \dots, u_n + \eta_n). \end{aligned}$$

It is clear that the functions V_j are holomorphic for sufficiently small $|x_s|$ and $|\eta_j|$, and their expansions in positive integral powers of $x_1, \dots, x_k, \eta_1, \dots, \eta_n$ do not contain terms that are linear in these quantities.

Let us consider the system of partial differential equations

$$\begin{aligned} \sum_{j=1}^n \frac{\partial x_s}{\partial \eta_j} \left(\sum_{i=1}^n p_{ij}\eta_i + V_j \right) &= \\ &= X_s(x_1, \dots, x_k, \eta_1 + u_1, \dots, \eta_n + u_n) \quad (s = 1, \dots, k). \end{aligned} \quad (2.72)$$

Let us make in the system (2.72) the substitution $x_s = c_s + f_s$, where the f_s are new unknown functions and the c_s ($s = 1, \dots, k$) are sufficiently small arbitrary constants. Then the system (2.72) goes into

$$\begin{aligned} \sum_{j=1}^n \frac{\partial f_s}{\partial \eta_j} \left(\sum_{i=1}^n (p_{ij} + c_{ij})\eta_i + V'_j \right) &= \\ &= \sum_{i=1}^n \gamma_{si}\eta_i + U_s(f_1, \dots, f_k, \eta_1, \dots, \eta_n, c_1, \dots, c_k) \end{aligned} \quad (2.73)$$

since

$$X_s(x_1, \dots, x_n, u_1, \dots, u_n) \equiv 0.$$

Here the c_{ij} and γ_{si} are holomorphic functions in $c_1, \dots, c_k$, having the properties

$$\begin{aligned} c_{ij} &= 0 \text{ for } c_1 = c_2 = \dots = c_k = 0 \\ \gamma_{si} &= 0 \text{ for } c_1 = c_2 = \dots = c_k = 0 \end{aligned} \quad \left(\begin{array}{l} j = 1, \dots, n \\ i = 1, \dots, n \\ s = 1, \dots, k \end{array} \right).$$

For sufficiently small $|c_s|$ ($s = 1, \dots, k$), there exists a system of holomorphic functions [1]

$$f_s = f_s(\eta_1, \dots, \eta_n, c_1, \dots, c_k),$$

satisfying the system (2.73). Each of these functions has the properties

$$f_s \equiv 0 \text{ for } \eta_1 = \dots = \eta_n = 0$$

and

$$f_s \equiv 0 \text{ for } c_1 = c_2 = \dots = c_k = 0 \quad (s = 1, \dots, k). \quad (2.74)$$

Returning to the previous unknown functions, we obtain

$$x_s = c_s + f_s(\eta_1, \dots, \eta_n; c_1, \dots, c_k) \quad (s = 1, \dots, k). \quad (2.75)$$

The system (2.75), by virtue of the properties (2.74), can be solved with respect to $c_1, \dots, c_k$, so that

$$c_s = x_s + \varphi_s(x_1, \dots, x_k; \eta_1, \dots, \eta_n) \quad (s = 1, \dots, k). \quad (2.76)$$

Setting

$$\eta_j = y_j - u_j(x_1, \dots, x_k) \quad (j = 1, \dots, n)$$

in (2.76), we obtain a system of holomorphic integrals for the system (2.69):

$$c_s = x_s + \varphi_s(x_1, \dots, x_k; y_1 - u_1, \dots, y_n - u_n). \quad (2.77)$$

This completes the proof of the theorem. [This theorem can also be derived directly from the results contained in Section 65 of reference [1].]

Let us proceed to solve the stability problem. Using relations (2.75), we eliminate from the second group of equations of the system (2.71) the quantities x_s ($s = 1, \dots, k$). We then obtain the following system of equations for the functions η_j ($j = 1, \dots, n$):

$$\frac{d\eta_s}{dt} = \sum_{i=1}^n (p_{si} + c_{si})\eta_i + V'_s(c_1, \dots, c_k, \eta_1, \dots, \eta_n). \quad (2.78)$$

The trivial solution of the system (2.78) is asymptotically stable uniformly with respect to the quantities c_s ($s = 1, \dots, k$), which means the following: for any $\varepsilon > 0$ we can find a $\delta > 0$ such that for all sufficiently small $|c_s|$ we have

$$\sum_{i=1}^n \eta_i^2(t, \eta_i^{(0)}, \dots, \eta_n^{(0)}) < \varepsilon^2 \text{ for } \sum_{i=1}^n \eta_i^{(0)^2} < \delta^2$$

and

$$\sum_{i=1}^n \eta_i^2 \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

In fact, let us find a positive definite quadratic form in the quantities $\eta_1, \dots, \eta_n$ as a solution of the equation

$$\sum_{j=1}^n \frac{\partial V}{\partial \eta_j} \left(\sum_{i=1}^n p_{ji} \eta_i \right) = - \sum_{i=1}^n \eta_i^2.$$

It is clear that for sufficiently small $|\eta_j|$ ($j = 1, \dots, n$) and $|c_s|$ ($s = 1, \dots, k$), the following inequality holds:

$$\frac{dV}{dt} \leq - \frac{1}{2} \sum_{i=1}^n \eta_i^2,$$

and this leads to the above statement. Thus, any solution of the system (2.71) for sufficiently small $|c_s|$ ($s = 1, \dots, k$) and $|\eta_j^{(0)}|$ ($j = 1, \dots, n$) has the following property:

$$\eta_j(t, \eta_1^{(0)}, \dots, \eta_n^{(0)}) \rightarrow 0 \text{ as } t \rightarrow +\infty \text{ and } x_s \rightarrow c_s.$$

This means that $y_j \rightarrow u_j(c_1, \dots, c_k)$ as $t \rightarrow +\infty$. Thus, if the conditions of the theorem are satisfied, the trivial solution of the system (2.69) is stable. Let us now choose sufficiently small values of $x_s^{(0)}$ ($s = 1, \dots, k$) and $\eta_j^{(0)}$ ($j = 1, \dots, n$); we then determine from (2.73) a unique system of quantities

$$c_s^{(0)} = x_s^{(0)} + \varphi_s(x_1^{(0)}, \dots, x_k^{(0)}; \eta_1^{(0)}, \dots, \eta_n^{(0)}).$$

From (2.76) we have

$$x_s^{(0)} = c_s^{(0)} + f_s(\eta_1^{(0)}, \dots, \eta_n^{(0)}, c_1^{(0)}, \dots, c_k^{(0)}) \quad (s = 1, \dots, k).$$

Thus, the quantities $x_1^{(0)}, \dots, x_k^{(0)}$ can be considered as functions of the quantities $\eta_1^{(0)}, \dots, \eta_n^{(0)}$. If, in this case, the quantities $c_s^{(0)}$ ($s = 1, \dots, k$) are sufficiently small, then any solution of the system (2.71) has the property that

$$\begin{aligned} x_s(t, x_1^{(0)}, \dots, x_k^{(0)}, \eta_1^{(0)}, \dots, \eta_n^{(0)}) &= \\ &= x_s(t, c_1^{(0)} + f_1(\eta_1^{(0)}, \dots, \eta_n^{(0)}, c_1^{(0)}, \dots, c_k^{(0)}), \dots, c_k^{(0)} + \\ &+ f_k(\eta_1^{(0)}, \dots, \eta_n^{(0)}, c_1^{(0)}, \dots, c_k^{(0)}), \eta_1^{(0)}, \dots, \eta_n^{(0)}) \rightarrow c_s^{(0)}, \\ \eta_j(t, \eta_1^{(0)}, \dots, \eta_n^{(0)}, c_1^{(0)}, \dots, c_k^{(0)}) &\rightarrow 0 \text{ as } t \rightarrow +\infty \end{aligned}$$

and, consequently,

$$y_j(t, \dots) \rightarrow u_j(c_1^{(0)}, \dots, c_k^{(0)}) \text{ as } t \rightarrow +\infty.$$

This proves Theorem 44.

If $X_s(x_1, \dots, x_k, u_1, \dots, u_n) \equiv 0$, then the trivial solution of the system (2.69) is stable in the sense of Lyapunov, and any solution of this system

$$x_s = c_s, \quad y_j = u_j(c_1, \dots, c_k)$$

is conditionally asymptotically stable.

We now consider the general case when

$$X_s(x_1, \dots, x_k, u_1, \dots, u_n) = X_s^{(0)}(x_1, \dots, x_n) \neq 0. \quad (2.79)$$

Let us make in the system (2.69) the substitution $y_j = \eta_j + u_j(x_1, \dots, x_k)$, ($j = 1, \dots, n$). The functions $u_j(x_1, \dots, x_k)$, which are the solution of the system (2.70), are expanded in convergent series in powers of the quantities $x_1, \dots, x_k$, and their expansions do not contain terms that are linear in $x_1, \dots, x_k$. Let us denote by μ the lowest power of the forms, with which the expansion of the function $X_s^{(0)}$ actually begins:

$$X_s^{(0)} = X_s^{(0)(\mu)} + X_s^{(0)(\mu+1)} + \dots$$

We denote by ν the lowest power of the forms, with which the expansion of the function

$$V_j^{(0)} = V_j(x_1, \dots, x_k, 0, \dots, 0)$$

actually begins. We then have

$$V_j^{(0)} = V_j^{(0)(\nu)} + V_j^{(0)(\nu+1)} + \dots$$

From the form of the function V_j it follows that $\nu \geq \mu + 1$. Let us set, furthermore, in the system (2.71)

$$x_s = \xi_s + \varphi_s(\eta_1, \dots, \eta_n),$$

where the functions $\varphi_s(\eta_1, \dots, \eta_n)$ are holomorphic solutions of the system of equations

$$\begin{aligned} \sum_{i=1}^n \frac{\partial x_s}{\partial \eta_j} \left(\sum_{i=1}^n p_{ji} \eta_i + V_j \right) &= \\ &= X_s(x_1, \dots, x_k, u_1 + \eta_1, \dots, u_n + \eta_n). \end{aligned}$$

It is clear that $\varphi_s = 0$ when $\eta_j = 0$ ($j = 1, \dots, n$). After this

substitution, the system (2.71) will have the form

$$\left. \begin{aligned} \frac{d\xi_s}{dt} &= U_s(\xi_1, \dots, \xi_k, \eta_1, \dots, \eta_n) \\ \frac{d\eta_j}{dt} &= \sum_{i=1}^n p_{ji}\eta_i + \bar{V}_j(\xi_1, \dots, \xi_k, \eta_1, \dots, \eta_n) \end{aligned} \right\} \quad (s = 1, \dots, k), \quad (j = 1, \dots, n) \quad (2.80)$$

It is clear that $U_s = 0$ when $\xi_1 = \dots = \xi_k = 0$. For the indicated substitution, the following equations hold:

$$U_s^{(0)} = X_s^{(0)}(\xi_1, \dots, \xi_k), \quad \bar{V}_j^{(0)} = V_j^{(0)}(\xi_1, \dots, \xi_k).$$

Let us set

$$U_s = \sum_{N=1}^{+\infty} U_s^{(N)} + U_s^{(0)}(\xi_1, \dots, \xi_k)$$

and

$$\bar{V}_j^{(0)} = \sum_{N=0}^{+\infty} V_j^{(N)} + V_j^{(0)}(\xi_1, \dots, \xi_k),$$

where the functions $U_s^{(N)}$ and $V_j^{(N)}$ represent homogeneous forms in the quantities $\xi_1, \dots, \xi_k$ of order N with coefficients which are analytic in $\eta_1, \dots, \eta_n$, which vanish when $\eta_1 = \eta_2 = \dots = \eta_n = 0$. Next, we find a transformation that reduces the system (2.79) to the following form:

$$\begin{aligned} \frac{dx_s}{dt} &= \bar{X}_s, \\ \frac{d\eta_j}{dt} &= \sum_{i=1}^n p_{ji}\eta_i + \bar{Y}_j, \text{ where } \bar{X}_s = \sum_{N=\mu+1}^{+\infty} X_s^{(N)} + \bar{X}_s^{(0)}(x_1, \dots, x_k), \\ \bar{Y}_j &= \sum_{N=1}^{+\infty} Y_j^{(N)} + V_j^{(0)}(x_1, \dots, x_k), \end{aligned} \quad (2.81)$$

where $X_s^{(N)}$, $Y_j^{(N)}$ ($s = 1, \dots, k$), ($j = 1, \dots, n$) are homogeneous forms of degree N with respect to the quantities $x_1, \dots, x_k$ with coefficients which are analytic in $\eta_1, \dots, \eta_n$ and vanish when $\eta_1 = \dots = \eta_n = 0$.

Let us replace the unknown functions, in the system (2.71),

without affecting the stability problem, by means of the formulas

$$\xi_s = x_s + \sum_{N=1}^{\mu} p_s^{(N)} \quad (s = 1, \dots, k). \quad (2.82)$$

The functions $U_s^{(N)}$ are forms in the quantities $x_1, \dots, x_k$ of order N with coefficients which are analytic in $\eta_1, \dots, \eta_n$, and which are to be determined. Substituting (2.82) into (2.80), we obtain

$$U'_s = \frac{dx_s}{dt} + \sum_{i=1}^k \sum_{N=1}^{\mu} \frac{\partial V_s^{(N)}}{\partial x_i} \frac{dx_i}{dt} + \\ + \sum_{N=1}^{\mu} \sum_{j=1}^n \frac{\partial V_s^{(N)}}{\partial \eta_j} \left(\sum_{i=1}^n p_{ji} \eta_i + V'_j \right), \quad (2.83)$$

$$\frac{d\eta_j}{dt} = \sum_{i=1}^n p_{ji} \eta_i + V'_j. \quad (2.84)$$

Here we denote by U'_s ($s = 1, \dots, k$) and V'_j ($j = 1, \dots, n$) the result obtained when the values of ξ_s from formulas (2.82) are substituted into the functions U_s and $\bar{V}_j$. Next, equating on the right and on the left in Eqs. (2.83) forms of equal power in the quantities $x_1, \dots, x_k$, which do not contain dx_s/dt , we obtain a system of equations from which we determine successively the coefficients of the forms

$$p_s^{(N)} (s = 1, \dots, k, N = 1, \dots, \mu - 1).$$

Let us first demonstrate this for $N = 1$. We set

$$p_s^{(1)} = \sum_{i=1}^k \varphi_{si}(\eta_1, \dots, \eta_n) x_i$$

and

$$U_s^{(1)} = \sum_{i=1}^k g_{si} \xi_i.$$

The functions g_{si} are known analytic functions in $\eta_1, \dots, \eta_n$, which vanish when $\eta_1 = \dots = \eta_n = 0$. We next set

$$V_j^{(0)'} = V'_j|_{x_1=\dots=x_k=0}.$$

Equating the forms of the first degree relative to the quantities

$(x_1, \dots, x_k)$ on the left and on the right of (2.83), we obtain

$$\sum_{s=1}^k (\varphi_{is} + \delta_{is}) g_{sj} = \sum_{l=1}^n \frac{\partial \varphi_{lj}}{\partial \eta_l} \left(\sum_{i=1}^n p_{li} \eta_i + V_j^{(0)} \right) \quad (2.85)$$

$$(i = 1, \dots, k; j = 1, \dots, k),$$

where δ_{is} is the Kronecker symbol.

The system (2.85) defines a unique set of k^2 holomorphic functions $\varphi_{ij}(\eta_1, \dots, \eta_n)$, which vanish when $\eta_i = 0$ ($i = 1, \dots, n$), as follows directly from the auxiliary theorem of Lyapunov.

Thus, the coefficients of the forms $P_s^{(1)}$, ($s = 1, \dots, k$) are completely determined. Equating forms of second order in Eqs. (2.83) on the left and on the right, we obtain a system of equations from which on the basis of the Lyapunov auxiliary theorem we can determine uniquely all the coefficients of the forms $P_s^{(2)}$ ($s = 1, \dots, k$), in the form of holomorphic functions with respect to the quantities $\eta_1, \dots, \eta_n$, which vanish when $\eta_1 = \dots = \eta_n = 0$. Proceeding further in a similar manner, we obtain in a unique manner the coefficients of all forms $P_s^{(N)}$, $N \leq \mu - 1$, in the form of holomorphic functions in $\eta_1, \dots, \eta_n$, which vanish when $\eta_1 = \dots = \eta_n = 0$. Let us transfer all the terms containing forms of the μ -th degree in $x_1, \dots, x_n$ into the left member of (2.83). The resultant μ -th degree in $x_1, \dots, x_k$ will be denoted by $\bar{U}_s^{(\mu)}$. It is clear that the coefficients of the forms $P_s^{(\mu)}$ can be chosen in the form of holomorphic functions, which vanish when $\eta_1 = \dots = \eta_n = 0$, in such a way that the coefficients of the forms $\bar{U}_s^{(\mu)}$ will be specified analytic functions of the quantities $\eta_1, \dots, \eta_n$, which vanish when $\eta_1 = \dots = \eta_n = 0$. Solving Eqs. (2.73) with respect to the quantities dx_s/dt , we obtain

$$\frac{dx_s}{dt} = \bar{X}_s(x_1, \dots, x_k, \eta_1, \dots, \eta_n) = \sum_{N=\mu}^{+\infty} X_s^{(N)} + \bar{X}_s^{(0)}(x_1, \dots, x_k).$$

$$\left[\bar{X}_s^{(0)} = X_s^{(0)} - \sum_{N=1}^{\mu} \sum_{j=1}^n \frac{\partial V_s^{(N)}}{\partial \eta_j} \left(\sum_{i=1}^n p_{ji} \eta_i + V_j \right) \text{ for } \eta_1 = \dots = \eta_n = 0. \right]$$

It is easy to verify that the coefficients of the functions $\bar{U}_s^{(\mu)}$ can be chosen such that $X_s^{(\mu)} \equiv 0$. Thus, we have shown that the transformation (2.82) can be chosen such that after it is applied to the system (2.80), we obtain the system (2.81).

THEOREM 45. *If the trivial solution of the system of k equations*

$$\frac{dx_s}{dt} = X_s^{(0)(\mu)}, \quad (2.86)$$

where the functions $X_s^{(0)(\mu)}$ are homogeneous forms of degree μ in the quantities $x_1, \dots, x_k$ and are first forms in the expansion of the functions $X_s(x_1, \dots, x_k, u_1, \dots, u_n)$ is asymptotically stable, then the trivial solution of the system (2.69) will also be asymptotically stable. In this connection, any solution of the system (2.81) beginning in the region

$$|X^{(0)}| = \sqrt{\sum_{s=1}^k x_s^{(0)2} + \sum_{j=1}^n \eta_j^{(0)2}} \leq h,$$

satisfies the inequalities

$$p_1 |X^{(0)}| e^{-p_2 t} \leq |X(t, X^{(0)})| \leq \frac{c_1 |X^{(0)}|}{\sqrt{\mu-1} \sqrt{1 + c_2 t |X^{(0)}|^{\mu-1}}}. \quad (2.87)$$

PROOF. The trivial solution of the system (2.86) is asymptotically stable. Then, as was shown in Section 3 of this chapter, there exist two homogeneous functions V and W , which have the following properties:

- 1) V is positive definite and W is negative definite;
- 2) W has the order m and V has the order $m + 1 - \mu$;
- 3) the function V is continuously differentiable in all its arguments, and the following equality holds:

$$\sum_{i=1}^k \frac{\partial V}{\partial x_i} X_i^{(0)(\mu)} = W.$$

Let us further choose a function V_1 in the form of a positive definite quadratic form, satisfying the equation

$$\sum_{i=1}^n \frac{\partial V_1}{\partial \eta_i} \left(\sum_{j=1}^n p_{ij} \eta_j \right) = - \sum_{i=1}^n \eta_i^2,$$

which can always be done on the basis of the Lyapunov theorem. We set

$$U = V + V_1, \quad V = V(x_1, \dots, x_k), \quad V_1 = V_1(\eta_1, \dots, \eta_n).$$

We calculate the total derivative of the function U by virtue of the system (2.81). We then obtain

$$\frac{dU}{dt} = \frac{dV}{dt} + \frac{dV_1}{dt} = \sum_{s=1}^k \frac{\partial V}{\partial x_s} \bar{X}_s + \sum_{j=1}^n \frac{\partial V}{\partial \eta_j} \left(\sum_{i=1}^n p_{ji} \eta_i + \bar{Y}_j \right),$$

from which it follows that

$$\begin{aligned} \frac{dU}{dt} = & W - \sum_{i=1}^n \eta_i^2 + \\ & + \sum_{s=1}^k \frac{\partial V}{\partial x_s} \left(\sum_{N=\mu+1}^{+\infty} (X_s^{(N)} + \bar{X}_s^{(0)(N)}) \right) + \sum_{j=1}^n \frac{\partial V_1}{\partial \eta_j} \bar{Y}_j. \end{aligned} \quad (2.88)$$

The function $\partial V / \partial x_s$ is homogeneous of order $m - \mu$, and therefore for sufficiently small $|x_s|$ and $|\eta_j|$ ($s = 1, \dots, k; j = 1, \dots, n$), we have

$$\left| \sum_{s=1}^k \frac{\partial V}{\partial x_s} \sum_{N=\mu+1}^{+\infty} (X_s^{(N)} + \bar{X}_s^{(0)(N)}) \right| \leq a \sum_{s=1}^k |x_s|^{m+1}.$$

Let us now estimate the second sum in the expression (2.88):

$$\begin{aligned} \left| \sum_{j=1}^n \frac{\partial V_1}{\partial \eta_j} \left(\sum_{N=1}^{+\infty} Y_j^{(N)} + Y_j^{(0)}(x_1, \dots, x_k) \right) \right| & \leq \\ & \leq a \sum_{s=1}^k |x_s|^v \cdot \sum_{i=1}^n |\eta_i| + b \sum_{i=1}^n \eta_i^2 \cdot \sum_{s=1}^k |x_s|. \end{aligned}$$

We now set $m = \mu + 1$. It was shown earlier that $v \geq \mu + 1$, and therefore the function

$$\frac{dU}{dt} = W_1$$

is negative definite, and consequently the trivial solution of the system (2.81) is asymptotically stable, and then the trivial solution of the system (2.69) will also be asymptotically stable because all the substitutions which reduce the system (2.69) to the form (2.81) do not change the stability problem.

Let us now show that the estimate (2.87) holds, in which we set

$$\begin{aligned} X(t, X^{(0)}) = & (x_1(t, X^{(0)}), \dots, x_k(t, X^{(0)}), \\ & \eta_1(t, X^{(0)}), \dots, \eta_n(t, X^{(0)})), \end{aligned}$$

where $X(t, X^{(0)})$ is a solution of the system (2.81). We set $V(p) = |X|$, where $V(p)$ is the functional referred to in Theorem 20. It is clear that the function U satisfies the following inequalities

$$a_1 |X|^2 \leq U \leq a_2 |X|^2,$$

since for a chosen m , V is a positive definite homogeneous function of degree two. The function W_1 satisfies, for sufficiently small $|X|$, the inequalities

$$-b_1 |X|^{\mu+1} \leq W_1 \leq -b_2 |X|^{\mu+1}. \quad (2.89)$$

Here

$$|X| = \sqrt{\sum_{i=1}^k x_i^2 + \sum_{j=1}^n \eta_j^2}.$$

Let us apply Theorem 20 to this case (Section 15). We set in it, as applied to this case,

$$l_1 = l_2 = 2, \quad k_1 = 2, \quad k_2 = \mu + 1,$$

and obtain

$$\lambda_1 = 1, \quad \lambda_2 = \frac{\mu + 1}{2} > 1,$$

from which we obtain, with the aid of the estimate (1.42), the estimate (2.87). Reverting back to the old variables, we find that the solution of the system (2.81) satisfies the estimate (2.87) for all $t \geq 0$ for all sufficiently small

$$|x_s^{(0)}|, \quad (s = 1, \dots, k), \quad |\eta_j^{(0)}|, \quad (j = 1, \dots, n).$$

Examples show that the estimate is exact in the sense that in each specific case one can change only the values of the constants $p_i > 0$, $c_i > 0$.

REMARK. The quantity m in the proof of Theorem 45 is chosen such that $m = \mu + 1$. This is due to the fact that $\nu \geq \mu + 1$, where ν is the order of the forms with which the expansion of the function $V_j^{(0)}$ in powers of $x_1, \dots, x_k$ actually begins. However, as was shown in Section 3 of this chapter, the function $V(x_1, \dots, x_k)$, which is referred to in the proof of Theorem 45, will be differentiable only for sufficiently large values of m .

The contradiction obtained is nevertheless actually only apparent, since the equality ν can in fact be made as large as desired. Let us demonstrate this. We define a system of analytic functions $u_j^{(1)}(x_1, \dots, x_k)$ such that they satisfy the system of equations

$$\sum_{i=1}^k p_{ji} \eta_i + \bar{Y}_j = 0 \quad (j = 1, \dots, n), \quad (2.90)$$

if we substitute the functions $u_i^{(1)}(x_1, \dots, x_k)$ instead of η_i . It is clear that the expansions of the functions $u_j^{(1)}$ in series in positive integral powers of the quantities $x_1, \dots, x_k$ begin with forms of order ν , as can be readily established when finding the functions $u_j^{(1)}$ in the form of arbitrary power series that satisfy the system (2.90). Let us make in the system (2.81) the substitution

$$y_j = y_j + u_j^{(1)}(x_1, \dots, x_k);$$

we then obtain

$$\left. \begin{aligned} \frac{dx_s}{dt} &= \bar{X}_s(x_1, \dots, x_k, y_1 + u_1^{(1)}, \dots, y_n + u_n^{(1)}) \\ \frac{dy_j}{dt} &= \sum_{i=1}^n p_{ji} y_i + Y_j \quad (j = 1, \dots, n) \end{aligned} \right\}, \quad (2.91)$$

where the functions Y_j have the form

$$\begin{aligned} Y_j &= \sum_{i=1}^n p_{ji} u_i^{(1)} + \bar{Y}_j(x_1, \dots, x_k, y_1 + u_1^{(1)}, \dots, y_n + u_n^{(1)}) - \\ &\quad - \sum_{s=1}^k \frac{\partial u_j^{(1)}}{\partial x_s} \bar{X}_s(x_1, \dots, x_k, u_1^{(1)} + y_1, \dots, u_n^{(1)} + y_n). \end{aligned} \quad (2.92)$$

Let us set

$$Y_j^{(0')} = Y_j' \quad \text{for } y_1 = y_2 = \dots = y_n = 0.$$

It follows from (2.92) that the expansion of the function $Y_j^{(0')}$ in non-negative integral powers of the quantities $x_1, \dots, x_k$ begins with a form of the order $\nu_1 = \nu - 1 + \mu$. Making this transformation in the system (2.91), we obtain a certain new system of differential equations, for which the quantity $\nu_2 = \nu + 2(\mu - 1)$. We shall assume that this transformation has been made in the

system (2.81) l times. Then the expansion of the functions $V_j^{(0)}$ in non-negative integral powers of the quantities $x_1, \dots, x_k$ will begin with forms of order $\nu_l = \nu + l(\mu - 1)$. This proves the foregoing statement. Thus, when proving Theorem 45, the quantity ν can be considered as large as desired. Accordingly, the quantity m can be then chosen to be sufficiently large.

Let us develop further a certain general theorem on instability, on the basis of which it is possible to formulate the conditions of instability of the trivial solution of the system (2.69) in terms of the coefficients of the forms $X_s^{(0)(\mu)}$, which will be given in the next chapter.

Let us consider the system of differential equations

$$\begin{aligned}\frac{dx_s}{dt} &= X_s(x_1, \dots, x_k, t) + \bar{X}_s(x_1, \dots, x_k, y_1, \dots, y_n, t), \\ \frac{dy_j}{dt} &= Y_j(x_1, \dots, x_k, y_1, \dots, y_n, t).\end{aligned}\tag{2.93}$$

We assume that the functions in the right members of the system (2.93) are defined in the region

$$|x_s| \leq h, |y_j| \leq h, t > 0 \quad (j = 1, \dots, n; s = 1, \dots, k) \tag{2.94}$$

and satisfy certain conditions there, for which there exist solutions of the system (2.93), passing through any point $(x_s^{(0)}, y_j^{(0)}, t_0)$ ($s = 1, \dots, k, j = 1, \dots, n$) of the region (2.94). We shall assume further that in the region

$$|x_s| \leq h_1, |y_j| \leq h_1, t > 0 \quad (s = 1, \dots, k, j = 1, \dots, n) \tag{2.95}$$

the functions $\bar{X}_s$ satisfy the inequalities

$$|\bar{X}_s(x_1, \dots, x_k, y_1, \dots, y_n, t)| \leq G_s(x_1, \dots, x_k, t). \tag{2.96}$$

Further, let us consider the system

$$\frac{dx_s}{dt} = X_s(x_1, \dots, x_n, t) + F_s(x_1, \dots, x_n, t). \tag{2.97}$$

THEOREM 46. *If the system (2.93) has the trivial solution*

$$x_1 = \dots = x_k = y_1 = \dots = y_n = 0,$$

then it will be unstable if the following conditions are satisfied:

- 1) for any choice of the functions F_s , satisfying the inequality

$$|F_s| \leq G_s(X, t) \text{ for } |x_s| \leq h_1, \quad (2.98)$$

the system (2.97) has an unstable trivial solution ($s = 1, \dots, k$);

- 2) there exists a set B in $(k+1)$ -dimensional space of points $(x_1^{(0)}, \dots, x_k^{(0)}, t_0)$ with the following properties:

- a) for any quantity $\delta > 0$ there can be found a point $(x_1^{(0)}, \dots, x_k^{(0)}, t_0) \in B$ such that

$$\sqrt{\sum_{s=1}^k x_s^{(0)^2} < \delta;$$

- b) the integral curve $x_s = x_s(t, x_1^{(0)}, \dots, x_k^{(0)}, t_0)$ of the system (2.97), for any choice of the functions F_s , satisfying the inequality

$$|F_s| \leq G_s \text{ for } |x_s| \leq h_1 \text{ and } t > 0,$$

has the following property: for any $\gamma > 0$ it is possible to find a quantity

$$T(x_1^{(0)}, \dots, x_k^{(0)}, t_0, \gamma)$$

such that

$$\sqrt{\sum_{s=1}^k x_s^2(t, X^{(0)}, t_0)} = \bar{\varepsilon} \text{ for } t = t_0 + T,$$

$$\text{for } |X^{(0)}| < \gamma \text{ and } (X^{(0)}, t_0) \in B,$$

where $\bar{\varepsilon}$ is a certain positive number.

PROOF. Let the trivial solution of the system (2.93) be stable. Then for the quantity $\varepsilon > 0$, it is possible to find a quantity $\delta > 0$ such that for

$$|x_s^{(0)}| < \delta, |y_j^{(0)}| < \delta \quad (s = 1, \dots, k, j = 1, \dots, n)$$

we have

$$|x_s(t, X^{(0)}, Y^{(0)}, t_0)| < \varepsilon \text{ and } |y_j(t, X^{(0)}, Y^{(0)}, t_0)| < \varepsilon \text{ for } t \geq t_0 \geq 0,$$

where

$$X^{(0)} = (x_1^{(0)}, \dots, x_k^{(0)}), \quad Y^{(0)} = (y_1^{(0)}, \dots, y_n^{(0)}).$$

Substituting into the first group of the system of equations (2.93) the functions

$$y_j(t, X^{(0)}, Y^{(0)}, t_0),$$

we obtain a system of differential equations of the form (2.97). If ε is chosen such that $\varepsilon < h_1$, then it follows from inequality (2.96) that inequality (2.98) is satisfied. The set of functions

$$x_s(t, X^{(0)}, Y^{(0)}, t_0) \quad (s = 1, \dots, k)$$

will satisfy the system (2.97) and, furthermore, the point $(X^{(0)}, t_0)$ can be chosen such as to make $(X^{(0)}, t_0) \in B$, and then, choosing $\varepsilon < \bar{\varepsilon}$, we obtain, by virtue of item "b" of Condition 2 of Theorem 46,

$$|X(t, X^{(0)}, Y^{(0)}, t_0)| > \varepsilon \quad \text{for } t = t_0 + T, T < +\infty.$$

Consequently, the foregoing assumption about the stability of the trivial solution of the system (2.93) is incorrect.

We note that Theorem 43, the complete proof of which is given here, can be directly obtained from the Lyapunov theorem, contained in Section 56 of reference [1].

5. Case of several pairs of pure imaginary roots

Let us consider the system

$$\begin{aligned} \frac{dx_s}{dt} &= -\lambda_s y_s + X_s(x_1, \dots, x_k, y_1, \dots, y_k, z_1, \dots, z_n), \\ &\hspace{15em} (s = 1, \dots, k) \\ \frac{dy_s}{dt} &= \lambda_s x_s + Y_s(x_1, \dots, x_k, y_1, \dots, y_k, z_1, \dots, z_n), \\ &\hspace{15em} (2.99) \\ \frac{dz_j}{dt} &= Z_j(x_1, \dots, x_k, y_1, \dots, y_k, z_1, \dots, z_n) + \sum_{i=1}^n r_{ji} z_i \\ &\hspace{15em} (j = 1, \dots, n). \end{aligned}$$

We assume that the functions X_s , Y_s , Z_j are expanded in convergent power series in $x_1, \dots, x_k, y_1, \dots, y_k, z_1, \dots, z_n$, which do not contain linear terms. We shall assume that the right members of the system (2.99) are real, the quantities $\lambda_1, \dots, \lambda_k$ are

positive and such that

$$\sum_{i=1}^k \mu_i \lambda_i \neq 0 \quad (2.100)$$

for all integral

$$\mu_1, \dots, \mu_k, \sum_{i=1}^k |\mu_i| \neq 0.$$

The system

$$\left. \begin{aligned} \sum_{j=1}^n \frac{\partial x_s}{\partial z_j} \left(\sum_{i=1}^n r_{ji} z_i + \bar{Z}_j \right) &= -\lambda_s y_s + \bar{X}_s \\ \sum_{j=1}^n \frac{\partial y_s}{\partial z_j} \left(\sum_{i=1}^n r_{ji} z_i + \bar{Z}_j \right) &= \lambda_s x_s + \bar{Y}_s \end{aligned} \right\} \quad (2.101)$$

has a solution

$$x_s = u_s(\bar{z}_1, \dots, \bar{z}_n), \quad y_s = v_s(\bar{z}_1, \dots, \bar{z}_n),$$

which is holomorphic in a neighborhood of the point $\bar{z}_1 = \dots = \bar{z}_n = 0$. Let us make the transformation

$$x_s = \bar{x}_s + u_s, \quad y_s = \bar{y}_s + v_s \quad (2.102)$$

in the system (2.99). We then obtain

$$\left. \begin{aligned} \frac{d\bar{x}_s}{dt} &= -\lambda_s \bar{y}_s + \bar{X}_s(\bar{x}_1, \dots, \bar{x}_k, \bar{y}_1, \dots, \bar{y}_k, \bar{z}_1, \dots, \bar{z}_n) \\ \frac{d\bar{y}_s}{dt} &= \lambda_s \bar{x}_s + \bar{Y}_s \\ \frac{d\bar{z}_j}{dt} &= \sum_{i=1}^n r_{ji} \bar{z}_i + \bar{Z}_j \end{aligned} \right\}. \quad (2.103)$$

It is clear that $\bar{X}_s$, $\bar{Y}_s$ and $\bar{Z}_j$ have the same properties as the initial X_s , Y_s and Z_j , and in addition $\bar{X}_s = \bar{Y}_s = 0$ when $\bar{x}_s = \bar{y}_s = 0$ ($s = 1, \dots, k$). Further, in the system (2.103), let us make the substitution

$$\left. \begin{aligned} \bar{x}_s &= r_s \cos \vartheta_s \\ \bar{y}_s &= r_s \sin \vartheta_s \end{aligned} \right\}, \quad (2.104)$$

as a result of which we obtain the system

$$\left. \begin{aligned} \frac{dr_s}{dt} &= R_s, & \frac{d\vartheta_s}{dt} &= \lambda_s + \theta_s \\ \frac{d\bar{z}_j}{dt} &= \sum_{i=1}^n r_{ji} \bar{z}_i + P_j(r_1, \dots, r_k, \vartheta_1, \dots, \vartheta_k, \bar{z}_1, \dots, \bar{z}_n) \end{aligned} \right\}, \quad (2.105)$$

(s = 1, \dots, k)

where

$$\begin{aligned} R_s &= \cos \vartheta_s \bar{X}_s(r_1 \cos \vartheta_1, \dots, r_k \cos \vartheta_k, r_1 \sin \vartheta_1, \dots, r_k \sin \vartheta_k, \\ &\quad \bar{z}_1, \dots, \bar{z}_n) + \sin \vartheta_s \bar{Y}_s(r_1 \cos \vartheta_1, \dots, r_k \sin \vartheta_k, \bar{z}_1, \dots, \bar{z}_n), \\ \theta_s &= \frac{\cos \vartheta_s \bar{Y}_s - \sin \vartheta_s \bar{X}_s}{r_s}, \quad P_j = \bar{Z}_j(r_1 \cos \vartheta_1, \dots, \bar{z}_1, \dots, \bar{z}_n). \end{aligned}$$

It is easy to see that the functions R_s and P_j are holomorphic in a neighborhood of $r_1 = \dots = r_k = \bar{z}_1 = \dots = \bar{z}_n = 0$. In this connection, $R_s = 0$ when $r_1 = \dots = r_k = 0$. The coefficients in the expansions of the functions R_s and P_j are periodic functions in ϑ_i ($i = 1, \dots, k$). We shall assume further that the functions θ_s have the same properties as R_s and P_j . We note that the conditions (2.100) enable us to carry out the transformation in the system (2.103), after which the functions θ_s will be formally expanded in series in powers of $r_1, \dots, r_k, \bar{z}_1, \dots, \bar{z}_n$. We shall seek a solution of the system (2.105) in the form of series

$$\begin{aligned} r_s &= c_s + \sum_{m=2}^{\infty} r_s^{(m)}(\vartheta_1, \dots, \vartheta_k, c_1, \dots, c_k), \\ \bar{z}_j &= \sum_{m=1}^{\infty} \bar{z}_j^{(m)}(\vartheta_1, \dots, \vartheta_k, c_1, \dots, c_k). \end{aligned} \quad (2.106)$$

The functions $r_s^{(m)}$ and $\bar{z}_j^{(m)}$ are homogeneous forms in $c_1, \dots, c_k$ of degree m with periodic coefficients in $\vartheta_1, \dots, \vartheta_k$, which are to be determined.

Let us eliminate the parameter t from the system (2.105). We then obtain

$$\begin{aligned} \sum_{i=1}^k \frac{\partial r_s}{\partial \vartheta_i} (\lambda_i + \theta_i) &= R_s \quad (s = 1, \dots, k), \\ \sum_{i=1}^k \frac{\partial \bar{z}_j}{\partial \vartheta_i} (\lambda_i + \theta_i) &= \sum_{i=1}^n r_{ji} \bar{z}_i + P_j \quad (j = 1, \dots, n). \end{aligned} \quad (2.107)$$

Substituting the series (2.106) into the system (2.107) and equating forms of equal degree in the quantities $c_1, \dots, c_k$, we obtain the system of equations

$$\sum_{i=1}^k \frac{\partial r_s^{(m)}}{\partial \vartheta_i} \lambda_i = R_s^{(m)} \quad (s = 1, \dots, k),$$

$$\sum_{i=1}^k \lambda_i \frac{\partial \bar{z}_j^{(m)}}{\partial \vartheta_i} = \sum_{i=1}^n r_{ji} \bar{z}_i^{(m)} + P_j^{(m)} \quad (j = 1, \dots, n). \quad (2.108)$$

The forms $R_s^{(m)}$ and $P_j^{(m-1)}$ can be defined if one can find the functions $r_s^{(m_1)}$ and

$$\bar{z}_j^{(m_1)}, m_1 < m, m_2 < m - 1.$$

Let us assume that the functions (2.106), defined from the system (2.108), are periodic in $\vartheta_1, \dots, \vartheta_k$ and represent a solution of the system (2.107) for sufficiently small $|c_s|$. Let the construction of the series (2.106) be such that $r_s^{(m)} \equiv 0$ for $m \geq 2$ ($\vartheta_1, \dots, \vartheta_k = 0$). Let us investigate in this particular case the problem of the stability of the trivial solution of the system (2.99). In the system (2.105), we make the substitution

$$r_s = \rho_s + \sum_{m=2}^{\infty} r_s^{(m)}(\vartheta_1, \dots, \vartheta_k, \rho_1, \dots, \rho_k), \quad (2.109)$$

where $\rho_1, \dots, \rho_k$ are new unknown functions, and obtain

$$\frac{d\rho_s}{dt} = \bar{R}_s, \quad \frac{d\vartheta_s}{dt} = \lambda_s + \bar{\theta}_s,$$

$$\frac{dz_j}{dt} = \sum_{i=1}^n r_{ji} z_i + \bar{P}_j. \quad (2.110)$$

The system (2.110) has a family of solutions $\rho_s = c_s, s = 1, \dots, k, z_j = 0, j = 1, \dots, k$. It can be shown next that the system

$$\sum_{i=1}^n \frac{\partial \rho_s}{\partial \bar{z}_j} \left(\sum_{i=1}^n r_{ji} \bar{z}_i + \bar{P}_j \right) + \sum_{i=1}^k \frac{\partial \rho_s}{\partial \vartheta_i} (\lambda_i + \bar{\theta}_i) = \bar{R}_s \quad (2.111)$$

has a family of solutions

$$\rho_s = c_s + F_s(\bar{z}_1, \dots, \bar{z}_n, \vartheta_1, \dots, \vartheta_k, c_1, \dots, c_k) \quad (2.112)$$

$$(s = 1, \dots, k).$$

The functions F_s are expanded in convergent power series in $\bar{z}_1, \dots, \bar{z}_n, c_1, \dots, c_k$ for sufficiently small $|\bar{z}_j|$ and $|c_s|$, and $F_s \equiv 0$ when $\bar{z}_1 = \dots = \bar{z}_n = 0$. Using equalities (2.112), we eliminate the quantities ρ_s from the third group of equations in (2.110). We then obtain a system for determining the functions $\bar{z}_j$, from which it follows that for all sufficiently small $|c_s|$ and for any choice of continuous real functions $\vartheta_1(t), \dots, \vartheta_k(t)$ we have $\bar{z}_j(t) \rightarrow 0$ as $t \rightarrow +\infty$ uniformly with respect to $c_1, \dots, c_k$ as long as the $|\bar{z}_j(0)|$ are sufficiently small. From this and (2.112) it follows that $\rho_s \rightarrow c_s$ as $t \rightarrow +\infty$, meaning that the trivial solution of the system (2.99) is stable. This proves the following theorem.

THEOREM 47. *If the system (2.105) has a family of bounded solutions (2.106), then the trivial solution of the system (2.99) is stable in the sense of Lyapunov.*

Let us solve the equalities (2.112) with respect to the quantities $c_1, \dots, c_k$; we obtain

$$c_s = \rho_s + \Phi_s(\vartheta_1, \vartheta_2, \dots, \vartheta_k, \rho_1, \dots, \rho_k, \bar{z}_1, \dots, \bar{z}_n) \quad (2.113) \\ (s = 1, \dots, k).$$

Let us change in (2.113) to the quantities $r_1, \dots, r_k$ and, by using (2.104), obtain

$$c_s^2 = \bar{x}_s^2 + \bar{y}_s^2 + \psi_s(\bar{x}_1, \dots, \bar{x}_k, \bar{y}_1, \dots, \bar{y}_k, \bar{z}_1, \dots, \bar{z}_n) \quad (2.114) \\ (s = 1, \dots, k).$$

It can be shown that the functions ψ_s are expandable in convergent series in positive integral powers of the quantities $\bar{x}_s, \bar{y}_s$ and $\bar{z}_i$. Thus, the Eq. (2.14) yields k holomorphic integrals of the system (2.103), and

$$\sqrt{\bar{x}_s^2 + \bar{y}_s^2} + \psi_s = r_s + \bar{\psi}_s(\vartheta_1, \dots, \vartheta_k, r_1, \dots, r_k, \bar{z}_1, \dots, \bar{z}_n),$$

where $\bar{\psi}_s$ is holomorphic in a neighborhood of the point

$$r_1 = r_2 = \dots = r_k = \bar{z}_1 = \dots = \bar{z}_n = 0. \quad (2.115)$$

THEOREM 48. *In order for the system (2.105) to have a family of bounded solutions (2.106), it is necessary and sufficient that there exist k holomorphic integrals of the system (2.103) of the form (2.114), having the property (2.115). ..*

PROOF. The necessity of the condition has been established above, and the sufficiency of the condition can be derived by performing the inverse transformations on the integrals (2.115).

Let us proceed to the analysis of the general case. We assume that $r_s^{(m)}$ ($m < \mu$) have been found to be periodic, and among the functions $r_s^{(\mu)}$, ($s = 1, \dots, k$) at least one of these is not determined to be periodic. In the system (2.107), we set

$$R_s^{(\mu)} = \sum_{m_1 + \dots + m_k = \mu} R_s^{(\mu, m_1, \dots, m_k)}(\vartheta_1, \dots, \vartheta_k) c_1^{m_1} \dots c_k^{m_k},$$

where

$$R_s^{(\mu, m_1, \dots, m_k)} = \sum_{\substack{k \\ \sum_{j=1}^k |\mu_j| \leq M}} R_s^{(\mu, m_1, \dots, m_k, \mu_1, \dots, \mu_k)} e^{i \sum_{j=1}^k \mu_j \vartheta_j};$$

it is clear that

$$R_s^{(\mu, m_1, \dots, m_k, \mu_1, \dots, \mu_k)} \neq 0 \quad \text{for} \quad \sum_{j=1}^k |\mu_j| = 0.$$

We denote by $R_s^{(0)(\mu)}$ the form

$$\sum_{m_1 + \dots + m_k = \mu} R_s^{(\mu, m_1, \dots, m_k, 0, \dots, 0)} c_1^{m_1} \dots c_k^{m_k}.$$

We denote by $r_s^{(\mu)}$ the periodic solution of the system

$$\sum_{i=1}^k \lambda_i \frac{\partial r_s^{(\mu)}}{\partial \vartheta_i} = R_s^{(\mu)} - R_s^{(0)(\mu)}. \quad (2.116)$$

In the system (2.105), let us make the substitution

$$\begin{aligned} r_s &= \rho_s + \sum_{m=2}^{\mu} r_s^{(m)}(\vartheta_1, \dots, \vartheta_k, \rho_1, \dots, \rho_k), \\ \bar{z}_j &= \sum_{m=1}^{\mu} z_j^{(m)}(\vartheta_1, \dots, \vartheta_k, \rho_1, \dots, \rho_k) + \xi_j \quad (j = 1, \dots, n), \end{aligned} \quad (2.117)$$

and obtain

$$\begin{aligned} \frac{d\rho_s}{dt} &= \bar{R}_s(\vartheta_1, \dots, \vartheta_k, \rho_1, \dots, \rho_k), \\ \frac{d\vartheta_s}{dt} &= \lambda_s + \bar{\vartheta}_s, \quad \frac{d\xi_j}{dt} = \sum_{i=1}^n r_{ji} \xi_i + \bar{P}_j. \end{aligned} \quad (2.118)$$

The expansion of the functions R_s in powers of $\rho_1, \dots, \rho_k$ begins with the forms $R_s^{(0)(\mu)}(\rho_1, \dots, \rho_k)$, and the expansion of the functions P_j for $\xi_2 = \dots = \xi_n = 0$ begins with the forms

$$P_j^{(0)(\nu)}(\vartheta_1, \dots, \vartheta_k, \rho_1, \dots, \rho_k), \quad \nu \geq \mu + 1.$$

THEOREM 49. *If the trivial solution of the system*

$$\frac{d\rho_s}{dt} = R_s^{(0)(\mu)}(\rho_1, \dots, \rho_k) \quad (s = 1, \dots, k) \quad (2.119)$$

is asymptotically stable, then the trivial solution of the system (2.99) is also asymptotically stable. With this, any integral curve of the system (2.118), beginning in a sufficiently small neighborhood $|x_s| \leq h, |y_s| \leq h, |z_j| \leq h$, satisfies the inequalities

$$p_1 |X^{(0)}| e^{-\nu t} \leq |X(t, X^{(0)})| \leq \frac{|X^{(0)}| C_1}{\sqrt{\mu-1} \sqrt{1 + C_2 t |X^{(0)}|^{\mu-1}}}, \quad (2.120)$$

for $t > 0$, where

$$\begin{aligned} X(t, X^{(0)}) &= (\rho_1, \dots, \rho_k, \bar{z}_1, \dots, \bar{z}_n), \\ X^{(0)} &= (\rho_1^{(0)}, \dots, \rho_k^{(0)}, z_1^{(0)}, \dots, z_n^{(0)}). \end{aligned}$$

PROOF. As a consequence of the asymptotic stability of the trivial solution of the system (2.119), there exist two homogeneous functions V and W having the following properties:

- 1) the function V is positive definite and W is negative definite;
- 2) the function V has the order $m + 1 - \mu$ and the function W has the order m ;

$$3) \quad \sum_{s=1}^k \frac{\partial V}{\partial \rho_s} R_s^{(0)(\mu)} = W. \quad (2.121)$$

Let us construct the positive definite quadratic form V_1 satisfying the equation

$$\sum_{j=1}^n \frac{\partial V_1}{\partial \xi_j} \cdot \sum_{i=1}^n r_{ji} \xi_i = - \sum_{i=1}^n \xi_i^2. \quad (2.122)$$

We set $U = V + V_1$. We calculate the total derivative of the

function U by virtue of the system (2.118) and obtain

$$\begin{aligned} \frac{dU}{dt} = \frac{dV}{dt} + \frac{dV_1}{dt} = W - \sum_{i=1}^n \xi_i^2 + \\ + \sum_{s=1}^k \frac{\partial V}{\partial \rho_s} (\tilde{R}_s - R_s^{(0)(\mu)}) + \sum_{j=1}^n \frac{\partial V_1}{\partial \xi_j} \tilde{P}_j. \end{aligned} \quad (2.123)$$

We estimate both sums in the expression (2.123), and obtain

$$\left| \sum_{s=1}^k \frac{\partial V}{\partial \rho_s} (\tilde{R}_s - R_s^{(0)(\mu)}) \right| \leq a \left[\sum_{s=1}^k |\rho_s| \right]^{m+1} \quad (2.124)$$

and

$$\left| \sum_{j=1}^n \frac{\partial V_1}{\partial \xi_j} \tilde{P}_j \right| \leq a \left[\sum_{i=1}^k |\rho_i| \right]^{\mu+1} \cdot \sum_{i=1}^n |\xi_i| + b \sum_{j=1}^n |\xi_j|^2 \cdot \sum_{s=1}^k |\rho_s|. \quad (2.125)$$

The inequalities (2.124) and (2.125) hold for sufficiently small $|\rho_s|$ and $|\xi_j|$, ($s = 1, \dots, k$), ($j = 1, \dots, n$) and all ϑ_s , $\vartheta_s \in (-\infty, +\infty)$. The function U is positive definite and its derivative dU/dt , calculated by virtue of the system (2.118), is a negative definite function for $m = \mu + 1$, and consequently the solution

$$\rho_s = 0, \quad (s = 1, \dots, k), \quad \xi_j = 0, \quad (j = 1, \dots, n), \quad \vartheta_s = \lambda_s t$$

of the system (2.118) is asymptotically stable. When $m = \mu + 1$, the function U satisfies the following inequalities

$$a_1 \left(\sum_{s=1}^k |\rho_s|^2 + \sum_{j=1}^n \xi_j^2 \right) \leq U \leq a_2 \left(\sum_{s=1}^k |\rho_s|^2 + \sum_{j=1}^n \xi_j^2 \right).$$

The function dU/dt in the region indicated above satisfies the inequalities

$$-b_1 \left(\sum_{s=1}^k \rho_s^2 + \sum_{j=1}^n \xi_j^2 \right) \leq \frac{dU}{dt} \leq -b_2 \left(\sum_{s=1}^k \rho_s^2 + \sum_{j=1}^n \xi_j^2 \right)^{(\mu+1)/2}.$$

Let us now apply Theorem 20 to this case, for which we set

$$V(p) = \sqrt{\sum_{s=1}^k \rho_s^2 + \sum_{j=1}^n \xi_j^2},$$

where $V(p)$ is the functional involved in that theorem. Applying furthermore the same arguments as in the proof of Theorem 45,

we obtain that the estimate (2.120) holds for the solutions of the system (2.118).

REMARK. In proving Theorem 49 we assumed $m = \mu + 1$, which was necessary because $\nu \geq \mu + 1$. However, we used in the proof the differential properties of the function $V(\rho_1, \dots, \rho_k)$, which hold at sufficiently large values of m (see Section 3 of this chapter). This contradiction can be eliminated by the same method as was used in Section 2, for the quantity ν can be made as large as desired.

In the next chapter we shall give the conditions for the coefficients of the forms $R_s^{(0)(\mu)}$ which, when satisfied, realize all the requirements of Theorem 46. Thus, we obtain the conditions for instability of the trivial solution of the system (2.99).

6. System of non-stationary differential equations

In the present section we shall apply Theorem 19 to the investigation of the stability of the trivial solution of non-stationary systems, and also to the investigation of the problem of the stability of periodic solutions. Let us consider the system of ordinary differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n), \quad (2.126)$$

the right members of which are defined for $x_s \in (-\infty, +\infty)$, $t \in (-\infty, +\infty)$, ($s = 1, \dots, n$) and which satisfy certain conditions for which there exists the unique solution

$$X = X(t, X^{(0)}, t_0), \quad X(t_0, X^{(0)}, t_0) = X^{(0)}, \\ t_0 \in (-\infty, +\infty), \quad X^{(0)} \in E_n.$$

Here we shall assume that the function $X(t, X^{(0)}, t_0)$ depends continuously on the initial data $X^{(0)}, t_0$. We assume further that the system (2.126) has a trivial solution $X = 0$, i.e.

$$f_s(0, \dots, 0, t) \equiv 0 \quad (s = 1, \dots, n).$$

DEFINITION 1. The trivial solution of the system (2.126) is called *stable in the sense of Lyapunov* if for any $\varepsilon > 0$ one can find

a quantity $\delta > 0$ such that when $|X^{(0)}| < \delta$ and $t_0 \in (-\infty, +\infty)$, we have

$$|X(t, X^{(0)}, t_0)| < \varepsilon \text{ for } t > t_0.$$

If, furthermore,

$$|X(t, X^{(0)}, t_0)| \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

then the trivial solution of the system (2.126) is called *asymptotically stable*.

DEFINITION 2. If the trivial solution of the system (2.126) is asymptotically stable, then the set of all the points $(X^{(0)}, t_0)$, having the property $|X(t, X^{(0)}, t_0)| \rightarrow 0$ as $t \rightarrow +\infty$, is called the *region A of asymptotic stability* ($|X^{(0)}| \neq 0$).

DEFINITION 3. An asymptotically stable trivial solution of the system (2.126) is called *uniformly asymptotically stable* if $|X(t, X^{(0)}, t_0)| \rightarrow 0$ as $t - t_0 \rightarrow +\infty$ uniformly with respect to $t_0 \in (-\infty, +\infty)$ and $|X^{(0)}| < \delta$, where $\delta > 0$ corresponds, on the basis of Definition 1, to a certain fixed $\varepsilon > 0$.

DEFINITION 4. An asymptotically stable trivial solution of the system (2.126) is called *uniformly attracting* if for any $h > 0$, $h < \delta$ one can find quantities $T > 0$ and $\alpha > 0$ such that $|X(t, X^{(0)}, t_0)| > \alpha$ when $t \in (t_0, t_0 + T]$ for any $t_0 \in (-\infty, +\infty)$ and for $h \leq |X^{(0)}| \leq \delta$.

If any solution of the system (2.126) is defined for all $t \in (-\infty, +\infty)$, then it is possible to define with its aid a dynamical system $X = X(\tau + t_0, X^{(0)}, t_0)$, $t = t_0 + \tau$.

In this case all the definitions given above are particular cases of those given in Sections 9 and 10 of Chapter I.

If any solution of the system (2.126) is defined for all $t \in (-\infty, +\infty)$, then to investigate the problem of stability of the trivial solution of the system one can employ the theorems of Sections 12, 13 and 14 of Chapter I, and particularly Theorem 19.

THEOREM 50. If any solution of the system (2.126) is defined for $t \in (-\infty, +\infty)$, then in order for the region A, which contains a sufficiently small neighborhood of the set $X = 0$, $t \in (-\infty, +\infty)$, to be a region of asymptotic stability of the uniformly asymptotically stable and uniformly attracting trivial solution of the system (2.126), it is necessary and sufficient that there exist two functions $V(X, t)$ and $\rho(X, t)$, having the following properties:

1) the function $V(X, t)$ is defined and continuous in A ; the function $\varphi(X, t)$ is defined and continuous in $t \in (-\infty, +\infty)$, $X \in E_n$;

2) $-1 < V(X, t) < 0$ for $(t, X) \in A$;

$\varphi(X, t) > 0$ for $X \in E_n$; $t \in (-\infty, +\infty)$ and $|X| \neq 0$;

3) for any quantity $\gamma_2 > 0$ it is possible to find quantities γ_1 and α_1 such that

$V(X, t) < -\gamma_1$ for $|X| \geq \gamma_2$, $t \in (-\infty, +\infty)$,

$\varphi(X, t) > \alpha_1$ for $|X| \geq \gamma_2$ and $t \in (-\infty, +\infty)$;

4) the functions $\varphi(X, t)$ and $V(X, t)$ approach 0 uniformly with respect to $t \in (-\infty, +\infty)$ as $|X| \rightarrow 0$;

5) if $(\bar{X}, \bar{t})$ is a point on the boundary of the region A , $|\bar{X}| \neq 0$, then $\lim V(X, t) = -1$ for $(X, t) \in A$, $|X - \bar{X}| \rightarrow 0$, $|t - \bar{t}| \rightarrow 0$;

6) the total derivative of the function V , calculated by virtue of the system (2.126), satisfies the relation

$$\frac{dV}{dt} = \varphi(X, t)(1 + V(X, t)).$$

PROOF. See Theorem 19.

Let us now make in the system (2.126) a change in the independent variable t in accordance with the formula

$$dt = \frac{ds}{\sqrt{1 + \sum_{s=1}^n f_s^2(X, t)}}$$

and obtain the system

$$\left. \begin{aligned} \frac{dx_k}{ds} &= \frac{f_k(x_1, \dots, x_n, t)}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \\ \frac{dt}{ds} &= \frac{1}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \end{aligned} \right\}. \quad (2.127)$$

The system of differential equations (2.127) defines in the space

(X, t) a dynamical system

$$X = X(s, X^{(0)}, t_0), \quad t = t(s, X^{(0)}, t_0). \quad (2.128)$$

If the invariant set $X = 0, t = s, s \in (-\infty, +\infty)$ is asymptotically stable, then it is at the same time uniformly attracting. In fact, the new independent variable s introduced into the system (2.126) is the length of arc of the integral curve of this system, considered in $(n+1)$ -dimensional space (X, t) , since

$$ds = dt \sqrt{1 + \sum_{i=1}^n f_i^2} = \sqrt{dt^2 + \sum_{i=1}^n dx_i^2}.$$

We assume that $|X^{(0)}| \geq h$. We take a quantity α such that $\alpha > 0$ and $\alpha < h$. We then have

$$|X(s, X^{(0)}, t_0)| \geq \alpha \quad \text{for } s \in [0, S],$$

where the quantity $S > 0$ is such that $S \leq h - \alpha$, since when the integral curve goes from the surface of the cylinder $|X| = h$ to the surface $|X| = \alpha$, it has a length not less than $h - \alpha$. Thus, if the solution $X = 0, t = s$ is asymptotically stable, then the trivial solution of the system (2.127) is uniformly attracting. If the solution $X = 0, t = s$ of the system (2.127) is uniformly asymptotically stable, then the trivial solution of the system (2.126) is also uniformly asymptotically stable, as follows from the inequality

$$t - t_0 = \int_0^s \frac{1}{\sqrt{1 + \sum_{i=1}^n f_i^2}} ds \leq s, \quad s > 0.$$

However, if the functions $f_s(x_1, \dots, x_n, t)$ are bounded for $t \in (-\infty, +\infty)$ and $|X| \leq H$, then it follows from the uniform asymptotic stability of the trivial solution of the system (2.126) that the solution $X = 0, t = s$ of the system (2.127) is uniformly asymptotically stable. In fact, let

$$\sum_{i=1}^n f_i^2 < M \quad (2.129)$$

for $t \in (-\infty, +\infty)$, $|X| < h$; we then have

$$s = \int_{t_0}^t \sqrt{1 + \sum_{i=1}^n f_i^2} dt,$$

from which $0 < s \leq (t - t_0)\sqrt{1 + M}$. From this follows the statement given above.

THEOREM 51. *In order for a certain region A , containing a sufficiently small neighborhood of the axis $X = 0$, $t \in (-\infty, +\infty)$, to be the region of asymptotic stability of the uniformly asymptotically stable trivial solution of the system (2.126), it is sufficient, and in the case*

$$\sum_{i=1}^n f_i^2 < M, t \in (-\infty, +\infty), |X| < h, M < +\infty, h > 0$$

it is also necessary, that there exist two functions $V(X, t)$ and $\varphi(X, t)$, having the following properties:

- 1) *Conditions 1-5 of Theorem 50 are satisfied.*
- 2) *The total derivative of the function $V(X, t)$, calculated by virtue of the system (2.126), satisfies the equality*

$$\frac{dV}{dt} = \varphi(1 + V) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

PROOF. *Necessity.* Let the trivial solution of the system (2.126) be uniformly asymptotically stable. Then, as shown above, when inequality (2.129) is satisfied, the solution $X = 0$, $t = s$ of the system (2.127) will be uniformly asymptotically stable and, in addition, it will be uniformly attracting. Then, on the basis of Theorem 19, there exist two functions $V(X, t)$ and $\varphi(X, t)$, satisfying the Conditions 1-5 of Theorem 50, and also the condition

$$\frac{dV}{ds} = \varphi(1 + V),$$

where dV/ds denotes the total derivative of the function V , calculated over the motion $X = X(s, X^{(0)}, t_0)$, $t = t(s, X^{(0)}, t_0)$ of the

dynamical system (2.128). Then

$$\frac{dV}{ds} = \frac{dV}{dt} \frac{1}{\sqrt{1 + \sum_{i=1}^n f_i^2}},$$

from which Condition 2 of Theorem 51 follows.

Sufficiency. Let there exist functions V and φ , satisfying conditions 1–2 of Theorem 51. Condition 2 of the theorem can be written in the form

$$\frac{dV}{ds} = \varphi(1 + V),$$

and then the sufficient conditions of Theorem 19 are satisfied for the invariant set $X = 0$, $t = s$, $s \in (-\infty, +\infty)$ of the dynamical system (2.128). Consequently, this invariant set is uniformly asymptotically stable, and then, as shown above, the trivial solution of the system (2.126) will also be uniformly asymptotically stable, which was to be proved.

REMARK. Let us assume that the function

$$m(t) = \sup_{|X| \leq h} \sum_{i=1}^n |f_i(X, t)|,$$

where the quantity $h > 0$ is sufficiently small. The function $m(t)$ is non-negative and continuous. Let us make a change of the independent variable t in the system (2.126) in accordance with the formula

$$d\tau = dt(1 + m(t))$$

and obtain

$$\frac{dx_s}{d\tau} = \frac{f_s(x_1, \dots, x_n, t)}{1 + m(t)} \quad (s = 1, \dots, n). \quad (2.130)$$

It is clear that if $\tau \in (-\infty, +\infty)$, then $t \in (-\infty, +\infty)$, and conversely. Furthermore, $d\tau/dt \geq 0$, and therefore $t = m_1(\tau)$, where m_1 is an implicit function of the variable τ , and one must replace t by $m_1(\tau)$. The right members of the system (2.130) satisfy the condition (2.129), since they are bounded for $\tau \in (-\infty, +\infty)$ and $|X| \leq h$, and therefore the following theorem holds.

THEOREM 52. *In order for the region A , which contains a sufficiently small neighborhood of the axis $X = 0$, $t \in (-\infty, +\infty)$, to be a region of asymptotic stability of the uniformly asymptotically stable trivial solution of the system (2.130), it is necessary and sufficient that there exist two functions $V(X, t)$ and $\varphi(X, t)$, satisfying the following conditions:*

- 1) *Conditions 1–5 of Theorem 50 are satisfied;*
- 2) *The total derivative of the function V , calculated by virtue of the system (2.126), satisfies the equality*

$$\frac{dV}{dt} = \varphi(1 + V) \sqrt{(1 + m(t))^2 + \sum_{i=1}^n f_i^2}.$$

PROOF. In the system (2.130) we make the following change of independent variable:

$$d\tau \sqrt{1 + \sum_{i=1}^n \left[\frac{f_i}{1 + m(m_1(\tau))} \right]^2} = ds.$$

Then the solution $X = 0$, $t = s$ of the system

$$\left. \begin{aligned} \frac{dx_k}{ds} &= \frac{f_k(x_1, \dots, x_n, m_1(\tau))}{[1 + m(m_1(\tau))] \sqrt{1 + \sum_{i=1}^n \frac{f_i^2}{(m+1)^2}}} \\ \frac{d\tau}{ds} &= \frac{1}{\sqrt{1 + \sum_{i=1}^n \frac{f_i^2}{(m+1)^2}}} \end{aligned} \right\} \quad (2.131)$$

will be uniformly attracting, if the trivial solution of the system (2.126) is asymptotically stable. In addition, from the fact of uniform asymptotic stability of the solution $X = 0$, $\tau = s$ of the system (2.131) follows the uniform asymptotic stability of the system (2.126), and conversely.

Applying Theorem 19 to the system (2.131), we find that in order for the set A to be the region of asymptotic stability of the uniformly asymptotically stable solution $X = 0$, $\tau = s$ of the system (2.131), it is necessary and sufficient that there exist two functions $V(X, t)$ and $\varphi(X, t)$, satisfying Conditions 1–5 of Theorem 50 and

the relation

$$\frac{dV}{ds} = \varphi(1 + V);$$

hence

$$\frac{dV}{d\tau} = \varphi(1 + V) \sqrt{1 + \sum_{i=1}^n \frac{f_i^2}{(m+1)^2}}.$$

Changing to the variable t , we obtain

$$\frac{dV}{dt} = \varphi(X, t)(1 + V) \sqrt{(1 + m(t))^2 + \sum_{i=1}^n f_i^2}.$$

This completes the proof of the theorem.

REMARK. If it is assumed that the right members of the system (2.126) are continuously differentiable with respect to all the arguments up to order ν inclusive, one can also show, as was done in the proof of Theorem 22, that the function $\varphi(X, t)$ can be chosen such that the function $V(X, t)$ will be continuously differentiable in all its arguments up to order ν inclusive. If $\nu \geq 1$, then the derivative of the function V , calculated by virtue of the system (2.126), will have the form

$$\frac{dV}{dt} = \sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(X, t) + \frac{\partial V}{\partial t}.$$

This remark pertains to all three theorems proved in this section. From this it follows that the function V can be found as the solution of the equation

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i + \frac{\partial V}{\partial t} = \varphi(1 + V)$$

(Theorem 50), defined by the condition $V(0, \dots, 0, t) = 0$. As regards the function φ , for which a similar solution exists, the remarks made in Section 1 of this chapter are true. Instead of the function $V(X, t)$ it is possible to consider the function $V_1 = \ln(1 + V)$. Then in all the theorems the condition $V \rightarrow -1$ is replaced by the requirement $V \rightarrow -\infty$ for $(X, t) \in A$ and $(X, t) \rightarrow (\bar{X}, \bar{t}) \in \bar{A} \setminus A$ and, in addition, $dV_1/dt = \varphi$ (Theorem 50).

If the region A coincides with the entire space (X, t) , with the exception of $X = 0$, $t \in (-\infty, +\infty)$, then the trivial solution of the system (2.126) is called asymptotically stable in the large.

The foregoing theorem includes also the case when a uniformly asymptotically stable trivial solution of the system (2.126) is asymptotically stable in the large. The conditions for asymptotic stability in the large were first formulated for continuously differentiable right members of the system (2.126) by Ye. A. Barabashin and N. N. Krasovsky (see Section 1, Chapter II).

Let us examine the system

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t) + r_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n). \quad (2.132)$$

The functions r_s will be assumed to be such that the conditions for which there exists a unique solution

$$X = X(t, X^{(0)}, t_0), \quad t_0 \in (-\infty, +\infty), \quad X^{(0)} \in E_n$$

are satisfied. If the functions f_s are continuously differentiable with respect to all the arguments, and if the trivial solution of the system (2.126) is uniformly asymptotically stable, one can find an upper limit for $r_s(x_1, \dots, x_n, t)$ such that when $|r_s| < r(x_1, \dots, x_n, t)$ the trivial solution of the system (2.132) will be asymptotically stable and will have the same region of asymptotic stability as the trivial solution of the system (2.126). This statement can be established by the same method as used to obtain the corresponding statement in Section 1 of Chapter II.

THEOREM 53. *If the trivial solution of the system (2.126) is uniformly asymptotically stable, the condition (2.129) is satisfied, and there exists asymptotic stability in the large, then in the presence of continuous derivatives of the right members of the system (2.126) one can find a function $R_1(X, t)$ in the variables (X, t) , such that for all functions $r_i(X, t)$ any solution of the system (2.132) will be bounded for $|r_i| \leq R_1(x_1, \dots, x_n, t)$ $|X| \geq H$ and $t \in (-\infty, +\infty)$.*

PROOF. The trivial solution of the system (2.126) is uniformly asymptotically stable and moreover asymptotic stability in the large holds. Therefore, there exist two functions $V_1(X, t)$ and $W(X, t)$,

having the properties

$$W = \varphi(X, t) \sqrt{1 + \sum_{i=1}^n f_i^2} \quad \text{and} \quad \frac{dV_1}{dt} = W.$$

If we form the total derivative of the function $V_1(X, t)$ by virtue of the system (2.127), we obtain $dV_1/ds = \varphi$, from which we obtain

$$V_1(X^{(0)}, t_0) = \int_0^{+\infty} -\varphi ds. \quad (2.133)$$

It follows from formula (2.133) that

$$\sup_{\substack{t \in (-\infty, +\infty) \\ |X|=R}} V_1(X, t) = M(R) \quad \text{and} \quad \inf_{\substack{t \in (-\infty, +\infty) \\ |X|=R}} V_1(X, t) = m(R)$$

have the properties

$$M(R) \rightarrow -\infty \quad \text{and} \quad m(R) \rightarrow -\infty \quad \text{as} \quad R \rightarrow +\infty.$$

Let us now form the total derivative of the function V_1 by virtue of the system (2.132). We then obtain

$$\frac{dV_1}{dt} = \sum_{i=1}^n \frac{\partial V_1}{\partial x_i} (f_i + r_i) + \frac{\partial V_1}{\partial t} = W + \sum_{i=1}^n \frac{\partial V_1}{\partial x_i} r_i = W + W_1.$$

Let us estimate the function W_1 :

$$|W_1| \leq \sqrt{\sum_{i=1}^n \left(\frac{\partial V_1}{\partial x_i} \right)^2} \cdot \sqrt{\sum_{i=1}^n r_i^2} \leq \sqrt{\sum_{i=1}^n \left(\frac{\partial V_1}{\partial x_i} \right)^2} \cdot \sqrt{n} R(X, t).$$

We choose the function $R(X, t)$ such that we obtain $W + W_1 > \alpha$ for $|X| \geq H_1$. Let us examine the surface $V_1(X, t) = m(H_1)$. It is included between two cylindrical surfaces, namely $|X| = H_1$ and $|X| = H_2$, where H_2 is chosen such that $M(H_2) = m(H_1)$. Any integral curve of the system (2.132) intersects the surface $V_1(X, t) = m(H_1)$ from the outside in, and therefore each integral curve of the system (2.132) beginning in the region $V_1(X, t) > m(H_1)$ remains bounded as $t \rightarrow +\infty$. Let us show now that any integral curve of the system (2.132), beginning inside the region $V_1(X, t) < m(H_1)$, enters into the region $V_1(X, t) > m(H_1)$ and remains there, and, consequently, it is bounded as $t \rightarrow +\infty$. Assume that

this is not so, i.e. that there exists a point

$$(X^0, t_0), V_1(X^{(0)}, t_0) < m(H_1)$$

such that the integral curve of the system (2.132)

$$X = X(t, X^{(0)}, t_0)$$

remains in the region $V(X, t) < m(H_1)$ for $t \geq t_0$. Then we have that

$$\frac{dV_1}{dt} > \alpha$$

along the integral curve under consideration; hence

$$V_1(X(t, X^{(0)}, t_0), t) > V_1(X^{(0)}, t_0) + \alpha(t - t_0),$$

which is impossible for all $t > t_0$, because

$$V_1(X(t, X^{(0)}, t_0), t) < 0 \text{ for } t \geq t_0,$$

which was to be proved.

Let us turn now to an examination of the stationary system of differential equations

$$\frac{dx_s}{dt} = f_s(X) \quad (s = 1, \dots, n). \quad (2.134)$$

We assume that the system (2.134) has a periodic solution $X = \bar{X}(t, X^{(0)})$. Let us examine the problem of stability of this periodic solution. If we make in the system (2.134) the substitution

$$x_s = \xi_s + \bar{x}_s(t, X^{(0)}) \quad (s = 1, \dots, n),$$

we obtain a system of equations in the form (2.126), the right members of which are functions which are periodic in the argument t , and, in addition, the resultant system will have the trivial solution. The problem of the stability of this trivial solution is equivalent to the problem of the stability of the periodic solution of the system (2.134), and we can therefore apply Theorems 50, 51 and 52. However, the theorems formulated in Sections 12, 13 and 14 make it possible to obtain directly the conditions for stability, asymptotic stability, and instability of the periodic solution of the system (2.134) without reference to the system (2.126).

Let us apply Theorem 19 to the investigation of the problem of the asymptotic stability of the periodic solution of the system (2.134).

THEOREM 54. *In order for the open set A of the space E_n , containing a sufficiently small neighborhood of the closed curve $X = \bar{X}(t, X^{(0)})$, to be a region of asymptotic stability of the periodic solution $X = \bar{X}(t, X^{(0)})$ of the system (2.134), it is necessary and sufficient that there exist two functions $V(X)$ and $\varphi(X)$, satisfying the following conditions:*

- 1) *the function $V(X)$ is defined and continuous in A , the function $\varphi(X)$ is defined and is continuous in E_n ;*
- 2) *$-1 < V(X) < 0$ for $X \in A$; $\varphi(X) > 0$ for $X \in E_n$ and $X \neq \bar{X}(t, X^{(0)})$;*
- 3) *for any quantity $\gamma_2 > 0$ it is possible to find quantities γ_1 and α_1 such that*

$$V(X) < -\gamma_1 \text{ as } \inf_{t \in [0, \Omega]} |X - \bar{X}(t, X^{(0)})| \geq \gamma_2,$$

$$\varphi(X) > \alpha_1 \text{ for } \inf_{t \in [0, \Omega]} |X - \bar{X}(t, X^{(0)})| \geq \gamma_2;$$

- 4) *$\varphi \rightarrow 0$ and $V \rightarrow 0$ as $\inf_{t \in [0, \Omega]} |X - \bar{X}(t, X^{(0)})| \rightarrow 0$, where Ω is the period of the solution $\bar{X}(t, X^{(0)})$;*

- 5) *If $\bar{X}$ is a point on the boundary $\bar{A}$ and $X \neq \bar{X}(t, X^{(0)})$, then $\lim_{|X - \bar{X}| \rightarrow 0} V(X) = -1$;*

- 6) *the total derivative of the function V , calculated by virtue of the system (2.134), satisfies the relation*

$$\frac{dV}{dt} = \varphi(X)(1 + V(X)) \sqrt{1 + \sum_{i=1}^n f_i^2(X)}.$$

PROOF. In the system (2.134), let us make a change of independent variable in accordance with the formula $ds = dt \sqrt{1 + \sum_{i=1}^n f_i^2}$; then the system of differential equations

$$\left. \begin{aligned} \frac{dx_k}{ds} &= \frac{f_k(X)}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \\ \frac{dt}{ds} &= \frac{1}{\sqrt{1 + \sum_{i=1}^n f_i^2}} \end{aligned} \right\} \quad (2.135)$$

determines in the space E_n a dynamical system, for which the closed curve $\bar{X} = \bar{X}(t, X^{(0)})$ will be a periodic motion and will represent an invariant set. A certain neighborhood of this invariant set is compact, and therefore, on the basis of Theorem 6, it will follow from the asymptotic stability that this set will be uniformly asymptotically stable and uniformly attracting; therefore Theorem 19 is applicable, by which there will exist functions V and φ , satisfying Conditions 1–5 of Theorem 54 and the relation $dV/ds = \varphi$, where dV/ds is the total derivative of the function V , calculated by virtue of (2.135). Going over in this relation to the parameter t , we find that the function V satisfies the condition 6 of Theorem 54, which was to be proved.

REMARK. If the functions $f_i(X)$ in the right members of the system (2.135) are continuously differentiable in all their arguments up to order ν inclusive, then the function φ can be chosen such that the function V will be continuously differentiable in all its arguments up to order ν inclusive. If $\nu \geq 1$, Condition 6 of Theorem 54 can be written in the form

$$\sum_{i=1}^n \frac{\partial V}{\partial x_i} f_i(X) = \varphi(1 + V) \sqrt{1 + \sum_{i=1}^n f_i^2}.$$

Let us give an example to clarify Theorem 54. We consider the system of two equations

$$\begin{aligned}\dot{x} &= x + y - x(x^2 + y^2) = f_1(x, y), \\ \dot{y} &= -x + y - y(x^2 + y^2) = f_2(x, y).\end{aligned}$$

We set

$$\varphi(x, y) \sqrt{1 + f_1^2 + f_2^2} = 2(1 - x^2 - y^2)^2.$$

Let us form the partial differential equation corresponding to this system:

$$\frac{\partial V}{\partial x} f_1(x, y) + \frac{\partial V}{\partial y} f_2(x, y) = \varphi(x, y) (1 + V(x, y)) \sqrt{1 + f_1^2 + f_2^2}.$$

It is easy to see that the function V , satisfying this partial differential equation, has the form

$$V = (x^2 + y^2) e^{1-x^2-y^2} - 1$$

under the condition $V = 0$ for $x^2 + y^2 = 1$. The circle $x^2 + y^2 = 1$ is a periodic integral curve. The constructed functions V and φ satisfy all the conditions of Theorem 54, from which it follows that the region of asymptotic stability is in this case the entire plane, with the exception of the points $x = y = 0$ and those lying on the circle $x^2 + y^2 = 1$. In this case, the region of asymptotic stability is an open disconnected set.

We note that with the aid of the method indicated in Section 1 of this chapter, we can construct an entire set of systems of the form (2.126), having uniformly asymptotic stable trivial solution with a specified region A of asymptotic stability, and that it is also possible to construct the set of all systems of type (2.135), having a given closed curve as its periodic asymptotically stable solution with a prescribed region A of asymptotic stability.

In reference [30] there are also contained necessary and sufficient conditions for uniform asymptotic stability. However, an analysis of these conditions shows that when these are satisfied the trivial solution of the system of differential equations is also uniformly attracting. In addition, these conditions are of local character. Consequently, summarizing everything said above, it can be stated that these conditions are particular cases of those proposed in the present section.

CHAPTER III

INVESTIGATION OF A NEIGHBORHOOD OF THE TRIVIAL SOLUTION OF THE DIFFERENTIAL EQUATIONS WITH THE AID OF LYAPUNOV'S FIRST METHOD

In this chapter we consider the following two problems.

1) The problem of analytic representation of the solutions of systems of partial differential equations and systems of ordinary differential equations in a neighborhood of the singular initial data.

2) The problem of stability of the trivial solution of the system of ordinary differential equations in several doubtful cases, mentioned in Sections 4-5 of Chapter II.

This chapter is connected with the results of Briot and Bouquet [20], Poincaré [21], Picard [22], Horn [23], Lyapunov [1], and also with the work of Yerugin [24] and Shestakov [25], and develops them further. The basic idea of almost all the results are the works of A. M. Lyapunov.

1. *Auxiliary theorems from the theory of partial differential equations*

Let us consider the system of equations

$$\begin{aligned} \frac{\partial z_j}{\partial t} + \sum_{s=1}^n \frac{\partial z_j}{\partial x_s} \left(\sum_{i=1}^n p_{si}(t)x_i + X_s(t, x_1, \dots, x_n, z_1, \dots, z_k) \right) = \\ = \sum_{i=1}^k q_{ji}(t)z_i + \sum_{i=1}^n r_{ji}(t)x_i + \\ + Z_j(t, x_1, \dots, x_n, z_1, \dots, z_k) \quad (j = 1, \dots, k). \end{aligned} \quad (3.1)$$

The functions X_s and Z_j are expanded in series of positive integral powers of the quantities $x_1, \dots, x_n, z_1, \dots, z_k$, which converge for all $t \in [0, +\infty)$ for sufficiently small $|x_s|$ and $|z_j|$.

The coefficients in the expansions of the functions X_s and Z_j ,

and also the coefficients

$$p_{st}(t), q_{jt}(t), r_{jt}(t),$$

are defined for $t \in [0, +\infty)$, real, continuous, and bounded.

We shall assume furthermore that the expansions of the functions X_s and Z_j do not contain terms which are linear in the quantities

$$x_1, \dots, x_n, z_1, \dots, z_k.$$

We consider a system of ordinary differential equations, corresponding to the system (3.1)

$$\left. \begin{aligned} \frac{dx_s}{dt} &= \sum_{i=1}^n p_{si}(t)x_i + X_s \quad (s = 1, \dots, n) \\ \frac{dz_j}{dt} &= \sum_{i=1}^k q_{ji}(t)z_i + \sum_{i=1}^n r_{ji}(t)x_i + Z_j \quad (j = 1, \dots, k) \end{aligned} \right\}. \quad (3.2)$$

We denote by $\mu_1, \dots, \mu_{n+k}$ the eigenvalues of the linear system, which forms the first approximation for the system (3.2).

We shall assume that $\mu_1, \dots, \mu_n$ are the eigenvalues of the system

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si}(t)x_i \quad (s = 1, \dots, n), \quad (3.3)$$

and that $\mu_{n+1}, \dots, \mu_{n+k}$ are the eigenvalues of the system

$$\frac{dz_j}{dt} = \sum_{i=1}^k q_{ji}(t)z_i \quad (j = 1, \dots, k). \quad (3.4)$$

THEOREM 55. *If the following conditions are satisfied:*

- 1) $\mu_i > 0$, $i \leq n$;
- 2) $\mu_i = \mu_{n+i}$, $i \leq l$;
- 3) the systems (3.3) and (3.4) are regular;

then the system of Eqs. (3.1) has a family of solutions, which depends on l arbitrary constants, represented in the form of series

$$z_j = \sum_{m=1}^{\infty} z_j^{(m)}(t, c_1, \dots, c_l, x_1, \dots, x_n) \quad (j = 1, \dots, k), \quad (3.5)$$

which converge when

$$|c_s| < c_0, \quad |x_s| < x_0(t),$$

where $c_0 > 0$ is a constant, and $x_0(t)$ is a positive function, which generally speaking tends quite rapidly to zero as $t \rightarrow +\infty$.

The functions $z_j^{(m)}$ are homogeneous forms of degree m in the quantities $x_1, \dots, x_n$, the coefficients of which are continuous functions of t , defined for $t \geq 0$ with non-negative eigenvalues, and are simultaneously polynomials in $c_1, \dots, c_l$.

PROOF. For the system (3.2), following A. M. Lyapunov [1], we construct a family of solutions, corresponding to the eigenvalues μ_i , $i \leq n + l$:

$$x_s = \sum_{m_1 + \dots + m_{n+l} \geq 1} L_s^{(m_1, \dots, m_{n+l})}(t) e^{-t \sum_{i=1}^{n+l} \mu_i m_i} \alpha_1^{m_1} \dots \alpha_{n+l}^{m_{n+l}},$$

$$(s = 1, \dots, n), \quad (3.6)$$

$$z_j = \sum_{m_1 + \dots + m_{n+l} \geq 1} M_j^{(m_1, \dots, m_{n+l})}(t) e^{-t \sum_{i=1}^{n+l} \mu_i m_i} \alpha_1^{m_1} \dots \alpha_{n+l}^{m_{n+l}},$$

$$(j = 1, \dots, k). \quad (3.7)$$

The series (3.6) and (3.7) converge when $t \geq 0$, if $|\alpha_i| \leq \alpha$, $j = 1, \dots, n + l$ (α is a sufficiently small number), and the eigenvalues of the functions

$$L_s^{(m_1, \dots, m_{n+l})}(t) \text{ and } M_j^{(m_1, \dots, m_{n+l})}(t)$$

are non-negative. Let us set

$$\alpha_{n+i} = c_i \alpha_i, \quad i \leq l$$

and

$$\beta_s = e^{-\mu_s t} \cdot \alpha_s, \quad s \leq n. \quad (3.8)$$

It is easy to see that the Jacobian of the functions (3.6) with respect to the quantities (3.8) coincides at $\beta_s = 0$ and $s = 1, \dots, n$ with the determinant of the fundamental system of solutions for the system (3.3), provided the latter is multiplied by $e^{t(\mu_1 + \dots + \mu_n)}$:

$$\frac{D(x_1, \dots, x_n)}{D(\beta_1, \dots, \beta_n)} \bigg|_{\beta_1 = \dots = \beta_n = 0} = C e^{\int_0^t \sum_{s=1}^n (\mu_s(t) + \mu_s) dt}, \quad (3.9)$$

where $C \neq 0$.

Let us solve (3.6) with respect to the quantities (3.8), which is possible by virtue of (3.9). We have the following:

$$\beta_s = \varphi_s(t, c_1, \dots, c_l, x_1, \dots, x_n) \quad (s = 1, \dots, n). \quad (3.10)$$

Using relation (3.10), we eliminate β_s , $s \leq n$, from the system (3.7). As a result, we obtain a family of solutions (3.5) for the system (3.1).

REMARK 1. Let us consider the case when all the coefficients in the expansions of the functions X_s and Z_j and the quantities p_{st} , q_{jt} , and r_{jt} are constants.

The system (3.1) corresponding to this assumption will be denoted by (3.1'). We denote by $\lambda_1, \dots, \lambda_n$ the roots of the equation

$$|P - \lambda E| = 0, \quad \{P\}_{ik} = p_{ik}, \quad i, k = 1, \dots, n,$$

and by $\kappa_1, \dots, \kappa_k$ the roots of the equation

$$|Q - \kappa E| = 0, \quad \{Q\}_{ij} = q_{ij}, \quad i, j = 1, \dots, k.$$

It is clear that

$$\mu_i = -\operatorname{Re} \lambda_i, \quad i \leq n, \quad \mu_{n+i} = -\operatorname{Re} \kappa_i, \quad i \leq k.$$

If the conditions of Theorem 55 are satisfied for the system (3.1'), then the coefficients of the forms

$$z_j^{(m)}(t, c_1, \dots, c_l, x_1, \dots, x_n)$$

can be represented in the form of polynomials in $t, c_1, \dots, c_l$, the coefficients of which are in turn trigonometric polynomials. These trigonometric polynomials will be periodic functions in t , if the quantities $\lambda_i - \kappa_i$, $i \leq l$ are commensurate. Otherwise, they will be almost periodic functions.

If $\lambda_i - \kappa_i = 0$, $i \leq l$, then the coefficients of the forms $z_j^{(m)}$ are polynomials in $c_1, \dots, c_l, t$.

REMARK 2. The system (3.1') can be reduced, by means of a non-singular linear transformation on the functions $z_1, \dots, z_k$ sought and on the independent variables $x_1, \dots, x_n$, to the following form

$$\begin{aligned} \frac{\partial \bar{z}_j}{\partial t} + \sum_{s=1}^n \frac{\partial \bar{z}_j}{\partial \bar{x}_s} (\varepsilon_{s-1} \bar{x}_{s-1} + \lambda_s \bar{x}_s + \bar{X}_s) = \delta_{j-1} \bar{z}_{j-1} + \\ + \kappa_j \bar{z}_j + \sum_{i=1}^n \bar{r}_{ji} \bar{x}_i + \bar{Z}_j \quad (j = 1, \dots, k), \end{aligned} \quad (3.11)$$

where $\varepsilon_0 = \delta_0 = 0$, and ε_s, δ_j are either 0 or 1.

If $\operatorname{Re} \lambda_s < 0$, $s \leq n$, and $\kappa_i = \lambda_i$, $i \leq l$, then the system (3.11) has a family of solutions

$$\bar{z}_j = \sum_{m=1}^{\infty} \bar{z}_j^{(m)}(t, c_1, \dots, c_l, \bar{x}_1, \dots, \bar{x}_n) \quad (j = 1, \dots, k). \quad (3.12)$$

Let us put in (3.12) $t = (1/\lambda_1) \ln \bar{x}_1$ and let us again denote the resultant functions by σ_j , $j = 1, \dots, k$. It is clear that the functions

$$\sigma_j = \sum_{m=1}^{\infty} \bar{z}_j^{(m)} \left(\frac{\ln \bar{x}_1}{\lambda_1}, c_1, \dots, c_l, \bar{x}_1, \dots, \bar{x}_n \right) \quad (j = 1, \dots, k) \quad (3.13)$$

will satisfy the system of equations

$$\begin{aligned} \sum_{s=1}^n \frac{\partial \sigma_j}{\partial \bar{x}_s} (\varepsilon_{s-1} \bar{x}_{s-1} + \lambda_s \bar{x}_s + \bar{X}_s(\bar{x}_1, \dots, \bar{x}_n, \sigma_1, \dots, \sigma_k)) = \\ = \delta_{j-1} \sigma_{j-1} + \kappa_j \sigma_j + \sum_{i=1}^n \bar{r}_{ji} \bar{x}_i + \bar{Z}_j(\bar{x}_1, \dots, \bar{x}_n, \sigma_1, \dots, \sigma_k). \end{aligned} \quad (3.14)$$

THEOREM 56. *If*

- 1) $\operatorname{Re} \lambda_i < 0$, $j \leq n$,
- 2) $\kappa_i = \lambda_i$, $i \leq l$,

then the system of equations (3.14) and (3.11) has a family of solutions (3.13), which depends on l arbitrary constants. With this, the coefficients of the forms $\bar{z}_j^{(m)}$ are polynomials in $c_1, \dots, c_l, \ln \bar{x}_1$.

The series (3.13) converge when

$$|c_j| < c_0, \quad |\bar{x}_s| < x_0 \left(\left| \frac{\ln |\bar{x}_1|}{\lambda_1} \right| \right), \quad j \leq l, \quad s \geq 2,$$

x_0 is the function referred to in Theorem 55, and $|\bar{x}_1|$ is sufficiently small.

Let us indicate further a case when the functions (3.13) are independent of $\ln \bar{x}_1$.

THEOREM 57. *If*

- 1) $\operatorname{Re} \lambda_i < 0$, $i \leq n$,
- 2) $\lambda_i = \kappa_i$, $i \leq l$,
- 3) $\varepsilon_i = \delta_i$, $i \leq l-1$,
- 4) $\varepsilon_l = 0$, $\bar{r}_{ij} = 0$, $i, j \leq l$,
- 5) *there do not exist relations of the form*

$$\sum_{i=1}^n m_i \lambda_i = \kappa_j, \quad j \leq k$$

for any non-negative integral

$$m_i, \sum_{i=1}^n m_i \geq 1$$

with the exception of the cases

$$\lambda_i = \kappa_i, \quad i \leq l,$$

then the system (3.1'), (3.11), (3.14) has a family of holomorphic solutions, which depend on l arbitrary constants, representable in the form of the series

$$z_j = \sum_{m=1}^{\infty} \tilde{z}_j^{(m)}(c_1, \dots, c_l, x_1, \dots, x_n) \quad (j = 1, \dots, k), \quad (3.15)$$

where $\tilde{z}_j^{(m)}$ are homogeneous forms of degree m in $\tilde{x}_1, \dots, \tilde{x}_n$, the coefficients of which are polynomials in $c_1, \dots, c_l$.

The series (3.15) converge when

$$|c_s| \leq c_0, \quad |x_s| \leq r, \quad r > 0, \quad c_0 > 0.$$

PROOF. Let us carry out the proof for the system (3.14). For this purpose, we substitute the series

$$\sigma_j = \sum_{m=1}^{\infty} \sigma_j^{(m)}(\tilde{x}_1, \dots, \tilde{x}_n), \quad j = 1, \dots, k, \quad (3.16)$$

into (3.14), where $\sigma_j^{(m)}$ are homogeneous forms of degree m in $\tilde{x}_1, \dots, \tilde{x}_n$, with coefficients that are to be determined.

Equating forms of equal degree, we obtain a system of equations for determining $\sigma_j^{(m)}$:

$$\sum_{s=1}^n \frac{\partial \sigma_j^{(m)}}{\partial \tilde{x}_s} (\varepsilon_{s-1} \tilde{x}_{s-1} + \lambda_s \tilde{x}_s) = \delta_{j-1} \sigma_{j-1}^{(m)} + \kappa_j \sigma_j^{(m)} + R_j^{(m)}. \quad (3.17)$$

The functions $R_j^{(m)}$ are homogeneous forms in degree m with respect to $\tilde{x}_1, \dots, \tilde{x}_n$, which depend on the expansions of the functions $\bar{X}_s$ and $\bar{Z}_j$ and on the forms $\sigma_j^{(\mu)}$, $j = 1, \dots, k$, $\mu = 1, \dots, m-1$.

If the function $R_j^{(m)}$ is defined at $m > 1$, it follows from Condition 5 of the present theorem that the system (3.17) has [1] a unique solution in the form of the form

$$\sigma_j^{(m)}(x_1, \dots, x_n) \quad j = 1, \dots, k.$$

Let us show that when $m = 1$ the system (3.17) has a solution in the form of a system of linear forms $\sigma_j^{(1)}$, which depend on l arbitrary constants.

$$X(t) = \begin{pmatrix} x_1 & 0 & 0 & \dots & 0 \\ x_2 & x_1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{k_1} & x_{k_1-1} & x_{k_1-2} & \dots & x_1 \\ & & & & \vdots \\ & & & & \vdots \\ & & & & x_{k_{\alpha}+1} & 0 & \dots & 0 \\ & & & & x_{k_{\alpha}+2} & x_{k_{\alpha}+1} & \dots & 0 \\ & & 0 & & \vdots & \vdots & \ddots & \vdots \\ & & & & x_l & x_{l-1} & \dots & x_{k_{\alpha}+1} \end{pmatrix}.$$

It is clear that the vector function

$$Y(t) = X(t)C, \quad (3.22)$$

where

$$C = \begin{pmatrix} c_1 \\ \vdots \\ c_l \end{pmatrix}, \quad Y = \begin{pmatrix} y_1^{(1)} \\ \vdots \\ y_l^{(1)} \end{pmatrix},$$

gives the general solution of the system (3.21), and at the same time the functions $y_j^{(1)}$, ($j = 1, \dots, l$), considered as linear forms in the quantities $\bar{x}_i$ ($i = 1, \dots, l$), are the solution of the system (3.19). Thus, the linear forms $\sigma_j^{(1)}$ for $j = 1, \dots, l$ have been determined.

The forms $\sigma_j^{(1)}$ for $j \geq 1 + l$ are determined uniquely from the system (3.17) for $m = 1$.

This determines the system of formal series (3.16) completely.

The convergence of the series (3.16) follows from Theorem 55.

2. Representation of solutions of systems of ordinary differential equations in a neighborhood of a singular point

Let us consider the system

$$z \frac{dy_s}{dz} = \sum_{i=1}^n p_{si}(z)y_i + p_s(z)z + Y_s(z, y_1, \dots, y_n) \quad (3.23)$$

$$(s = 1, \dots, n).$$

The functions Y_s are expanded in series

$$Y_s = \sum_{m_1 + \dots + m_n \geq 2} P_s^{(m_1, \dots, m_n)}(z) z^m y_1^{m_1} \dots y_n^{m_n} \quad (3.24)$$

$$(s = 1, \dots, n),$$

which converge when $|z| < z_1$, where $z_1 > 0$ is a constant and $|y_j| < y_0$ ($j = 1, \dots, n$).

The functions

$$p_{si}(z), p_s(z), P_s^{(m_1, \dots, m_n)}(z)$$

are defined for $z \in (0, 1]$, real, continuous, and bounded.

At the point $y_1 = \dots = y_n = 0$, $z = 0$, the system (3.23) does not satisfy the condition for the existence of a solution. [Generally

speaking, the conditions of the Peano theorem are not satisfied.] The question therefore arises, under what conditions does a solution of the system (3.23) $y_s = y_s(z)$ ($s = 1, \dots, n$) exist, such that $y_s \rightarrow 0$ as $z \rightarrow 0$. These conditions will be derived later on and an analytic representation of such solutions will be given.

We denote by $1, \mu_1, \dots, \mu_n$ the eigenvalues of the system

$$\left. \begin{aligned} \frac{dz}{dt} &= -z \\ \frac{dy_s}{dt} &= - \left[\sum_{i=1}^n p_{si}(e^{-t})y_i + p_s(e^{-t})z \right] \end{aligned} \right\} \quad (s = 1, \dots, n). \quad (3.25)$$

THEOREM 58. *If*

- 1) $\mu_i > 0$ for $i \leq l$;
- 2) the system (3.25) is regular;

then the system of Eqs. (3.23) has a family of solutions, which depends on l arbitrary constants, represented in the form of series

$$y_s = \sum_{m+m_1+\dots+m_l \geq 1} K_s^{(m, m_1, \dots, m_l)}(z) z^{m+m_1\mu_1+\dots+m_l\mu_l} \cdot c_1^{m_1} \dots c_l^{m_l} \quad (s = 1, \dots, n). \quad (3.26)$$

This series converges for $|z| \leq z_0$, $|c_i| \leq c_0$ ($i = 1, \dots, l$) and $z_0 < \beta$, where β is sufficiently small constant and c_0, z_0 are constants greater than 0.

The functions $K_s^{(m, m_1, \dots, m_l)}(z)$ have the property

$$K_s^{(m, m_1, \dots, m_l)}(z) z^\alpha \rightarrow 0$$

as $z \rightarrow 0$, where $\alpha > 0$ is constant.

PROOF. Let us consider the system of equations

$$\begin{aligned} - \sum_{j=1}^l \frac{\partial y_s}{\partial x_j} \mu_j x_j - z \frac{\partial y_s}{\partial z} + \frac{\partial y_s}{\partial t} &= \\ &= - \sum_{i=1}^n p_{si}(e^{-t})y_i - p_s(e^{-t})z - Y_s. \end{aligned} \quad (3.27)$$

For the system (3.27), the conditions of the Theorem 55 are

satisfied. Therefore there exists a system of functions

$$\begin{aligned} y_s(t, c_1, \dots, c_l, x_1, \dots, x_l, z) = \\ = \sum_{m=1}^{+\infty} y_s^{(m)}(t, c_1, \dots, c_l, x_1, \dots, x_l, z) \quad (s = 1, \dots, n), \end{aligned} \quad (3.28)$$

which satisfies the system (3.27).

It is clear that the forms $y_s^{(m)}$ have the form

$$\begin{aligned} y_s^{(m)} = \sum_{m+m_1+\dots+m_l=m} \bar{K}_s^{(m, m_1, \dots, m_l)} z^m x_1^{m_1} \dots x_l^{m_l} c_1^{m_1} \dots c_l^{m_l} \\ (s = 1, \dots, n), \end{aligned} \quad (3.29)$$

where $m \geq 1$.

In (3.28), let us set

$$x_j = z^{\mu_j}, \quad t = -\ln z. \quad (3.30)$$

We then obtain a family of solutions (3.26), satisfying the system (3.23), as can be verified by direct substitution.

REMARK 1. Let the functions (3.24) be real constants. In this case the system (3.23) will be denoted by (3.23').

We denote by $\lambda_1, \dots, \lambda_n$, as before, the roots of the equation

$$|P - \lambda E| = 0 \quad \{P\}_{ik} = p_{ik}.$$

It is easy to see that $\mu_i = \operatorname{Re} \lambda_i$ for the system (3.23'). Thus, when $\operatorname{Re} \lambda_i > 0$, $i \leq l$, the system (3.23') has the family of solutions (3.26). Here, the functions $\bar{K}_s^{(m, m_1, \dots, m_l)}(z)$ are polynomials in $\ln z$, the coefficients of which are trigonometric polynomials.

REMARK 2. We introduce into the system (3.23') the matrix P in the canonical form. We then obtain

$$z \frac{d\bar{y}_s}{dz} = \varepsilon_{s-1} \bar{y}_{s-1} + \lambda_s \bar{y}_s + \bar{p}_s z + \bar{Y}_s \quad (s = 1, \dots, n), \quad (3.31)$$

where $\varepsilon_0 = 0$.

We consider a system of partial differential equations corresponding to (3.31):

$$\begin{aligned} - \sum_{j=1} \frac{\partial \bar{y}_s}{\partial x_j} [\varepsilon_{j-1} x_{j-1} + \lambda_j x_j] - z \frac{\partial \bar{y}_s}{\partial z} = \\ = - (\varepsilon_{s-1} \bar{y}_{s-1} + \lambda_s \bar{y}_s + \bar{Y}_s + \bar{p}_s z). \end{aligned} \quad (3.32)$$

By Theorem 56, this system has a solution in the form of convergent series

$$\bar{y}_s = \sum_{m+m_1+\dots+m_l=1}^{\infty} \tilde{K}_s^{(m, m_1, \dots, m_l)}(z, c_1, \dots, c_l) z^m x_1^{m_1} \dots x_l^{m_l} \quad (3.33)$$

$$s = 1, \dots, n,$$

where the $\tilde{K}_s^{(m, m_1, \dots, m_l)}$ are polynomials in powers of $\ln z$.

If $m + \sum_{i=1}^l m_i \lambda_i \neq \lambda_j$, $j = 1, \dots, l$ for any non-negative integral $m, m_1, \dots, m_l$ such that $m + m_1 + \dots + m_l \geq 2$, then the quantities $\tilde{K}_s^{(m, m_1, \dots, m_l)}$ are constants. We set

$$x_j = \frac{(\ln z)^{l_j \lambda_j}}{l_j!}, \quad (3.34)$$

where the constants l_j are chosen such that the functions (3.34) satisfy the system

$$z \frac{dx_j}{dz} = \varepsilon_j - 1 x_{j-1} + \lambda_j x_j.$$

If the functions (3.34) are substituted into the series (3.33), we obtain a family of solutions of the system (3.31).

THEOREM 59. *If among the eigenvalues of the matrix P in the system (3.23') there are l with positive real parts, then there exists a family of solutions of the system (3.23'), representable in the form of series*

$$y_s = \sum_{m+m_1+\dots+m_l \geq 1} N_s^{(m, m_1, \dots, m_l)} (\ln z) z^{m + \sum_{i=1}^l m_i \lambda_i} (\ln z)^{\sum_{i=1}^l m_i \lambda_i} \times \\ \times c_1^{m_1} \dots c_l^{m_l} \quad (s = 1, \dots, n), \quad (3.35)$$

which converge when

$$|z| \leq z_0; \quad |c_j| \leq c_0 \quad (j = 1, \dots, l); \quad z_0 < \beta,$$

where the $N_s^{(m, m_1, \dots, m_l)}(\ln z)$ are polynomials in $\ln z$.

On the other hand, if $m + \sum_{i=1}^l m_i \lambda_i \neq \lambda_j$, $j = 1, \dots, l$ for $m + \sum_{i=1}^l m_i \geq 2$, and $\lambda_j \neq 1$ at $j \leq l$, where l is the number of all the quantities λ_i such that $\operatorname{Re} \lambda_i > 0$, then the quantities $N_s^{(m, m_1, \dots, m_l)}$ are constants.

Let us examine the system

$$z^N \frac{dy_s}{dz} = \sum_{i=1}^n p_{si} y_i + p_s z + Y_s(z, y_1, \dots, y_n), \quad (3.36)$$

$$(s = 1, \dots, n).$$

We assume further that the right members of the system (3.36) are the same as in the system (3.23').

Let $z_1, z_2, \dots, z_k, \dots$ be a sequence such that $|z_j| \rightarrow 0$ as $j \rightarrow +\infty$, and $|\arg z_j| \leq c < +\infty$.

The question arises, does there exist a solution $y_s = y_s(z)$, $s = 1, \dots, n$, of the system (3.36), having the property

$$|y_s(z_j)| \rightarrow 0 \text{ as } j \rightarrow +\infty. \quad (3.37)$$

If such a solution exists, we raise the question of finding the solution itself and of its analytical representation for $N \neq 1$.

I. Let $N < 1$, $N = p/q$ where p, q are natural numbers.

THEOREM 60. *The system (3.36), in the case $N = p/q$, has a unique solution, which has the property (3.37), which is representable in the form of a series*

$$y_s(z) = \sum_{k=1}^{\infty} a_s^{(k)} z^{k/q}, \quad (s = 1, \dots, n). \quad (3.38)$$

PROOF. In the system (3.36) we set $z = \xi^q$. We then have

$$\xi^{p-q+1} \frac{dy_s}{d\xi} = f_s(\xi^q, y_1, \dots, y_n), \quad (s = 1, \dots, n);$$

hence

$$\frac{dy_s}{d\xi} = \xi^{q-p-1} f_s(\xi^q, y_1, \dots, y_n). \quad (3.39)$$

Applying the Cauchy theorem to Eq. (3.39), we obtain the required proof.

II. Let $N > 1$.

We consider the case when $p_s = 0$, $\partial Y_s / \partial z \equiv 0$ ($s = 1, \dots, n$).

We choose a certain function ψ , satisfying the equation

$$z^N d\psi - \psi dz = 0. \quad (3.40)$$

THEOREM 61. *If in the system (3.36) among the eigenvalues of the matrix P , $\{P\}_{ij} = p_{ij}$, there are l eigenvalues with positive real parts, namely $\lambda_1, \dots, \lambda_l$, then there exists a solution in the form of series*

$$\gamma_s(z) = \sum_{m+m_1+\dots+m_l \geq 1} G_s^{(m, m_1, \dots, m_l)} (\ln \psi) \psi^{m + \sum_{i=1}^l m_i \lambda_i} \times \\ \times c_1^{m_1} \dots c_l^{m_l} \quad (s = 1, \dots, n), \quad (3.41)$$

which converge for $|c_i| \leq c_0$, $|\psi| \leq \psi_0$, where c_0 and ψ_0 are certain positive constants.

The proof follows from Theorem 59.

The region $|\psi(z)| \leq \psi_0$ in the plane of the complex variable Z does not contain, generally speaking, any circle with center at the origin. Therefore the series (3.41) satisfy conditions (3.37) only for certain specially selected sequences

$$z_1, z_2, \dots, z_k, \dots; \quad |\psi(z_i)| < \psi_0.$$

Let us solve the problem now for a particular case.

Let $\operatorname{Re} \lambda_i \leq 0$, $i = 1, \dots, n$; then the method given above for constructing solutions that have the property of interest to us is no longer suitable. Even when $\operatorname{Re} \lambda_i > 0$ for $i = 1, \dots, l$, it may turn out that the solution with the necessary initial data does not occur in the family of solutions constructed in the form of series.

THEOREM 62. *If the system (3.36) has for $N = 1$ a solution $\gamma_s = \gamma_s(z)$ ($s = 1, \dots, n$) having the properties*

$$\gamma_s(z) \rightarrow 0 \text{ as } z \rightarrow 0, z > 0, \gamma_s(z_0) = \gamma_s^{(0)}, z_0 > 0, \quad (3.42)$$

then the functions $\gamma_s(z)$ are analytic in the circle $|z| < \eta(z_0)$ with a cut along the sector, containing a section of the negative real half-axis $-\eta(z_0) < z < 0$, where $\eta(z_0)$ is a sufficiently small positive quantity.

The proof can be obtained by modifying somewhat the proof of Theorem 25 [12].

The preceding theorem implies the following statement.

THEOREM 63. *If for $N = 1$, the system (3.36) has a solution $\gamma_s = \gamma_s(z)$ having the property (3.42), then it can be constructed in the form of series*

$$\gamma_s(z) = \sum_{m=0}^{\infty} \Psi_s^{(m)}(z_0, \gamma_1^{(0)}, \dots, \gamma_n^{(0)}) V^m(z) \quad (s = 1, \dots, n),$$

which converge for $|V(z)| < 1$, where $\Psi_s^{(m)}$ are certain known functions and $V(z)$ is a function which maps the circle $|z| < \eta(z_0)$ with the cut indicated above on the circle $|V| < 1$.

3. Analytic representation of the O -curves of a special system of differential equations

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = \sum_{m=\mu}^{+\infty} X_s^{(m)}(t, x_1, \dots, x_n) \quad (s = 1, \dots, n), \quad (3.43)$$

where the functions $X_s^{(m)}$ are homogeneous forms of degree m in $x_1, \dots, x_n$ with real, continuous and bounded coefficients defined for $t \in (-\infty, +\infty)$. [The integral curve $X(t, X^{(0)})$ of the system (3.43) is called an O -curve, if $X(t, X^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$ ($t \rightarrow -\infty$).] We shall assume that the series in the right members of the system (3.43) converge for sufficiently small $|x_s|$ and for all $-\infty < t < +\infty$, while the coefficients of the forms $X_s^{(\mu)}$ are real constants.

Let us consider the system of equations

$$\frac{dx_s}{dt} = X_s^{(\mu)} \quad (s = 1, \dots, n).$$

The right members of this system are homogeneous forms of order μ . Consequently, the entire theory developed in Section 3 of Chapter II is applicable to this system. Let us recall that the trivial solution of this system can be asymptotically stable for $\mu = 2k + 1$ only for real perturbations, and when $\mu = 2k$ the trivial solution of this system cannot be asymptotically stable. In Section 3 of Chapter II we gave the necessary and sufficient conditions for asymptotic stability of the trivial solution of this system with the aid of the Lyapunov function. In the present section we derive the conditions for asymptotic stability, connected with certain algebraic properties of the forms $X_s^{(\mu)}$, and also the conditions for instability, which we shall use to investigate the doubtful cases, analyzed in Secs. 4–5 of Chapter II.

Let us examine the system

$$\frac{dx_s}{d\varphi} = x_s + aX_s^{(\mu)}, \quad s = 1, \dots, k, \quad (3.44)$$

where $a = +1$ or $a = -1$.

We denote by

$$x_s = x_s(\varphi, a) \quad (s = 1, \dots, n) \quad (3.45)$$

the solution of the system (3.44), which is defined when $\varphi \in (-\infty, +\infty)$. We denote by $\mu_1, \dots, \mu_n$ the eigenvalues of the system

$$\frac{dx_s}{d\varphi} = \sum_{i=1}^n p_{si}(\varphi, a)x_i \quad s = 1, \dots, n, \quad (3.46)$$

where

$$p_{st}(\varphi, a) = \delta_{st} + a \frac{\partial X_s^{(\mu)}}{\partial x_t} \bigg|_{\substack{x_1 = x_1(\varphi, a) \\ \dots \\ x_n = x_n(\varphi, a)}}$$

Let $1, \mu_1, \dots, \mu_n$ be the eigenvalues of the system

$$\left. \begin{aligned} \frac{dz}{d\varphi} &= -z \\ \frac{dx_s}{d\varphi} &= \sum_{i=1}^n p_{si}(\varphi, a)x_i + zp_s(\varphi, a) \quad s = 1, \dots, n \end{aligned} \right\}. \quad (3.47)$$

The form of the function $p_s(\varphi, a)$ will be given later.

THEOREM 64. *If*

1) *the system (3.44) has a solution (3.45) which is bounded for* $\varphi \in (-\infty, +\infty)$;

2) *the system (3.46) is regular;*

3) *among the numbers $\mu_1, \dots, \mu_n$ there are l positive ones: $\mu_i > 0$ when $i \leq l$;*

then the system (3.43) has two families of solutions, each of which depends on l arbitrary constants and can be represented in the form of series

$$\begin{aligned} x_s^{(j)} &= z(at)[(-1)^{j+1}x_s(\varphi, a) + \sum_{m+m_1+\dots+m_l \geq 1} F_s^{(j, m, m_1, \dots, m_l)}(z) \times \\ &\times c_{1j}^{m_1} \dots c_{lj}^{m_l} z^{m+m_1\mu_1+\dots+m_l\mu_l}], \quad s = 1, \dots, n, \quad j = 1, 2, \end{aligned} \quad (3.48)$$

which converge when

$$|z| \leq z_0; |c_{ij}| \leq c_0, z_0 < \beta,$$

where $z(at)$ is a certain positive function, satisfying the equation

$$\frac{dz}{dt} = -az^\mu, \quad (3.49)$$

$$\varphi = -\ln z. \quad (3.50)$$

The functions $F_s^{(j,m,m_1,\dots,m_l)}(z)$ are such that $z^\alpha F_s^{(j,m,m_1,\dots,m_l)} \rightarrow 0$ as $z \rightarrow 0$, for any $\alpha > 0$.

PROOF. With the aid of Eq. (3.49) we make a change of independent variable in the system (3.43), and then change the unknown functions in accordance with the formula

$$\begin{aligned} x_s^{(j)} &= z(at)[x_s(\varphi, a)(-1)^{j+1} + y_s^{(j)}], \\ s &= 1, \dots, n; j = 1, 2, \end{aligned} \quad (3.51)$$

where $y_s^{(j)}$ are new unknown functions.

For the determination of $y_s^{(j)}$ we obtain a system

$$\begin{aligned} z \frac{dy_s^{(j)}}{dz} &= - \left[\sum_{i=1}^n p_{si}(\varphi, a)y_i + p_s(\varphi, a)z \right] + Y_s^{(j)}, \\ s &= 1, \dots, n; j = 1, 2. \end{aligned} \quad (3.52)$$

We can apply Theorem 58 to the system (3.52).

As a result we obtain two families of solutions, each of which depends on l arbitrary constants and can be represented in the form of series, which, thanks to the substitution (3.51), determine the family (3.48). This completes the proof of the theorem.

REMARK 1. It follows from (3.49) that

$$z(at) = z_0(1 + 2katz_0^{2k})^{-(1/2k)}.$$

Therefore, for a sufficiently small value of $z_0 > 0$, the series (3.48) will converge for all $at \in [0, +\infty)$.

Thus, any integral curve which occurs in the family (3.48) will be an O -curve, and moreover it will be a O^+ -curve for $a = 1$ and a O^- -curve for $a = -1$. In the latter case instability takes place.

REMARK 2. Let the functions (3.45) be constant:

$$x_s(\varphi, a) = a_s, \quad s = 1, \dots, n.$$

Then any integral curve which occurs in the family (3.48) has the property

$$\frac{x_s^{(j)}(t)}{z(at)} \rightarrow a_s(-1)^{j+1} \text{ as } at \rightarrow +\infty. \quad (3.53)$$

In this case, the quantities $p_{st}(\varphi, a) = p_{st}(a, a_1, \dots, a_n)$ are constants.

Let us proceed to a more detailed analysis of a neighborhood of the equilibrium position of the system (3.43).

We examine the system of algebraic equations

$$x_s + X_s^{(\mu)}(x_1, \dots, x_n) = 0 \quad (s = 1, \dots, n), \mu = 2k + 1, \quad (3.54)$$

where k is a positive integer.

Let us examine the matrix $P(x_1, \dots, x_n)$:

$$P(x_1, \dots, x_n) = \left\| -\frac{\partial X_s^{(\mu)}}{\partial x_i} - \delta_{st} \right\| \quad (i, s = 1, \dots, n).$$

The eigenvalues of this matrix will be denoted by $\lambda_i(x_1, \dots, x_n)$. If the system (3.54) has a solution $x_s = a_s$ ($s = 1, \dots, n$), then it also has the solution $x_s = -a_s$ ($s = 1, \dots, n$). In this connection, $\lambda_i(a_1, \dots, a_n) = \lambda_i(-a_1, \dots, -a_n)$. In fact, the system (3.54) does not change when x_s is replaced by $-x_s$. Therefore, along with $x_s = a_s$ it has a solution $x_s = -a_s$. The functions $\partial X_s^{(\mu)} / \partial x_i$ are forms of degree $2k$ in $x_1, \dots, x_n$, and therefore the elements of the matrix P do not change when x_s is replaced by $-x_s$, from which it follows that

$$\lambda_i(a_1, \dots, a_n) = \lambda_i(-a_1, \dots, -a_n) \quad (i = 1, \dots, n).$$

THEOREM 65. If the system (3.54) has a real solution $x_s = a_s$ ($s = 1, \dots, n$), and if among the quantities $\lambda_i(a_1, \dots, a_n)$, ($i = 1, \dots, n$) there are l with positive real parts: $\text{Re } \lambda_i > 0$ when $i \leq l$, then there exist two families of solutions of the system (3.43), each of which depends on l arbitrary constants and is represented in

the form of series

$$x_s^{(1)} = z(a_s + \sum_{M+M_1+\dots+M_n \geq 1} K_s^{(M, M_1, \dots, M_n)} (\ln z) \times \\ \times c_1^{M_1} \dots c_n^{M_n} z^{M + \sum_{i=1}^n M_i \lambda_i}) \quad (s = 1, \dots, n) \quad (3.55)$$

$$x_s^{(2)} = z(-a_s + \sum_{M+M_1+\dots+M_n \geq 1} L_s^{(M, M_1, \dots, M_n)} (\ln z) \times \\ \times \gamma_1^{M_1} \dots \gamma_n^{M_n} z^{M + \sum_{i=1}^n M_i \lambda_i}), \quad (3.56)$$

which converge when $|z| \leq z_0$, $|c_i| \leq c_0$, $|\gamma_i| \leq \gamma_0$, where the functions

$$K_s^{(M, M_1, \dots, M_n)} \text{ and } L_s^{(M, M_1, \dots, M_n)} \quad (s = 1, \dots, n)$$

are generally speaking polynomials in $\ln z$, if the coefficients of the forms $X_s^{(M)}$, $M \geq \mu$ are constant. The quantity z is a function of the argument t , satisfying the equation

$$\frac{dz}{dt} = -z^\mu; \quad (3.57)$$

c_i and γ_i are arbitrary constants, while c_0 , γ_0 , and z_0 are greater than 0.

The proof follows from Theorem 64, for Theorem 65 is a particular case of Theorem 64.

THEOREM 66. *If system (3.54) has a real solution $x_s = a_s$ ($s = 1, \dots, n$), then among the quantities λ_i ($i = 1, \dots, n$) there is at least one with a positive real part.*

PROOF. Consider the system of linear forms

$$y_s = \sum_{i=1}^n p_{si}(a_1, \dots, a_n) x_i \quad (s = 1, \dots, n).$$

We set $x_s = a_s$; then

$$y_s = \sum_{i=1}^n - \left(\frac{\partial X_s^{(\mu)}(a_1, \dots, a_n)}{\partial x_i} + \delta_{si} \right) a_i = \\ = -a_s - \sum_{i=1}^n \frac{\partial X_s^{(\mu)}(a_1, \dots, a_n)}{\partial x_i} a_i.$$

By Euler's theorem,

$$\sum_{i=1}^n \frac{\partial X_s^{(\mu)}}{\partial x_i} (a_1, \dots, a_n) a_i = \mu X_s^{(\mu)}(a_1, \dots, a_n).$$

From (3.54) we get $X_s^{(\mu)} = -a_s$. Comparing the last equations, we obtain

$$y_s = 2ka_s,$$

from which it follows that the vector $(a_1, \dots, a_n)$ is the eigenvector of the matrix P , and $2k$ is the eigenvalue. Setting $\lambda_1 = 2k$, we prove the theorem.

COROLLARY. *If the system (3.54) has a real solution $x_s = a_s$, then there exist two families of integral curves of the system (3.43) each of which depends on at least one arbitrary constant and can be represented in the form of series*

$$x_s^{(1)} = z(a_s + \sum_{M+M_1 \geq 1} K_s^{(M, M_1)} (\ln z) c^{M_1} z^{M+2kM_1}) \quad (3.58)$$

$$x_s^{(2)} = z(-a_s + \sum_{M+M_1 \geq 1} L_s^{(M, M_1)} (\ln z) \gamma^{M_1} z^{M+2kM_1}), \quad (3.59)$$

which converge when $|c| \leq c_0$, $|\gamma| \leq \gamma_0$, $t \in [0, +\infty)$. Each curve of the families (3.58) and (3.59) has the property $|x_s^{(i)}| \rightarrow 0$ as $t \rightarrow +\infty$, $i = 1, 2$.

REMARK. In the case considered above, any integral curve of the family (3.55) enters in the point $X = 0$ as $t \rightarrow +\infty$ and is tangent to the ray $x_s = \tau a_s$, ($s = 1, \dots, n$, $\tau > 0$). Any curve of the family (3.56) enters in the point $X = 0$ as $t \rightarrow +\infty$, and is tangent to the ray $x_s = -\tau a_s$, ($s = 1, \dots, n$, $\tau > 0$).

Let us consider further a system of algebraic equations

$$x_s = X_s^{(\mu)}. \quad (3.60)$$

We introduce the matrix

$$Q = \left\| \frac{\partial X_s^{(\mu)}}{\partial x_i} - \delta_{si} \right\|.$$

The eigenvalues of Q will be denoted by $\mu_i(x_1, \dots, x_n)$, ($i = 1, \dots, n$). If the system (3.60) has the solution

$$x_s = b_s \quad (s = 1, \dots, n),$$

then it also has the solution $x_s = -b_s$. With this

$$\mu_i(b_1, \dots, b_n) = \mu_i(-b_1, \dots, -b_n).$$

THEOREM 67. *If the system (3.60) has a real solution $x_s = b_s$ and among the quantities $\mu_i(b_1, \dots, b_n)$ there are l with positive real part: $\operatorname{Re} \mu_i > 0$ when $i \leq l$, then the system (3.43) has two families of solutions, each of which depends on l arbitrary constants and can be represented in the form of a series*

$$x_s^{(1)} = y(b_s + \sum_{M+M_1+\dots+M_l \geq 1} \bar{K}_s^{(M, M_1, \dots, M_l)} (\ln y) \times \\ \times c_1^{M_1} \dots c_l^{M_l} y^{M + \sum_{i=1}^l M_i \mu_i}) \quad (3.61)$$

$$x_s^{(2)} = y(-b_s + \sum_{M+M_1+\dots+M_l \geq 1} \bar{L}_s^{(M, M_1, \dots, M_l)} (\ln y) \times \\ \times \gamma_1^{M_1} \dots \gamma_l^{M_l} y^{M + \sum_{i=1}^l M_i \mu_i}), \quad (3.62)$$

which converge when $|y| \leq y_0$, $|c_i| \leq c_0$, $|\gamma_i| \leq \gamma_0$. The quantities $\bar{K}_s^{(M, \dots, M_l)}$, $\bar{L}_s^{(M, \dots, M_l)}$, are, generally speaking, polynomials in $\ln y$, where y is the positive function of the argument t , which is the solution of the equation

$$\frac{dy}{dt} = y^\mu. \quad (3.63)$$

The quantities c_0 , γ_0 , and y_0 are positive constants.

PROOF. Let us replace t in the system (3.43) by $-t$; then the proof of this theorem reduces to the proof of Theorem 65.

COROLLARY. The general solution of Eq. (3.63) has the form

$$y = \frac{\bar{y}}{\sqrt[2k]{1 - 2tk\bar{y}^{2k}}},$$

from which it follows that the positive number $\bar{y}$ can be chosen so small that $|y(t)| \leq y_0$ when $t < 0$. Thus, if the system (3.60) has a real solution and among the quantities μ_i there are l quantities with positive real parts, $\operatorname{Re} \mu_i > 0$, $i \leq l$, then the system (3.43) has two families of solutions, represented in the form of series (3.61) and (3.62), which converge when $|c_i| \leq c_0$, $|\gamma_i| \leq \gamma_0$, $t \in [0, -\infty)$.

Any integral curve of the family (3.61) has the property

$$|x_s^{(1)}| \rightarrow 0 \quad (s = 1, \dots, n) \text{ as } t \rightarrow -\infty.$$

Moreover, it enters the point $X=0$, tangent to the ray $x_s = \tau b_s$ ($s = 1, \dots, n$). Any integral curve of the family (3.62) has the property $|x_s^{(2)}| \rightarrow 0$ ($s = 1, \dots, n$) as $t \rightarrow -\infty$. With this, it enters the point $X=0$ tangent to the ray $x_s = -\tau b_s$, $s = 1, \dots, n$. If the system (3.60) has a real solution $x_s = b_s$ ($s = 1, \dots, n$), then, as before (see Theorem 66), we can state that among the quantities $\mu_i(b_1, \dots, b_n)$ there is at least one with a positive real part, $\mu_1 = 2k$.

THEOREM 68. *If the algebraic system (3.60) has a real solution, then the trivial solution of the system (3.43) will be unstable for any choice of forms $X_s^{(M)}$, $M > \mu$.*

[It is naturally understood that this solution is different from the trivial solution.]

PROOF. If the system (3.60) has a real solution, then $\mu_1(b_1, \dots, b_n) = 2k$. Then, on the basis of Theorem 68, we can state that there exist two families of solutions, each of which depends on one arbitrary constant and can be represented in the form of the series

$$x_s^{(1)} = y(b_s + \sum_{M+M_1 \geq 1} \bar{K}_s^{(M, M_1)}(\ln y) c^{M_1} y^{M+2kM_1}), \quad (3.64)$$

$$x_s^{(2)} = y(-b_s + \sum_{M+M_1 \geq 1} \bar{L}_s^{(M, M_1)}(\ln y) \gamma^{M_1} y^{M+2kM_1}), \quad (3.65)$$

which converge when $|c_1| \leq c_0$, $|\gamma| \leq \gamma_0$, $t \in (-\infty, 0]$.

According to the corollary to Theorem 67, any integral curve of the family (3.64) has the property

$$|x_s^{(1)}| \rightarrow 0, \quad t \rightarrow -\infty, \quad s = 1, \dots, n.$$

Analogously

$$|x_s^{(2)}| \rightarrow 0, \quad t \rightarrow -\infty, \quad s = 1, \dots, n.$$

And then, on the basis of Theorem 10, we can state that the trivial solution of the system (3.43) is unstable for any choice of $X_s^{(M)}$, $M > \mu$. [This follows from the definitions of Section 6 of Chapter II, and from the proof of the necessity in Theorem 7, in which no

use was made of the compactness of a neighborhood of the invariant set.]

EXAMPLE. Let us consider the system

$$\frac{dx_s}{dt} = \sum_{M=2k+1}^{+\infty} X_s^{(M)}, \quad s = 1, \dots, n, \quad (3.66)$$

where the functions $X_s^{(M)}$ are homogeneous forms of degree M in the quantities $x_1, \dots, x_n$ with real coefficients. We assume that

$$X_s^{(\mu)} = 0 \text{ for } x_i = 0 \quad \begin{matrix} s = 2, \dots, n \\ i = 2, \dots, n; \end{matrix}$$

this means that these forms do not contain the term x_1^μ . We assume that the series in the right members of the system (3.66) converge for sufficiently small $|x_s|$, $s = 1, \dots, n$. We denote by ϕ the coefficient of x_1^μ in the form X_1^μ . We construct for the system (3.66) the algebraic Eqs. (3.54) and (3.60). If $\phi < 0$, then the system (3.54) has the solution

$$x_1 = \sqrt[2k]{\frac{-1}{\phi}}; \quad x_i = 0, \quad i = 2, \dots, n.$$

In this case the system (3.66) has two families of solutions, each of which depends on one arbitrary constant and can be represented in the form of series (3.58) and (3.59), where we must set

$$a_1 = \sqrt[2k]{\frac{-1}{\phi}}; \quad a_i = 0, \quad i = 2, \dots, n.$$

Moreover, each curve of the family (3.58) enters the point $X \rightarrow 0$ tangent to the positive half-axis x_1 , and each curve of the family (3.59) enters tangent to the negative half-axis x_1 .

If $\phi > 0$, then the system (3.60) has a solution

$$x_1 = \sqrt[2k]{\frac{1}{\phi}}; \quad x_i = 0, \quad i = 2, \dots, n.$$

Then there exist two families of integral curves, each of which depends on one arbitrary constant and can be represented in the

form of the series (3.64) and (3.63), where

$$b_1 = \sqrt[2k]{\frac{1}{p}}; \quad b_i = 0, \quad i = 2, \dots, n.$$

Each integral curve of the family (3.64) enters the point $X = 0$ tangent to the positive half-axis x_1 as $t \rightarrow -\infty$. Analogously, any integral curve of the family (3.65) enters the point $X = 0$ tangent to the negative half-axis $x_1 = 0$ as $t \rightarrow -\infty$.

On the basis of Theorem 68, for $p > 0$ the trivial solution of the system (3.66) is unstable for any choice of the coefficients of the forms

$$X_s^{(M)}; \quad M = \mu + 1, \dots$$

THEOREM 69. *If the system (3.54) has no solution in the field of complex numbers, then the system (3.60) will also have no solution in the field of complex numbers.*

PROOF. If the system (3.60) has a solution

$$x_s = b_s \quad (s = 1, \dots, n),$$

then the system (3.54) has a solution

$$x_s = b_s \sqrt[2k]{-1} \quad (s = 1, \dots, n),$$

as can be verified by direct substitution.

COROLLARY. *If the system (3.54) has no solutions of the form*

$$x_s = b_s \sqrt[2k]{-1} \quad (s = 1, \dots, n),$$

where b_s is real, then the system (3.60) has no real solutions.

DEFINITION. We shall say that an integral curve of the system (3.43) $x_s = x_s(t)$, $(s = 1, \dots, n)$ has a *definite direction* if $\lim [x_s(t)/z(at)] = a_s$ as $at \rightarrow +\infty$, $(s = 1, \dots, n)$, where $a_1, \dots, a_n$ are real constants. It is clear that the quantities $a_1, \dots, a_n$ are a real solution of the algebraic system

$$x_s + aX_s^{(\mu)} = 0 \quad (s = 1, \dots, n). \quad (3.67)$$

Any real solution of the system (3.67), different from the trivial solution, will be called a *direction*.

DEFINITION. The direction $a_1, \dots, a_n$ will be called *asymptotically stable* if $\operatorname{Re} \lambda_s(a, a_1, \dots, a_n) > 0$ ($s = 1, \dots, n$). Here, as above, either $a = +1$ [case of the system (3.54)] or $a = -1$ [case of the system (3.60)].

Let us make in the system

$$\frac{dx_s}{dt} = X_s^{(\mu)} \quad (s = 1, \dots, n)$$

the substitution $x_s = z(t)[a_s + y_s(t)]$, ($s = 1, \dots, n$), where $y_s(t)$ are the new unknown functions, and the quantities $a_1, \dots, a_n$ are an asymptotically stable direction, corresponding to $a = 1$.

Setting $\varphi = -\ln z(t)$, we obtain the following system for the determination of the functions $y_s(t)$:

$$\begin{aligned} \frac{dy_s}{d\varphi} &= \gamma_s + a_s + X_s^{(\mu)}(y_1 + a_1, y_2 + a_2, \dots, y_n + a_n) = \\ &= \sum_{i=1}^n p_{si}(a_1, a_2, \dots, a_n)y_i + \\ &+ \sum_{m_1 + \dots + m_n = 2}^{\mu} P_s^{(m_1, \dots, m_n)} y_1^{m_1} \dots y_n^{m_n} \quad (s = 1, \dots, n). \end{aligned} \quad (3.68)$$

The trivial solution of the system (3.68) is asymptotically stable as $t \rightarrow +\infty$.

We denote by $A(a_1, a_2, \dots, a_n)$ the region of asymptotic stability of the trivial solution of the system (3.68). Let us set

$$B(a_1, \dots, a_n) = (a_1, a_2, \dots, a_n) + (d_1, \dots, d_n), \quad (3.69)$$

where the point $(d_1, d_2, \dots, d_n) \in A$.

We consider the case when the system (3.43) is asymptotically stable and moreover each O^+ -curve has a definite direction. Then there are two possibilities:

- 1) there exists a finite number of real solutions of the system (3.67) for $a = 1$,
- 2) there exists an infinite number of real solutions of the system (3.67) for $a = 1$.

In these cases the system (3.67) has no real solutions for $a = -1$.

THEOREM 70. *In order for the trivial solution of the system (3.43) to be asymptotically stable for any choice of the forms $X_s^{(M)}$, $M > \mu$,*

and in order for any integral curve to have a direction, where the number of such directions is to be finite, it is necessary and sufficient that the following conditions be satisfied:

- 1) the system (3.67) has a finite number of real solutions for $a = 1$;
- 2) the closure of the sum of regions $\bar{B}(a_1, \dots, a_n)$, extended over all the asymptotically stable directions, covers the n -dimensional spherical neighborhood of the point $x_1 = x_2 = \dots = x_n = 0$.

The proof follows from Theorem 65 and the preceding arguments.

REMARK. If the functions (3.45) are bounded for $\varphi \in (-\infty, +\infty)$, then there exists an O -curve of the system (3.43), and, furthermore, among the quantities $\mu_1, \dots, \mu_n$ there exists at least one positive one: $\mu_1 = 2k$. [$\mu_1, \dots, \mu_n$ are the eigenvalues of the system (3.46) which is considered here to be reducible.]

We give here a solution of the problem of the stability for a system of two equations.

THEOREM 71. *If the system of Eqs. (3.67) has for $a = +1$ and $n = 2$ a real non-trivial solution, then for asymptotic stability of the trivial solution of the system (3.43) for $n = 2$ it is necessary and sufficient that the system (3.67) have for $n = 2$ and $a = -1$ no real solutions other than the trivial solution. Furthermore, the trivial solution of the system (3.43) must be isolated, i.e. the system*

$$\begin{aligned}\frac{dx_1}{dt} &= X_1^{(\mu)}(x_1, x_2), \\ \frac{dx_2}{dt} &= X_2^{(\mu)}(x_1, x_2),\end{aligned}$$

must have a unique equilibrium position $x_1 = x_2 = 0$.

The proof of this theorem can be obtained from Theorems 65 and 11.

Let us consider the system

$$\frac{dx_i}{dt} = X_i^{(\mu)} \quad (i = 1, 2). \quad (3.70)$$

By assumption, this system has, as follows from Theorem 64, a family of integral curves which enter the point $x_1 = x_2 = 0$ as $t \rightarrow +\infty$, and each integral curve of this family has a definite direction.

If the system (3.70) has integral curves that enter at the point $x_1 = x_2 = 0$ as $t \rightarrow -\infty$, then there are only two possibilities: either these integral curves have a definite direction, or else they are spiral curves. The first is impossible in view of the assumption made, and the second in view of the uniqueness of the solutions of the system (3.70), for the spiral curves would intersect the integral curves which enter the point $x_1 = x_2 = 0$ as $t \rightarrow +\infty$. And then, as shown by Theorem 11, the trivial solution of the system (3.70) is asymptotically stable. It therefore follows that the trivial solution of the system (3.43) for $n = 2$ will also be asymptotically stable (see Section 3 of Chapter II).

Let us return now to the study of the stability problem in doubtful cases, treated in Sections 4 and 5 of Chapter II.

Let us assume that the algebraic system (3.67) has a real solution $x_s = b_s$ ($s = 1, \dots, n$) which is an asymptotically stable direction. We shall assume everywhere from now on that the right members of the system (3.43) are defined for $t \in (-\infty, +\infty)$. Under these assumptions the system (3.43), as shown by Theorem 69, has a family of integral curves which can be represented in the form of convergent series (3.61) and (3.62) for $l = n$.

Let us put, for example, in (3.61) $c_1 = c_2 = \dots = c_n = 0$. The functions obtained as a result will be denoted by $\bar{x}_s(t)$. Let us make in the system (3.43) the substitution $x_s = \xi_s + \bar{x}_s$. Then we obtain the following system of equations for the determination of the functions ξ_s :

$$\frac{d\xi_s}{dt} = \sum_{i=1}^n q_{si} \xi_i + \sum_{m_1 + \dots + m_n = 2}^{+\infty} Q_s^{(m_1, \dots, m_n)}(t) \xi_1^{m_1} \dots \xi_n^{m_n}. \quad (3.71)$$

The system (3.71) has a trivial solution which is asymptotically stable as $t \rightarrow -\infty$, uniformly with respect to $t_0 \leq 0$.

The functions $Q_s^{(m_1, \dots, m_n)}(t)$ are defined, generally speaking, only for $t \in (-\infty, \tau]$, $0 < \tau < +\infty$. The moduli of these functions undergo changes which are small as desired, if the coefficients of the forms $X_s^{(m)}$, ($m > \mu$), which are located in the right members of the system (3.43), are changed by sufficiently small quantities. Therefore, independently of small changes of the coefficients of the forms $X_s^{(\mu)}$, one can indicate a region $|\mathcal{E}| \leq r$, $t \leq 0$, $r > 0$, such that any integral curve of the system (3.71), beginning

in this region, has the property

$$\mathcal{E}(t, \mathcal{E}^{(0)}, t_0) \rightarrow 0 \text{ as } t \rightarrow -\infty.$$

Consequently, no matter what quantity $\varepsilon < r$ we take, it is possible to find, for the quantity $t_0 < 0$, a $\delta_{t_0} > 0$ such that in the region

$$|\mathcal{E}| < \delta_{t_0}$$

there exist points such that

$$|\mathcal{E}(t, \mathcal{E}^{(0)}, t_0)| = \varepsilon \text{ for } t = 0.$$

In this connection, $\delta_{t_0} \rightarrow 0$ as $t_0 \rightarrow -\infty$. Let us set

$$\bar{\varepsilon} = \sqrt{\sum_{s=1}^n \bar{x}_s^2} \text{ for } t = 0.$$

From the arguments given above, it follows that for any $\varepsilon > 0$, $\varepsilon < \bar{\varepsilon}$, $\varepsilon < r$, and for any $t_0 < 0$, in the region

$$|X - \bar{X}| < \delta_{t_0}$$

one can find points such that

$$|X(t, X^{(0)}, t_0)| \geq \bar{\varepsilon} - \varepsilon,$$

where $X(t, X^{(0)}, t_0)$ is the solution of the system (3.43). It follows from this that no matter what $\delta_1 > 0$ is taken, one can find points $X^{(0)}$ such that

$$|X^{(0)}| < \delta_1$$

and values of t_0 so as to make

$$X(t, X^{(0)}, t_0) > \frac{\bar{\varepsilon}}{2} \text{ for } t = 0.$$

In this connection, the set of such points $X^{(0)}$ and values of the parameter t_0 can be considered invariant under sufficiently small variations of the coefficients of the forms

$$X_s^{(m)}, \quad m > \mu.$$

Let us now consider again the system (2.69). It was reduced in Section 4 of Chapter II, by means of a transformation which did not affect the stability problem, to a system of the form (2.81).

From the foregoing arguments and from Theorem 46 we obtain the following statement.

COROLLARY. *If the system of algebraic equations*

$$x_s = X_s^{(0)(\mu)} \quad (s = 1, \dots, k)$$

has a real solution, which is an asymptotically stable direction, then the trivial solution of the system (2.69) is unstable.

The same result holds for the system (2.99), if the system of equations

$$\rho_s = R_s^{(0)(\mu)}(\rho_1, \dots, \rho_k) \quad (s = 1, \dots, k)$$

has a real solution, which is an asymptotically stable direction.

Let us now formulate several general remarks, pertaining to the solution of the problem of the stability of the trivial solution of the systems (2.69) and (2.99).

Let us examine again the system (2.69). We shall assume that it has been reduced to the form (2.71). Let us now subject the system (2.71) to a transformation which does not affect the stability:

$$x_s = z_s + \sum_{m=0}^N Z_s^{(m)} \quad (s = 1, \dots, k), \quad (3.72)$$

where the functions $Z_s^{(m)}$ are homogeneous forms of degree m relative to the new unknown functions $z_1, \dots, z_k$ with arbitrary analytic coefficients with respect to $\eta_1, \dots, \eta_n$, which are to be determined and which vanish when $\eta_1 = \dots = \eta_n = 0$. As follows from the arguments given in Section 4 of Chapter II, the system (2.71) can be reduced, by suitable choice of the coefficients of the forms $Z_s^{(m)}$, to the following form:

$$\begin{aligned} \frac{dz_s}{dt} &= \tilde{X}_s^{(0)}(z_1, \dots, z_k) + \sum_{m=N+1}^{+\infty} \bar{X}_s^{(m)} \quad (s = 1, \dots, k), \\ \frac{d\eta_j}{dt} &= \sum_{i=1}^n p_{ji}\eta_i + \bar{V}_j(z_1, \dots, z_k, \eta_1, \dots, \eta_n) \quad (j = 1, \dots, n). \end{aligned} \quad (3.73)$$

If we set $N = +\infty$, we find that the first group of equations in the system (3.71) becomes

$$\frac{dz_s}{dt} = \tilde{X}_s^{(0)}(z_1, \dots, z_k) \quad (s = 1, \dots, k). \quad (3.74)$$

The functions

$$\bar{V}_j^{(0)} = \bar{V}_j(z_1, \dots, z_k, 0, \dots, 0)$$

are formally analytic with respect to the quantities $z_1, \dots, z_k$. Their expansions actually begin with forms of order $\nu \geq \mu + 1$. In Section 4 of Chapter II it was shown that by a series of transformations it is possible to obtain from the system (3.73) a system in which $\nu_l = \nu + l(\mu - 1)$. Setting $l = +\infty$, we obtain the transformation

$$\eta_j = \bar{y}_j + \sum_{l=1}^{+\infty} u_{jl}(z_1, \dots, z_k), \quad (3.75)$$

which transforms the second group of equations in the system (3.73) into

$$\frac{d\bar{y}_j}{dt} = \sum_{i=1}^n p_{ji}\bar{y}_i + \sum_{m=1}^{+\infty} \bar{Y}_j^{(m)}(\bar{y}_1, \dots, \bar{y}_n, z_1, \dots, z_k), \quad (3.76)$$

where the functions $\bar{Y}_j^{(m)}$ are homogeneous forms of degree m with respect to the quantities $\bar{y}_1, \dots, \bar{y}_n$, the coefficients of which are series in powers of the quantities $z_1, \dots, z_k$. The functions u_{jl} in the transformation (3.75) are analytic with respect to $z_1, \dots, z_k$ and satisfy certain algebraic equations, indicated in Section 4 of Chapter II. If the transformations (3.72) ($N = +\infty$) and (3.75) are not formal, i.e. the series converge in a certain sufficiently small neighborhood of the point $z_1 = \dots = z_k = \eta_1 = \dots = \eta_n = 0$, then the series in the right members of the system (3.76) also converge. In this case the trivial solution of the system (2.69) will be stable, asymptotically stable, or unstable if and only if the trivial solution of the system (3.74) will be stable, asymptotically stable, or unstable, respectively.

A similar transformation can be indicated for the system (2.99), which reduces the solution of the problem of stability of the trivial solution of this system to the solution of the same problem for a system of type (3.74). The question now arises of demonstrating that the transformations (3.72) and (3.75) are not formal, or else to indicate those conditions, under which these transformations will not be formal. Let us note that A. M. Lyapunov was the first to call attention to the fact that the solution of the problem of the

stability of the trivial solution of the systems (2.69) and (2.99) depends on the solution of the analogous problem for the system (3.74). Many general theorems were formulated in this direction by Professor I. G. Malkin [27]. However, as noted by N. P. Yerugin [28], an error crept into their proof, and this error has not yet been eliminated. More general propositions in this direction are formulated in reference [29]. A careful analysis of the proofs of these propositions, contained in reference [29], has shown that they are faulty. Thus, the general principles expressed in references [27 and 29] still need a rigorous justification. [This principle can be justified on the basis of the work by I. G. Malkin (PMM, Vol. VI, No. 6, 1942).] Therefore in the investigation of doubtful cases we did not make use of these references.

CHAPTER IV

INVESTIGATION OF THE STABILITY OF INVARIANT SETS OF GENERAL SYSTEMS

1. General systems. Basic definitions

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n), \quad (4.1)$$

the right members of which are defined for $t \geq 0$ and $X \in E_n$, and satisfy certain conditions under which a solution exists:

$$X = X(t, X^{(0)}, t_0) \quad \text{for } t \geq t_0 \geq 0, \quad X^{(0)} \in E_n.$$

We shall assume that each such solution is defined for all $t \geq t_0$ and is such that

$$X(t, X^{(0)}, t_0) \rightarrow X^{(0)} \quad \text{as } t \rightarrow t_0 + 0.$$

Moreover, an infinite set of solutions of the system (4.1), having the indicated properties, can pass through the point $(X^{(0)}, t_0)$. We shall further denote by $F_{t_0}^t(X^{(0)})$ the set of points belonging to all such solutions at some fixed instant of time $t \geq t_0$. Let

$$F_{t_0}^t(X^{(0)}) \rightarrow X^{(0)} \quad \text{as } t \rightarrow t_0 + 0.$$

We shall call $F_{t_0}^t(X^{(0)})$ a *motion*, and the set of all the points belonging to this motion for $t \geq t_0$ will be called the *trajectory* of this motion. Let us note the principal properties which are possessed by a motion.

1. For any $X^{(0)} \in E_n$, the set of elements

$$F_{t_0}^t(X^{(0)}) \in E_n$$

is defined for any $t \geq t_0$.

2. $F_{t_0}^t(X^{(0)}) \rightarrow X^{(0)}$ as $t \rightarrow t_0 + 0$.

3. Let $X^{(1)}$ be a certain point of the set $F_{t_0}^{t_1}(X^{(0)})$; then according to property 1, the motion $F_{t_1}^t(X^{(1)})$ is defined, and moreover

$$\cup F_{t_1}^t(X^{(1)}) = F_{t_0}^t(X^{(0)}) \text{ when } t \geq t_1 \text{ for } X^{(1)} \in F_{t_0}^{t_1}(X^{(0)}).$$

The third property of a motion follows from the definition of a motion itself as the set of endpoints of integral curves, considered at the instant t , each of which passes through the point $(X^{(0)}, t_0)$.

It is clear that the set of the motions $F_{t_0}^t(X^{(0)})$ maps a two-parameter family of transformations of E_n onto itself, defined for all $t \geq t_0 \geq 0$ and having Properties 1–3. Such a two-parameter family of transformations of E_n onto itself will be called a *general system* in E_n . If the right members of the system (4.1) are continuous, then it has a solution $X = X(t, X^{(0)}, t_0)$ for any $X^{(0)} \in E_n$ and defined for certain $t \geq t_0 \geq 0$. If each solution of the system (4.1) is defined for all $t \geq t_0$, then (4.1) defines a general system in E_n . If it is found that for continuous right members of the system (4.1) not all of its solutions are defined for $t \geq t_0 \geq 0$, then by making a change of independent variable by means of the formula

$$s = \int_{t_0}^t \sqrt{1 + \sum_{i=1}^n f_i^2(X(\tau, X^{(0)}, t_0))} d\tau,$$

we obtain the system

$$\frac{dx_k}{ds} = f_k(x_1, \dots, x_n, t), \quad (4.2)$$

the right members of which are defined for $t \geq 0$ and $X \in E_n$. Moreover, any solution of the system (4.2) will be defined for all $s \geq s_0 \geq 0$. Thus, the system (4.2) defines in the space E_n a two-parameter family of transformations $F_{s_0}^s(X^{(0)})$, having properties 1–3. Thus, if the right members of the system (4.1) are continuous it is always possible to define with its aid a general system in E_n .

Let us give now an example of a two-parameter transformation of a functional space onto itself. We consider for this purpose a

system of partial differential equations

$$\frac{\partial u_s}{\partial t} = f_s(x_1, \dots, x_k, u_1, \dots, u_n, \frac{\partial u_i}{\partial x_j}, \dots, t), \quad (4.3)$$

$$(s = 1, \dots, n; i = 1, \dots, n; j = 1, \dots, k),$$

the right members of which are defined for $t \geq 0$ and for

$$\left(x_1, \dots, x_k, u_1, \dots, u_n, \frac{\partial u_i}{\partial x_j}\right) \in G,$$

where G is a certain region in N -dimensional space. We assume that there exists a certain functional space Φ of the vector functions

$$\varphi = (\varphi_1(x_1, \dots, x_k), \dots, \varphi_n(x_1, \dots, x_k))$$

such that when $\varphi \in \Phi$ the system (4.3), for any $t_0 \geq 0$, has a solution $U(t, \varphi, t_0)$ which is defined for $t \geq t_0$ and $U(t, \varphi, t_0) \in \Phi$ for $t > t_0$, $U(t, \varphi, t_0) \rightarrow \varphi$ as $t \rightarrow t_0 + 0$. It is clear that the element φ of the space Φ and the positive quantity t_0 can define more than one solution $U(t, \varphi, t_0)$ which has the above properties. Each such solution at a certain instant of time $t \geq t_0$ assumes a definite value from Φ . The set of values assumed by all such solutions will be denoted by $F_{t_0}^t(\varphi)$, i.e. no matter what the solution of the system (4.3) $U(t, \varphi, t_0)$, defined by the condition

$$U(t, \varphi, t_0) \rightarrow \varphi \text{ as } t \rightarrow t_0 + 0,$$

we take, it will be found that

$$U(t, \varphi, t_0) \in F_{t_0}^t(\varphi) \text{ for any } t \geq t_0.$$

We shall call $F_{t_0}^t(\varphi)$ a *motion*, and the set of all the points $F_{t_0}^t(\varphi)$, $t \in [t_0, +\infty)$ will be called the *trajectory* of this motion. Let us note the following principal properties of a motion:

1) For any $\varphi \in \Phi$ and any $t_0 \geq 0$, the set

$$F_{t_0}^t(\varphi) \in \Phi \text{ is defined for all } t \geq t_0;$$

2) $F_{t_0}^t(\varphi) \rightarrow \varphi$ as $t \rightarrow t_0 + 0$;

3) Let $\bar{\varphi}$ be a point in the space Φ such that $\bar{\varphi} \in F_{t_0}^{t_1}(\varphi)$; then

there is defined a motion $F_{t_0}^t(\bar{\varphi})$ such that

$$\cup F_{t_1}^t(\bar{\varphi}) = F_{t_0}^t(\varphi) \text{ for } t \geq t_1 \text{ and } \bar{\varphi} \in F_{t_0}^{t_1}(\varphi).$$

Let us show that Property 3 is satisfied.

Actually, the set $F_{t_0}^t(\bar{\varphi})$ represents an aggregate of points of the space Φ , located on the solutions of the Cauchy problem for the system (4.3), defined by the following conditions:

$$U(t, \varphi, t_0) \rightarrow \varphi \text{ as } t \rightarrow t_0 + 0,$$

$$U(t, \varphi, t_0) \rightarrow \bar{\varphi} \text{ as } t \rightarrow t_1 + 0.$$

The set $F_{t_0}^t(\varphi)$ represents the set of points of the space Φ , belonging to the solutions of the Cauchy problem for the system (4.3), defined only by the condition

$$U(t, \varphi, t_0) \rightarrow \varphi \text{ as } t \rightarrow t_0 + 0.$$

Therefore the following relation holds:

$$\cup F_{t_1}^t(\bar{\varphi}) = F_{t_0}^t(\varphi) \text{ for } t \geq t_1 \text{ and } \bar{\varphi} \in F_{t_0}^{t_1}(\varphi).$$

It is clear that the set of motions $F_{t_0}^t(\varphi)$ represents a two-parameter family of transformations of the space Φ onto itself. This family of transformations, having the Properties 1–3, will be called a *general system* in the space Φ . We proceed now to develop the theory for the general case.

DEFINITION 1. We shall say that a general system is *defined* in the metric space R if there is defined $F_{t_0}^t$, which is a two-parameter family of transformations of R onto itself, with the following properties:

- 1) For any $p \in R$ and $t_0 \geq 0$, the set $F_{t_0}^t(p) \subset R$ is defined for $t \geq t_0$, with $F_{t_0}^{t_0}(p)$ not empty;
- 2) $F_{t_0}^t(p) \rightarrow p$ as $t \rightarrow t_0 + 0$;

[This must be understood in the following manner: for each $\varepsilon > 0$ we can find a $\delta > 0$ such that when $t - t_0 < \delta$ we have $F_{t_0}^t(p) \subset S(p, \varepsilon)$.]

- 3) For any element $p_1 \in F_{t_0}^{t_1}(p)$ there is defined a set $F_{t_1}^t(p_1)$ such that

$$\cup F_{t_1}^t(p_1) = F_{t_0}^t(p) \text{ for } t \geq t_1 \text{ and } p_1 \in F_{t_0}^{t_1}(p).$$

We shall henceforth call $F_{t_0}^t(p)$, for a fixed p , a *motion*, and the set of all the points belonging to the motion for $t \geq t_0$ will be called

the *trajectory* of this motion. The set $M \subset R$ will be called an *invariant set* of the general system if M consists of trajectories of the general system. Let us give a more accurate definition of an invariant set.

DEFINITION 2. The set M is *invariant* with respect to a general system if it follows from $p \in M$ that $F(t_0, p) \subset M$ for any $t_0 \geq 0$. Here we denote by $F(t_0, p)$ the trajectory of the motion $F_{t_0}^t(p)$.

Let us set further

$$\rho(t, p_0, t_0) = \sup_{p \in F_{t_0}^t(p_0)} \rho(p, M).$$

We introduce the concept of stability of an invariant set of a general system, located in metric space.

DEFINITION 3. The invariant set M of a general system in R is called *stable*, if for any $\varepsilon > 0$ one can find a quantity $\delta > 0$ such that when $\rho(p_0, M) < \delta$ we have

$$\rho(t, p_0, t_0) < \varepsilon \quad \text{for } t \geq t_0 \geq 0.$$

If, furthermore, $\rho(t, p_0, t_0) \rightarrow 0$ as $t \rightarrow +\infty$, then the invariant set M is called *asymptotically stable*.

DEFINITION 4. An asymptotically stable invariant set M of a general system in R is called *uniformly asymptotically stable* if there exists a quantity δ_1 , corresponding to a certain ε_1 (Definition 3), such that $\rho(t, p_0, t_0) \rightarrow 0$ as $t - t_0 \rightarrow +\infty$ uniformly with respect to $t_0 \geq 0$ and $\rho(p_0, M) < \delta_1$.

REMARK. In exactly the same way as was done in Section 10, we can show that there exists a continuous function $L(\tau)$, which decreases strictly monotonically from $+\infty$ to 0, defined for $\tau \in (-\infty, +\infty)$, and such that

$$\rho(t, p_0, t_0) < L(t - t_0) \quad \text{for } \rho(p_0, M) < \delta_1.$$

Actually, let

$$\lambda(t - t_0) = \sup_{\rho(p_0, M) < \delta_1} \rho(t, p_0, t_0).$$

It is clear that $\lambda(t - t_0) \rightarrow 0$ as $t - t_0 \rightarrow +\infty$. Using the same procedure as in Section 10, we can readily construct a function L , having the aforementioned properties.

DEFINITION 5. An asymptotically stable invariant set M is called *uniformly attracting* if for any $h > 0$ it is possible to find quantities $T > 0$ and $\alpha > 0$ such that $\rho(t, p_0, t_0) > \alpha$ when $0 \leq t - t_0 \leq T$ and $h \leq \rho(p_0, M) \leq \delta_2$, where δ_2 is a positive quantity corresponding to a certain $\varepsilon_2 > 0$ (Definition 3).

DEFINITION 6. An invariant set M is called *unstable* if there exists a certain quantity $\varepsilon > 0$ such that for any $\delta > 0$ it is possible to find a point p_0 and a number $t_0 > 0$ such that for a certain $t > t_0$ the inequalities $\rho(p_0, M) < \delta$ and $\rho(t, p_0, t_0) \geq \varepsilon$ hold.

The task of this chapter is to determine the conditions under which the invariant set M has a certain stability or instability property, and also to develop a method that makes it possible to estimate the function $\rho(t, p_0, t_0)$.

The first property of the general system is generally speaking quite stringent, for when solving, for example, the Cauchy problem for systems of partial differential equations the following fact is sometimes encountered: the Cauchy problem can be solved, for example, for the system (4.3) at $\varphi \in R_1$, where φ is a vector function in the variables $x_1, \dots, x_k$, to which corresponds the solution $U = U(\varphi, t, t_0)$ of the system (4.3), having the property $U \rightarrow \varphi$ as $t \rightarrow t_0 + 0$. However, $U(\varphi, t, t_0) \in R$ when $t > t_0$, where R is a certain functional space, containing the set of functions R_1 . In the case of a general system, we have $R = R_1$. From these considerations we shall consider from now on also such two-parameter families of operators $F_{t_0}^t$, which are defined for $t \geq t_0$ on any element $p_0 \in R_1$, and whose values fall into R , where R is a certain metric space, and R_1 is a subset of R . Such two-parameter families of operators will be called henceforth an *incomplete general system*, so that an incomplete general system has the second and third properties of a general system. For the case of an incomplete general system the invariant set M will be considered to be a subset of R_1 . All the definitions given in this section of the stability of an invariant set can be directly transferred to the case of an incomplete general system. In these definitions, it is necessary to specify in each case that the motions $F_{t_0}^t(p_0)$ are defined only for $p_0 \in R_1$ or else $p_0 \in F_{t_0}^t(p_1)$.

2. Conditions for stability and instability of an invariant set M of a general system in metric space

In the present section we shall give the conditions under which an invariant set M will have the property of stability or instability, and propose a method which permits, for certain general systems, to estimate the function $\rho(t, p_0, t_0)$ with the aid of a family of functionals.

Let us consider a one-parameter family of functionals V_t , which depend on the parameter $t \geq 0$. We assume that for any $p \in G \subset R$ there is defined a function $V_t(p)$ of the real variable t , defined for $t \geq 0$. Furthermore we shall denote

$$\sup_{p \in F_{t_0}^t} V_t(p) \quad (4.4)$$

by $V(t, p_0, t_0)$.

THEOREM 74. *In order for an invariant set M of a general system in R to be stable, it is necessary and sufficient that there exist a one-parameter family of functionals V_t , having the following properties:*

1) *on any element p of a certain neighborhood $S(M, r)$ of the set M , there is defined a function of the real argument t , namely $V_t(p)$, defined for $t \geq 0$;*

2) *for any sufficiently small $c_1 > 0$ it is possible to find a quantity $c_2 > 0$ such that $V_t(p) > c_2$ for $\rho(p, M) > c_1$ and all $t \geq 0$;*

3) *$V_t(p) \rightarrow 0$ uniformly with respect to $t \geq 0$ as*

$$\rho(p, M) \rightarrow 0;$$

4) *the function $V(t, p_0, t_0)$ does not increase for all $t \geq t_0$, for which it is defined.*

PROOF. Necessity. We assume that the set M is stable. Then for any $\varepsilon > 0$ we can find a quantity $\delta > 0$ such that when $\rho(p_0, M) < \delta$ we have $\rho(t, p_0, t_0) < \varepsilon$ for $t \geq t_0$. We set

$$V_{t_0}(p_0) = \sup_{t \geq t_0} \rho(t, p_0, t_0) \text{ for } \rho(p_0, M) < \delta. \quad (4.5)$$

It is clear that the equality (4.5) defines the family of functionals, which depends on one parameter $t \geq 0$, defined on any element $p_0 \in S(M, \delta)$.

It follows from (4.5) that $V_{t_0}(p_0) \leq \varepsilon_1$ for $\rho(p_0, M) < \delta_1$, where $\delta_1 < \delta$ is a certain positive quantity, corresponding by Definition 3

of Section 1 of Chapter IV to the chosen $\varepsilon_1 > 0$. From this it follows that the family of functionals V_t , defined by Eq. (4.5), has Property 3. In addition, $V_{t_0}(p_0) \geq \rho(p_0, M)$, and consequently the family of functionals V_t also has Property 2. Let us show now that Property 4 holds. We take a certain quantity $t_1 > t_0$. We consider the expression $\rho(t, p, t_1)$ where $p \in F_{t_0}^{t_1}(p_0)$. By virtue of the Property 3 of a general system, we have

$$F_{t_1}^t(p) \subset F_{t_0}^t(p_0) \quad \text{for } t \geq t_1$$

and for any $p \in F_{t_0}^{t_1}(p_0)$. Therefore

$$\rho(t, p, t_1) \leq \rho(t, p_0, t_0) \quad \text{for } t \geq t_1$$

for any $p \in F_{t_0}^{t_1}(p_0)$, from which we obtain that

$$V_{t_1}(p) = \sup_{t \geq t_1} \rho(t, p, t_1) \leq \sup_{t \geq t_0} \rho(t, p_0, t_0) \leq V_{t_0}(p_0).$$

Thus, for any element $p \in F_{t_0}^{t_1}(p_0)$, we have

$$V_{t_1}(p) \leq V_{t_0}(p_0);$$

hence

$$\sup_{p \in F_{t_0}^{t_1}(p_0)} V_{t_1}(p) = V(t_1, p_0, t_0) \leq V_{t_0}(p_0)$$

for any $t_1 \geq t_0$. From this it follows that the constructed family of functionals has also the property 4.

Sufficiency. We assume that there exists a family of functionals V_t which depends on the parameter $t \geq 0$ and has Properties 1–4, indicated in Theorem 74.

Let us show that the invariant set M is stable. We take $\varepsilon > 0$ and $\varepsilon < r/2$. We set

$$\lambda = \inf V_t(p) \quad \text{for } \rho(p, M) \geq \varepsilon \quad \text{and} \quad \rho(p, M) \leq \frac{r}{2}.$$

Let us choose δ such that when $\rho(p, M) < \delta$ we have

$$V_t(p) < \lambda \quad \text{for } t \geq 0.$$

Let us show that the quantity δ corresponds by Definition 3 of Section 1, Chapter IV to the selected ε . By virtue of Condition 4, we have

$$V_{t_0}(p_0) \geq V(t, p_0, t_0),$$

from which it follows that when $\rho(p_0, M) < \delta$ we have

$$V(t, t_0, p_0) < \lambda \text{ for } t \geq t_0.$$

Assume that there exists an instant $t > t_0$ such that $\rho(t, p_0, t_0) \geq \varepsilon$. Then, moreover we have $V(t, p_0, t_0) > \lambda$ for a certain value of $t > t_0$, which is impossible, for the opposite inequality holds for any choice of the quantity $t \geq t_0$ and for any $p_0 \in S(M, \delta)$. This proves the theorem.

REMARK. Generally speaking, the function of three arguments $\rho(t, p_0, t_0)$ may not be continuous for $t \geq t_0$, and therefore there may not exist values of t for which

$$\varepsilon \leq \rho(t, p_0, t_0) \leq \frac{r}{2}$$

for a certain $p_0 \in S(M, \delta)$. In this case the proof of the theorem remains in force, if we consider that the family of functionals is also defined outside the set $S(M, r/2)$. For example, we can set

$$V_t(p) = 1 \text{ for } p \in S\left(M, \frac{r}{2}\right) \text{ and } p \in \bar{M} \text{ and } V_t(p) = 0 \text{ for } p \in \bar{M}.$$

In the same way we can extend the family of functionals, constructed in the proof of necessity of Conditions 1–4 of Theorem 74, over all the space R .

THEOREM 75. *In order for an invariant set M of a general system in R to be asymptotically stable, it is necessary and sufficient that there exist a one-parameter family of functionals V_t , having the following properties:*

- 1) *all the Conditions 1–4 of the theorem are satisfied for the family V_t ;*
- 2) *the function $V(t_0, p, t_0) \rightarrow 0$ as $t \rightarrow +\infty$ for any $p_0 \in S(M, \delta)$, where δ is a certain positive quantity.*

PROOF. *Necessity.* Let the invariant set M be asymptotically stable. Then it is simply stable. Therefore, as follows from Theorem 74, there exists a family of functionals, defined by Eq. (4.5), having Properties 1–4 of Theorem 74. Let us show that the family of functionals constructed with the aid of Eq. (4.5) satisfies Condition 2 of the present theorem.

For the number $\varepsilon > 0$ we can find a $\delta > 0$ such that when $\rho(p_0, M) < \delta$ we have

$$\rho(t, p_0, t_0) < \varepsilon \text{ for } t \geq t_0,$$

and moreover

$$\rho(t, p_0, t_0) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Therefore, for any δ_1 , corresponding to a certain ε_1 , we can find on the basis of Definition 3 of Section 1 of Chapter IV a quantity $T > 0$ such that $\rho(t, p_0, t_0) < \delta_1$ for $t = T$. We then have

$$\rho(t, p, T) < \varepsilon_1 \text{ for } t \geq T \text{ for any } p \in F_{t_0}^T(p_0),$$

and then

$$V_T(p) = \sup_{t \geq T} \rho(t, p, T) \leq \varepsilon_1$$

for any $p \in F_{t_0}^T(p_0)$; therefore

$$V(t, p_0, t_0) = \sup V_T(p) \leq \varepsilon_1 \text{ for } p \in F_{t_0}^T(p_0) \text{ and } t = T;$$

this inequality will hold a fortiori when $t \geq T$, since it has been established that the function $V(t, p_0, t_0)$ does not increase. In other words, Condition 2 is satisfied.

Sufficiency. Let the conditions of the theorem be satisfied; then, as follows from Theorem 74, the set M is stable. As a result of this one can find, for the quantity $\varepsilon > 0$, a quantity $\delta > 0$ such that when $\rho(p_0, M) < \delta$ we have

$$\rho(t, p_0, t_0) < \varepsilon \text{ for } t \geq t_0.$$

Further, there exist two possibilities:

- 1) either $\rho(t, p_0, t_0) \rightarrow 0$ as $t \rightarrow +\infty$;
- 2) or $\rho(t, p_0, t_0) > \gamma(p_0) > 0$ when $t \geq t_0$. Actually, assume that the latter does not hold. We shall show then that $\rho(t, p_0, t_0) \rightarrow 0$ as $t \rightarrow +\infty$. For the quantity ε_1 we find a δ_1 such that when $\rho(p_0, M) < \delta_1$ we have

$$\rho(t, p_0, t_0) < \varepsilon_1 \text{ for } t \geq t_0.$$

For the quantity δ_1 we can find a quantity $T > 0$ such that $\rho(T, p_0, t_0) < \delta_1$, and then $\rho(t, p, T) < \varepsilon_1$ when $t \geq T$ for any

$$p \in F_{t_0}^T(p_0).$$

Then $\rho(t, p_0, t_0) \leq \varepsilon_1$ when $t \geq T$. Thus, $\rho \rightarrow 0$ as $t \rightarrow +\infty$. We now assume that there exists a motion of a general system

$$F_{t_0}^1(p_0)$$

such that

$$\rho(t, p_0, t_0) > \gamma(p_0) > 0.$$

Then $V(t, p_0, t_0) > \gamma_1 > 0$, which contradicts Condition 2 of the present theorem.

THEOREM 76. *In order for the invariant set M to be uniformly asymptotically stable, it is necessary and sufficient that there exist a one-parameter family of functionals V_t , having the following properties:*

- 1) *all the conditions of Theorem 75 are satisfied;*
- 2) *the function $V(t, p_0, t_0) \rightarrow 0$ as $t - t_0 \rightarrow +\infty$ uniformly with respect to $t_0 \geq 0$ and for $\rho(p_0, M) \leq \delta$.*

PROOF. We assume that the conditions of the theorem are satisfied. Then the set M is stable. Therefore there exists a family of functionals, defined by means of Eq. (4.5). Let us show that

$$V(t, p_0, t_0) \rightarrow 0 \text{ as } t - t_0 \rightarrow +\infty$$

uniformly with respect to

$$t_0 \geq 0 \text{ and } \rho(p_0, M) \leq \delta.$$

We recall that

$$V(t, p_0, t_0) = \sup_{p \in F_{t_0}^1(p_0)} V_t(p).$$

In accordance with the condition of the theorem, the function

$$\rho(t, p_0, t_0) \rightarrow 0 \text{ as } t - t_0 \rightarrow +\infty$$

uniformly with respect to

$$t_0 \geq 0 \text{ and } \rho(p_0, M) < \delta.$$

By virtue of this, we can find for the quantity $\varepsilon_1 > 0$, a quantity $\delta_1 > 0$, corresponding to ε_1 by Definition 3 of Section 1, Chapter IV, such that

$$\rho(t, p_0, t_0) < \delta_1 \text{ for } t = T + t_0.$$

Then

$$\rho(t, p, T + t_0) < \varepsilon_1 \text{ for } t \geq T + t_0 \text{ and any } p \in F_{t_0}^{T+h}(p_0).$$

By virtue of this,

$$\rho(t, p_0, t_0) < \varepsilon_1 \text{ for } t \geq T + t_0.$$

But

$$V(t, p_0, t_0) = \sup_{p \in F_{t_0}^t(p_0)} V_t(p) = \sup_{p \in F_{t_0}^t(p_0)} \sup_{\tau \geq t \geq T+t_0} \rho(\tau, p, t).$$

Hence $V(t, p_0, t_0) < \varepsilon_1$ when $t \geq T + t_0$. Thus Condition 2 of this theorem is satisfied.

Sufficiency. Let the conditions of the theorem be satisfied. We shall show that the set M is uniformly asymptotically stable.

Let the function $\rho(t, p_0, t_0) \rightarrow 0$ as $t - t_0 \rightarrow +\infty$, but not uniformly with respect to

$$t_0 \geq 0 \text{ for } \rho(p_0, M) \leq \delta_1,$$

where δ_1 is a sufficiently small positive number. Then there exists at least one quantity $h > 0$ such that it is possible to find a sequence of points $p_{0k} \in S(M, \delta_1)$, a sequence of quantities $t_{0k} \geq 0$, and a sequence $t_k > t_{0k} \geq 0$ such that

$$\rho(t, p_{0k}, t_{0k}) \geq h \text{ for } t \in [t_{0k}, t_k], t_k - t_{0k} \rightarrow +\infty \text{ as } k \rightarrow +\infty;$$

then

$$V(t, p_{0k}, t_{0k}) \geq h_1 > 0 \text{ for } t \in [t_0, t_k],$$

which is impossible because

$$V(t, p_0, t_0) \rightarrow 0$$

uniformly with respect to $t_0 \geq 0$ and $\rho(p_0, M) < \delta$.

THEOREM 77. *In order for the invariant set M to be uniformly attracting, it is necessary and sufficient that there exist a one-parameter family of functionals, having the following properties:*

- 1) *all the conditions of Theorem 75 are satisfied;*
- 2) *for any sufficiently small $h > 0$ it is possible to find quantities $T > 0$ and $\alpha > 0$ such that $V(t, p_0, t_0) > \alpha$ when $t \in [t_0, t_0 + T]$ for $t_0 \geq 0$ and $h \leq \rho(p_0, M) \leq \delta$, where $\delta > h$ is a sufficiently small quantity.*

PROOF. Necessity. We assume that the set M is uniformly attracting. Then the set M is asymptotically stable. Therefore the equality (4.5) defines a family of functionals, having all the properties indicated in Theorem 75. In addition, for the quantity h , by virtue of Definition 5 of Section 1, Chapter IV, it is possible to find quantities $T > 0$ and $\alpha > 0$ such that

$$\rho(t, t_0, p_0) > \alpha \text{ for } t \in [t_0, t_0 + T] \text{ and } \rho(p_0, M) \geq h.$$

From (4.1) it follows that

$$V(t, p_0, t_0) = \sup_{p \in F_{t_0}^t(p_0)} \sup_{\tau \geq t \geq t_0} \rho(\tau, p, t) \geq \alpha$$

when $t - t_0 \leq T$, from which follows the satisfaction of Condition 2 of Theorem 77.

Sufficiency. Let there exist a one-parameter family of functionals V_t satisfying all the conditions of Theorem 77.

It follows from Condition 1 that the set M is asymptotically stable. Let us show that it is also uniformly attracting. Let this not be so; then there exists at least one sufficiently small positive quantity h such that it is possible to find a sequence of points p_{0k} , $\rho(p_{0k}, M) \geq h$ and sequences of numbers t_{0k} and T_k such that

$$\rho(T_k + t_{0k}, p_{0k}, t_{0k}) \rightarrow 0 \text{ as } k \rightarrow +\infty.$$

In this connection, $T_k \leq T$, where T may be any positive quantity, which is as small as convenient. Then, by virtue of the fact that $V_t(p) \rightarrow 0$ uniformly with respect to $t \geq 0$, when $\rho(p, M) \rightarrow 0$, we have

$$V(t_{0k} + T_k, p_{0k}, t_{0k}) \rightarrow 0 \text{ as } k \rightarrow +\infty,$$

which contradicts Condition 2 of Theorem 77 because

$$\rho(p_{0k}, M) \geq h \text{ and } T_k < T.$$

This proves Theorem 77.

Now let the general system, defined in R , be such that $F_{t_0}^t(p)$ is a single-point set and continuous in t for any $t \geq t_0$.

THEOREM 78. *In order for the invariant set M to be uniformly asymptotically stable and uniformly attracting, it is necessary and sufficient that there exist two families of functionals V_t and Φ_t , having the following properties:*

1) the quantities $V_t(p)$ and $\Phi_t(p)$, which are functions of the real argument $t \geq 0$, are defined for any $p \in S(M, r)$, $r > 0$ sufficiently small;

2) for any sufficiently small quantity $c_1 > 0$ it is possible to find positive quantities c_2 and c_3 such that $V_t(p) < -c_2$, and

$$\Phi_t(p) > c_3 \text{ for } \rho(p, M) > c_1 \text{ and } t \geq 0;$$

3) $V_t(p)$ and $\Phi_t(p) \rightarrow 0$ as $\rho(p, M) \rightarrow 0$ uniformly with respect to $t \geq 0$;

$$4) (d/dt)V(t, p_0, t_0) = \Phi(t, p_0, t_0). \quad (4.6)$$

PROOF. *Necessity.* Let the set M be uniformly asymptotically stable and uniformly attracting. Then, as shown, there exists a function $L(\tau)$, defined for $\tau \in (-\infty, +\infty)$, continuous, strictly monotonically varying from $+\infty$ to 0, and such that when $\rho(p, M) < \delta_1$ we have

$$L(t - t_0) > \rho(t, p_0, t_0) \text{ for } t \geq t_0.$$

Let us construct the functional

$$\Phi = \rho(p, M) e^{-L^{-1}(\rho(p, M))}.$$

This functional has all the properties indicated in the theorem. Furthermore, we set

$$V(F_{t_0}^t(p_0), t) = - \int_{t_0}^{+\infty} \Phi(F_{t_0}^\tau(p_0)) d\tau \text{ and } t \geq t_0. \quad (4.7)$$

Formula (4.7) determines along the motion $F_{t_0}^t(p_0)$ a functional which becomes, when $t = t_0$,

$$V_{t_0}(p_0) \text{ for } \rho(p_0, M) \leq \delta_1.$$

In fact,

$$V(F_{t_0}^t(p_0), t) \geq -\varepsilon_1 e^{t_0 - t},$$

where ε_1 is a certain positive quantity to which δ_1 corresponds according to the Definition 3 of Section 1, Chapter IV. From the foregoing inequality it follows that

$$V_{t_0}(p_0) > -\varepsilon_1 \text{ for } t_0 \geq 0 \text{ and } \rho(p_0, M) < \delta_1,$$

which means that Condition 3 of Theorem 78 is satisfied.

It follows from (4.7) that

$$\frac{d}{dt} V(F_{t_0}^t(p_0), t) = \Phi(F_{t_0}^t(p_0)).$$

Let us now show that the family of functionals V_t , defined by means of (4.7), satisfies Condition 2 of Theorem 78.

$$V_{t_0}(p_0) = - \int_{t_0}^{+\infty} \Phi(F_{t_0}^\tau(p_0)) d\tau \leq - \int_{t_0}^{t_0+T} \Phi(F_{t_0}^\tau(p_0)) d\tau.$$

For the quantity h , by virtue of the fact that set M is uniformly attracting, it is possible to find quantities T and α such that

$$\rho(t, p_0, t_0) > \alpha \text{ for } t \in [t_0, t_0 + T] \text{ and } \rho(p_0, M) \geq h.$$

Then

$$\Phi(F_{t_0}^t(p_0)) > \beta \text{ for } t \in [t_0, t_0 + T],$$

where β is a sufficiently small positive constant. We then have

$$V_{t_0}(p_0) \leq -\beta T \text{ for } \rho(p_0, M) > h$$

and for any $t \geq 0$. This proves the necessity of the conditions.

Sufficiency. Let there exist two families of functionals, satisfying the conditions of Theorem 78. The family of functionals $-V_t$ satisfies all the conditions of Theorem 74, and therefore the set M is stable.

Assume that there exists a motion

$$F_{t_0}^t(p_0), \quad \rho(p_0, M) < \delta,$$

where $\delta > 0$ is sufficiently small and corresponds to a certain $\varepsilon > 0$, by virtue of Definition 3 of Section 1, Chapter IV, such that $\rho(t, p_0, t_0) > \gamma > 0$. We shall then have on this motion the inequality

$$\Phi(t, p_0, t_0) > \beta > 0 \text{ for } t \geq t_0.$$

Then

$$V(t, p_0, t_0) > (t - t_0)\beta + V_{t_0}(p_0),$$

which is impossible because

$$V(t, p_0, t_0) < 0 \text{ for } t \geq t_0.$$

It follows from this that any motion

$$F_{t_0}^t(p_0)$$

has, for $\rho(p_0, M) < \delta$, the property that

$$\rho(t, p_0, t_0) \rightarrow 0 \text{ as } t - t_0 \rightarrow +\infty$$

and, furthermore, uniformly with respect to

$$t_0 \geq 0 \text{ and } \rho(p_0, M) < \delta.$$

Let us show now that the set M is uniformly attracting. Assume that this is not so. Then there exists a sufficiently small number $h > 0$ such that for $\rho(p_0, M) > h$ and for any $T > 0$ we have

$$\inf_{t \in [t_0, t_0 + T]} \rho(t, p_0, t_0) = 0, \quad t_0 \geq 0, \quad \rho(p_0, M) \geq h,$$

and then there exist sequences t_{0k}, t_k, p_{0k} such that

$$\rho(t_k, p_{0k}, t_{0k}) \rightarrow 0 \text{ as } k \rightarrow +\infty.$$

From formula (4.7) we have

$$V_{t_0}(p_0) = \int_{t_0}^{+\infty} -\Phi(\tau, p_0, t_0) d\tau.$$

Let us set here $t_0 = t_{0k}, p_0 = p_{0k}$. We now break the integral in the right member up into three integrals:

$$V_{t_0}(p_0) = \int_{t_0}^{t_0+T} -\Phi(\tau, p_0, t_0) d\tau + \int_{t_0+T}^{T_1} + \int_{T_1}^{+\infty} -\Phi(\tau, p_0, t_0) d\tau.$$

The first of these integrals can be made as small as desired by a proper choice of the number T . The third integral can be made as small as desired by choosing the quantity T_1 sufficiently large. The middle integral can be made as small as desired by choosing p_{0k} and t_{0k} with sufficiently large index k . Then we obtain that there exists a quantity $k_0(\varepsilon)$, which has the property that

$$-V_{t_{0k}}(p_{0k}) < \varepsilon \text{ for } k \geq k_0(\varepsilon) \text{ and } \rho(p_{0k}, M) > h,$$

thus contradicting Property 2. This proves the theorem.

REMARK. In the case when the general system in a metric space has only Properties 1-3 (see Definition 1 of this chapter), it is also possible to formulate a theorem analogous to Theorem 78.

THEOREM 79. *In order for an invariant set M to be unstable, it is necessary and sufficient that there exist two families of functionals V_t and W_t , having the following properties:*

- 1) *on any element $p \in S(m, r)$, $r > 0$, there are defined functions $V_t(p)$ and $W_t(p)$ for $t \geq 0$;*
- 2) *the function $V_t(p)$ is bounded when $t \in [0, +\infty)$ and $p \in S(M, r)$;*
- 3) *for any $\delta > 0$ it is possible to find a point p_0 and a value of the parameter t_0 such that $V_{t_0}(p_0) > 0$, $\rho(p_0, M) < \delta$;*
- 4) *$(d/dt)V(t, p_0, t_0) = \lambda V(t, p_0, t_0) + W(t, p_0, t_0)$, $W_t(p) \gg 0$, $\lambda = \text{const.} > 0$.*

PROOF. *Necessity.* Let us take $\bar{\epsilon} > 0$, referred to in the definition of the instability of the set M . Let the point $p \in S(M, \bar{\epsilon})$. If $\rho(t, p, t_0) < \bar{\epsilon}$ when $t \geq t_0$, then we define $V_t(p)$ so as to make

$$V_{t_0}(p) = 0.$$

If there exists a quantity t_p such that $\rho(t, p, t_0) \geq \bar{\epsilon}$ when $t = t_p$ and $\rho(t, p, t_0) < \bar{\epsilon}$ when $t \in [t_0, t_p)$, then we set

$$V_t(F_{t_0}^t(p_0)) = e^{t-t_p} = V(t, p_0, t_0).$$

It is clear that in either case we have $dV/dt = V$.

In addition, the family of functionals V_t is bounded. Furthermore, by virtue of the definition of instability, for any $\delta > 0$ we can find a point p and a value of the parameter t_0 such that

$$\rho(t, p, t_0) \geq \bar{\epsilon} \text{ for } t = t_p.$$

Therefore

$$V_{t_0}(p) > 0 \text{ for } \rho(p, M) > \delta.$$

Thus the family of functionals constructed satisfies all the conditions of the theorem.

Sufficiency. Let the conditions of the theorem be satisfied. Then no matter what the quantity $\delta > 0$ is taken, it is possible to find a p_0 , $\rho(p_0, M) < \delta$ and a value of the parameter t_0 such that $V_{t_0}(p_0) > 0$. The function $V(t, p_0, t_0)$ satisfies the linear differential

equation

$$\frac{dV}{dt} = \lambda V + W$$

and the initial condition

$$V(t, p_0, t_0) = V_{t_0}(p_0) \text{ for } t = t_0.$$

We then have that

$$V(t, p_0, t_0) \geq V_{t_0}(p_0) e^{\lambda(t-t_0)}. \quad (4.8)$$

If we now assume that the set M is stable, it is found that the on one hand the motion $F_{t_0}^t(p_0)$ lies in its entirety when $t \geq t_0$ in the region where the family V_t is defined, and consequently $V(t, p_0, t_0)$ is bounded when $t \geq t_0$; on the other hand, along this motion inequality (4.8) holds, which contradicts the stability assumption made. Thus, the set M is unstable.

REMARK. An analogous theorem can be proved also in the general case, when the set $F_{t_0}^t(p_0)$ is not a single-point set.

Let us now describe a method which makes it possible, for certain general systems, to estimate the function

$$\rho(t, p_0, t_0) = \sup_{p \in F_{t_0}^t(p_0)} \rho(p, M).$$

Let us consider the two functions $M_1(x)$ and $M_2(x)$, defined for $x \geq 0$, continuous, and varying strictly monotonically for $x \in [0, +\infty)$. We consider a one-parameter family of functionals R_t , defined in a certain neighborhood $S(M, r)$ of the invariant set M , which depends on the parameter $t \geq 0$.

Let us assume that the following inequalities hold:

$$M_1(\rho(p, M)) \leq R_t(p) \leq M_2(\rho(p, M)) \quad (4.9)$$

when $t \in [0, +\infty)$. Then the function

$$\rho(t, p_0, t_0) = \rho(F_{t_0}^t(p_0), M)$$

can always be estimated from above and from below, provided the function $R_t(F_{t_0}^t(p_0)) = R(t, p_0, t_0)$ is known, or else some estimate is known for this function. In fact, we have, from inequality (4.9), that

$$\rho(t, p_0, t_0) \leq M_1^{-1}(R(t, p_0, t_0))$$

and

$$M_2^{-1}(R(t, p_0, t_0)) \leq \rho(t, p_0, t_0),$$

where M_1^{-1} and M_2^{-1} are the inverse functions of M_1, M_2 respectively.

Combining the last inequalities, we obtain the required two-sided estimate for the function $\rho(t, p_0, t_0)$. Let us assume that in a certain neighborhood $S(M, r)$ of the invariant set M there are defined two one-parameter families of functionals V_t and W_t , which have the following properties:

1) for any element $p \in S(M, r)$ there are defined functions $V_t(p)$ and $W_t(p)$ for all $t \in [0, +\infty)$;

2) the family of functionals V_t satisfies the inequality

$$\Phi_1(t)R_t^{l_1} \leq V_t \leq \Phi_2(t)R_t^{l_2}; \quad (4.10)$$

3) the family of functionals W_t satisfies the inequality

$$-\Psi_1(t)R_t^{k_1} \leq W_t \leq -\Psi_2(t)R_t^{k_2}; \quad (4.11)$$

4) the function $V(t, p_0, t_0)$ is continuously differentiable for all $t \geq t_0$, provided it is defined, and satisfies the relation

$$\frac{d}{dt} V(t, p_0, t_0) = W(t, p_0, t_0). \quad (4.12)$$

In short, $dV_t/dt = W_t$. Here $\Phi_i(t)$ and $\Psi_i(t)$ ($i = 1, 2$) are continuous positive functions, defined for $t \geq 0$, and k_i, l_i ($i = 1, 2$) are positive constants. However, the functions $\Psi_i(t)$ can also assume zero values. Furthermore, let us set

$$\frac{k_i}{l_i} = \lambda_i, f_i(\lambda_i, t) = \Psi_i(t)\Phi_i^{-\lambda_i}(t).$$

Let $\lambda > 0$ be a certain number, and let $t \geq t_0$ be such that $V(t, p_0, t_0) \neq 0$. We multiply both members of (4.12) by $V^{-\lambda}(t, p_0, t_0)$ and integrate between the limits t_0 to $t > t_0$, and then for $\lambda \in (0, 1)$ we obtain

$$V(t, p_0, t_0) = \sqrt[1-\lambda]{V_{t_0}(p_0)^{1-\lambda} + (1-\lambda) \int_{t_0}^t W(\tau, p_0, t_0) V^{-\lambda}(\tau, p_0, t_0) d\tau}. \quad (4.13)$$

When $\lambda = 1$, we obtain

$$V(t, p_0, t_0) = V_{t_0}(p_0) e^{\int_{t_0}^t W V^{-1} d\tau}; \quad (4.14)$$

when $\lambda > 1$, we obtain

$$V(t, p_0, t_0) = \frac{V_{t_0}(p_0)}{\sqrt[\lambda-1]{1 + V_{t_0}(p_0)^{\lambda-1}(1-\lambda) \int_{t_0}^t W V^{-\lambda} d\tau}}. \quad (4.15)$$

Let us consider several of the following cases:

1. $\lambda_i > 1$ ($i = 1, 2$). Applying in this case inequalities (4.10) and (4.11) to Eq. (4.15), we obtain

$$\begin{aligned} \frac{V_{t_0}(p_0)}{\sqrt[\lambda_1-1]{1 + (\lambda_1 - 1)V_{t_0}(p_0)^{\lambda_1-1} \int_{t_0}^t f_1(\lambda_1, \tau) d\tau}} &\leq V(t, p_0, t_0) \leq \\ &\leq \frac{V_{t_0}(p_0)}{\sqrt[\lambda_2-1]{1 + (\lambda_2 - 1)V_{t_0}(p_0)^{\lambda_2-1} \int_{t_0}^t f_2 d\tau}}. \end{aligned} \quad (4.16)$$

Applying inequality (4.10) to inequality (4.16), we obtain

$$\begin{aligned} \sqrt[\lambda_1]{\frac{\Phi_1(t_0)\Phi_2^{-1}(t)R_{t_0}(p_0)^{l_1}}{\sqrt[\lambda_1-1]{1 + (\lambda_1 - 1)(\Phi_2(t_0)R_{t_0}(p_0)^{l_1})^{\lambda_1-1} \int_{t_0}^t f_1(\lambda_1, \tau) d\tau}}} &\leq R(t, t_0, p_0) \leq \\ &\leq \sqrt[\lambda_2]{\frac{\Phi_2(t_0)\Phi_1^{-1}(t)R_{t_0}(p_0)^{l_2}}{\sqrt[\lambda_2-1]{1 + (\lambda_2 - 1)(\Phi_1(t_0)R_{t_0}(p_0)^{l_2})^{\lambda_2-1} \int_{t_0}^t f_2(\lambda_2, \tau) d\tau}}}. \end{aligned} \quad (4.17)$$

Here and everywhere in the sequel, R_t is a family of functionals having property (4.9). We denote furthermore by $N_1(t, p_0, t_0)$ the left member of inequality (4.17), while $N_2(t, p_0, t_0)$ denotes the right member of this inequality.

2. Let $\lambda_1 = \lambda_2 = 1$. Applying inequalities (4.10) and (4.11) to the relation (4.14), we obtain

$$V_{t_0}(\rho_0) e^{-\int_{t_0}^t f_1(\tau, 1) d\tau} \leq V(t, \rho_0, t_0) \leq V_{t_0}(\rho_0) e^{-\int_{t_0}^t f_2(\tau, 1) d\tau} \quad (4.18)$$

Applying again inequality (4.10) to inequalities (4.18), we obtain

$$\begin{aligned} \sqrt[1]{V_{t_0}(\rho_0) e^{-\int_{t_0}^t f_1(\tau, 1) d\tau}} &\leq R(t, \rho_0, t_0) \leq \\ &\leq \sqrt[1]{\Phi_1(t)^{-1} \Phi_1(t_0) R_{t_0}(\rho_0)^{1_1} e^{-\int_{t_0}^t f_1(\tau, 1) d\tau}} \\ &\leq \sqrt[1]{\Phi_1(t)^{-1} \Phi_2(t_0) R_{t_0}(\rho_0)^{1_2} e^{-\int_{t_0}^t f_2(\tau, 1) d\tau}}. \end{aligned} \quad (4.19)$$

We denote the left member of the inequality (4.19) by $P_1(t, \rho_0, t_0)$ and the right member by $P_2(t, \rho_0, t_0)$.

3. Let $0 < \lambda_i < 1$ ($i = 1, 2$). Applying inequalities (4.10) and (4.11) to (4.13), we obtain

$$\begin{aligned} \sqrt[1-\lambda_1]{V_{t_0}(\rho_0)^{1-\lambda_1} - (1-\lambda_1) \int_{t_0}^t f_1(\tau, \lambda_1) d\tau} &\leq V(t, \rho_0, t_0) \leq \\ &\leq \sqrt[1-\lambda_2]{V_{t_0}(\rho_0)^{1-\lambda_2} - (1-\lambda_2) \int_{t_0}^t f_2(\tau, \lambda_2) d\tau}. \end{aligned} \quad (4.20)$$

It is naturally understood that the last inequality holds when the expression under the root is positive and $V(t, \rho_0, t_0) \neq 0$. Applying again inequality (4.10) to (4.20), we obtain

$$\begin{aligned} \sqrt[1-\lambda_1]{\Phi_2(t)^{-1} \sqrt[1-\lambda_2]{(\Phi_1(t_0) R_{t_0}(\rho_0)^{1_1})^{1-\lambda_1} (1-\lambda_1) \int_{t_0}^t f_1(\tau, \lambda_1) d\tau}} &\leq \\ &\leq R(t, \rho_0, t_0) \leq \sqrt[1-\lambda_2]{\Phi_2(t)^{-1} \sqrt[1-\lambda_1]{(\Phi_2(t_0) R_{t_0}(\rho_0)^{1_2})^{1-\lambda_2} -}} \\ &- (1-\lambda_2) \int_{t_0}^t f_2(\tau, \lambda_2) d\tau. \end{aligned} \quad (4.21)$$

We denote the left member of inequality (4.21) by $Q_1(t, p_0, t_0)$ and the right member by $Q_2(t, p_0, t_0)$.

4. Let $\lambda_2 = 1$, $\lambda_1 > 1$; then, applying successively inequalities (4.10) and (4.11) to relations (4.14) and (4.15), we obtain

$$N_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq P_2(t, p_0, t_0). \quad (4.22)$$

5. Let $\lambda_1 = 1$, $\lambda_2 > 1$. Applying the inequalities (4.10) and (4.11) to relations (4.14) and (4.15) we obtain

$$P_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq N_2(t, p_0, t_0). \quad (4.23)$$

6. When $\lambda_1 > 1$ and $\lambda_2 < 1$, we have

$$N_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq Q_2(t, p_0, t_0). \quad (4.24)$$

7. When $\lambda_1 < 1$ and $\lambda_2 > 1$, we have

$$Q_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq N_2(t, p_0, t_0). \quad (4.25)$$

8. When $\lambda_1 = 1$, $\lambda_2 < 1$, we have

$$P_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq Q_2(t, p_0, t_0). \quad (4.26)$$

9. When $\lambda_2 = 1$, $\lambda_1 < 1$, we have

$$Q_1(t, p_0, t_0) \leq R(t, p_0, t_0) \leq P_2(t, p_0, t_0). \quad (4.27)$$

REMARK. All the inequalities obtained here hold only as long as:

- 1) the function $V(t, p_0, t_0)$ is defined;
- 2) for all $t \geq t_0$ as long as $V(t, p_0, t_0) \neq 0$;
- 3) the expressions under the roots in the estimates are not negative.

In those cases when Q_2 is on the right side of the inequalities or Q_1 on the left side, the motion

$$F'_{i_0}(p_0)$$

can enter into the invariant set M in a finite time. We denote by $S_j(t, p_0, t_0)$ the function in the right member of the inequality in the j -th case ($j = 1, \dots, 9$), i.e., $S_1 = N_2$, $S_2 = P_2$, $S_3 = Q_2$, etc.

THEOREM 80. *If for any $\varepsilon > 0$ it is possible to find a number $\delta > 0$ such that for a certain j the inequality $S_j(t, p_0, t_0) < \varepsilon$ holds when $t \geq t_0 \geq 0$ and $\rho(p_0, M) < \delta$, then the invariant set M is*

stable, and any motion

$$F_{t_0}^l(p_0)$$

for sufficiently small $\rho(p_0, M)$, namely, one making $\rho(p_0, M) < \delta(\varepsilon)$ where $\varepsilon \leq \bar{r}$, will satisfy the estimates corresponding to the j -th case as long as $V(t, p_0, t_0) \neq 0$ and the expressions under the roots are not negative. The quantity $\bar{r}$ can be found effectively, by using inequalities (4.9).

The proof follows from the foregoing arguments. We give now the necessary and sufficient conditions for stability and asymptotic stability, under which estimates of a specific type are satisfied. We assume that there exist two one-parameter families of functionals V_t and W_t , having Properties 1-4 for $l_i = 1$ ($i = 1, 2$), $k_i = l$ ($i = 1, 2$), $\Phi_i(t) = a_i \Phi(t)$, $\Psi_i(t) = b_i \Psi(t)$, where a_i and b_i are positive constants, and $\Phi(t)$ and $\Psi(t)$ are continuous positive functions defined for $t \geq 0$. However, the function $\Psi(t)$ can assume also zero values. Along any motion $F_{t_0}^l(p_0)$, as long as

$$F_{t_0}^l(p_0) \in S(M, r) \text{ and } V(t, p_0, t_0) \neq 0,$$

the following inequalities will be satisfied for $l \in (0, 1)$:

$$\begin{aligned} \Phi(t)^{-1} \sqrt[1-l]{\alpha_1(\Phi(t_0)R_{t_0}(p_0))^{1-l} - \alpha_2 \int_{t_0}^t \Psi(\tau)\Phi(\tau)^{-l} d\tau} &\leq R(t, p_0, t_0) \leq \\ &\leq \Phi(t)^{-1} \sqrt[1-l]{\beta_1(\Phi(t_0)R_{t_0}(p_0))^{1-l} - \beta_2 \int_{t_0}^t \Psi(\tau)\Phi(\tau)^{-l} d\tau} \quad (t \geq t_0); \quad (4.28) \end{aligned}$$

for $l > 1$:

$$\begin{aligned} \sqrt[l-1]{\frac{\gamma_1 \Phi(t)^{-1} \Phi(t_0) R_{t_0}(p_0)}{1 + \gamma_2 (\Phi(t_0) R_{t_0}(p_0))^{l-1} \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-l} d\tau}} &\leq R(t, p_0, t_0) \leq \\ &\leq \sqrt[l-1]{\frac{\delta_1 \Phi(t)^{-1} \Phi(t_0) R_{t_0}(p_0)}{1 + \delta_2 (\Phi(t_0) R_{t_0}(p_0))^{l-1} \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-l} d\tau}} \quad (t \geq t_0); \quad (4.29) \end{aligned}$$

and for $l = 1$:

$$\begin{aligned} \varepsilon_1 \Phi(t)^{-1} \Phi(t_0) R_{t_0}(p_0) e^{-\varepsilon_2 \int_{t_0}^t \Phi(\tau) \Phi(\tau)^{-1} d\tau} &\leq R(t, p_0, t_0) \leq \\ &: \sigma_1 \Phi(t)^{-1} \Phi(t_0) R_{t_0}(p_0) e^{-\sigma_2 \int_{t_0}^t \Phi(\tau) \Phi(\tau)^{-1} d\tau} \quad (t \geq t_0). \end{aligned} \quad (4.30)$$

We assume furthermore that

$$\int_{t_0}^t \Psi(t) \Phi(\tau)^{-1} d\tau \rightarrow +\infty \text{ as } t \rightarrow +\infty \text{ for a certain } l > 0. \quad (4.31)$$

THEOREM 81. *If the condition (4.31) is satisfied for a certain l , then in order for two families of functionals V_t and W_t to exist, having the following properties:*

1) *for any element $p \in M$ the functions $V_t(p)$ and $W_t(p)$ are defined for $t \geq 0$;*

2) *the family V_t satisfies the inequalities*

$$a_1 \Phi(t) R_t \leq V_t \leq a_2 \Phi(t) R_t;$$

3) *the family W_t satisfies the inequalities*

$$-b_1 \Psi(t) R_t^l \leq W_t \leq -b_2 \Psi(t) R_t^l;$$

4) *the function $V(t, p_0, t_0)$ is continuously differentiable, and the following equality*

$$\frac{d}{dt} V_t = W_t$$

holds, it is necessary and sufficient that there be satisfied the inequalities (4.30) if $l = 1$, or (4.29) if $l > 1$, or (4.28) if $l \in (0, 1)$, where $\alpha_i, \beta_i, \gamma_i, \delta_i, \sigma_i, (i = 1, 2)$ are certain positive constants.

PROOF. Necessity. Let there exist two families of functionals V_t and W_t , satisfying Conditions 1–4 of the present theorem. Then, as shown, one of the inequalities (4.28)–(4.30) will be satisfied, depending on whether $l > 1$, $l = 1$, or $l < 1$.

Sufficiency. Let the indicated inequalities be satisfied. We set

$$W_t = -\Psi(t) R_t^l \text{ and } V_{t_0}(p_0) = - \int_{t_0}^{+\infty} W(\tau, p_0, t_0) d\tau. \quad (4.32)$$

Let us assume that $l > 1$. Substituting inequalities (4.29) into relation (4.32), we obtain

$$\begin{aligned} & \int_{t_0}^{+\infty} \frac{\gamma_1^l \Phi(t)^{-l} \Phi(t_0)^l R_{t_0}^l(p_0) \Psi(t)}{\left[1 + \gamma_2(\Phi(t) R_{t_0}(p_0))^{l-1} \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-l} d\tau \right]^{l(l-1)}} dt \leq V_{t_0}(p_0) \leq \\ & \leq \int_{t_0}^{+\infty} \frac{\delta_1^l \Phi(t)^{-l} \Phi(t_0)^l R_{t_0}^l(p_0) \Psi(t)}{\left[1 + \delta_2(\Phi(t_0) R_{t_0}(p_0))^{l-1} \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-l} d\tau \right]^{l(l-1)}} dt. \end{aligned} \quad (4.33)$$

The integrals in both members of (4.33), taken over the interval $[t_0, +\infty)$, can be calculated by making the substitution

$$\theta = \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-l} d\tau.$$

When these integrals are evaluated, we obtain that $a_1 \Phi(t_0) R_{t_0} \leq \leq V_{t_0} \leq a_2 \Phi(t_0) R_{t_0}$, where a_1 and a_2 are positive constants, connected in a definite manner with the quantities γ_t and δ . Let us show now that $dV_t/dt = W_t$. In fact,

$$V(t, p_0, t_0) = \int_{t_0}^{+\infty} -W(\tau, p_0, t_0) d\tau,$$

which follows from (4.32), from which we obtain that

$$\frac{dV_t}{dt} = W_t.$$

Thus, the family of functionals V_t defined with the aid of Eqs. (4.32) satisfy all the conditions of the theorem. Let now $l = 1$. Substituting into (4.32) the inequalities (4.30), we obtain

$$\begin{aligned} & \varepsilon_1^l \int_{t_0}^{+\infty} \Phi(t)^{-l} \Phi(t_0)^l \Psi(t) R_{t_0}^l(p_0) e^{-\varepsilon_2^l \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-1} d\tau} dt \leq \\ & \leq V_{t_0}(p_0) \leq \sigma_1^l \int_{t_0}^{+\infty} \Phi(t)^{-l} \Phi(t_0)^l \Psi(t) R_{t_0}^l(p_0) e^{-\sigma_2^l \int_{t_0}^t \Psi(\tau) \Phi(\tau)^{-1} d\tau} dt. \end{aligned} \quad (4.34)$$

After evaluating the integrals in inequality (4.34), we obtain

$$a_1\Phi(t_0)R_{t_0} \leq V_{t_0} \leq a_2\Phi(t_0)R_{t_0},$$

where a_1 and a_2 are positive constants, which are related in a definite manner with ε_i and σ_i ($i = 1, 2$). As above, it is easy to show in this case, too, that

$$\frac{dV_t}{dt} = W_t.$$

Now let $0 < l < 1$. Substituting into (4.33) the inequalities (4.28), we obtain

$$\begin{aligned} \int_{t_0}^{t_1} \Phi(t)^{-l} \left[\alpha_1(\Phi(t_0)R_{t_0}(p_0))^{1-l} - \alpha_2 \int_{t_0}^t \Psi(\tau)\Phi(\tau)^{-l}d\tau \right]^{l/(1-l)} \Psi(t)dt &\leq \\ &\leq V_{t_0}(p_0) \leq \int_{t_0}^{t_2} \Phi(t)^{-l} \left[\beta_1(\Phi(t_0)R_{t_0}(p_0))^{1-l} - \right. \\ &\quad \left. - \beta_2 \int_{t_0}^t \Psi(\tau)\Phi(\tau)^{-l}d\tau \right]^{l/(1-l)} \Psi(t)dt. \end{aligned} \quad (4.35)$$

The quantities t_1 and t_2 are chosen such that the left and right members of inequality (4.28) vanish respectively at $t = t_1$ and $t = t_2$ ($t_2 \geq t_1$).

In this case the integral in Eq. (4.32) is taken only within finite limits, corresponding to $R(t, p_0, t_0) \neq 0$. After evaluating the integrals in inequality (4.35), we obtain

$$a_1\Phi(t_0)R_{t_0} \leq V_{t_0} \leq a_2\Phi(t_0)R_{t_0},$$

where a_1 and a_2 are positive constants, related in a definite manner with α_i and β_i ($i = 1, 2$).

COROLLARY 1. We denote by $U(t, p_0, t_0)$ the right member of one of inequalities (4.28)–(4.30).

If for any $\varepsilon > 0$ we can find a $\delta > 0$ such that when $\rho(p_0, M) < \delta$ we have

$$U(t, p_0, t_0) < \varepsilon \text{ for } t \geq t_0 \geq 0,$$

then the invariant set M is stable. If furthermore

$$U(t, p_0, t_0) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

then the invariant set M is asymptotically stable.

COROLLARY 2. *If for $l \in (0, 1)$ the set M is stable, then any motion enters into set M in a finite time and remains there.*

COROLLARY 3. *If the set M is stable (see Corollary 1), then in the presence of two families of functionals V_t and W_t , defined in the region $S(M, r)$, $r > 0$, and satisfying the condition of the Theorem 81, for any motion*

$$F_{t_0}^l(p_0),$$

beginning at sufficiently small $\rho(p_0, M)$, one of the estimates (4.28)–(4.30) holds for any $t \geq t_0$ as long as $V(t, p_0, t_0)$ does not vanish when $l \in (0, 1)$ and $t \leq t_1$.

Let us make now several general remarks.

REMARK 1. The first Lyapunov theorem on instability allows us to formulate sufficient conditions for instability of an invariant set M in the case of a general system.

THEOREM. *If there exist two one-parameter families of functionals V_t and W_t , having the following properties:*

1) *for any $p \in S(M, r)$, $r > 0$, M is a certain invariant set of a general system or an incomplete general system, the functions $V_t(p)$ and $W_t(p)$ of the real variable t are defined;*

2) *for any $c_1 > 0$ it is possible to find a $c_2 > 0$ such that when $\rho(p, M) > c_1$ we have $W_t(p) > c_2$, $t \in [0, +\infty)$;*

3) *$V_t(p) \rightarrow 0$ when $\rho(p, M) \rightarrow 0$ uniformly with respect to $t \geq 0$;*

4) *for any quantity $\delta > 0$ it is possible to find a point $p_0 \in R_1$ or in the case of a general system a $p_0 \in R$ and also a value of the parameter $t_0 \geq 0$, such that $V_{t_0}(p_0) > 0$ and $p_0 \in S(M, \delta)$;*

5) *the function $V(t, p_0, t_0)$ is continuously differentiable, and the following equality holds*

$$\frac{dV(t, p_0, t_0)}{dt} = W(t, p_0, t_0) \text{ for all } t \geq t_0,$$

as long as the function $V(t, p_0, t_0)$ is defined, then the invariant set M is unstable.

The proof of this theorem can be obtained directly by employing the same method as in reference [1].

REMARK 2. Theorem 74 also remains in force in the case of an incomplete general system. In this case the proof of sufficiency of the conditions, formulated in the theorem, is retained verbatim with mention only of the fact that the motion $F_{t_0}(p_0)$ is defined when $p_0 \in R_1$. In proving the necessity in the case of an incomplete general system, formula (4.5) does not define the family of functionals sought in every neighborhood of the set M . However, the functional defined with the aid of (4.5) can be extended to a certain sufficiently small neighborhood of the set M , for example, in the following manner. On the set of points p , such that $\rho(p, M) = \varepsilon'$, we set

$$V_{t_0}(p) = \inf_{\rho(p_0, M) = \varepsilon'} V_{t_0}(p_0),$$

where the *inf* is taken over the same points p_0 in which $V_{t_0}(p_0)$ has already been defined. By varying ε' continuously to 0, we define in this manner a family of functionals V_{t_0} , defined in a certain neighborhood of the set M .

It is easy to see that the family of functionals constructed in this manner will satisfy all the conditions of the Theorem 74.

REMARK 3. Theorems 75, 76 and 77 also remain valid for the case of incomplete general systems. Moreover, the proofs of sufficiency of the conditions formulated in these theorems remain in force. When proving the necessity one must take into account Remark 2.

REMARK 4. Theorems 78 and 79, like the preceding ones, can be carried over to the case of incomplete general systems. In this connection, the proof of the sufficiency of the conditions, as above, remains unchanged. The functionals constructed when proving the necessity in these theorems, in the case of an incomplete general system, are not defined in every sufficiently small neighborhood of the set M . However, they can be redefined in such a way as to satisfy all the conditions of these theorems.

Theorems 80 and 81 are based on the knowledge of the behavior of the functionals only along the motions, and therefore they retain their force also in the case of an incomplete general system. We note that many analogous results, connected with the estimates, have also been obtained in reference [38].

3. Certain applications to non-stationary systems of ordinary differential equations

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = f_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n), \quad (4.36)$$

the right members of which are defined for $t \geq 0$ and $X \in E_n$. Let $X = 0$ be a solution of the system (4.36) and let the functions $f_s(X, t)$ be continuous. We assume that any solution $X = X(t, X^{(0)}, t_0)$ is defined when $t \geq t_0 \geq 0$. Then, as was already shown, the system of Eqs. (4.36) defines in E_n a general system. Let $X = 0$ be an invariant set of this system, i.e. any integral curve $X = X(t, X^{(0)}, t_0)$ of the system (4.36), defined by the condition $X^{(0)} = 0$, has the property $X(t, X^{(0)}, t_0) \equiv 0$ when $t \geq t_0$.

THEOREM 82. *In order for the invariant set $X = 0$ to be stable, it is necessary and sufficient that there exist a function $V(X, t)$, defined for $t \geq 0$, $|X| \leq r$, having the following properties:*

- 1) $V(X, t)$ is positive definite;
- 2) $V(X, t) \rightarrow 0$ as $|X| \rightarrow 0$ uniformly with respect to $t \geq 0$;
- 3) the function $V(X(t, X^{(0)}, t_0), t)$ does not increase for $t \geq t_0$ where

$$|X(t, X^{(0)}, t_0)| \leq r.$$

If, furthermore, $V \rightarrow 0$ as $t \rightarrow +\infty$ when $|X^{(0)}| \leq \delta$, then the set $X = 0$ is asymptotically stable.

The proof follows from Theorems 74 and 75, for Theorem 82 is a particular case of these theorems.

We note that the sufficiency of the Theorem 82 can be obtained from [1], and the necessity from [7].

REMARK. If the conditions of Theorem 82 are satisfied, then in order for $X = 0$ to be uniformly asymptotically stable and uniformly attracting, it is necessary and sufficient that

$$V(X(t, X^{(0)}, t_0), t) \rightarrow 0 \text{ as } t - t_0 \rightarrow +\infty$$

uniformly with respect to $t_0 \geq 0$ and $|X^{(0)}| \leq \delta$, and that for any $h > 0$ (h sufficiently small) it is possible to find quantities $T > 0$

and $\alpha > 0$ such that

$$V(X(t, X^{(0)}, t_0), t) > \alpha \text{ for } t_0 \leq t \leq t_0 + T \text{ and } |X^{(0)}| > h.$$

This statement follows from Theorems 76 and 77.

We note that Theorem 78 makes it possible to formulate the necessary and sufficient conditions under which the point $X = 0$ is uniformly asymptotically stable and uniformly attracting.

Let us give now several applications of the method which makes it possible to estimate the quantity $|X(t, X^{(0)}, t_0)|$.

Let us consider the system of differential equations

$$\frac{dx_s}{dt} = p_s(x_1, \dots, x_n, t) \quad (s = 1, \dots, n). \quad (4.37)$$

The functions $p_s(X, t)$ are defined for $t \geq 0$, $|X|^2 \leq H$ and satisfy there certain conditions under which the system (4.37) has a solution $X = X(t, X^{(0)}, t_0)$ for any finite $(X^{(0)}, t_0)$, belonging to the region $t_0 \geq 0$, $|X^{(0)}|^2 < H$. We furthermore assume that $p_s(X, t) \rightarrow 0$ when $X = 0$ and for all $t > 0$. Let us consider two continuous positive functions $\varphi(t)$ and $\psi(t)$. We assume that they are defined for $t \geq 0$ and are such that the quantity

$$\kappa(t, t_0, k_2) = \sqrt[k+1]{\varphi(t_0)\varphi^{-1}(t)} e^{-\int_{t_0}^t k_2 \psi \varphi^{-1} dt} \rightarrow 0, \quad \int_{t_0}^t \psi \varphi^{-1} dt \rightarrow +\infty$$

as $t \rightarrow +\infty$ and is bounded for all $t \geq t_0 \geq 0$, where k and k_2 are certain positive constants.

THEOREM 83. *In order for the trivial solution of the system (4.37) to be asymptotically stable, and for any solution beginning in a sufficiently small neighborhood of the half-axis $X = 0$, $t \geq 0$ to satisfy the inequalities*

$$L_1 \kappa(t, t_0, k_1) |X^{(0)}|^2 \leq |X(t, X^{(0)}, t_0)|^2 \leq L_2 \kappa(t, t_0, k_2) |X^{(0)}|^2,$$

it is necessary and sufficient that there exist two functions

$$V = (|X|^2)^k \varphi(t) \sum_{i,l=1}^n a_{il}(X, t) x_i x_l,$$

$$W = -\psi(t) (|X|^2)^k \sum_{i,l=1}^n b_{il}(X, t) x_i x_l,$$

having the following properties:

1) V and W are defined and continuous when $|X|^2 \leq \delta_1$, $t > 0$, where $\delta_1 > 0$ is a sufficiently small quantity;

2)

$$a_1|X|^2 \leq \sum_{i,l=1}^n a_{il}x_i x_l \leq a_2|X|^2,$$

$$b_1|X|^2 \leq \sum_{i,l=1}^n b_{il}x_i x_l \leq b_2|X|^2,$$

where a_i and b_i ($i = 1, 2$) are certain positive constants;

3) the function W is continuous and $W = dV/dt$.

The proof of this theorem can be obtained from Theorem 81, in which one must set $R_l(p) = |X|^{2k+2}$, at $l = 1$, where R_l is the family of functionals referred to in Section 2 of Chapter IV.

EXAMPLE 1. Let us assume that for the system (4.3) there exist functions $V(X, t)$ and $W(X, t)$ satisfying the conditions of the Theorem 83, and, in this connection,

$$\varphi(t) = (t + 1)^\alpha, \quad \psi(t) = (t + 1)^\beta,$$

where $\alpha \geq 0$, $\beta \geq 0$ (they can also be negative). Then

$$\kappa(t, t_0, k_2) = \left(\frac{t_0}{t}\right)^{\alpha/(k+1)} e^{-(k_2/\gamma)(t^\gamma - t_0^\gamma)},$$

where $\gamma = \beta - \alpha + 1 \neq 0$. If $\gamma = 0$, then

$$\kappa'(t, t_0, k_2) = \left(\frac{t_0}{t}\right)^{[\alpha/(k+1)] + k_2}$$

Thus, for $\gamma \neq 0$ any solution of the system (4.37), beginning in a sufficiently small neighborhood of the half-axis $t \geq 0$, $X = 0$, satisfies the inequalities

$$L_1(X^{(0)})^2 \kappa(t, t_0, k_1) \leq X^2 \leq L_2(X^{(0)})^2 \kappa(t, t_0, k_2),$$

and when $\gamma = 0$ it satisfies the inequalities

$$L_1(X^{(0)})^2 \kappa'(t, t_0, k_1) \leq X^2 \leq L_2(X^{(0)})^2 \kappa'(t, t_0, k_2).$$

Let us investigate in greater detail the character of the asymptotic stability in this case.

For this purpose we consider the functions

$$\left(\frac{t_0}{t}\right)^{\alpha_1} \quad \text{and} \quad e^{-k_1(t^\gamma - t_0^\gamma)/\gamma}.$$

Let us specify a sufficiently small $H > 0$ and consider the equations

$$\left(\frac{t_0}{t}\right)^{\alpha_1} = H, \quad (4.38)$$

$$e^{-k_1(t^\gamma - t_0^\gamma)/\gamma} = H. \quad (4.39)$$

From (4.38), we find that

$$t - t_0 = t_0 \left(\frac{1}{\alpha_1 \sqrt[\alpha_1]{H}} - 1 \right),$$

from which it follows that $t - t_0 \rightarrow +\infty$ as $t_0 \rightarrow +\infty$. Thus, for $\alpha_1 > 0$, the function

$$\left(\frac{t_0}{t}\right)^{\alpha_1}$$

tends to 0 as $t \rightarrow +\infty$ non-uniformly with respect to t_0 . From Eq. (4.39), we obtain

$$t - t_0 = t_0 \left[\left(1 - \frac{\gamma \ln H}{k_2 t_0^\gamma} \right)^{1/\gamma} - 1 \right] \quad (4.40)$$

for $\gamma \in (0, 1)$; $t - t_0 \rightarrow +\infty$ as $t_0 \rightarrow +\infty$, which follows directly from (4.40).

When $\gamma > 1$, we have that $t - t_0 \rightarrow 0$ as $t_0 \rightarrow +\infty$ and for $\gamma = 1$,

$$t - t_0 = -\frac{\ln H}{k_2}.$$

Thus, the function

$$e^{-(k_1/\gamma)(t^\gamma - t_0^\gamma)} \xrightarrow[t - t_0 \rightarrow +\infty]{} 0$$

non-uniformly for $\gamma \in (0, 1)$.

It follows from the foregoing discussion that when $\gamma < 1$ the trivial solution of the system (4.37) will not be uniformly asymptotically stable, and when $\gamma > 1$, the trivial solution of the system (4.37) will not be uniformly attracting. Only when $\gamma = 1$ is the

trivial solution uniformly asymptotically stable and uniformly attracting. Thus, the stability of such a property is in a certain sense exclusive.

Let us consider the system of equations

$$\frac{dx_s}{dt} = t^\beta \sum_{i=1}^n p_{si} x_i + \sum_{i=1}^n p'_{si}(X, t) x_i \quad (s = 1, \dots, n), \quad (4.41)$$

where $\beta > -1$.

The quantities p_{si} in the system (4.41) are constant and real, and the functions $p'_{si}(X, t)$ are such that

$$p'_{si}(X, t) t^{-\beta} \rightarrow 0 \text{ as } t \rightarrow +\infty \text{ and } X \rightarrow 0.$$

We assume also that the functions

$$p'_{si}(X, t), \quad t \geq 0, \quad X^2 \leq H$$

have certain properties that insure the existence of a solution $X = X(t, X^{(0)}, t_0)$ of the system (4.41) and all $(X^{(0)}, t_0)$ from the indicated region. Let the eigenvalues of the matrix $P = ||p_{si}||$ have negative real parts.

We find the solution of the equation

$$\sum_{s=1}^n \frac{\partial V}{\partial x_s} \left(\sum_{s=1}^n p_{si} x_i \right) = - \sum_{i=1}^n x_i^2 \quad (4.42)$$

in the form of a quadratic form

$$V = \sum_{i,k=1}^n a_{ik} x_i x_k.$$

It is possible to find quantities $T > 0$ and $\delta_1 > 0$ such that when $t \geq T$ and $|X|^2 \leq \delta_1$ we have the inequality

$$t^\beta b_1 |X|^2 \leq - \frac{dV}{dt} \leq t^\beta b_2 |X|^2.$$

In this case $k = 0$ and $\alpha = 0$ and consequently any solution that starts out from a sufficiently small neighborhood of the half-axis $X = 0$, $t \geq 0$ will satisfy the inequalities

$$\begin{aligned} L_1(X^{(0)})^2 e^{-k_1(t^{\beta+1} - t_0^{\beta+1})/(\beta+1)} &\leq \\ &\leq |X|^2(t, X^{(0)}, t_0) \leq L_2(X^{(0)})^2 e^{-[k_2/(\beta+1)](t^{\beta+1} - t_0^{\beta+1})}. \end{aligned}$$

As follows from the above, the trivial solution of the system (4.41) is non-uniformly asymptotically stable for $\beta \in (-1, 0)$ and is not uniformly attracting when $\beta > 0$, and only for $\beta = 0$ is it both uniformly asymptotically stable and uniformly attracting.

Let us set

$$\kappa_2(t, t_0, c) = \frac{t^{-k} \sqrt[k+1]{\varphi(t_0) \varphi^{-1}(t)}}{\sqrt[k+1]{\left(1 + c \int_{t_0}^t \psi \sqrt[k+1]{\varphi^{-(l+1)}(t')} dt'\right)}}, \quad (4.43)$$

where k, l , and c are certain positive constants, and $l \geq k$. We assume that $\kappa_2 \rightarrow 0$ as $t \rightarrow +\infty$ and is bounded when $t \geq t_0 > 0$ and $c \geq 0$. We also assume that the quantity

$$\theta = \int_{t_0}^t \psi \sqrt[k+1]{\varphi^{-(l+1)}(t')} dt' \quad (4.44)$$

has the property that

$$\theta \rightarrow +\infty \text{ as } t \rightarrow +\infty.$$

THEOREM 84. *In order for the trivial solution of the system (4.37) to be asymptotically stable in the sense of Lyapunov and in order that the inequality*

$$M_1 |X^{(0)}|^2 \kappa_2(t, t_0, M_2 c) \leq |X|^2(t, X^{(0)}, t_0) \leq N_2 |X^{(0)}|^2 \kappa_2(t, t_0, N_2 c), \quad (4.45)$$

where

$$c = \left[(|X^{(0)}|)^2 \sqrt[k+1]{\varphi(t_0)} \right]^{l-k},$$

be satisfied for any solution, beginning in a sufficiently small neighborhood of the half-axis $X = 0$, $t \geq 0$ it is necessary and sufficient that there exist two functions $V(X, t)$ and $W(X, t)$ having the following properties:

1) the functions $V(X, t)$ and $W(X, t)$ are defined and continuous in the region $t \geq 0$ and $|X|^2 \leq \delta_1 < H$;

$$2) \quad V = (|X|^2)^{k\varphi(t)} \sum_{i,j=1}^n a_{ij}(X, t) x_i x_j,$$

$$W = -(|X|^2)^{l\psi(t)} \sum_{i,j=1}^n b_{ij}(X, t) x_i x_j,$$

where

$$a_{ij}(X, t) \text{ and } b_{ij}(X, t)$$

are such that there exist positive constants a_i and b_i ($i = 1, 2$) for which

$$a_1 |X|^2 \leq \sum a_{ij} x_i x_j \leq a_2 |X|^2 \text{ and } b_1 |X|^2 \leq \sum b_{ij} x_i x_j \leq b_2 |X|^2;$$

3) the total derivative of the function $V(|X|, t)$, calculated by virtue of the system (4.37), satisfies the equation

$$\frac{dV}{dt} = W.$$

The proof of this theorem can be obtained from Theorem 81, by setting $R_t(\phi) = |X|^{2k+2}$, where R_t is the family of functionals referred to in Section 2 of Chapter IV.

Let us consider the positive definite function $V_1(X)$, defined in a certain neighborhood of the point $X = 0$. We construct positive functions which decrease strictly monotonically to 0 (with decreasing $|X|$):

$$Y_1(|X|^2), \quad Y_2(|X|^2)$$

such that $Y_1 \leq V_1(X) \leq Y_2$. If the system (4.37) has a solution

$$X(t, X^{(0)}, t_0)$$

such that $|X|^2 \rightarrow 0$ as $t \rightarrow +\infty$, then also

$$V_1(X(t, X^{(0)}, t_0)) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Then, knowing the rate of decrease of the function V_1 along the solution under consideration, we can estimate the decrease of

$$|X^2(t, X^{(0)}, t_0)|.$$

In fact, along this solution we have the inequalities

$$Y_1(|X|^2) \leq V_1(X(t, X^{(0)}, t_0)) \leq Y_2(|X|^2);$$

hence

$$Y_2^{-1}(V_1(X(t, X^{(0)}, t_0))) \leq |X(t, X^{(0)}, t_0)|^2 \leq Y_1^{-1}(V_1(X(t, X^{(0)}, t_0))).$$

We set

$$\kappa_3(t, t_0, c, l) = \varphi(t_0)\varphi^{-1}(t) \sqrt[1-l]{1 + c \int_{t_0}^t \psi\varphi^{-l} dt'}, \quad (4.46)$$

where $l > 1$ and $c \geq 0$. We assume that κ_3 has the following properties:

- 1) $\kappa_3 \rightarrow 0$ as $t \rightarrow +\infty$;
- 2) κ_3 is bounded when $t > t_0 > 0$ and $c \geq 0$. We set

$$\tau = \int_{t_0}^t \psi\varphi^{-l} dt' \rightarrow +\infty \text{ as } t \rightarrow +\infty. \quad (4.47)$$

THEOREM 85. *In order for the trivial solution of the system (4.37) to be asymptotically stable and in order for any solution beginning in the region $|X^{(0)}|^2 \leq \delta$, $t_0 \geq 0$, where $\delta > 0$ is a sufficiently small number, to satisfy the inequalities*

$$M_1 V_1(X^{(0)}) \kappa_3(t, t_0, M_2 c(l), l) \leq V_1(X(t, X^{(0)}, t_0)) \leq \\ \leq N_1 V_1(X^{(0)}) \kappa_3(t, t_0, N_2 c(l), l),$$

where

$$c(l) = [\varphi(t_0) V_1(X^{(0)})]^{l-1},$$

it is necessary and sufficient that there exist two functions $V(X, t)$ and $W(X, t)$, satisfying the following conditions:

1) the functions $V(X, t)$ and $W(X, t)$ are defined and continuous in a sufficiently small neighborhood of the half-axis $t \geq 0$, $X = 0$;

$$2) a_1 \varphi(t) V_1(X) \leq V(X, t) \leq a_2 \varphi(t) V_1(X), \quad (4.48)$$

$W = -\psi W_1$, where W_1 satisfies the inequalities

$$b_1 V_1^{l_1} \leq W_1 \leq b_2 V_1^{l_1}. \quad (4.49)$$

The quantities a_i and b_i are positive;

$$3) \frac{dV}{dt} = W.$$

The proof follows from Theorem 81 if we put

$$R_t(p) = V_1(X).$$

THEOREM 86. *If there exist two functions $V(X, t)$ and $W(X, t)$, having the following properties:*

1) *the functions $V(X, t)$ and $W(X, t)$ are defined and continuous in the region $t \geq 0$, $|X|^2 < \delta$, $\delta > 0$ sufficiently small,*

2) *$V(x, t)$ and $W(x, t)$ satisfy the inequalities*

$$\begin{aligned} a_1 V_1(X) &\leq V(X, t)[\varphi(t)]^{-1} \leq a_2 V_1(X), \\ b_1 V_1^h(X)\psi(t) &\leq -W(X, t) \leq b_2 V_1^h(X)\psi(t), \end{aligned} \quad (4.50)$$

where

$$1 < l_1 \leq l_2;$$

$$3) \frac{dV}{dt} = W,$$

then the trivial solution of the system (4.37) is asymptotically stable and any solution, beginning in a sufficiently small neighborhood of the half-axis $X = 0$, $t \geq 0$, satisfies the inequalities

$$\begin{aligned} V_1(X^{(0)})M_{1\kappa_3}(t, t_0, M_{2c}(l_1), l_1) &\leq V_1(X(t, X^{(0)}, t_0)) \leq \\ &\leq N_1 V_1(X^{(0)})\kappa_3(t, t_0, N_{2c}(l_2), l_2), \end{aligned} \quad (4.51)$$

where

$$c(t) = [\varphi(t_0)V_1(X^{(0)})]^{t-1}.$$

The proof of this theorem follows from Theorem 80 if we set

$$R_t(p) = V_1(X).$$

REMARK. When $l_1 = 1$, $l_2 > 1$ the statement of the theorem remains in force, if inequalities (4.51) are replaced by

$$\begin{aligned} M_1 V_1(X^{(0)})\varphi(t_0)[\varphi(t)]^{-1} e^{-M_2 \int_{t_0}^t [\psi(t)/\varphi(t)] dt} &\leq V_1(X(t, X^{(0)}, t_0)) \leq \\ &\leq N_1 V_1(X^{(0)})\kappa_3(t_0, N_{2c}(l_2), l_2). \end{aligned}$$

When $l_1 = l_2 = 1$, the theorem remains valid if the inequalities (4.51) are replaced by

$$\begin{aligned} M_1 V_1(X^{(0)})\varphi(t_0)[\varphi(t)]^{-1} e^{-M_2 \int_{t_0}^t [\psi(t)/\varphi(t)] dt} &\leq V_1(X(t, X^{(0)}, t_0)) \leq \\ &\leq N_1 V_1(X^{(0)})\varphi(t_0)[\varphi(t)]^{-1} e^{-N_2 \int_{t_0}^t [\psi(t)/\varphi(t)] dt}. \end{aligned} \quad (4.52)$$

EXAMPLE 2. In the investigation of the general case, when there is one zero root, A. M. Lyapunov [1] has constructed for the system

$$\frac{dx}{dt} = X(x, x_1, \dots, x_n),$$

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si} x_i + X_s(x_1, \dots, x_n) \quad (s = 1, \dots, n),$$

where the functions X, X_s are holomorphic and their expansion in positive integral powers of the quantities $x, x_1, \dots, x_n$ does not contain terms linear in these quantities

$$X(x, 0, \dots, 0) = gx^m + \dots,$$

$$X_s(x, 0, \dots, 0) = g_s x^{m_s} + \dots,$$

where $m_s \geq m$, the functions $V(X), W(X)$ in the form

$$V = \sum_{i,j=1}^n a_{ij} x_i x_j + \frac{1}{2} x^2 + U,$$

$$W = gx^{m+1} + W_1(x_1, \dots, x_n) + S(x_1, \dots, x_n, x),$$

where $W_1(x_1, \dots, x_n)$ is a negative definite quadratic form.

Let $g < 0$, $m = 2\mu + 1$, where μ is a positive integer. Under these conditions the trivial solution of the system under consideration is asymptotically stable. For sufficiently small $|x_1|, \dots, |x_n|$, the functions V and W satisfy the inequalities

$$a_1(x^2 + \sum_{i=1}^n x_i^2) \leq V \leq a_2(x^2 + \sum_{i=1}^n x_i^2),$$

$$b_1(x^2 + \sum_{i=1}^n x_i^2)^{\mu+1} \leq -W \leq b_2(x^2 + \sum_{i=1}^n x_i^2).$$

Assuming $l_1 = 1$ and $l_2 = \mu + 1$, we obtain that any solution beginning with a sufficiently small neighborhood of the point $x_1 = \dots = x_n = x = 0$, satisfies the inequalities

$$\begin{aligned} M_1 e^{-M_2 t} (x^{(0)^2} + \sum_{i=1}^n x_i^{(0)^2}) &\leq x^2(t, x^{(0)}, x_1^{(0)}, \dots, x_n^{(0)}) + \\ &+ \sum_{i=1}^n x_i^2(t, x^{(0)}, x_1^{(0)}, \dots, x_n^{(0)}) \leq \\ &\leq N_1 (x^{(0)^2} + \sum_{i=1}^n x_i^{(0)^2})^{\frac{\mu+1}{2}} \sqrt{1 + N_2 (x^{(0)^2} + \sum_{i=1}^n x_i^{(0)^2})^{\mu} t}. \end{aligned}$$

To judge the accuracy of this estimate, let us consider the system

$$\frac{dx}{dt} = -x^{2\mu+1}, \quad \frac{dx_s}{dt} = \sum_{i=1}^n p_{si} x_i.$$

Its direct integration yields

$$x = \frac{x^{(0)}}{\sqrt[2\mu]{1 + 2\mu x^{(0)2\mu} t}}, \quad |x_s| \leq \left[\sum_{i=1}^n x_i^{(0)s} \right]^{\frac{1}{s}} \alpha e^{-\beta t}$$

and

$$|x_s| \geq \left[\sum_{i=1}^n x_i^{(0)s} \right]^{\frac{1}{s}} \alpha_1 e^{-\beta_1 t},$$

which is in line with the estimate given above.

EXAMPLE 3. Let us consider the system

$$\frac{dx_s}{dt} = \sum_{i=1}^n p_{si}(X, t) x_i \quad (s = 1, \dots, n).$$

We take the definite positive quadratic form

$$V = \sum_{i,j=1}^n c_{ij} x_i x_j.$$

Let us assume that the function

$$W = \sum_{s=1}^n \frac{\partial V}{\partial x_s} \left(\sum_{i=1}^n p_{si}(X, t) x_i \right)$$

satisfies the following inequalities

$$b_1 \left(\sum_{i=1}^n x_i^2 \right)^{l_1} \psi(t) \leq -W \leq b_2 \left(\sum_{i=1}^n x_i^2 \right)^{l_2} \psi(t);$$

then any solution of the system under consideration will satisfy the estimate (4.51).

In the case $l_1 = l_2 = 1$ we have the estimate (4.52). The case $l_1 = l_2 = 1$ was considered in reference [18].

SOLUTION OF THE STABILITY PROBLEM FOR A SYSTEM OF PARTIAL DIFFERENTIAL EQUATIONS

1 Some general propositions

We consider the system of partial differential equations

$$\frac{\partial u_s}{\partial t} = f_s(x_1, \dots, x_k, u_1, \dots, u_n, \frac{\partial u_i}{\partial x_j}, \dots), \quad (5.1)$$

$$(i, s = 1, \dots, n) \quad (j = 1, \dots, k)$$

and assume that the right members of the system (5.1) are defined and continuous in a certain region G of variation of all its arguments, contained in N -dimensional Euclidean space. We consider the functional space Φ , the elements of which are the vector functions $\varphi = (\varphi_1, \dots, \varphi_n)$, where $\varphi_i(x_1, \dots, x_k)$ is a certain function defined in E_k . The functional space Φ will be considered metric or normed linear. We assume that for any element $\varphi \in \Phi$ we can construct a solution of the system (5.1), namely $U = U(\varphi, t)$ where $U = (u_1, \dots, u_n)$ such that $U(\varphi, t)$ is defined for all $t \in (-\infty, +\infty)$ and $U(\varphi, t) \in \Phi$ when $t \in (-\infty, +\infty)$, $U(\varphi, 0) = \varphi$ and, in addition, $U(\varphi, t)$ is continuous in both its arguments. Let, furthermore, $U(\varphi, t) = 0$ when $\varphi = 0$. Thus, the system (5.1) has the trivial solution. Let us discuss the stability of this solution. If the assumptions made above with respect to the system (5.1) are satisfied, then the dynamical system $U(\varphi, t)$ is defined in the functional space and the point $\varphi = 0$ is a closed invariant set of this system. Let us recall the basic definitions that apply in this case.

DEFINITION 1. The solution $U = 0$ of the system (5.1) is called *stable* if for any $\varepsilon > 0$ we can find a $\delta > 0$ such that when

$\rho(\varphi, 0) < \delta$ we have

$$\rho(U(\varphi, t), 0) < \varepsilon \text{ for } t \geq 0.$$

If, furthermore,

$$\rho(U(\varphi, t), 0) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

then the solution $U = 0$ is called *asymptotically stable*. Here, as everywhere else in the sequel, $\rho(\varphi, 0)$ is the metric distance between the elements φ and 0 .

DEFINITION 2. The set of all points $\varphi \in \Phi$ having the property

$$\rho(U(\varphi, t), 0) \rightarrow 0 \text{ as } t \rightarrow +\infty$$

will be denoted by A . If the solution $U = 0$ is asymptotically stable, then the set A is called the *region of asymptotic stability of the trivial solution* of the system (5.1), $0 \in A$.

DEFINITION 3. An asymptotically stable solution $U = 0$ of the system (5.1) is called *uniformly asymptotically stable* if

$$\rho(U(\varphi, t), 0) \rightarrow 0 \text{ as } t \rightarrow +\infty$$

uniformly with respect to φ ,

$$\rho(\varphi, 0) < \delta_1.$$

DEFINITION 4. The point $U = 0$ is called, in the case of asymptotic stability, *uniformly attracting*, if for any sufficiently small $h > 0$ it is possible to find quantities $T > 0$ and $\alpha > 0$ such that

$$\rho(U(\varphi, t), 0) \geq \alpha$$

when

$$t \in [0, T] \text{ and } \rho(\varphi, 0) \geq h.$$

THEOREM 87. In order for an open invariant set A , containing a sufficiently small neighborhood of the point $\varphi = 0$, to be the region of asymptotic stability of a uniformly asymptotically stable and uniformly attracting trivial solution of the system (5.1), it is necessary and sufficient that there exist two functionals $V(\varphi)$ and $W(\varphi)$, with the following properties:

- 1) the functional $V(\varphi)$ is defined and continuous in A , the functional $W(\varphi)$ is defined and continuous in Φ ;
- 2) $-1 < V(\varphi) < 0$ for $\varphi \in A$; $W(\varphi) > 0$, $\varphi \in \Phi$ and $\rho(\varphi, 0) > 0$;

3) for any quantity $\gamma_2 > 0$ it is possible to find quantities γ_1 and α_1 such that $V(\varphi) < -\gamma_1$ when $\rho(\varphi, 0) \geq \gamma_2$; $W(\varphi) > \alpha_1$ when $\rho(\varphi, 0) \geq \gamma_2$;

4) the functionals $W(\varphi)$ and $V(\varphi) \rightarrow 0$ as $\rho(\varphi, 0) \rightarrow 0$;

5) if $\bar{\varphi}$ is a point on the boundary of the region A , $\rho(\varphi, 0) \neq 0$, then

$$\lim_{\varphi \in A, \rho(\varphi, \bar{\varphi}) \rightarrow 0} V(\varphi) = -1;$$

6) the total derivative of the functional $V(\varphi)$, calculated on the motion $U(\varphi, t)$, satisfies the relation

$$\frac{dV(U(\varphi, t))}{dt} = W(U(\varphi, t))(1 + V(U(\varphi, t))),$$

or more briefly

$$\frac{dV}{dt} = W(1 + V).$$

The proof of this theorem follows from Theorem 19, in view of the fact that the system (5.1) defines a dynamical system having $\varphi = 0$ as an invariant set.

REMARK. The set of elements $\varphi \in \Phi$, satisfying the relation

$$V(\varphi) = \lambda, \quad \lambda \in (-1, 0),$$

forms a section of A in the sense that any solution $U(\varphi, t)$, $\varphi \in A$ has a common point with that set, and only one such point. We note that any theorem proved in Chapter I and pertaining to a dynamical system $f(p, t)$ in R can be reformulated for the case considered here.

Let us examine the system of partial differential equations

$$\frac{\partial u_s}{\partial t} = f_s \left(t, x_1, \dots, x_k, u_1, \dots, u_n, \frac{\partial u_i}{\partial x_j}, \dots \right), \quad (5.2)$$

$$(s, i = 1, \dots, n; j = 1, \dots, k)$$

the right members of which are defined and continuous at $t \geq 0$ and in a certain region G of variation of its arguments $x_1, \dots, x_k, u_1, \dots, u_n, \partial u_i / \partial x_j, \dots$ of N -dimensional Euclidean space.

We assume that for any element $\varphi \in \Phi$ and $t_0 \geq 0$ it is possible to construct a solution of the system (5.2) $U = U(\varphi, t, t_0)$, having the following properties:

1. for any $\varphi \in \Phi$, $U(\varphi, t, t_0)$ is defined for all $t > t_0$ and $U(\varphi, t, t_0) \in \Phi$ for all $t \geq t_0 \geq 0$.

2. $U(\varphi, t, t_0) = \varphi$ when $t = t_0$.

It is natural that more than one solution with the foregoing properties will correspond in general to the selected element $\varphi \in \Phi$ and to the quantity $t_0 \geq 0$. We shall consider, however, only those solutions of the system (5.2) which have these properties. If any solution of the system (5.2) is defined for $t \geq t_0$, then the system of Equations (5.2) defines a general system in the space Φ .

Let us assume that the system (5.2) has a trivial solution $U \equiv 0$, which is an invariant set of this general system, i.e., it follows from $\varphi = 0$ that

$$U(\varphi, t, t_0) = 0 \text{ for } t \geq t_0. \quad (5.3)$$

We recall the basic definitions pertaining to this case.

DEFINITION 5. An invariant set $U = 0$ is called *stable* if for any $\varepsilon > 0$ one can find a $\delta > 0$ such that when $\rho(\varphi, 0) < \delta$ one obtains

$$\rho(U(\varphi, t), t_0, 0) < \varepsilon \text{ for } 0 \leq t_0 \leq t.$$

If, in addition,

$$\rho(U(\varphi, t, t_0), 0) \rightarrow 0 \text{ as } t \rightarrow +\infty,$$

then $U = 0$ is called *asymptotically stable*.

THEOREM 88. In order for the solution $U = 0$ of the system (5.2), having the property (5.3), to be stable it is necessary and sufficient that there exist a one-parameter family of functionals V_t , having the following properties:

1) on any element $\varphi \in S(0, r)$ there is defined a function $V_t(\varphi)$ of the real argument t , defined for $t \geq 0$, where $S(0, r)$ is the set of the functions φ , having the property

$$0 < \rho(\varphi, 0) < r;$$

2) for any sufficiently small $c_1 > 0$ it is possible to find a quantity $c_2 > 0$ such that $V_t(\varphi) > c_2$ for $\rho(\varphi, 0) > c_1$ and all $t \geq 0$;

3) $V_t(\varphi) \rightarrow 0$ uniformly relative to $t \geq 0$ as $\rho(\varphi, 0) \rightarrow 0$;

4) the function $V_t(U(\varphi, t, t_0))$ does not increase for all $t \geq t_0$ for which it is defined;

5) if, furthermore, the function $V_t(U(\varphi, t, t_0)) \rightarrow 0$ as $t \rightarrow +\infty$ for all $t_0 > 0$ and $\rho(\varphi, 0) < \delta_1$, where $\delta_1 > 0$ is sufficiently small, then the solution $U = 0$ of the system (5.2), satisfying condition (5.3), will be asymptotically stable, and, conversely, if the solution $U = 0$ is asymptotically stable, then item 5 holds.

The proof of Theorem 88 follows from Theorems 74 and 75.

DEFINITION 6. An asymptotically stable trivial solution of the system (5.2) is called *uniformly asymptotically stable* if

$$\rho(U(\varphi, t, t_0), 0) \rightarrow 0 \text{ as } t - t_0 \rightarrow +\infty$$

uniformly with respect to $t_0 \geq 0$ when $\rho(\varphi, 0) < \delta_2$, where $\delta_2 > 0$ is sufficiently small.

DEFINITION 7. An asymptotically stable trivial solution of the system (5.2) is called *uniformly attracting* if for any sufficiently small quantity $h > 0$ it is possible to find quantities $T > 0$ and $\alpha > 0$ such that

$$\rho(U(\varphi, t, t_0), 0) > \alpha \text{ when } 0 \leq t_0 \leq t \leq t_0 + T$$

and

$$\rho(\varphi, 0) > h$$

(for all $t_0 \geq 0$).

THEOREM 89. Let the solution of the Cauchy problem for the system (5.2) be unique in the space Φ .

In order for the trivial solution of the system (5.2), possessing property (5.3), to be uniformly asymptotically stable and uniformly attracting, it is necessary and sufficient that there exist two families of functionals V_t and W_t with the following properties:

1) for any $\varphi \in S(0, r)$ ($r > 0$ sufficiently small), there are defined quantities $V_t(\varphi)$ and $W_t(\varphi)$, which are functions of the real argument $t \geq 0$;

2) for any sufficiently small quantity $c_1 > 0$ it is possible to find positive quantities c_2 and c_3 such that $V_t(\varphi) < -c_2$, $W_t(\varphi) > c_3$ for $\rho(\varphi, 0) > c_1$ and $t \geq 0$;

3) $V_t(\varphi)$ and $W_t(\varphi) \rightarrow 0$ as $\rho(\varphi, 0) \rightarrow 0$ uniformly with respect to $t \geq 0$;

$$4) \frac{d}{dt} V_t(U(\varphi, t, t_0)) = W_t(U(\varphi, t, t_0)).$$

The proof of the theorem follows from Theorem 78.

Let us give still another theorem on the instability of the trivial solution of the system (5.2). Let the system (5.2) satisfy the conditions of uniqueness of the solution of the Cauchy problem in the space Φ .

THEOREM 90. *In order for the trivial solution of the system (5.2) to be unstable, it is necessary and sufficient that there exist two families of functionals V_t and W_t having the following properties:*

- 1) *on any element $\varphi \in S(0, r)$, $r > 0$, there are defined functions $V_t(\varphi)$ and $W_t(\varphi)$ for $t \geq 0$;*
- 2) *the function $V_t(\varphi)$ is bounded for $t \in [0, +\infty)$ and $\varphi \in S(0, r)$;*
- 3) *for any $\delta > 0$ it is possible to find a point φ_0 and a value of the parameter t_0 such that $V_{t_0}(\varphi_0) > 0$, $\rho(\varphi_0, M) < \delta$;*

$$4) \quad \frac{d}{dt} V_t(U(\varphi, t, t_0)) = \lambda V_t(U(\varphi, t, t_0)) + W_t(U(\varphi, t, t_0)), \quad W_t(\varphi) \geq 0,$$

$\lambda = \text{constant} > 0$.

The proof follows from Theorem 79.

REMARK. The sufficiency of the conditions in Theorems 89 and 90 is established without using the condition of uniqueness of the solution of the Cauchy problem for the system (5.2) in the space Φ .

The necessity of the conditions can also be proved without using the uniqueness by slightly complicating those arguments, which were carried out in proving Theorems 78 and 79. Therefore, the requirement of uniqueness, stated in Theorems 89 and 90 is inessential and can be lifted.

We note that the general theory developed in Chap. IV allows us to raise the question of stability of solutions of systems of partial differential equations not only in the case of the Cauchy problem but also in the case of various mixed problems for systems of higher order equations.

In the following sections we shall give several specific examples of the stability problem for systems of partial differential equations.

2. Stability of solutions of quasi-linear systems of special type

Let us consider the system of partial differential equations

$$\frac{\partial u_s}{\partial t} = f_s(u_1, \dots, u_n) + \sum_{i=1}^k b_i \frac{\partial u_s}{\partial x_i} \quad (s = 1, \dots, n), \quad (5.4)$$

where the $f_s(u_1, \dots, u_n)$ are continuous bounded functions, defined in E_n ; b_i ($i = 1, \dots, k$) are real constants.

Let us consider the normed linear space Φ , the elements of which are the vector functions $\varphi = (\varphi_1, \dots, \varphi_n)$. The components of the vector φ are continuously differentiable functions of the variables $x_1, \dots, x_k$, defined in E_k and bounded there together with their derivatives. Let

$$\|\varphi\| = \sqrt{\sup_{E_k} \sum_{i=1}^n \varphi_i^2}.$$

We assume that the system (5.4) has the solution $U = 0$, $U = (u_1, \dots, u_n)$. Let us analyze the problem of the stability of this solution.

THEOREM 91. *If the functions f_i are continuously differentiable in E_n , then for any element $\varphi \in \Phi$ there exists a family of functions $U(\varphi, t)$, having the following properties:*

- 1) *the family $U(\varphi, t)$ is defined for $t \in (-\infty, +\infty)$ and for any t we have $U(\varphi, t) \in \Phi$;*
- 2) *$U(\varphi, t)$ is continuous in both its arguments, i.e. $U(\varphi_1, t_1) - U(\varphi_2, t_2) \rightarrow 0$ as $|t_1 - t_2| \rightarrow 0$, $|t_2| < +\infty$ and $\|\varphi_1 - \varphi_2\| \rightarrow 0$;*
- 3) *$U(\varphi, t) = \varphi$ for $t = 0$;*
- 4) *the function $U(\varphi, t)$ satisfies the system (5.4).*

PROOF. Let us consider the system of differential equations

$$\frac{du_s}{dt} = f_s(u_1, \dots, u_n). \quad (5.5)$$

The right members of this system are defined in E_n , are bounded, and are continuously differentiable in all their arguments. Therefore there exists a unique vector function $U(t, U^{(0)})$ of two arguments $(t, U^{(0)})$, having the following properties:

- 1) *the function $U(t, U^{(0)})$ is defined for $U^{(0)} \in E_n$ and $t \in (-\infty, +\infty)$, and $U(t, U^{(0)}) \in E_n$;*
- 2) *the function $U(t, U^{(0)})$ is continuous in both arguments, i.e.,*

$$|U(t_1, U_1^{(0)}) - U(t_2, U_2^{(0)})| \rightarrow 0$$

when $|t_1 - t_2| \rightarrow 0$, and

$$|U_1^{(0)} - U_2^{(0)}| = \sqrt{\sum_{s=1}^n (U_{s1}^{(0)} - U_{s2}^{(0)})^2} \rightarrow 0$$

and is continuously differentiable with respect to the parameter t and the components of the vector $U^{(0)}$;

3) $U(t, U^{(0)})$ satisfies the system (5.5) and $U(t, U^{(0)}) = U^{(0)}$ when $t = 0$.

We note that the properties 2) and 3) follow from the existence and uniqueness theorems for ordinary differential equations. The first property means that any integral curve of the system (5.5) is defined for all $t \in (-\infty, +\infty)$; this follows from the fact that the functions f_s are bounded, which is a sufficient condition for the continuability of any solution of the system (5.5). Let us consider the vector function

$$U = U(t, \varphi(x_1 + b_1 t, \dots, x_k + b_k t)).$$

When $\varphi \in \Phi$ this function is continuously differentiable with respect to the variables $x_1, \dots, x_k, t$. Let us show that for any finite value of the parameter t this function is bounded together with its first order derivatives with respect to the variables $x_1, \dots, x_k$. In fact, we take the s -component of this function:

$$u_s = u_s(t, \varphi_1, \dots, \varphi_n(x_i + t b_i)) \quad (s = 1, \dots, n), (i = 1, \dots, k).$$

Differentiating with respect to the variable x_j , we obtain

$$\frac{\partial u_s}{\partial x_j} = \sum_{i=1}^n \frac{\partial u_s}{\partial \varphi_i} \frac{\partial \varphi_i}{\partial x_j}.$$

The values of the vector function $\varphi(x_i + t b_i)$ belong to the bounded region G of the space E_n . Therefore, for fixed t , each set of the quantities $x_1, \dots, x_k$ can be identified with a vector $U^{(0)} \in G$ so as to obtain

$$U(t, U^{(0)}) = U(t, \varphi(x_i + t b_i)),$$

from which, by virtue of the continuability of any solution of the system (5.5) to all values of $t \in (-\infty, +\infty)$, this assertion follows, by virtue of the boundedness of the region G and the boundedness of the functions $\partial \varphi_i / \partial x_j$. Thus,

$$U(t, \varphi(x_i + t b_i)) \in \Phi$$

for all $t \in (-\infty, +\infty)$. We note that $U(t, \varphi) = \varphi$ when $t = 0$ by virtue of Property 3), which is satisfied by the function $U(t, U^{(0)})$.

Let us show now that $U(t, \varphi(x_i + tb_i))$ is continuous in both its arguments.

From Property 2) for the function $U(t, U^{(0)})$, we obtain that

$$|U(t_1, \varphi_1(x_i + t_1 b_i)) - U(t_2, \varphi_2(x_i + t_2 b_i))| \rightarrow 0$$

$$\text{as } |t_2 - t_1| \rightarrow 0$$

and

$$|\varphi_1(x_i + t_1 b_i) - \varphi_2(x_i + t_2 b_i)| \rightarrow 0,$$

from which we obtain

$$\begin{aligned} |\varphi_1(x_i + t_1 b_i) - \varphi_2(x_i + t_2 b_i)| &\leq |\varphi_1(x_i + t_1 b_i) - \varphi_1(x_i + t_2 b_i)| + \\ &+ |\varphi_1(x_i + t_2 b_i) - \varphi_2(x_i + t_2 b_i)| \leq \|\varphi_1 - \varphi_2\| + \\ &+ \sqrt{\sum_{s=1}^n [\varphi_{s1}(x_i + t_1 b_i) - \varphi_{s1}(x_i + t_2 b_i)]^2}. \end{aligned}$$

Applying Lagrange's theorem on finite increments, we obtain

$$|\varphi_1(x_i + t_1 b_i) - \varphi_2(x_i + t_2 b_i)| \leq M |t_2 - t_1| + \|\varphi_1 - \varphi_2\|,$$

where M is a quantity defined by the maximum modulus of the quantities $\partial \varphi_{s1} / \partial x_i$. From this it follows that

$$\|U(t_1, \varphi_1(x_i + t_1 b_i)) - U(t_2, \varphi_2(x_i + t_2 b_i))\| \rightarrow 0$$

$$\text{as } |t_2 - t_1| \rightarrow 0 \text{ and } \|\varphi_1 - \varphi_2\| \rightarrow 0.$$

Let us show now that the function $U(t, \varphi(x_i + tb_i))$ is a solution of the system (5.4). In fact,

$$\frac{\partial u_s}{\partial t} = \left(\overline{\frac{\partial u_s}{\partial t}} \right) + \sum_{i=1}^k \frac{\partial u_s}{\partial \xi_i} \frac{\partial \xi_i}{\partial t},$$

where $\overline{(\partial u_s / \partial t)}$ is the derivative calculated under the assumption that $\varphi = \text{const.}$, and $\xi_i = x_i + tb_i$. From this relation we have

$$\frac{\partial u_s}{\partial t} = f_s(u_1, \dots, u_n) + \sum_{i=1}^k b_i \frac{\partial u_s}{\partial x_i} \quad (s = 1, \dots, n).$$

This proves the theorem.

REMARK 1. The requirement of the boundedness of the functions $f_s(u_1, \dots, u_n)$ has been used only in order to be able to continue

the solutions of the system (5.5) to all values $t \in (-\infty, +\infty)$. Therefore, the requirement of boundedness can be replaced by the requirement of continuability of the solutions of the system (5.5).

REMARK 2. Theorem 91 shows that the system (5.4) defines a dynamical system in the space Φ .

Let us proceed now to solving the problem of the stability of the trivial solution of the system (5.4). We assume that the trivial solution of the system (5.5) is asymptotically stable. Then, as follows from Theorem 22, there exist two functions $V(u_1, \dots, u_n)$ and $W(u_1, \dots, u_n)$, having the following properties:

- 1) the function V is defined in the region A of asymptotic stability of the trivial solution of the system (5.5) and satisfies there the inequality $-1 < V < 0$; the function W is defined in E_n and is such that for any $\alpha > 0$ one can find a $\beta > 0$ such that when $|U| > \alpha$ we have $W > \beta$, $V(0, \dots, 0) = W(0, \dots, 0) = 0$;
- 2) the total derivative of the function V , calculated by virtue of the system (5.5), satisfies the relation $dV/dt = W(1 + V)$;
- 3) $V \rightarrow -1$ as $|\bar{U} - U| \rightarrow 0$, $U \in A$, $\bar{U} \in \bar{A} \setminus A$, $|\bar{U}| \neq 0$.

As follows from Theorems 19 and 22, the condition that W is defined in E_n is not necessary. It is sufficient to have W defined in $\bar{A}$.

The set of the points of region A having the property $1 + V(u_1, \dots, u_n) = \lambda$ will be denoted by S_λ , $\lambda \in (0, 1)$. The set of points S_λ forms a closed surface, bounding the region G_λ , which contains a sufficiently small neighborhood of the point $u_1 = \dots = u_n = 0$.

Let us proceed to constructing the region of asymptotic stability of the trivial motion $\varphi = 0$ of a dynamical system, defined by the system of equations (5.4). The function $\varphi \in \Phi$ will be assigned to the class φ_λ if all the values of the function φ belong to the region G_λ . Moreover, there is at least one point $(\xi_1, \dots, \xi_k)$ such that $U^{(0)} = \varphi(\xi_1, \dots, \xi_k) \in S_\lambda$. [We shall include in the class φ_λ also those functions, the closure of the set of values of which is strictly contained in $\bar{G}_\lambda$.] Thus, a correspondence is established between the functions of class φ_λ and the points of the closed surface S_λ , i.e., to each $U^{(0)} \in S_\lambda$ there corresponds a function $\varphi_0 \in \varphi_\lambda$ such that $\varphi_0(\xi_1, \dots, \xi_k) = U^{(0)}$, and, conversely, to any function $\varphi_0 \in \varphi_\lambda$ there corresponds at least one point $U^{(0)} \in S_\lambda$. Thanks to this corres-

pondence, it is possible to define, by means of the function $V(u_1, \dots, u_n)$, a functional

$$V_1(\varphi) = V(u_1^{(0)}, \dots, u_n^{(0)}),$$

where $\varphi \in \varphi_\lambda$, $U^{(0)} \in S_\lambda$ is a point corresponding to the function φ . Let us put $\lambda(t) = V(U(t), U^{(0)}) + 1$ and let us show that $\varphi \in \varphi_\lambda$ implies that

$$U(t, \varphi(x_t + tb_t)) \in \varphi_{\lambda(t)}.$$

In fact, for any fixed value of t we set $x_t = \xi_t - tb_t$. Then

$$U(t, \varphi) = U(t, U^{(0)}),$$

from which it follows that the function

$$U(t, \varphi(x_t + tb_t))$$

assumes values that lie on the surface $S_{\lambda(t)}$. It is easy to show also that the remaining values of this function lie in the region $\bar{G}_{\lambda(t)}$. Thus, it is shown that

$$U(t, \varphi(x_t + tb_t)) \in \varphi_{\lambda(t)},$$

if $\varphi \in \varphi_\lambda$. Thanks to this, the functional defined on the functions of class φ_λ can be calculated along the entire motion

$$U(t, \varphi(x_t + tb_t))$$

with the aid of the formula

$$V(U(t, U^{(0)})) = V_1(U(t, \varphi(x_t + tb_t))).$$

From this it follows that

$$\frac{dV_1}{dt} = \frac{dV}{dt} = W(U(t, U^{(0)}))(1 + V_1).$$

Let us set

$$W(U^{(0)}) = W_1(\varphi).$$

We then have

$$\frac{dV_1}{dt} = W_1(1 + V_1).$$

We denote by A_φ the set-theoretical sum of the φ_λ :

$$A_\varphi = \sum_{0 < \lambda < 1} \varphi_\lambda.$$

THEOREM 92. *In order for the trivial motion $\varphi = 0$ of a dynamical system, defined by the system of equations (5.4), to be asymptotically stable, it is necessary and sufficient that the trivial solution of the system (5.5) be asymptotically stable. In this connection, the region of asymptotic stability A_φ of the motion $\varphi = 0$ of the system*

$$U = U(t, \varphi(x_i + tb_i))$$

consists of those functions φ , the closure of the set of values of which falls in A — the region of asymptotic stability of the trivial solution of the system (5.5). And, conversely, the region A consists of those and only those points $U^{(0)}$ which are values of the functions $\varphi \in A_\varphi$.

PROOF. *Necessity.* Let the trivial motion be asymptotically stable. We note that $E_n \subset \Phi$, and therefore $U(t, U^{(0)}) \rightarrow 0$ as $t \rightarrow +\infty$ for all sufficiently small $|U^{(0)}|$. [The vector $(\varphi_1, \dots, \varphi_n) \in E_n$ if $\varphi_i = \text{constant}$ for $X \in E_k$.] Here we have set $\varphi(x_1, \dots, x_k) = U^{(0)}$. From this it follows that the trivial solution of the system (5.5) is asymptotically stable. We denote by A the region of asymptotic stability of the trivial solution of the system (5.5), and let $\varphi \in A_\varphi$. We assume that there exists a point $\xi_1, \dots, \xi_k$, such that $\varphi(\xi_1, \dots, \xi_k) = U^{(0)}$ and $U^{(0)} \in A$. We then have for $x_i = \xi_i - tb_i$ that

$$U(t, U^{(0)}) = U(t, \varphi(x_i + tb_i)) \rightarrow 0 \text{ as } t \rightarrow +\infty.$$

Consequently, $\varphi \in A_\varphi$. The converse is obvious.

Sufficiency. Let the trivial solution of the system (5.5) be asymptotically stable. Then, as was shown, there exists a set $A_\varphi = \sum_{0 < \lambda < 1} \varphi_\lambda$, which has the following properties:

- 1) A_φ is an open set;
- 2) if $\varphi \in A_\varphi$, then $U(t, \varphi(x_i + tb_i)) \in A_\varphi$ when $t \in (-\infty, +\infty)$ and, in addition, there are defined in the set A_φ two functionals $V_1(\varphi)$ and $W_1(\varphi)$ such that:
 - a) $-1 < V_1(\varphi) < 0$ when $\varphi \in A_\varphi$. For any $\alpha > 0$ one can find a $\beta > 0$ such that when $\|\varphi\| > \alpha$ we have $\|W_1\| > \beta$;

$$\text{b) } \frac{dV_1}{dt} = W_1(1 + V_1).$$

From this we can establish, on the basis of Theorem 88, that the set A_φ is the region of asymptotic stability of the trivial motion $\varphi = 0$.

Let us consider the system of partial differential equations

$$\frac{\partial u_s}{\partial t} = U_s^{(\mu)} + \sum_{i=1}^k b_i \frac{\partial u_s}{\partial x_i}, \quad (5.6)$$

where the $U_s^{(\mu)}$ are homogeneous functions of order $\mu \geq 1$ in the variables $u_1, \dots, u_n$ ($\mu = p/q$, q odd, $p \neq 2k$). As follows from Theorem 3 of Chap. II, a necessary and sufficient condition for the trivial solution of the system

$$\frac{du_s}{dt} = U_s^{(\mu)} \quad (5.7)$$

to be asymptotically stable is the existence of two positive-homogeneous functions $V(u_1, \dots, u_n)$ of order $m - \mu + 1$ and $W(u_1, \dots, u_n)$ of order m , where $W < 0$ when $U \neq 0$, whereas $V > 0$ when $U \neq 0$, and $dV/dt = W$. We denote by Σ_λ the surface $V = \lambda$, by Ψ_λ the set of functions φ such that all the values of the functions φ fall in the region G_λ , bounded by the surface Σ_λ such that there exists at least one value of the functions

$$\varphi(\xi_1, \dots, \xi_n) = U^{(0)}$$

such that

$$U^{(0)} \in \Sigma_\lambda.$$

We set

$$V_1(\varphi) = \lambda \text{ for } \varphi \in \Psi_\lambda, \quad V_1(\varphi) = V(U^{(0)}).$$

Then, as above, we have

$$V(U(t, U^{(0)})) = V_1(U(t, \varphi(x_i + tb_i))) \quad (i = 1, \dots, k);$$

hence

$$\frac{dV_1}{dt} = W(U(t, U^{(0)})) = W_1(U(t, \varphi)).$$

It is easy to see that the functionals V_1 and W_1 satisfy the following inequalities:

$$\begin{aligned} a_1 \|\varphi\|^{m-\mu+1} &\leq V_1(\varphi) \leq a_2 \|\varphi\|^{m-\mu+1}, \\ -b_1 \|U(t, \varphi)\|^m &\leq W_1(U(t, \varphi)) \leq -b_2 \|U(t, \varphi)\|^m. \end{aligned} \quad (5.8)$$

Applying Theorem 20, we obtain the following inequalities for $\mu > 1$:

$$\frac{c_1 \|\varphi\|}{\sqrt[\mu-1]{1 + c_2 t \|\varphi\|^{\mu-1}}} \leq \|U(t, \varphi)\| \leq \frac{d_1 \|\varphi\|}{\sqrt[\mu-1]{1 + d_2 t \|\varphi\|^{\mu-1}}}$$

when $t > 0$, where c_i and d_i are positive constants, connected in a fixed manner with the positive constants a_i and b_i ($i = 1, 2$). If $\mu = 1$, then the estimate will have an exponential form:

$$p_1 \|\varphi\| e^{-p_2 t} \leq \|U(t, \varphi)\| = q_1 \|\varphi\| e^{-q_2 t}, \quad p_i, q_i > 0.$$

3. Stability of the trivial solution of a system of partial differential equations

In the present section we apply the theory developed in Chapter IV to an investigation of systems of linear partial differential equations. We consider the system of linear partial differential equations

$$\frac{\partial u_s}{\partial t} = \sum_{i=1}^n \sum_{(\alpha_i)} a_{si}^{(\alpha_1, \dots, \alpha_n)}(t) \frac{\partial^{a_1 + \dots + a_n} u_i}{\partial x_1^{\alpha_1} \dots \partial x_k^{\alpha_k}} \quad (s = 1, \dots, n). \quad (5.9)$$

The summation is over all $\sum_{j=1}^k \alpha_j$ from 0 to m_1 .

Let us assume that the coefficients $a_{si}^{(\alpha_1, \dots, \alpha_n)}(t)$ are real functions of the variable t , defined for $t \geq 0$ and continuous there. This system of equations can be written in a more compact form in the following manner. Let us put $U = (u_1, \dots, u_n)$ and $s_j = \partial / \partial x_j$, so that $s_j^{(0)} = 1$. We then have

$$\frac{\partial U}{\partial t} = \sum_{\substack{k \\ \sum_{j=1}^k \alpha_j = 0}}^{m_1} A_{\alpha_1, \dots, \alpha_n} s_1^{\alpha_1} \dots s_k^{\alpha_k} U,$$

where the $A_{\alpha_1, \dots, \alpha_n}$ represent square matrices with the elements $a_{si}^{(\alpha_1, \dots, \alpha_n)}$.

Let us consider a linear system of vector functions $\varphi = (\varphi_1, \dots, \varphi_n)$, where $\varphi_i(x_1, \dots, x_k)$ is a function defined in E_k and having there all the generalized derivatives up to the order m_2 inclusive and square-summable in E_k . This linear system of vector functions will be henceforth denoted by $W_2^{(m_2)}$. In this

linear system we can introduce in some manner a norm such that the linear system $W_2^{(m_2)}$ becomes a normed linear functional space [3].

We consider also a linear system of vector functions

$$\Psi = (\psi_1, \dots, \psi_n),$$

where $\psi_i(x_1, \dots, x_k)$ is defined in E_k , and is continuously differentiable there with respect to all its arguments up to order m_3 inclusive and differs from zero only within a certain finite region G_ψ . This linear system will be denoted by C_{m_3} . It is clear that when $m_3 \geq m_2$ we have

$$C_{m_3} \subset W_2^{(m_2)}.$$

In the space $W_2^{(m_2)}$, we define a family of functionals of the form

$$V_t(\varphi) = \int_{E_k} \sum_{\beta_1 + \dots + \beta_k = 0}^N V_{\beta_1, \dots, \beta_k} d\Omega, \quad (5.10)$$

where $V_{\beta_1, \dots, \beta_k}$ is a quadratic form in the quantities

$$s_1^{\beta_1} \dots s_k^{\beta_k} u_t = y_t$$

so that

$$V_{\beta_1, \dots, \beta_k} = Y^* C_{\beta_1, \dots, \beta_k} Y, \quad (5.11)$$

where $C_{\beta_1, \dots, \beta_k}$ is a symmetric square matrix, the elements of which are continuously differentiable functions of the argument t , defined for $t \geq 0$. Let us find a solution of the system (5.9) $U = U(\psi, t, t_0)$ such that $U(\psi, t, t_0) = \psi$ when $t = t_0$.

Let us calculate on this solution the family of functionals chosen by us, and let us then find the derivative with respect to t of the resultant function in the variable t . We then have

$$\begin{aligned} \frac{d}{dt} V_t(U(\psi, t, t_0)) &= \int_{E_k} \sum_{\beta_1 + \dots + \beta_k = 0}^N W_{\beta_1, \dots, \beta_k} d\Omega + \\ &+ \int_{E_k} \sum_{\substack{k \\ \sum_{i=1}^k \alpha_i > 0}}^{m_1} \sum_{\substack{k \\ \sum_{j=1}^k \beta_j = 0}}^N [W_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^{(1)} + W_{\beta_1, \dots, \beta_k, \alpha_1, \dots, \alpha_k}^{(2)}] d\Omega, \end{aligned} \quad (5.12)$$

where $W_{\beta_1, \dots, \beta_k}$ is a quadratic form in the expressions

$$s_1^{\beta_1} \dots s_k^{\beta_k} u_t = y_t$$

so that

$$W_{\beta_1, \dots, \beta_k} = Y^* D_{\beta_1, \dots, \beta_k} Y,$$

where $D_{\beta_1, \dots, \beta_k}$ is a square symmetric matrix, expressed by the equality

$$D_{\beta_1, \dots, \beta_k} = A^* C_{\beta_1, \dots, \beta_k} + C_{\beta_1, \dots, \beta_k} A + \frac{d}{dt} C_{\beta_1, \dots, \beta_k}. \quad (5.13)$$

Here $A = A_{\alpha_1, \dots, \alpha_k}$ when $\sum_{j=1}^k \alpha_j = 0$.

The functions

$$W_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^{(1)}$$

represent bilinear forms in the quantities

$$s_1^{\alpha_1 + \beta_1} s_2^{\alpha_2 + \beta_2} \dots s_k^{\alpha_k + \beta_k} u_t = y_t \text{ and } s_1^{\beta_1} \dots s_k^{\beta_k} u_t = z_t,$$

i.e.

$$W_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^{(1)} = Y^* B_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} Z, \quad (5.14)$$

where

$$B_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} = A_{\alpha_1, \dots, \alpha_k} C_{\beta_1, \dots, \beta_k}.$$

The functions $W_{\beta_1, \dots, \beta_k, \alpha_1, \dots, \alpha_k}^{(2)}$ are bilinear forms of the same quantities, i.e.

$$W_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^{(2)} = Z^* B_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^* Y. \quad (5.15)$$

Eqs. (5.14) and (5.15) lead to the equality

$$W_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}^{(1)} = W_{\beta_1, \dots, \beta_k, \alpha_1, \dots, \alpha_k}^{(2)}.$$

We thus have

$$\frac{dV_t}{dt} = W_t + W_t^{(1)},$$

where W_t is the family of functionals, defined by the equality

$$W_t = \int_{E_k} \sum_{\beta_1 + \dots + \beta_k = 0}^N W_{\beta_1, \dots, \beta_k} d\Omega. \quad (5.16)$$

The family of functionals $W_t^{(1)}$ is defined by the equation

$$W_t^{(1)} = \int_{E_k} \sum_{j=1}^k \sum_{\alpha_j=1}^k \sum_{j=1}^N \sum_{\beta_j=0}^k s_1^{\beta_1} \dots s_k^{\beta_k} U^* C_{\beta_1, \dots, \beta_k} A_{\alpha_1, \dots, \alpha_k} s_1^{\alpha_1 + \beta_1} \dots \dots s_k^{\alpha_k + \beta_k} U d\Omega. \quad (5.17)$$

The family of functionals $W_t^{(1)}$ contains derivatives of the functions u_t up to order $(N + m_1)$ inclusive. Let us assume that the functions u_t and all their derivatives up to order $(N + m_1)$ inclusive, are, for any finite $t \geq t_0$, such that the rule for integration by parts can be applied:

$$\int_{E_k} s_1^{\beta_1} \dots s_k^{\beta_k} u_t s_1^{\alpha_1 + \beta_1} \dots s_k^{\alpha_k + \beta_k} u_j d\Omega = (-1)^{\sum_{j=1}^k \alpha_j} \times \\ \times \int_{E_k} s_1^{\alpha_1 + \beta_1} \dots s_k^{\alpha_k + \beta_k} u_t s_1^{\beta_1} \dots s_k^{\beta_k} u_j d\Omega. \quad (5.18)$$

The application of (5.18) to the family of functionals (5.17) yields

$$W_t^{(1)} = \int_{E_k} \sum_{j=1}^k \sum_{\alpha_j=1}^k \sum_{j=1}^N \sum_{\beta_j=0}^k (-1)^{\sum_{j=1}^k \alpha_j} s_1^{\alpha_1 + \beta_1} \dots s_k^{\alpha_k + \beta_k} U^* C_{\beta_1, \dots, \beta_k} \times \\ \times A_{\alpha_1, \dots, \alpha_k} s_1^{\beta_1} \dots s_k^{\beta_k} U d\Omega = \int_{E_k} \sum_{j=1}^k \sum_{\alpha_j=1}^k \sum_{j=1}^N \sum_{\beta_j=0}^k (-1)^{\sum_{j=1}^k \alpha_j} \times \\ \times s_1^{\beta_1} \dots s_k^{\beta_k} U^* A_{\alpha_1, \dots, \alpha_k}^* C_{\beta_1, \dots, \beta_k} s_1^{\alpha_1 + \beta_1} \dots s_k^{\alpha_k + \beta_k} U d\Omega. \quad (5.19)$$

From (5.19) and (5.17), we obtain

$$\frac{dV_t(U(\psi, t, t_0))}{dt} = W_t + \int_{E_k} \sum_{j=1}^k \sum_{\alpha_j=1}^k \sum_{j=1}^N \sum_{\beta_j=0}^k s_1^{\beta_1} \dots \dots s_k^{\beta_k} U^* G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} s_1^{\alpha_1 + \beta_1} \dots s_k^{\alpha_k + \beta_k} U d\Omega, \quad (5.20)$$

where

$$G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} = \\ = \frac{(-1)^{\sum_{j=1}^k \alpha_j} A_{\alpha_1, \dots, \alpha_k}^* C_{\beta_1, \dots, \beta_k} + C_{\beta_1, \dots, \beta_k} A_{\alpha_1, \dots, \alpha_k}}{2}. \quad (5.21)$$

If the matrices $C_{\beta_1, \dots, \beta_k}$ are chosen such that

$$G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} = 0, \quad (\text{V})$$

then the derivative of the family of functionals V_t , calculated on the solution $U(\psi, t, t_0)$ of system (5.9), will have the form

$$\frac{dV_t}{dt} = W_t. \quad (5.22)$$

Let us consider the system of ordinary differential equations

$$\frac{dU}{dt} = AU. \quad (5.23)$$

Let $X(t, t_0)$ be the matrix of the fundamental system of solutions of the system (5.23), such that $X(t, t_0) = E$ when $t = t_0$.

It is easy to establish that the vector function $U = X(t, t_0)U^{(0)}$, $U^{(0)} \in E_n$ satisfies the system (5.9). Therefore, for the asymptotic stability of the trivial solution of the system (5.9) in terms of some sort of norm it is necessary that the trivial solution of the system (5.23) also be asymptotically stable (the same statement pertains also to the case of stability and instability).

Let us assume that the trivial solution of the system (5.23) is asymptotically stable, and with this there exist two positive functions $\varphi_1(t)$ and $\psi_1(t)$, defined for $t \geq 0$ and continuous there, such that the following inequality holds:

$$\begin{aligned} p_1 |U^{(0)}| \varphi_1(t)^{-1} \varphi_1(t_0) e^{-p_1 \int_{t_0}^t \varphi_1(\tau)^{-1} d\tau} &\leq |U(t, t_0, U^{(0)})| \leq \\ &\leq q_1 |U^{(0)}| \varphi_1(t)^{-1} \varphi_1(t_0) e^{-q_1 \int_{t_0}^t \varphi_1(\tau)^{-1} d\tau}, \end{aligned} \quad (5.24)$$

where $U = U(t, t_0, U^{(0)})$ is a solution of the system (5.23). Then, as shown in Theorem 83, there exist two quadratic forms

$$V_1 = \sum_{i,j=1}^n c_{ij}(t) x_i x_j \quad \text{and} \quad W_1 = \sum_{i,j=1}^n d_{ij}(t) x_i x_j,$$

where $c_{ij}(t)$ and $d_{ij}(t)$ are continuously differentiable functions of the argument t , defined for $t \geq 0$, and such that the following

inequalities hold:

$$a_1\varphi_1(t)|X|^2 \leq V_1 \leq a_2\varphi_1(t)|X|^2 \quad (5.25)$$

$$-b_1\psi_1(t)|X|^2 \leq W_1 \leq -b_2\psi_1(t)|X|^2 \quad (5.26)$$

and

$$\frac{dV_1}{dt} = W_1, \quad (5.27)$$

where dV_1/dt is the total derivative of the quadratic form V_1 , calculated by virtue of the system (5.23). From (5.27) we obtain

$$X^* \left[A^*C + CA + \frac{dC}{dt} \right] X = X^*DX,$$

which yields the equality

$$A^*C + CA + \frac{dC}{dt} = D \quad (5.28)$$

It follows from the foregoing discussion that for any square symmetric matrix $D = \|d_{ij}(t)\|$ such that inequalities (5.26) are satisfied, the equation (5.28) has a solution $C = \|c_{ij}\|$, with simultaneous satisfaction of the inequality (5.25), where $a_i, b_i > 0$ are constants.

We shall choose henceforth the matrices $C_{\beta_1, \dots, \beta_n}$ such that they satisfy the equation (5.28), where $D = D_{\beta_1, \dots, \beta_n}$ is a matrix satisfying inequalities (5.26). We shall also choose the matrices $C_{\beta_1, \dots, \beta_n}$ in such a way as to satisfy if possible the relations (V). Furthermore, we set

$$R = \int \sum_{i=1}^n \sum_{\substack{k \\ \sum_{j=1}^k \beta_j = 0}}^N (s_1^{\beta_1} \dots s_k^{\beta_k} u_i)^2 d\Omega,$$

where R is the functional referred to in Theorems 80 and 81.

From Eq. (5.22), we have

$$\varphi_1(t)c_1R(U(\psi, t, t_0)) \leq V(U(\psi, t, t_0)) \leq \varphi_1(t)c_2R(U(\psi, t, t_0)) \quad (5.29)$$

$$\begin{aligned} -d_1\psi_1(t)R(U(\psi, t, t_0)) &\leq W(U(\psi, t, t_0)) \leq \\ &\leq -d_2\psi_1(t)R(U(\psi, t, t_0)), \end{aligned} \quad (5.30)$$

from which we obtain

$$\begin{aligned} p_1 R(\psi) \varphi_1(t)^{-1} \varphi_1(t_0) e^{-p_2 \int_{t_0}^t \varphi_1(\tau) \varphi_1(\tau)^{-1} d\tau} &\leq R(U(\psi, t, t_0)) \leq \\ &\leq q_1 R(\psi) \varphi_1(t)^{-1} \varphi_1(t_0) e^{-q_2 \int_{t_0}^t \varphi_1(\tau) \varphi_1(\tau)^{-1} d\tau} \end{aligned} \quad (5.31)$$

for $t \geq t_0$ (see Theorems 80 and 81), where c_i , d_i , p_i and q_i ($i = 1, 2$) are certain positive constants interrelated in a definite manner.

It is easy to see that we can set [31]

$$\|U\|_{W_2^{(m_2)}} = \sqrt{R(U)},$$

where U is any function from $W_2^{(m_2)}$, $m_2 \geq N$. It follows from this that inequalities (5.31) show the presence of asymptotic stability of the trivial solution of the system (5.9) with norm $W_2^{(N)}$. This yields an estimate of the rate of decrease of the norm of any solution. As follows from Sobolev's embedding theorem, when $k < 2N$ it is possible to obtain from inequalities (5.31) an estimate of the rate of decrease of

$$\|U(\psi, t, t_0)\|_C \text{ for } t \geq t_0,$$

$$\|U\|_C = \max_{(x_1, \dots, x_n) \in E_k} |U|.$$

In this investigation we have made two important assumptions: one pertains to the existence of a solution of the system (5.9), $U(\psi, t, t_0)$ for $\psi \in C_{m_1}$, and the other is that the integration by parts rule (5.18) is applicable. We note that both these requirements are satisfied under certain assumptions regarding the coefficients of the system (5.9) [32], namely for a sufficiently large value of m_2 it is possible to guarantee the existence of a solution $U(\psi, t, t_0)$, having all derivatives up to order $(N + m_1)$ inclusive. This satisfies formula (5.18). From the foregoing results we obtain the following:

THEOREM 93. *If the trivial solution of the system (5.23) is asymptotically stable and at the same time inequalities (5.24) are satisfied, then, subject to satisfaction of conditions (V), the trivial solution of the system (5.9) is asymptotically stable in the sense of the norm in*

$W_2^{(N)}$, and with this inequalities (5.31) are satisfied. Furthermore, the quantity m_2 can be chosen so large that together with (5.31) the following inequality will be satisfied:

$$\|U(\psi, t, t_0)\|_C \leq q_1 \|\psi\|_{W_2^{(N)}} \varphi_1(t)^{-1} \varphi_1(t_0) e^{-\alpha_2 \int_{t_0}^t \varphi_1(\tau) \varphi_1^{-1}(\tau) d\tau}$$

for $t > t_0$, where $\psi \in C_{m_2}$. It is assumed here that the conditions for the existence of the solutions, formulated in [32], are satisfied.

It should be noted here that the stability of the solutions for linear partial differential equations was also investigated in Ref. [33].

Let us show now that the conditions (V) can be replaced by more general conditions, Theorem 93 still remaining valid. Let us consider those matrices $G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}$ for which $2\alpha_i + \beta_i = 2\gamma_i$, where the γ_i are integers. We denote by $F_{\gamma_1, \dots, \gamma_k}$ the diagonal matrix, consisting of diagonal elements of the matrix $G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k}$ for which $2\alpha_i + \beta_i = 2\gamma_i$. We set

$$D_{\gamma_1, \dots, \gamma_k}^1 = D_{\gamma_1, \dots, \gamma_k} + (-1)^{\sum_{j=1}^k (\beta_j/2)} F_{\gamma_1, \dots, \gamma_k}.$$

If the matrices $D_{\gamma_1, \dots, \gamma_k}^1$ satisfy the condition (5.28) and

$$G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} = 0 \text{ for } 2\alpha_i + \beta_i \neq 2\gamma_i,$$

and, for at least one value $i = 1, \dots, k$,

$$G_{\alpha_1, \dots, \alpha_k, \beta_1, \dots, \beta_k} = F_{\gamma_1, \dots, \gamma_k} \text{ for } 2\alpha_i + \beta_i = 2\gamma_i \quad (i = 1, \dots, k),$$

then the conditions of Theorem 93 remain valid. In this connection, as follows from [32], upon the satisfaction of these conditions there exists a solution of the system (5.9) $U = U(\psi, t, t_0)$, $\psi \in C_{m_2}$, satisfying the conditions (5.18).

General systems serve also as an abstract model of the mixed problem for systems of partial differential equations. Therefore, the theory of stability of invariant sets, developed in Chap. IV, can be applied to the investigation of the stability of solutions of the mixed problem. Let us illustrate this with several simple examples.

Let us consider the equation

$$\frac{\partial U}{\partial t} = \sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(a_{ij}(x_1, \dots, x_n, t) \frac{\partial U}{\partial x_j} \right) + \sum_{i=1}^n a_i \frac{\partial U}{\partial x_i} + a(X, t)U, \quad (5.32)$$

where the functions a_{ij} and a are defined for $X \in D$, $t \in [0, +\infty)$, the a_i are real constants, and for all the values of the arguments $X \in D$ the condition

$$\sum_{i,j=1}^n a_{ij}(X, t) \xi_i \xi_j \geq \alpha(t) \sum_{i=1}^n \xi_i^2, \quad (5.33)$$

is satisfied, where $\alpha(t)$ is a positive continuous function, defined for $t \geq 0$. We assume that Equation (5.32) has a solution $U(X, t)$, having the properties $U(X, t) = \varphi(X)$ when $t = t_0 \geq 0$, $U(X, t) = 0$ when $X \in S$, where S is the boundary of the region $\Omega \subset D$, and $\varphi(X)$ is an element of a certain space Φ_Ω . Moreover, shall assume that $U(X, t) \in L_2(\Omega)$ when $t \geq t_0$ and $\partial U(X, t)/\partial t$ exists in the sense of $L_2(\Omega)$.

Let us take the functional $V = \int_\Omega U^2 d\Omega$. It is clear that this functional is defined in $L_2(\Omega)$. We calculate the total derivative of this functional by virtue of Eq. (5.32), and obtain

$$\frac{dV}{dt} = 2 \int_\Omega U \left[\sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(a_{ij}(X, t) \frac{\partial U}{\partial x_j} \right) + \sum_{i=1}^n a_i \frac{\partial U}{\partial x_i} + a(X, t)U \right] d\Omega.$$

Integrating by parts, we obtain

$$\frac{dV}{dt} = -2 \int_\Omega \left[\sum_{i,j=1}^n a_{ij}(X, t) \frac{\partial U}{\partial x_i} \frac{\partial U}{\partial x_j} - aU^2 \right] d\Omega.$$

Applying inequality (5.33), we obtain

$$\frac{dV}{dt} \leq -2\alpha(t) \int_\Omega \left[\sum_{i=1}^n \left(\frac{\partial U}{\partial x_i} \right)^2 \right] d\Omega + \int_\Omega aU^2 d\Omega.$$

If the function $a(X, t)$ is nonpositive, then $dV/dt < 0$ and the

trivial solution of the Equation (5.32) is stable. Using the well known inequality

$$c_1 \int_{\Omega} U^2 d\Omega \leq \int_{\Omega} \sum_{i=1}^n \left(\frac{\partial U}{\partial x_i} \right)^2 d\Omega,$$

we obtain

$$\frac{dV}{dt} \leq -2\alpha(t) \int_{\Omega} b(X, t) U^2 d\Omega,$$

where

$$b(X, t) = c_1 - \alpha^{-1}(t) a(X, t).$$

When $b(X, t) > c_2 > 0$, the trivial solution of the system (5.32) will be asymptotically stable and will satisfy an estimate of the form

$$\|U\|_{L_2(\Omega)}^2 \leq \|\varphi\|_{L_2(\Omega)}^2 \cdot e^{-2c_2 \int_0^t \alpha(\tau) d\tau},$$

where

$$\int_0^t \alpha(\tau) d\tau \rightarrow +\infty \text{ as } t \rightarrow +\infty.$$

Let us turn now to systems of equations. It was shown in Ref. [35] that the mixed problem in the region $\Omega \times [0 \leq t < +\infty)$, for the strongly parabolic system

$$\frac{\partial U}{\partial t} = -L\left(x, \frac{\partial}{\partial x}\right) U,$$

is solvable if U and all its derivatives over all possible combinations of x up to $(m-1)$ -th order inclusive vanish on the boundary of the region Ω for any $t \in [0, +\infty)$ and $U(X, t)|_{t=0} = \varphi(X)$, $\varphi(X)$ is a vector function such that $\varphi(X) \in L_2(X)$.

It was shown in Ref. [36] that the trivial solution of this system is asymptotically stable in $L_2(\Omega)$. Here the basis was taken to be the functional

$$V = \int_{\Omega} \sum_{i=1}^n u_i^2 d\Omega$$

and the condition

$$(U, LU) \geq \alpha(U, U); \quad \alpha = \text{constant} > 0,$$

where

$$(U, U) = \int_{\Omega} \sum_{i=1}^n u_i^2 d\Omega$$

was used. We note further that a certain progress in the problem of the stability of solutions of the mixed problem for partial differential equations was made by D. M. Volkov [37].

His results, like the examples given above, can also be obtained from the point of view of the general theory treated in Chap. IV.

In conclusion we note that the theory developed in Chap. IV is of great orienting significance, but cannot serve as a universal means for solving all possible problems of this kind.

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INDEX

- Abehan group 13
- Additive group 12
- α -limit point 21
- Analytic representation of O -curves 179
 - right members 87
- Asymptotically stable direction 189
 - — set 22, 200, 238
 - — solution 2, 153, 236
- Banach space 13
- Barbashin 72, 160
- Boundary 10, 27
- Bounded set 12
- Closed set 10
- Closure of a set 10
- Compact in itself 12
- Compactness 12
- Complete space 10
- Continuity of a functional 10
- Continuous operator 10
- Convergent in itself sequence 10
- Direction 188
- Distance 9, 11
 - from a motion to a set 61
- Domain 10
 - of definition 10
- Doubtful cases 6
- Dynamical systems 9, 14, 16, 17, 20
- ϵ -net 12
- Functionals 10
- General systems 197, 199
- Hausdorff theorem 12
- Homogeneous right members 104
- Incomplete general system 201
- Infinitesimally small upper limit 2
- Invariant set 9, 21, 200
- Krasovsky 72, 129, 160
- Lefschetz 34
- Limit of a sequence 9
 - point of a set 9
- Linear system 13
- Lyapunov 1
 - function 3
- Lyapunov's first method 166
 - second method 1
- Metric distance 9
 - space 9
- Motion 20, 196, 198
- Negative definite function 2
- Neighborhood of a set 11
- Non-stationary systems 152, 224
- Norm 13
- Normed linear space 12, 13
- ω -limit point 21
- O -curves 179
- Open set 9
- Operators 10
- Periodic motion 20
- Poincaré 166
- Points of a space 9
- Positive definite function 2
- Pure imaginary roots 144
- Quasi-linear systems 240
- Region of asymptotic stability 26, 153, 236
- Representation of solutions 173
- Rest point 20
- r -neighborhood 9

INDEX

- Separable metric space 12
- Stability in the sense of Lyapunov 1, 22, 152
 - of invariant sets 9, 196
 - problem for ordinary differential equations 67
 - for partial differential equations 235
- Stable set 200, 238
 - solution 235
- Stationary systems 67
- Systems of ordinary differential equations 67
 - of partial differential equations 235
- Trajectory 20, 196, ~~198~~
- Type B space 13
- Uniformly asymptotically stable
 - 52, 200
 - — — solution 153, 236
 - attracting sets 29, 52, ~~501~~
 - — — solution 153, 236, ~~236~~
- Unstable invariant set 24
- Yerugin 195

